A PRACTICAL REPORT

ON

SERVER VIRTUALIZATION ON VMWARE

SUBMITTED BY

MISS. KHAN UZMA JAMIL

UNDER THE GUIDANCE OF

PROF. MEHDI REZAEI

Submitted in fulfilment of the requirements for qualifying MSc. IT Part II Semester - III Examination 2026-2027

University of Mumbai Department of Information Technology

R.D. & S.H National College of Arts, Commerce & S.W.A. Science College Bandra (West), Mumbai – 400050

SERVER VIRTUALIZATION ON VMWARE

R. D. & S. H. National & S. W. A. Science College

Bandra (W), Mumbai - 40050

Department of Information Technology

M.Sc. (IT – SEMESTER III)

Certificate

This is to certify that Server Virtualization on VMWare Practical's performed at R.D & S.H National & S.W.A. Science College By *Miss. Khan Uzma Jamil* holding Seat No._____studying Master of Science in Information Technology Semester – III has been satisfactorily completed as prescribed by the University of Mumbai, during the year 2026 – 2027.

Subject In-Charge Coordinator In-Charge External Examiner

College Stamp

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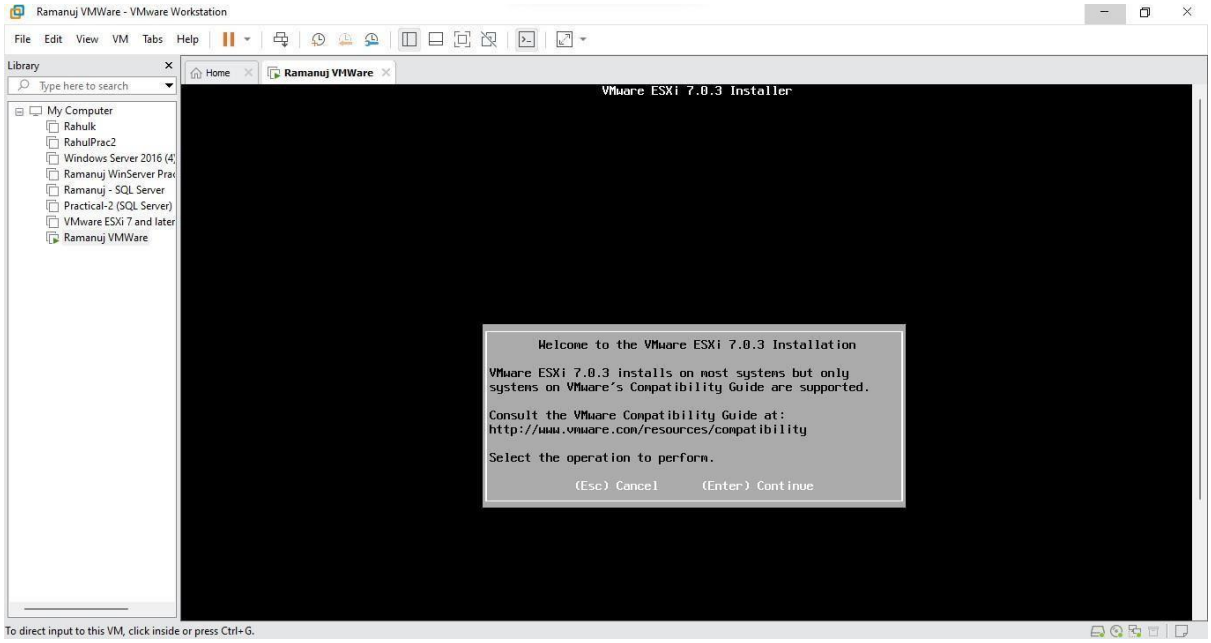
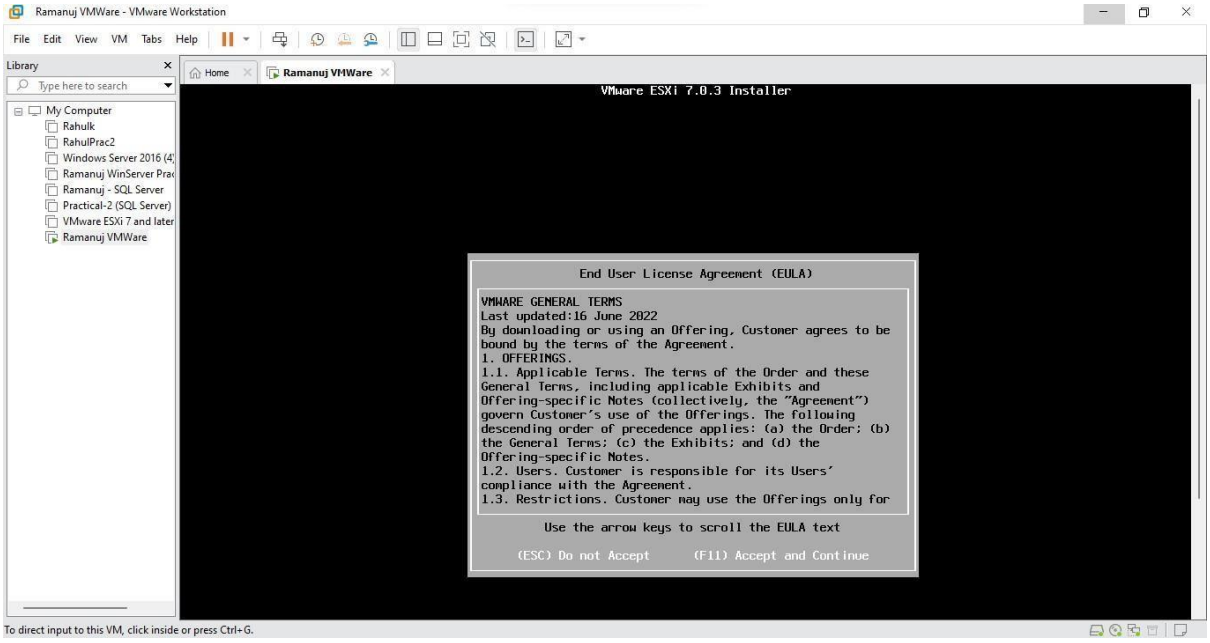
PRACTICAL 1

AIM:- Deploying and Configuring Virtual Machines

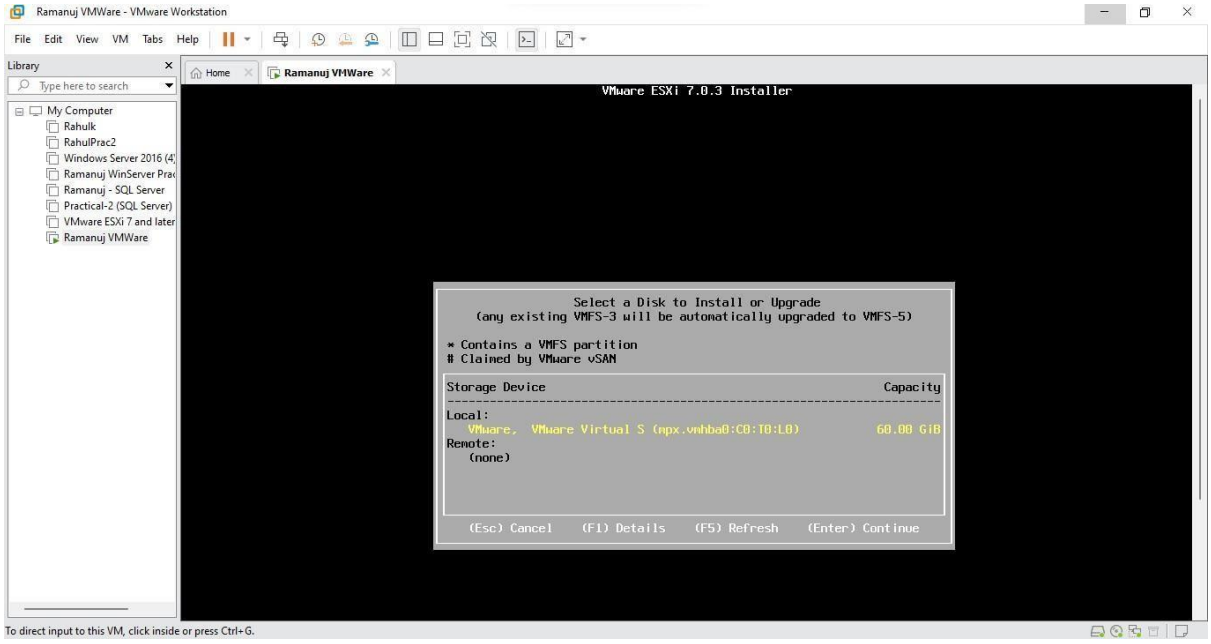
- Now it will enter the ESXi Setup, Press **Enter**



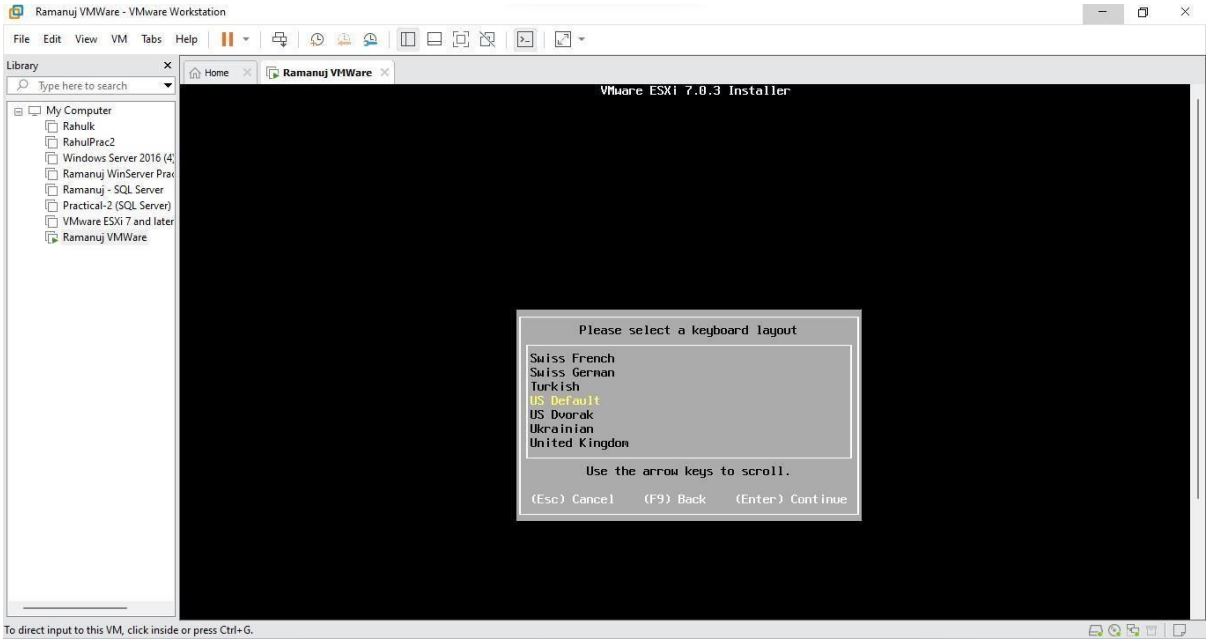
- Press **F11** to Accept and Continue



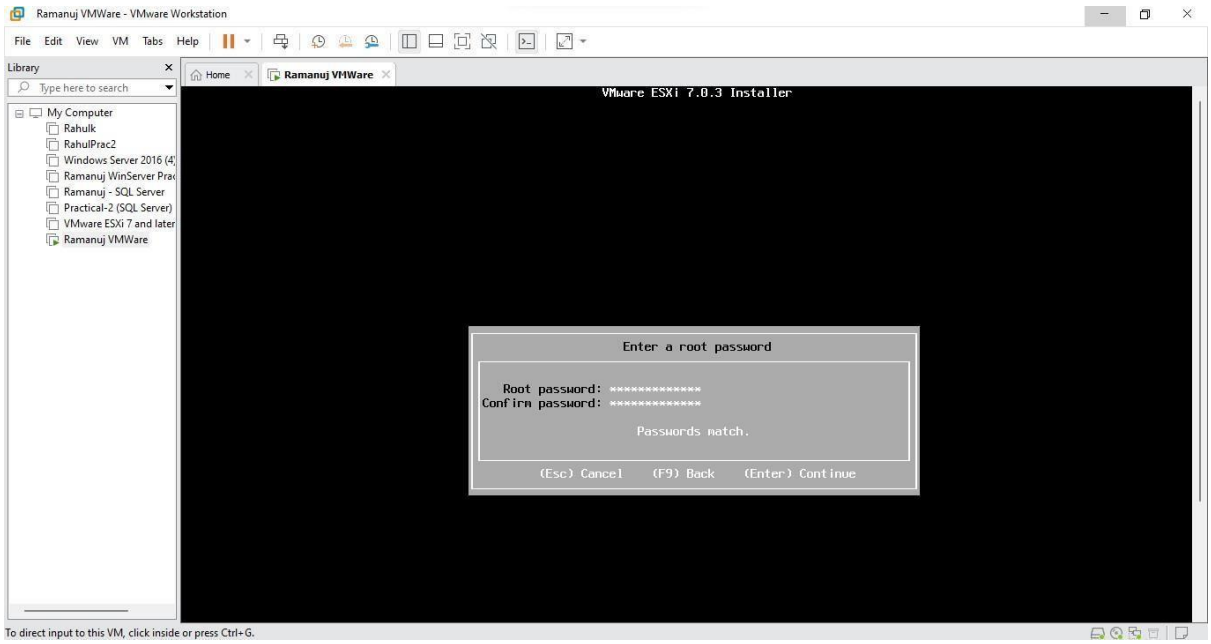
Press **Enter**



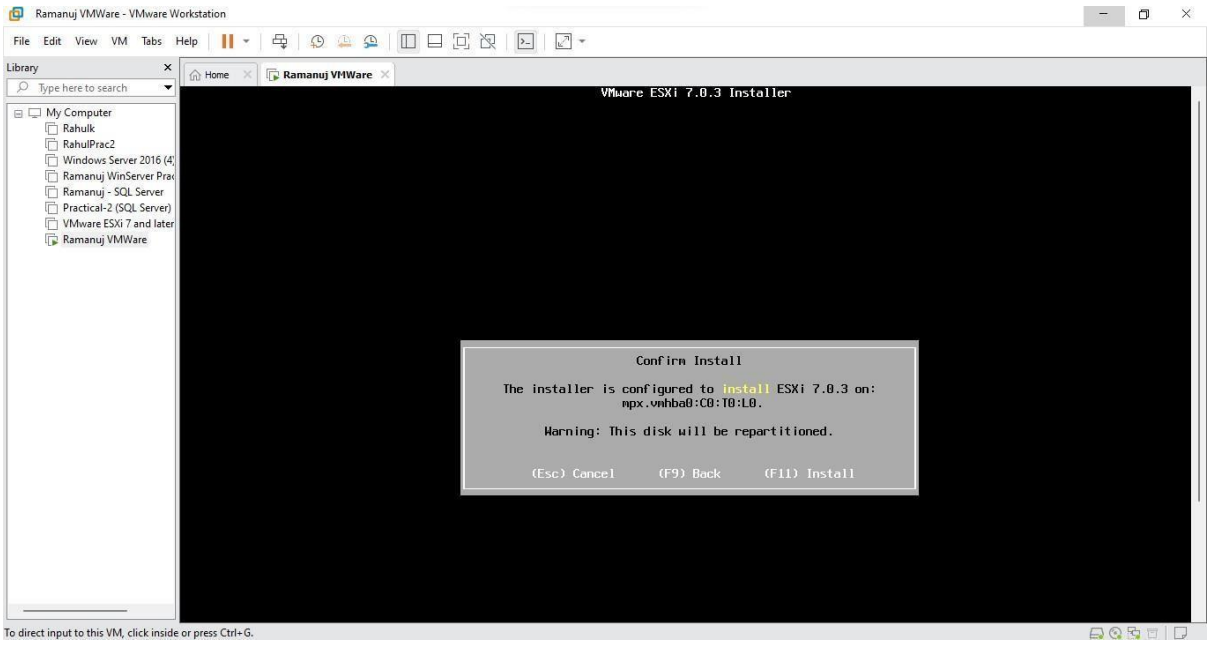
- Press **Enter**



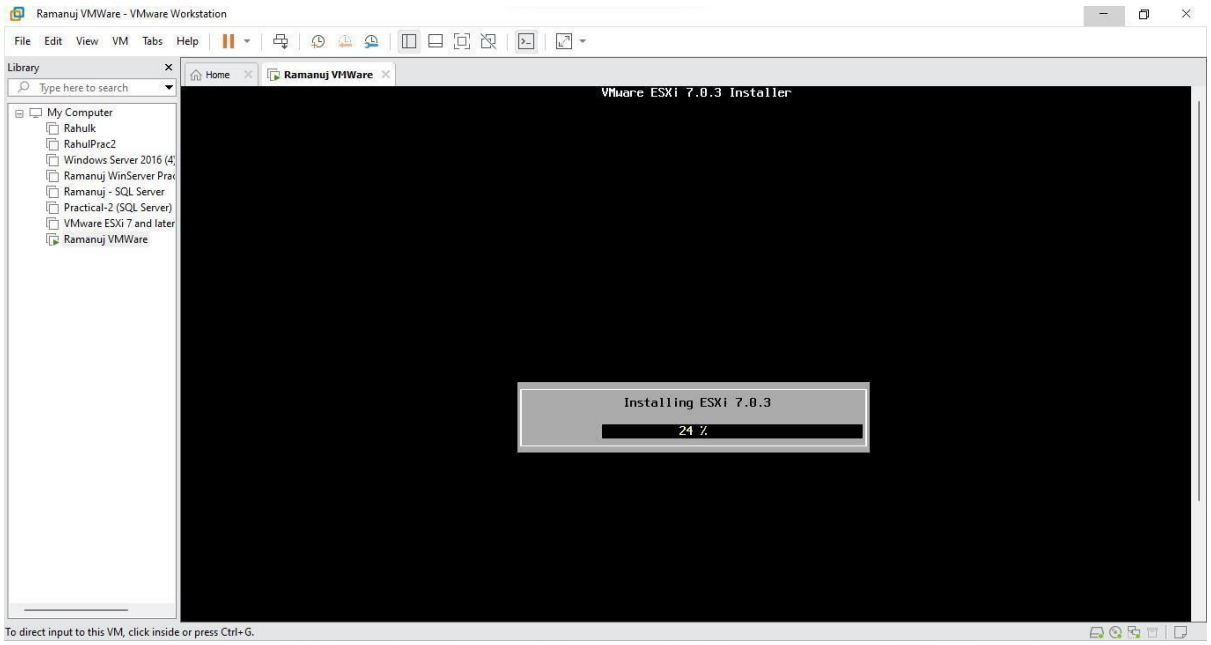
- Type in a root password and Press **Enter**



- Press **F11** to Install ESXi



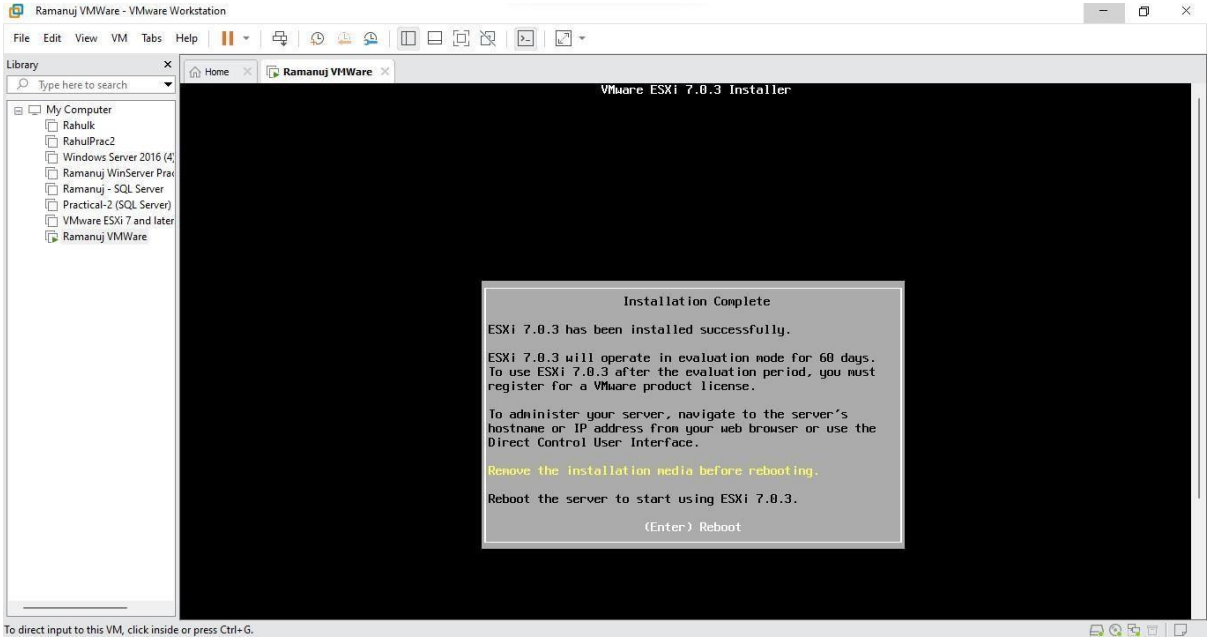
- ESXi will now install onto the Virtual Machine which will prompt a restart



- Press **Enter** to Reboot

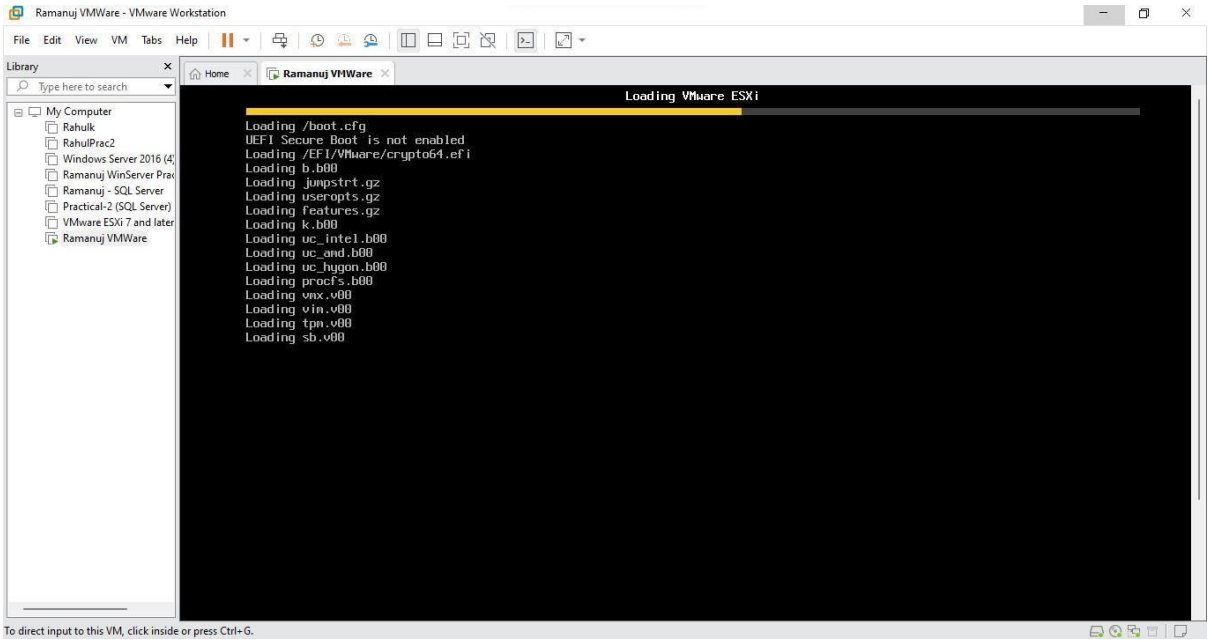
VMware ESXi has been successfully installed onto your Virtual Machine

Note: we will use the same Virtual Machine

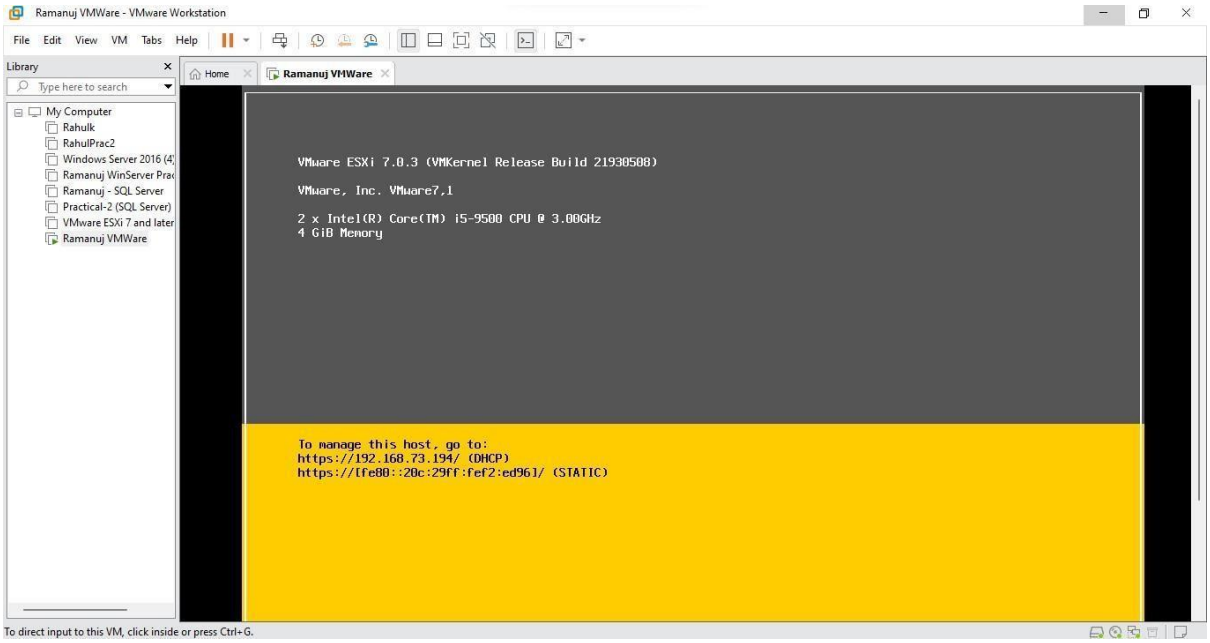


- In VMware Workstation Click on Power on this Virtual Machine

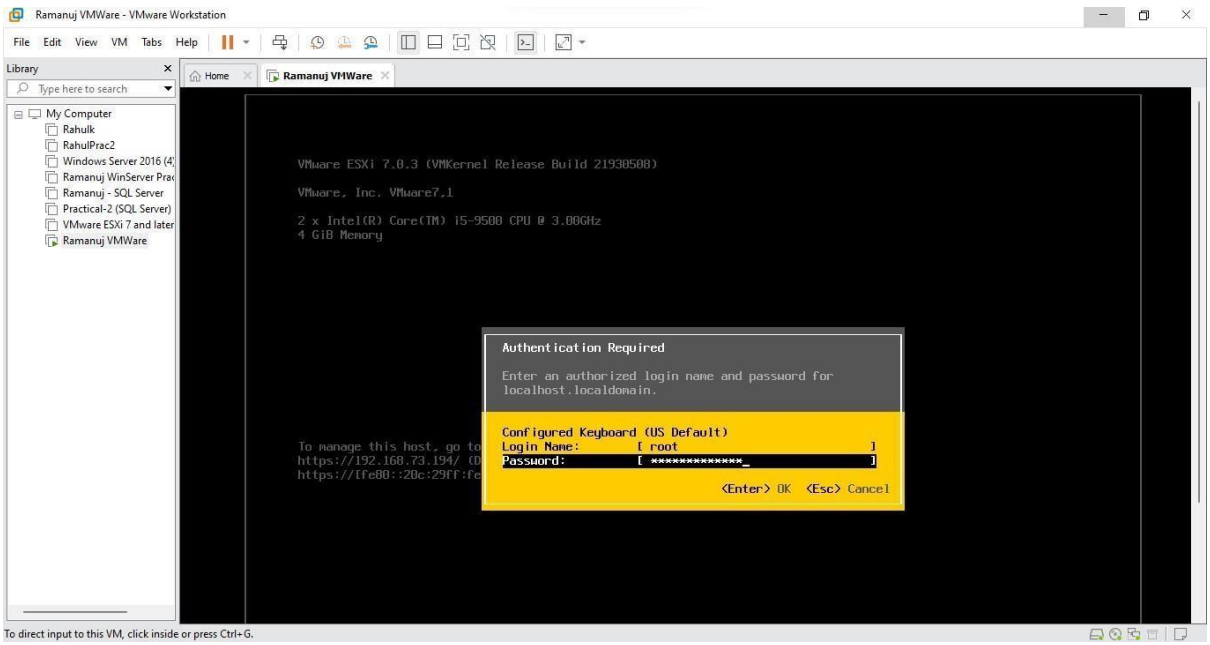
- The virtual machine is starting



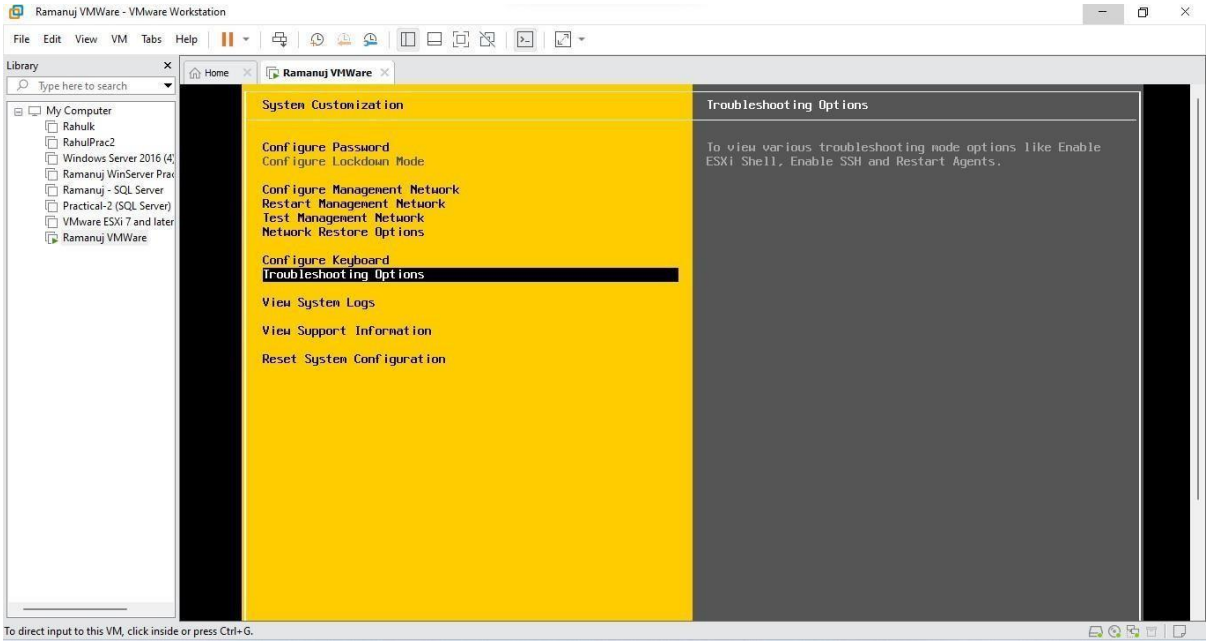
- VMware ESXi is **ON**



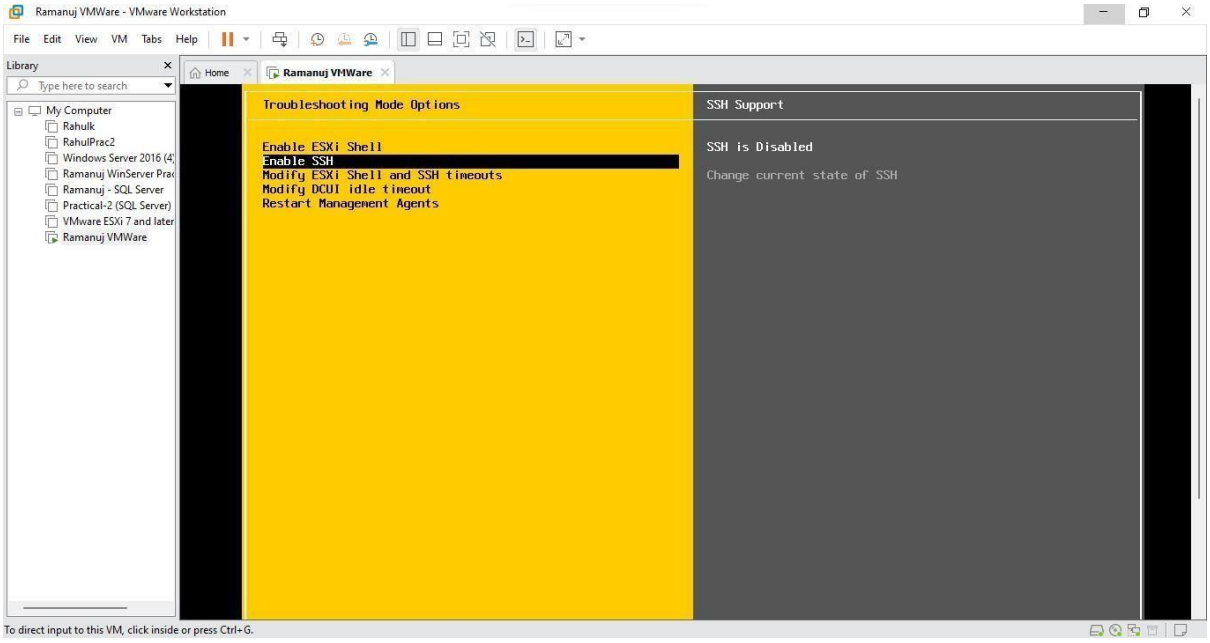
- To configure SSH press **F2** and enter your password then Press **Enter**



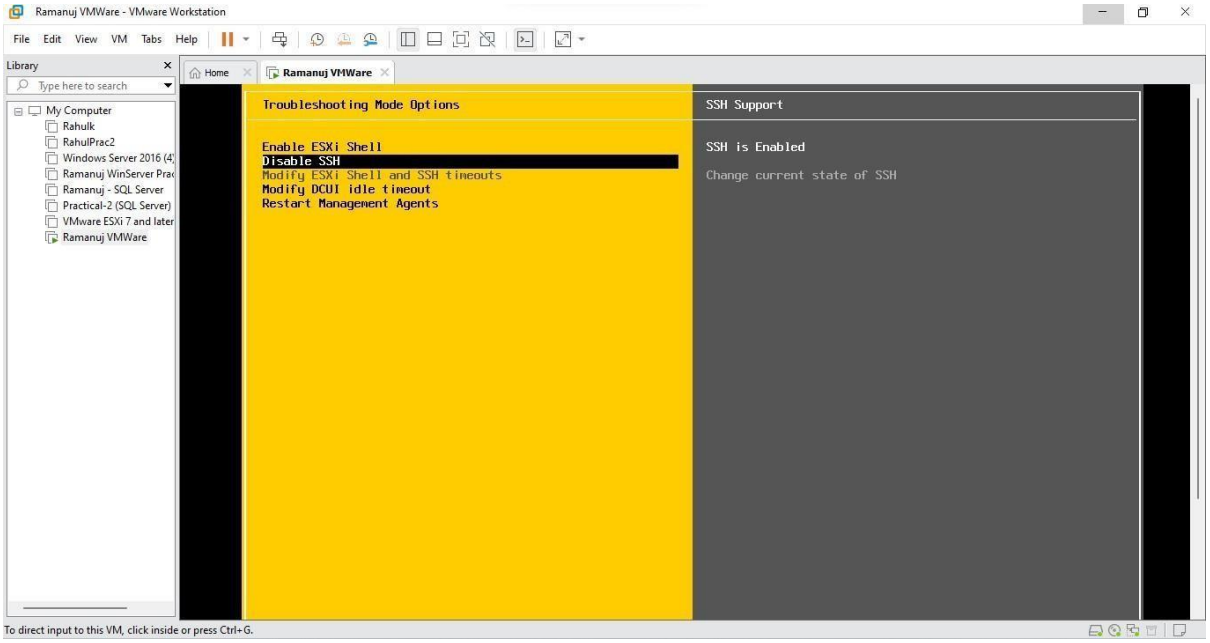
- ou will enter System Customization, within it Select **Troubleshooting Options** and Press **Enter**



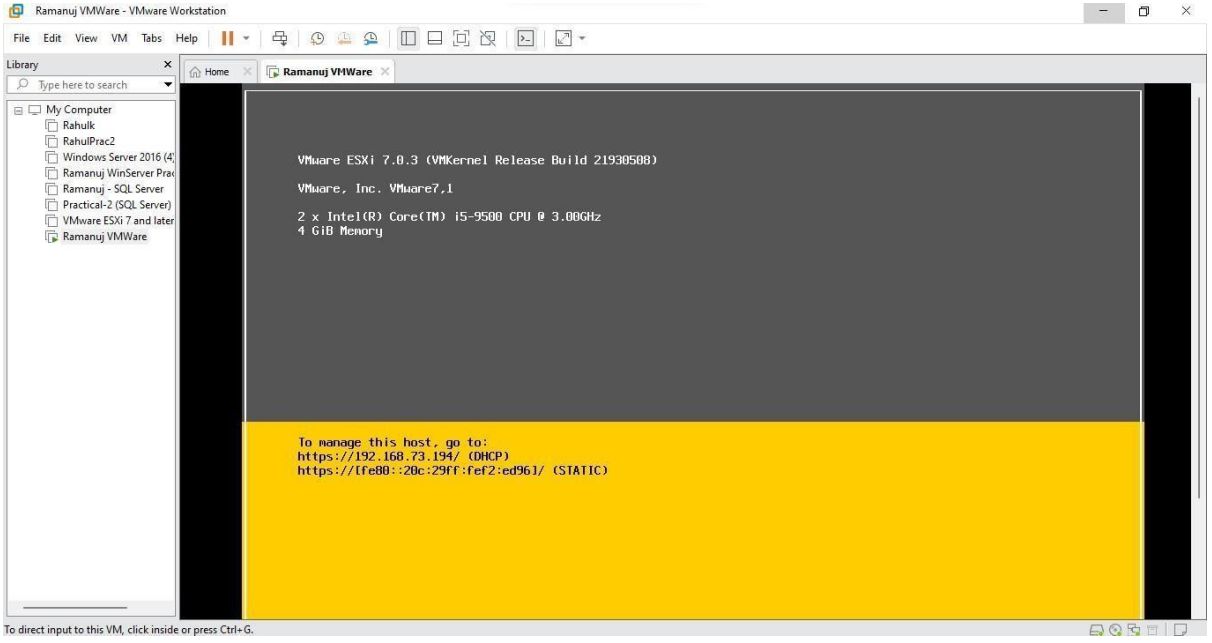
- Within Troubleshooting Options Select **Enable SSH** and Press **Enter**



- After pressing Enter you will see under SSH Support it will say " **SSH is enabled**"

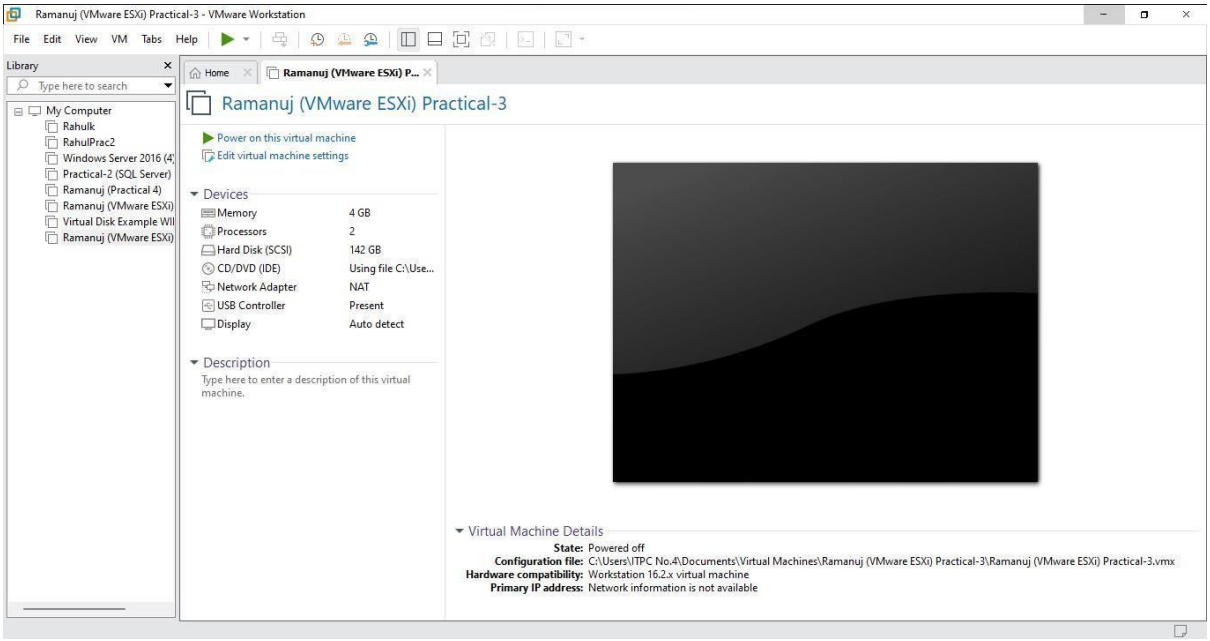


- After this press **Esc** twice to save and exit from settings

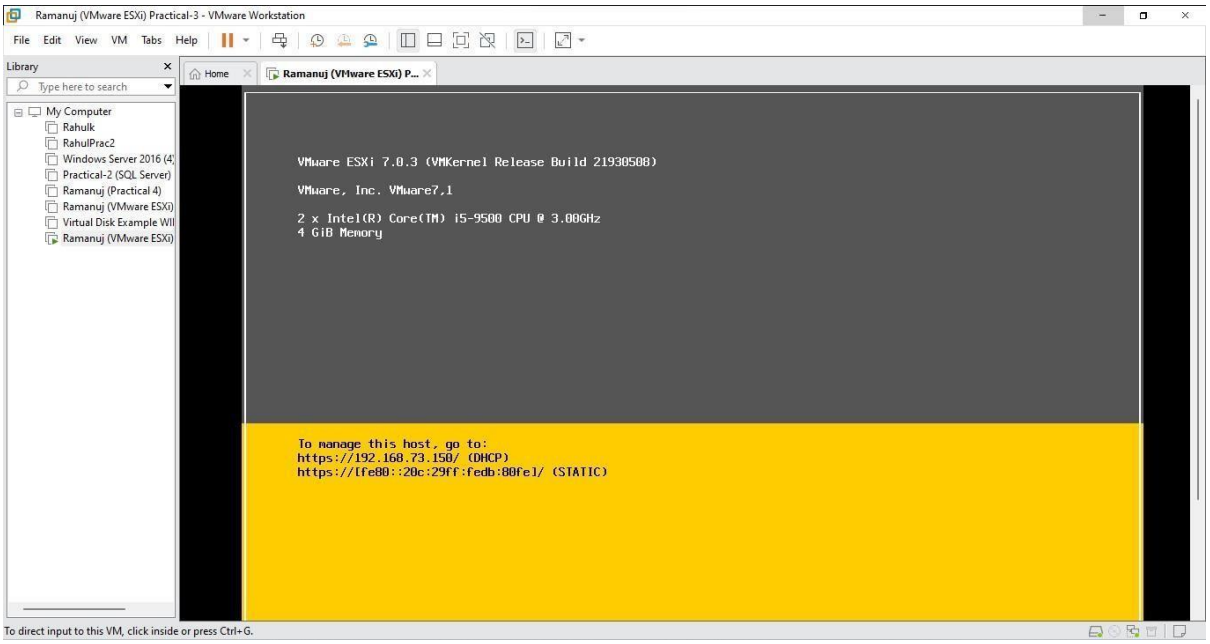


Step 1: Configuring ESXi Host Client

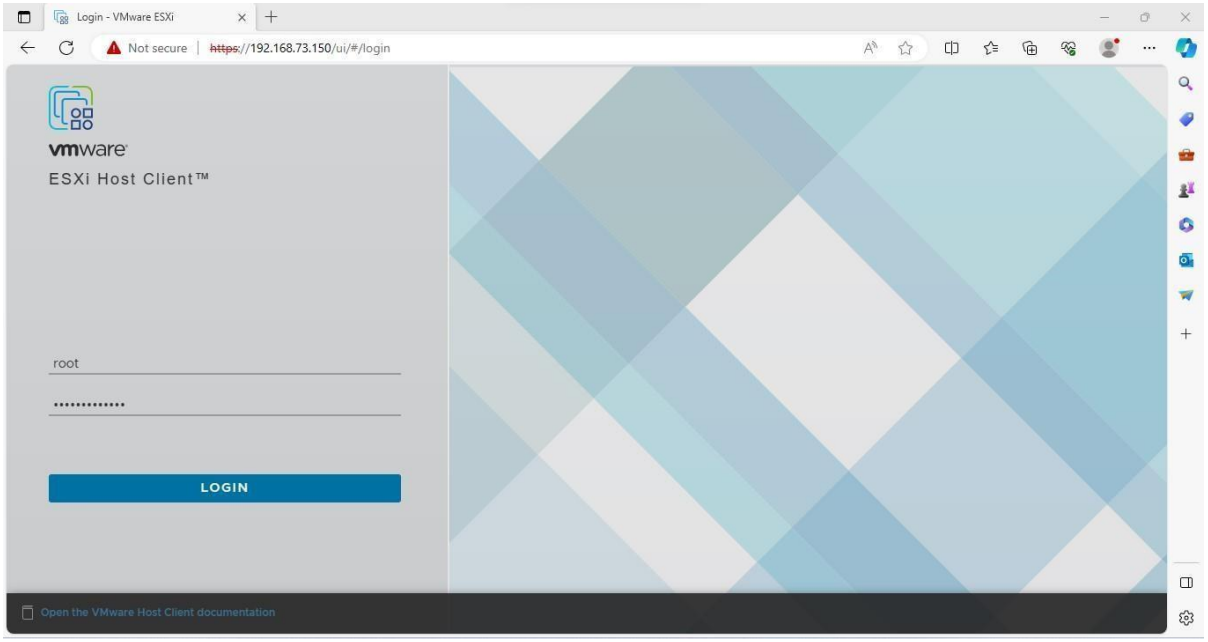
- We will be using the same ESXi server • Power On the virtual machine



- Copy the IP Address of the server and Paste it into your Web Browser

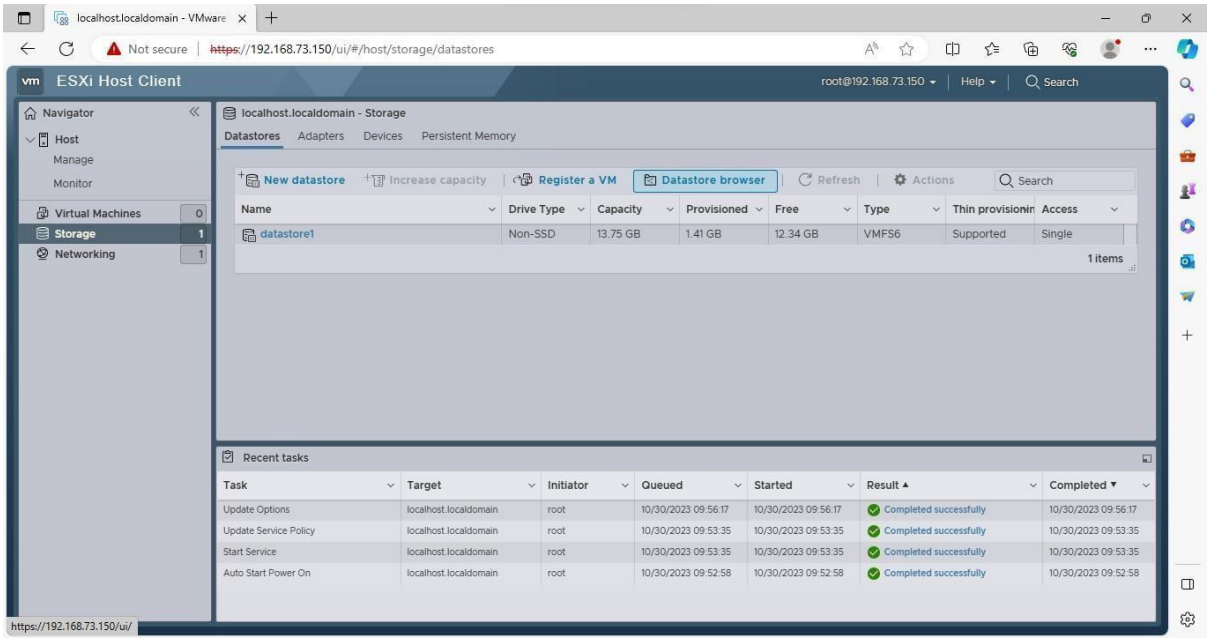


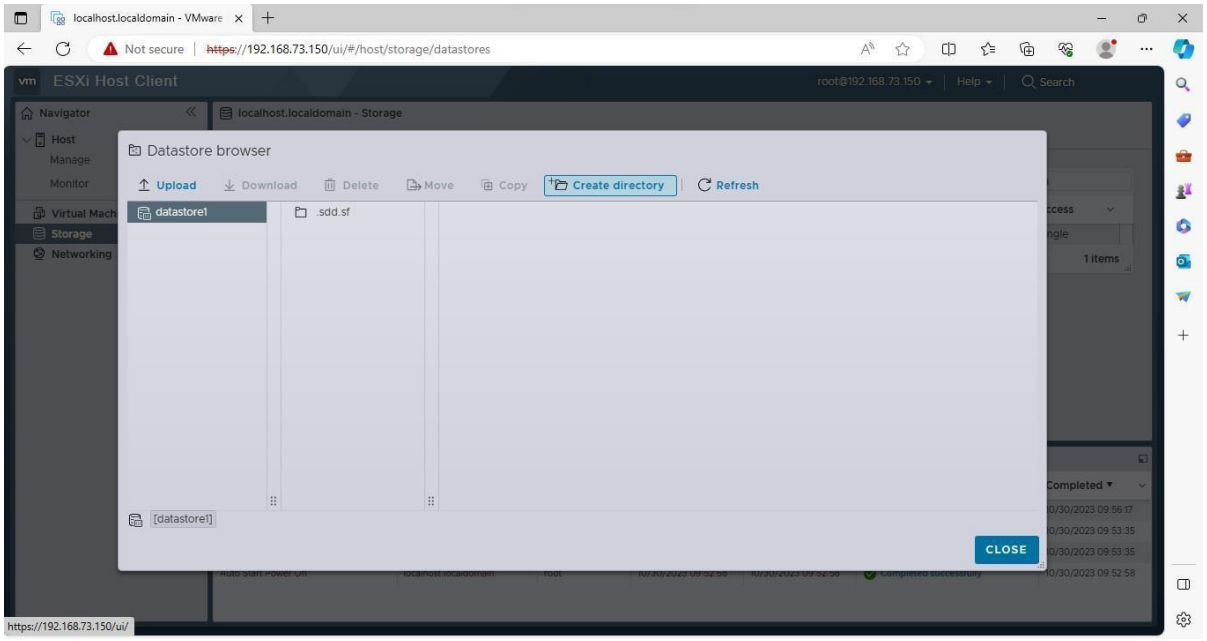
t will show you the login page for the ESXi Host Client, enter your ESXi server credentials to access the Host Client



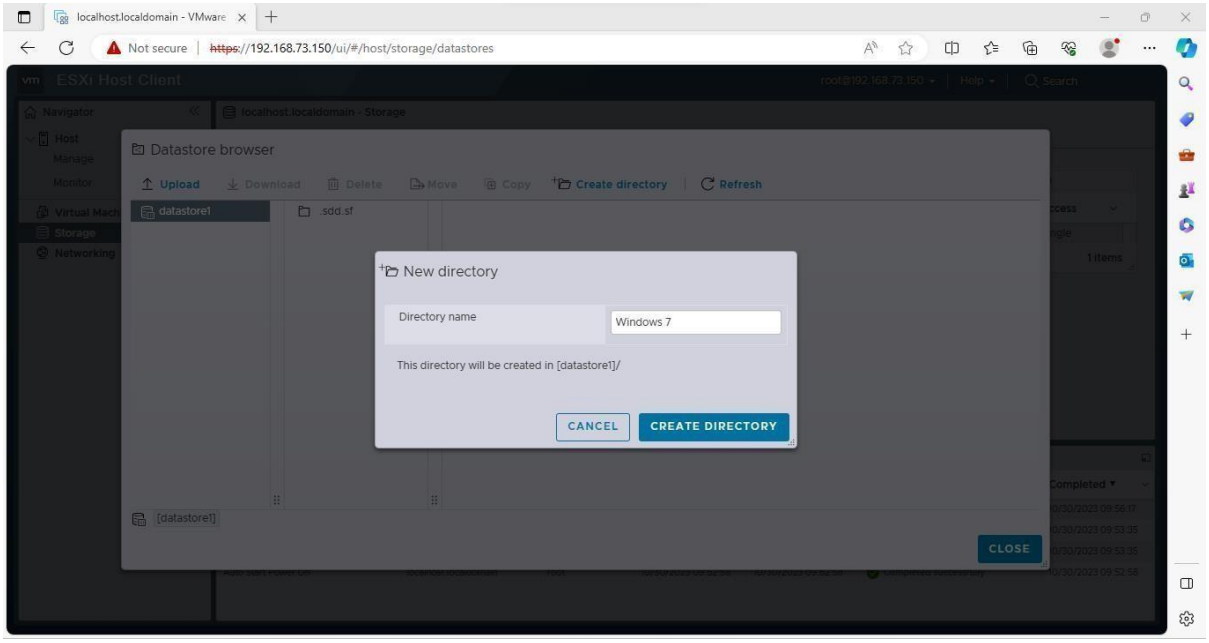
- Now before Creating a Virtual Machine we have to upload an Operating System iso file (Here we will use Windows 7)
- Within the ESXi Host Client Click on **Storage** and within Storage Click on **Datastore Browser**

Within Datastore Browser Click on **Create Directory**

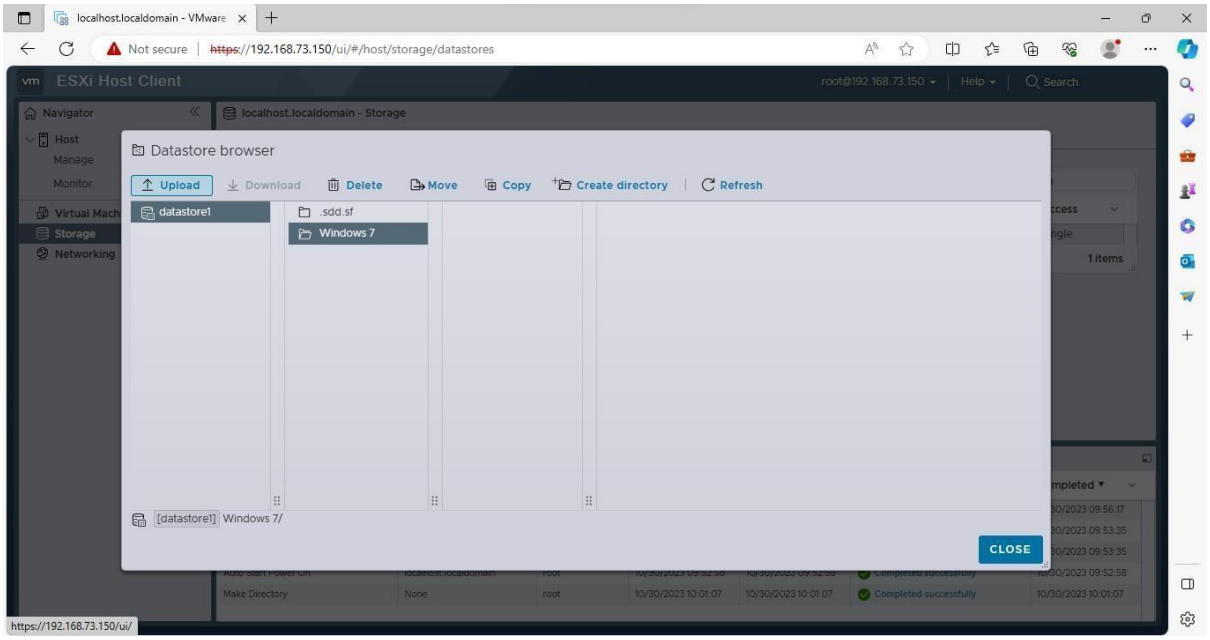




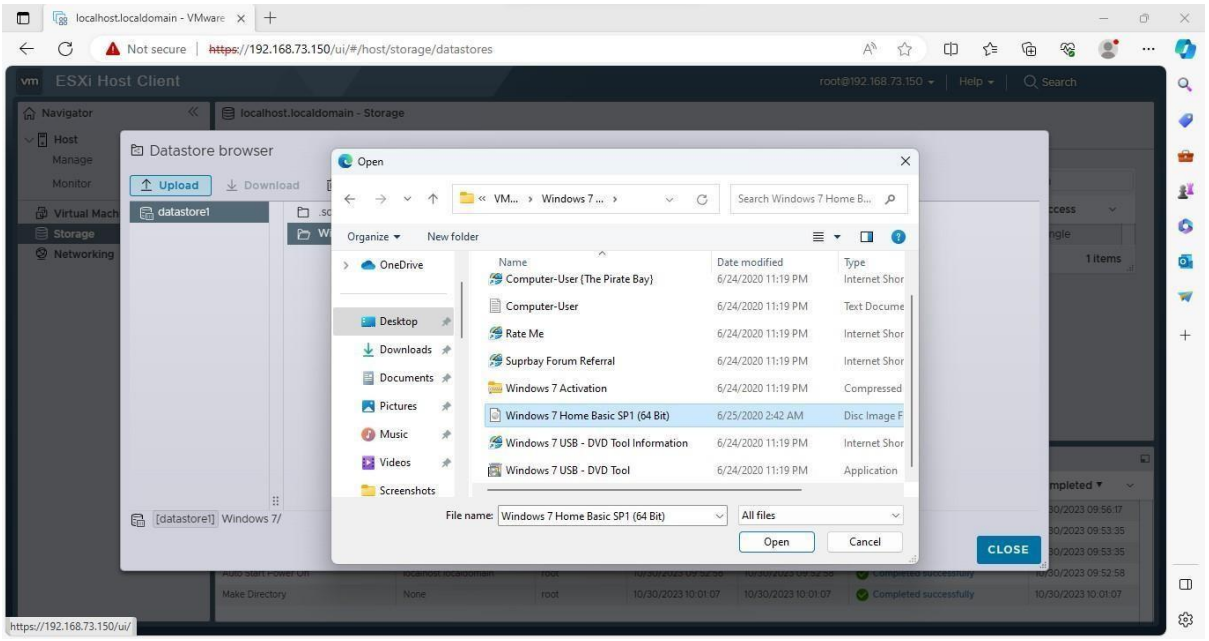
- Enter a Directory Name (Here it is Windows 7) and Click on **Create Directory**



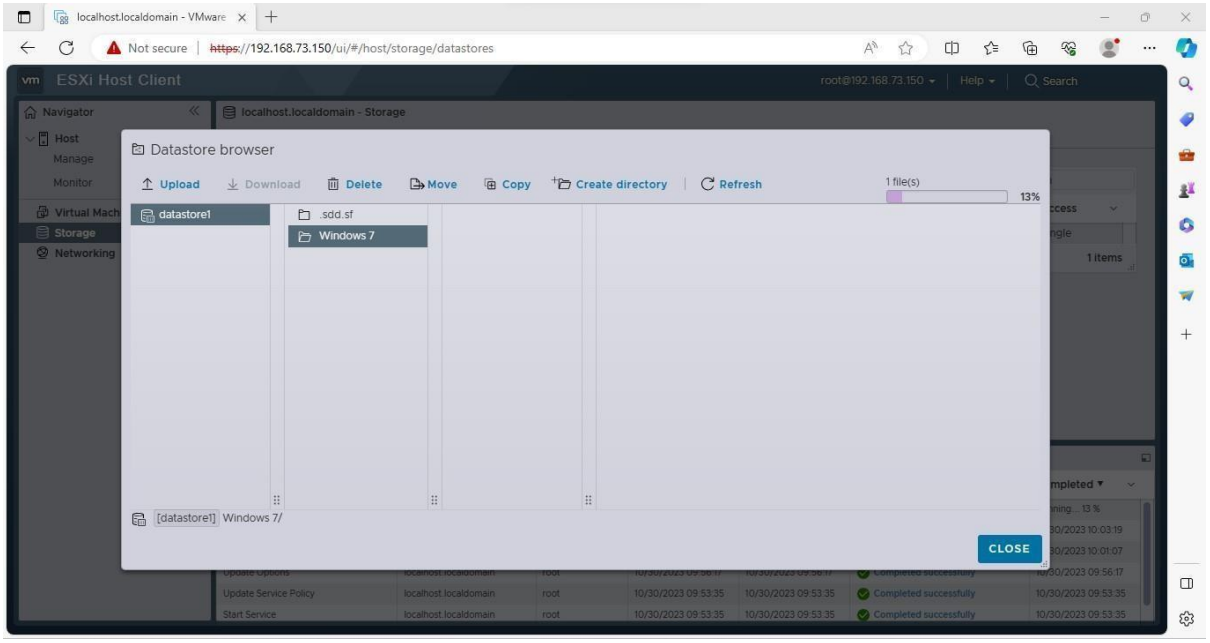
- After the new directory is created Click on **Windows 7** and Select the **Upload** option



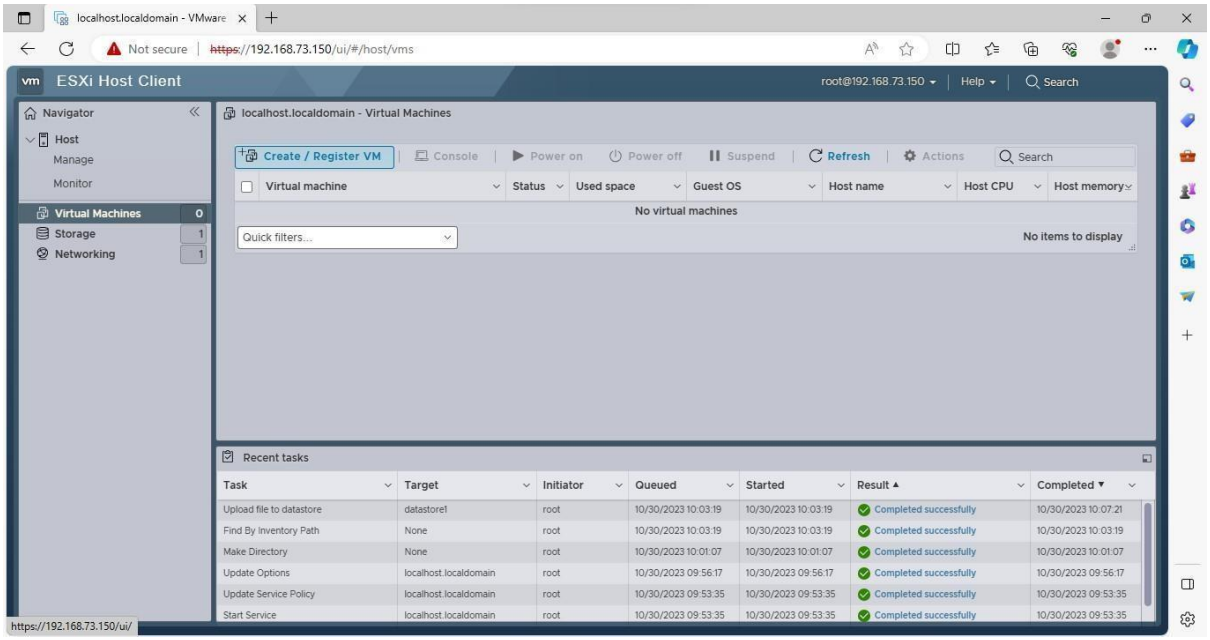
- Here browse your Host Directory for the Windows iso file and Click **Open**



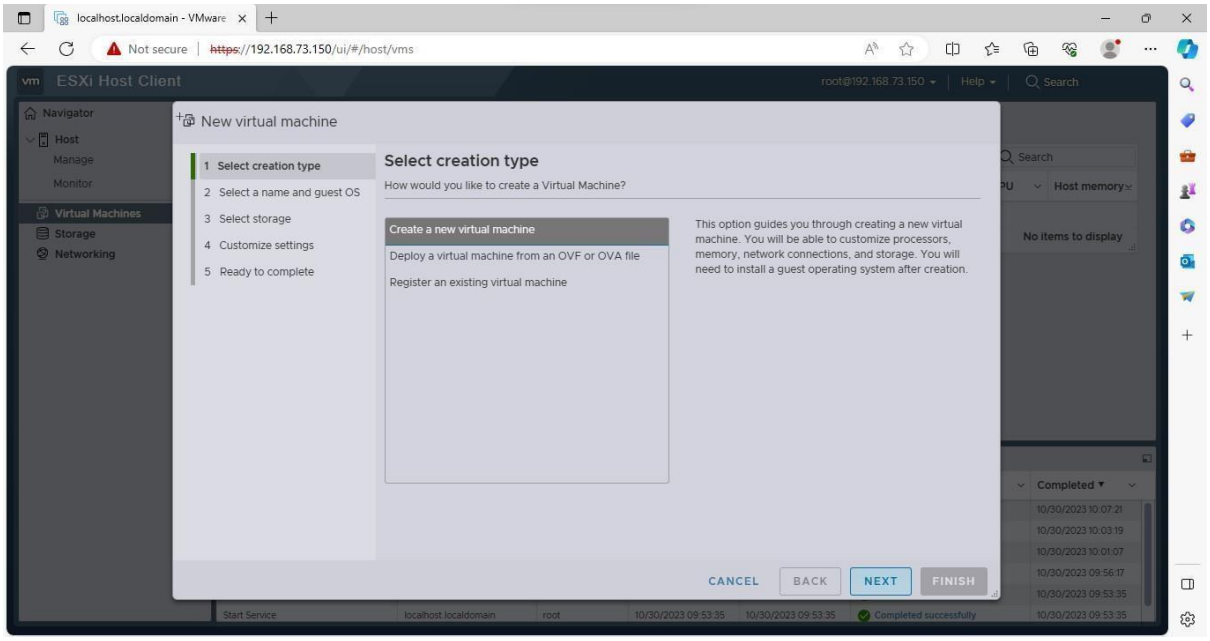
- The file will now upload to your Windows 7 Directory in the Datastore



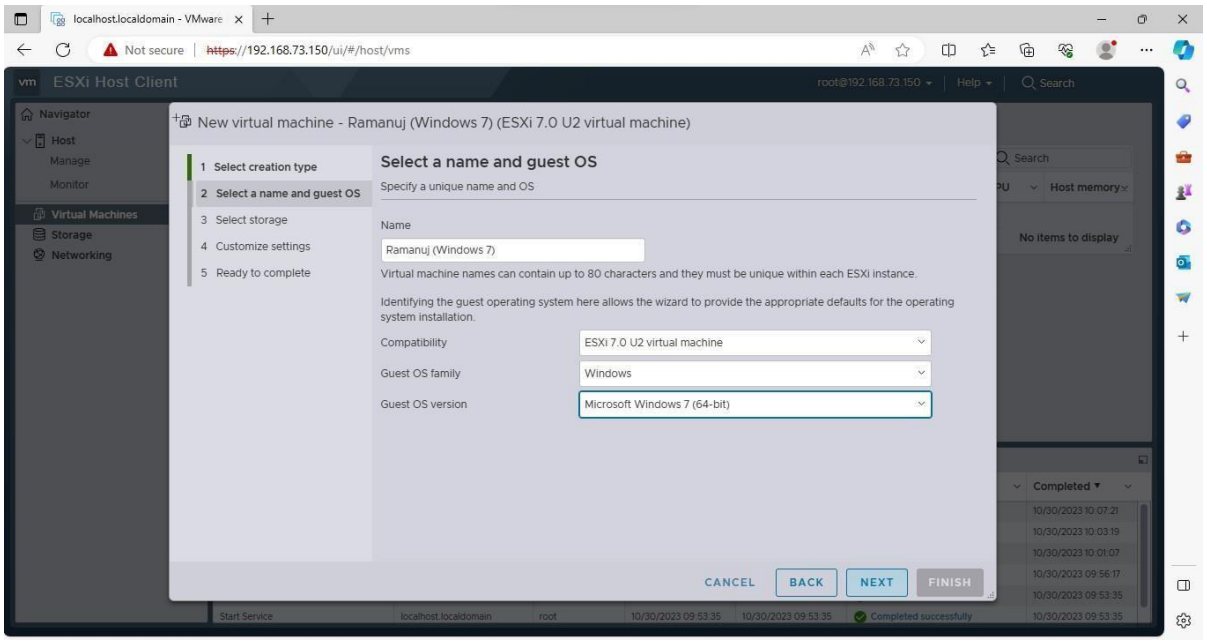
- After the upload is complete Click on **Close**
- On the Host Client Dashboard Select **Virtual Machines** and Click on **Create / Register a new VM**



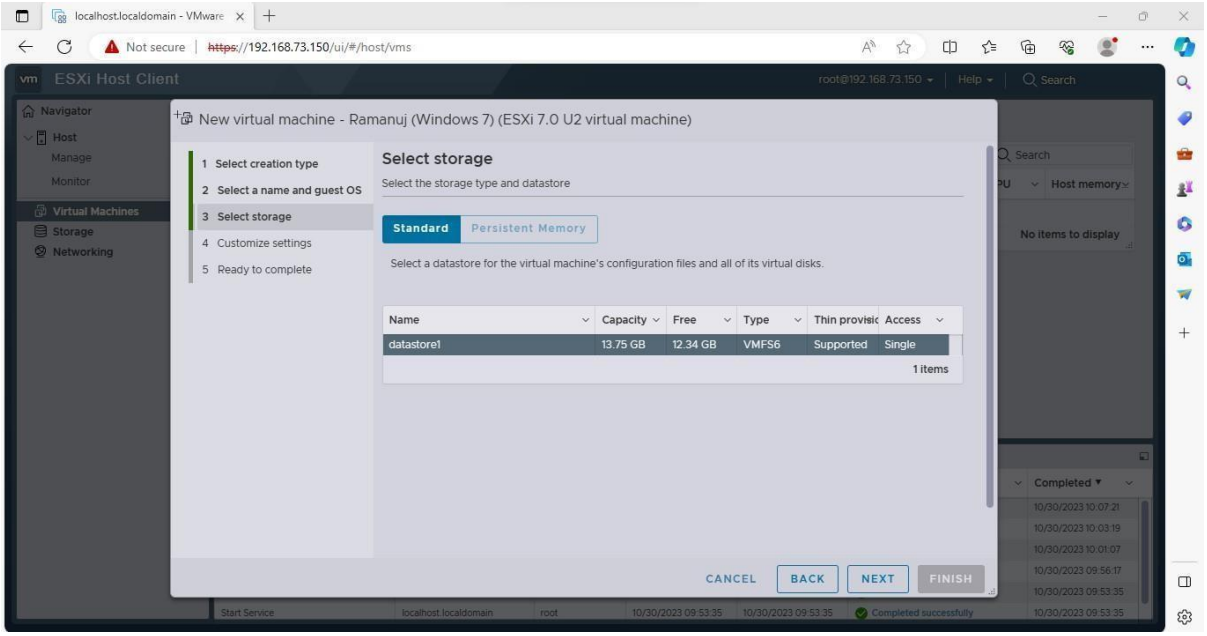
Select **Create a new virtual machine** and Click **Next**



- Enter a **Name** for the virtual machine
- Keep the Compatibility as its **default** value
- Select Guest OS Family as **Windows**
- Select Guest OS Version as **Microsoft Windows 7 (64-bit)** and Click **Next**

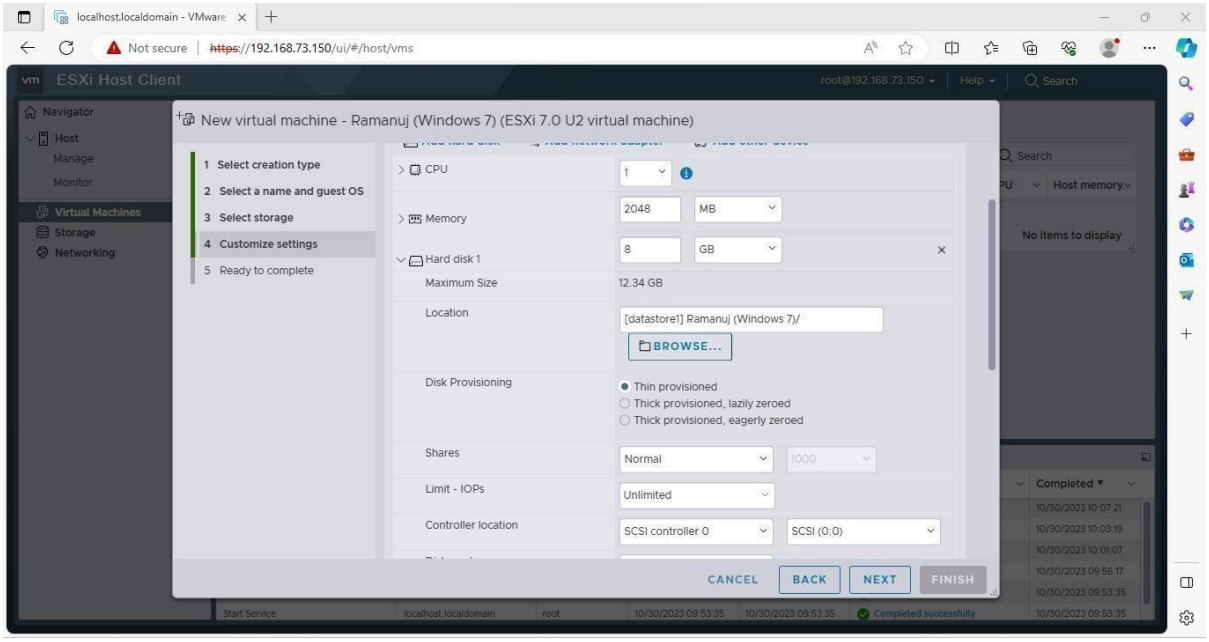


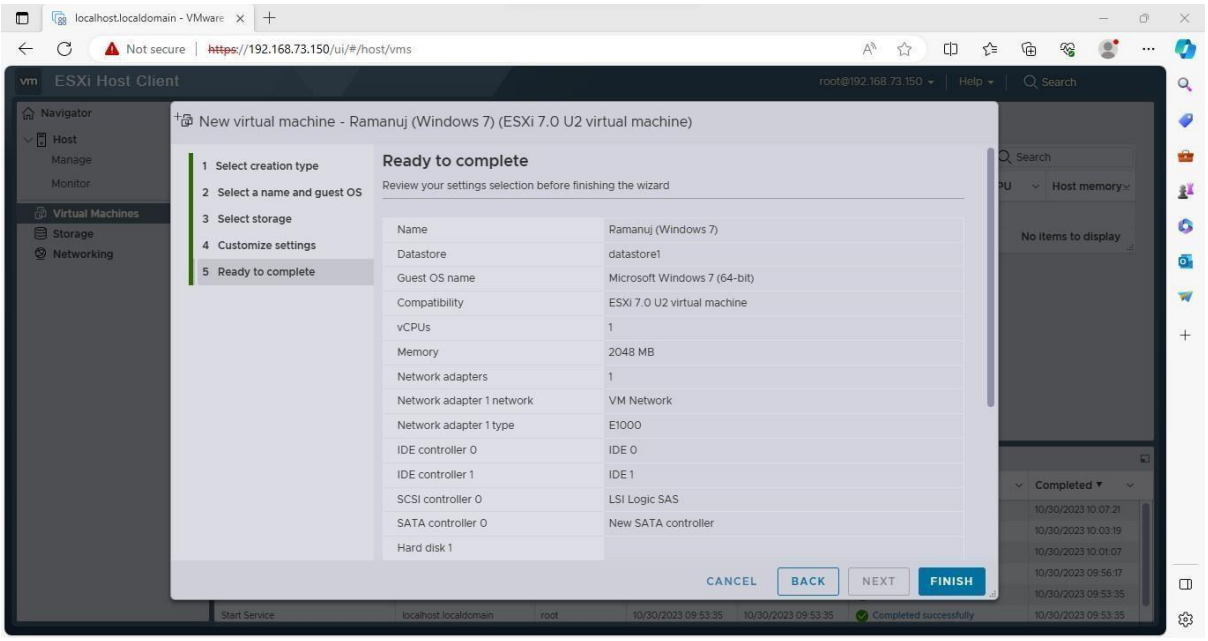
- Select the Standard datastore and Click **Next**



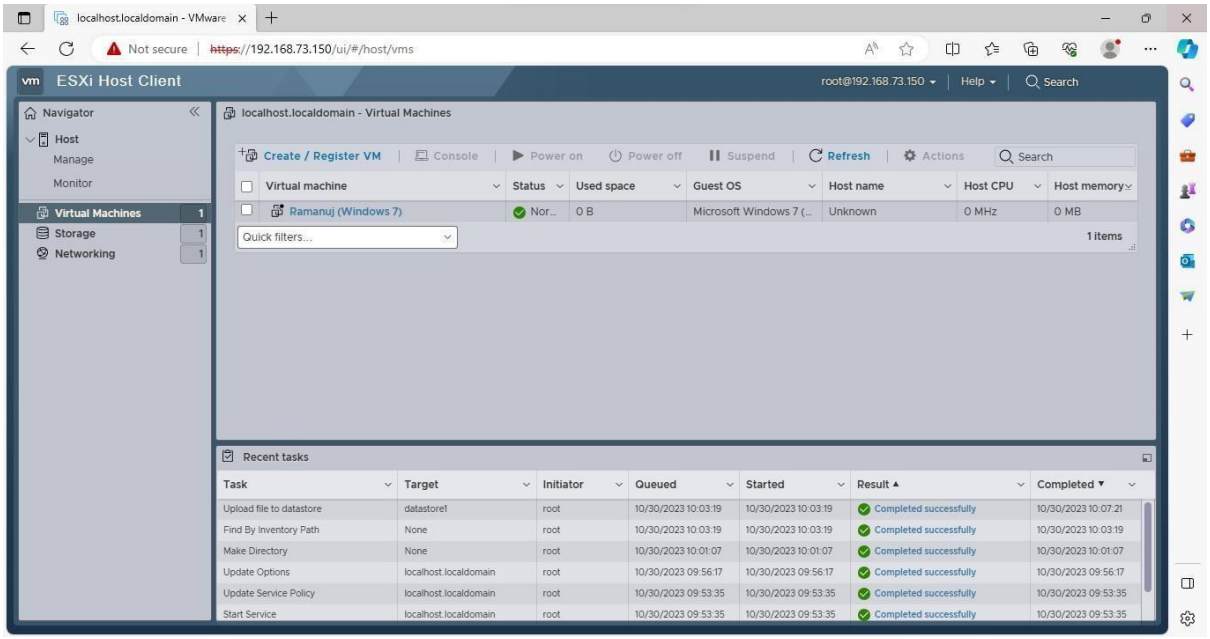
- Keep CPU at **1** core
- Memory at **2048 MB**
- Hard Disk at **8 GB**
- Disk Provisioning as **Thin provisioned**
- The remaining can have their default value, only exception being: **CD/DVD Drive 1** where you have to Select **Datastore iso file** and Select the **Windows 7 iso** file from the directory and ensure on Status the
  Connect on Power on option is selected and Click **Next**

  View the summary of your virtual machine and Click on **Finish**

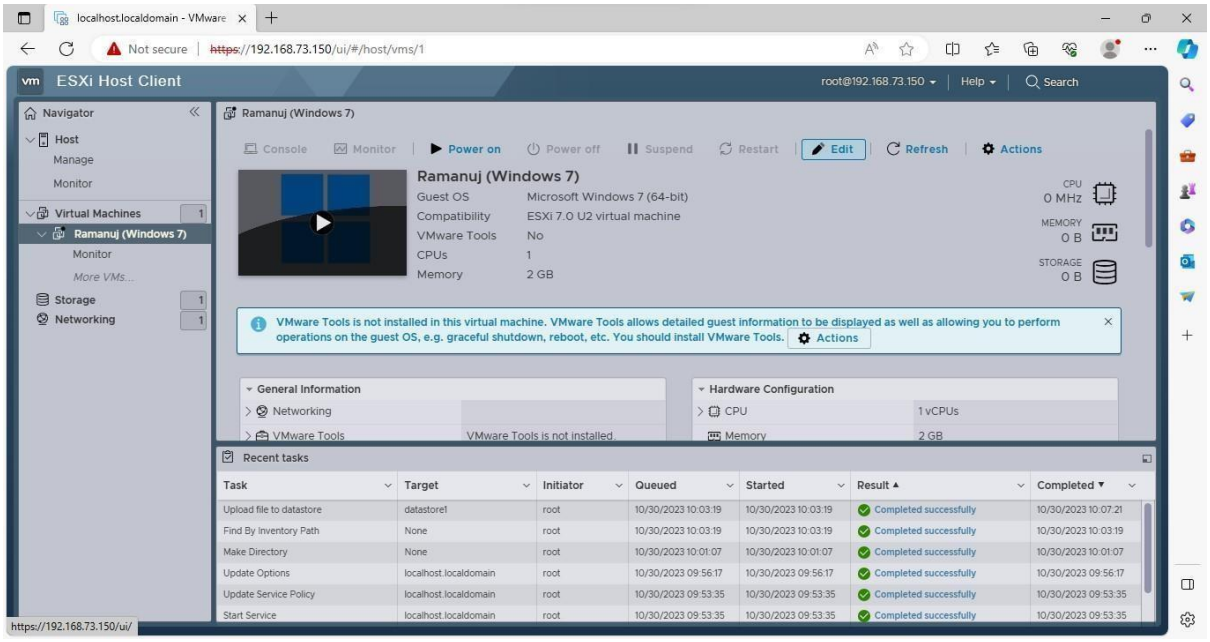




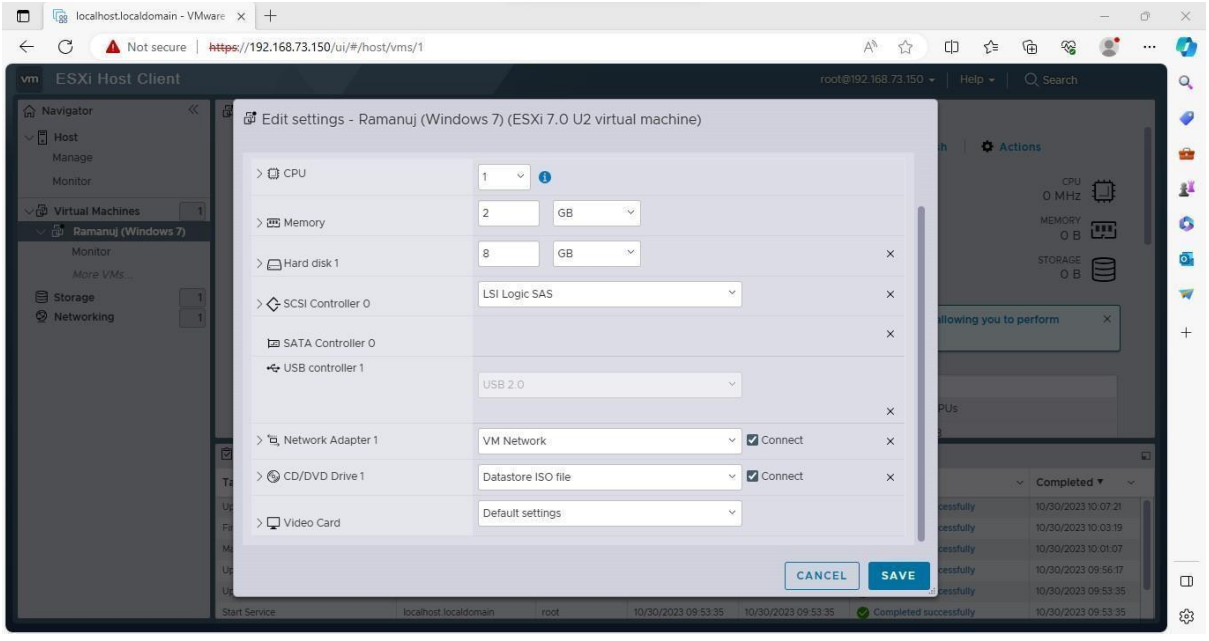
- Your Virtual Machine has now been created and will reflect on the ESXi Host Client



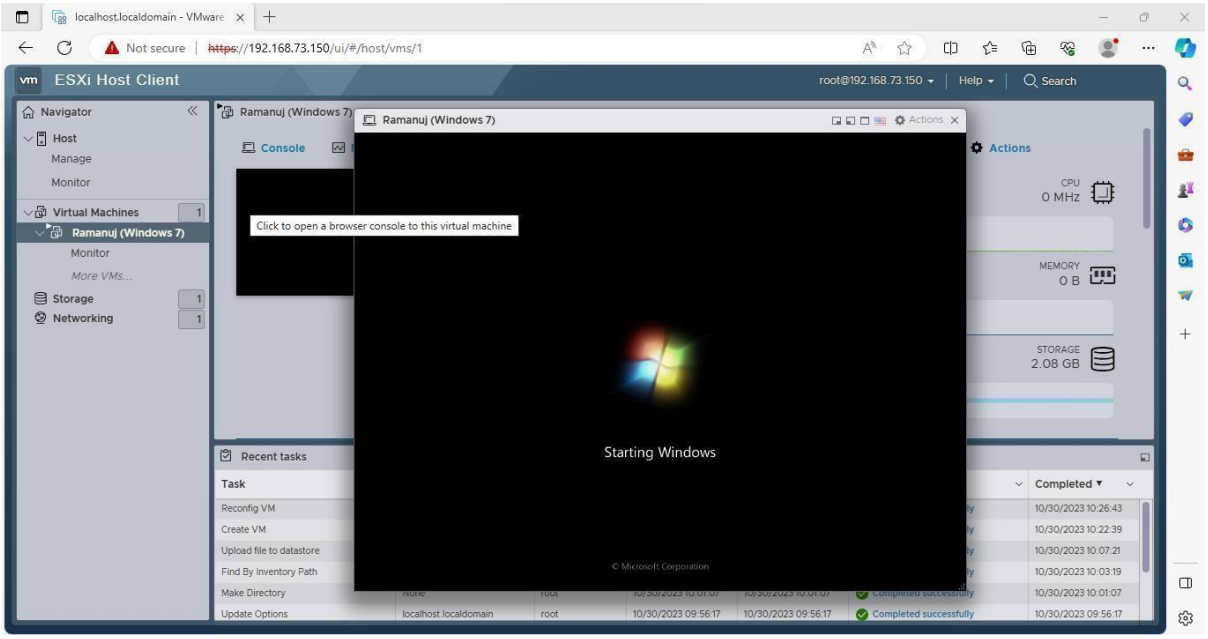
- Click on your virtual machine and before powering on Click on **Edit**



- Within Edit find the **CD/DVD Drive 1** option and Select **Connect** and Click **Save**

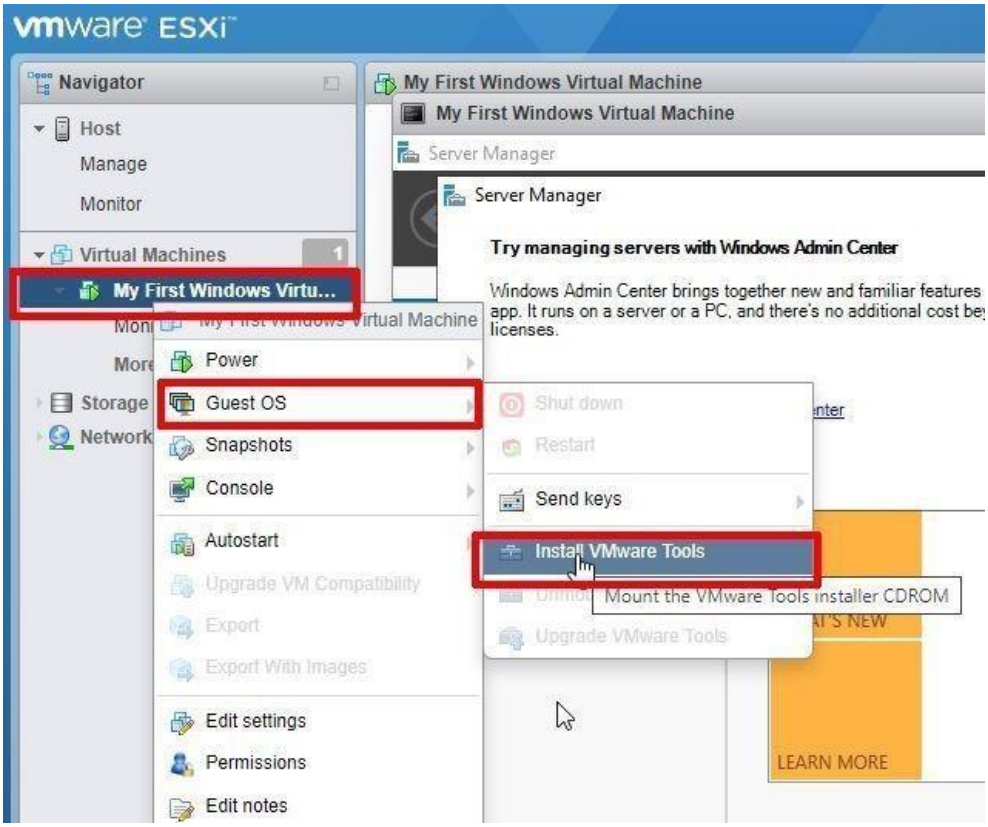


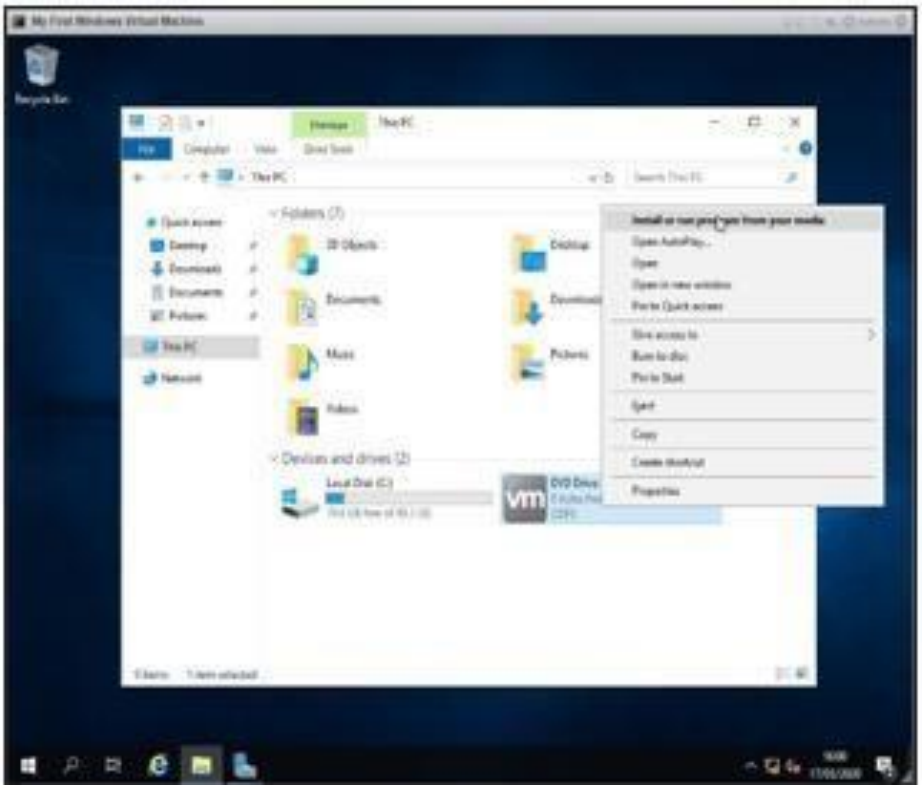
Now Power On your Virtual Machine, it has been successfully configured and deployed on an ESXi server using ESXi Host Client



Install VMware Tools

Select Actions > Guest OS > Install VMware Tools





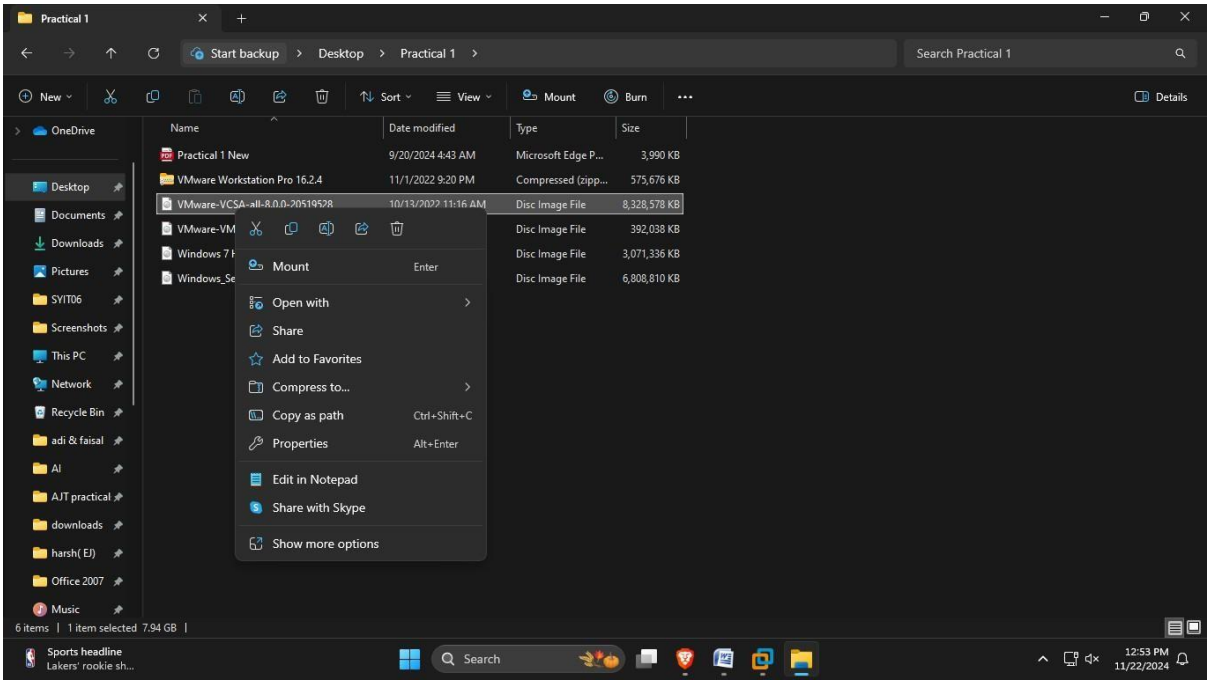
Copy Files to Desktop

Drag drop a desired file to the VM's Desktop after installing VMWare Too

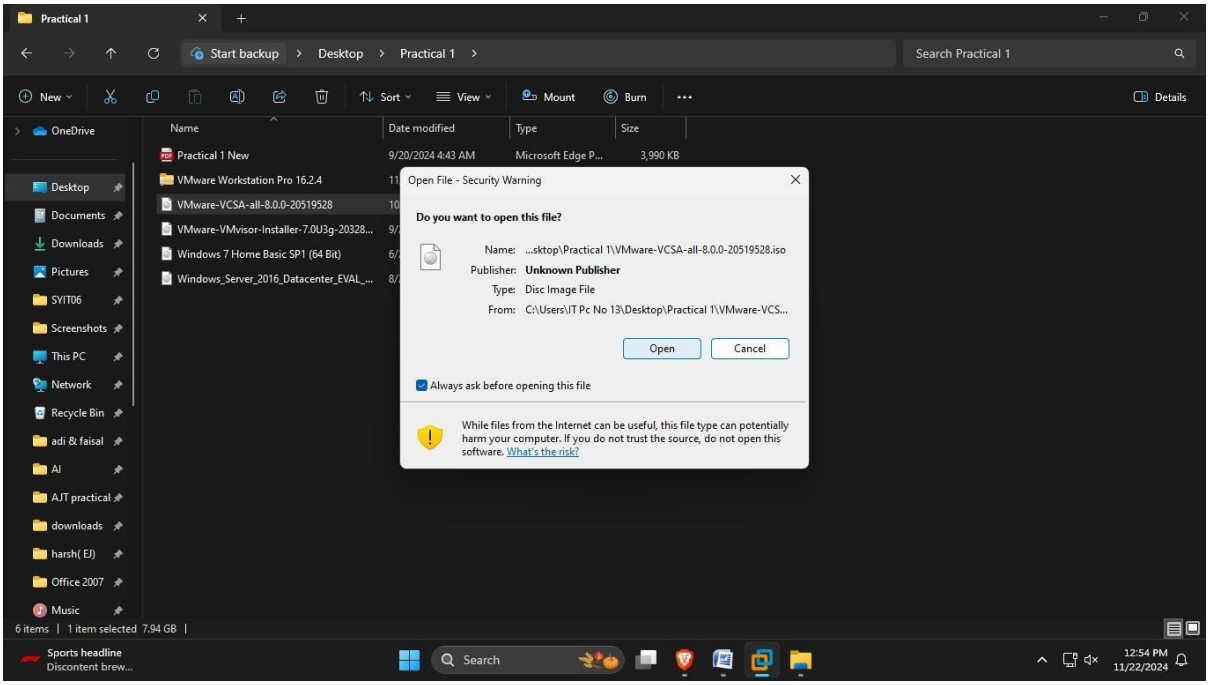
PRACTICAL 2

AIM:- Working with vCenter Server Appliance

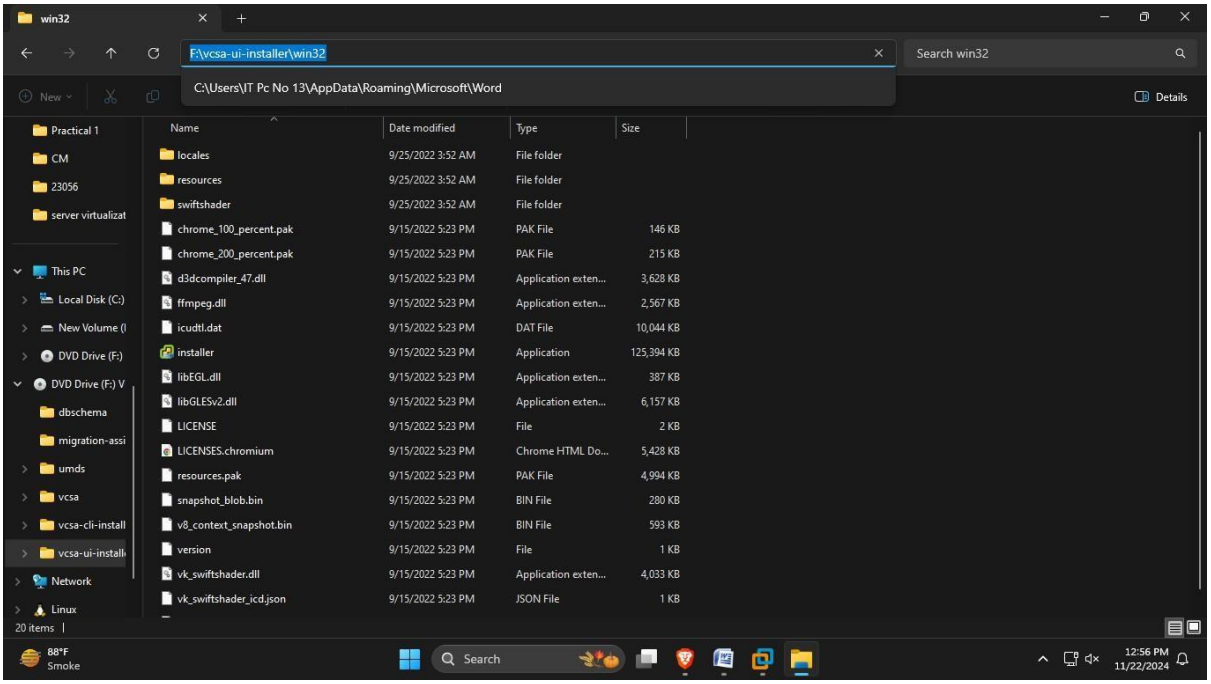
- Installing vcenter.



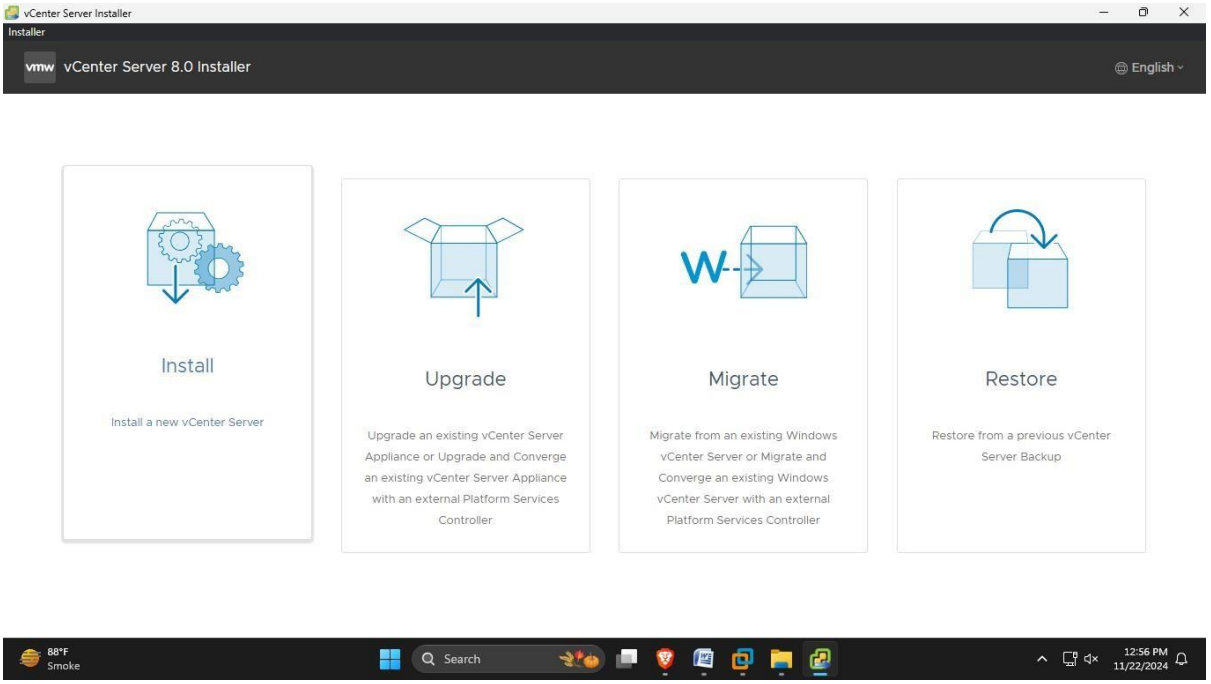
- Click on run.



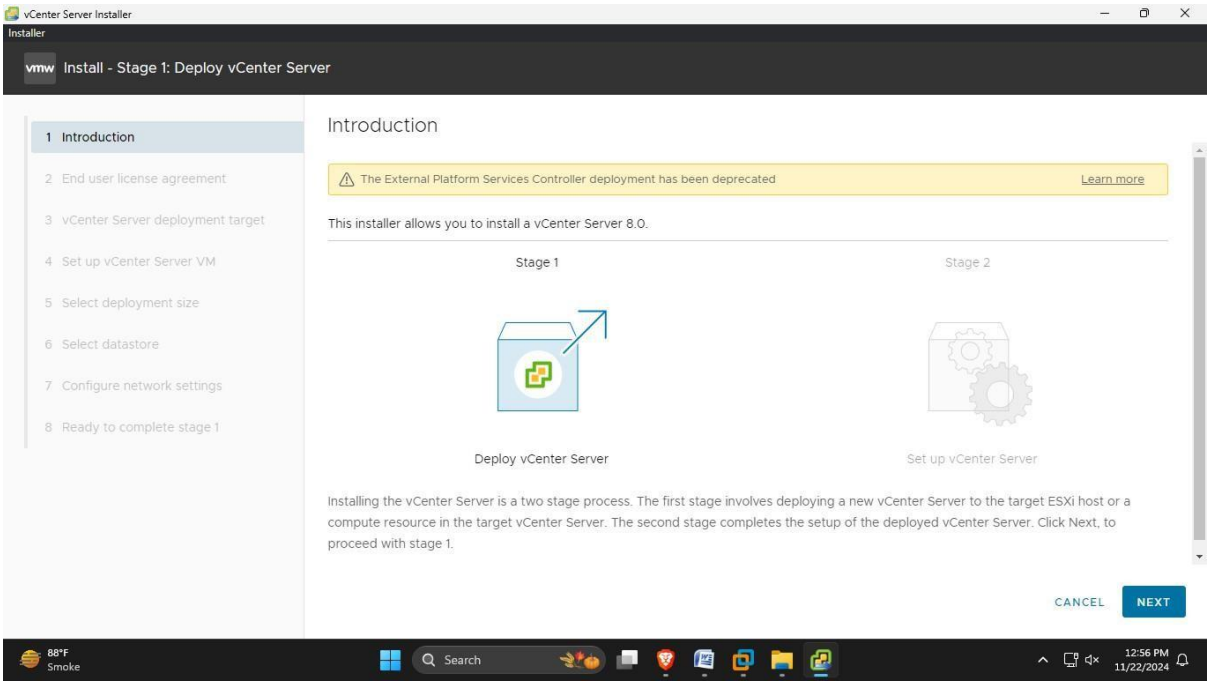
- Navigate to this file location after clicking on vcsa installer ans click on installer.



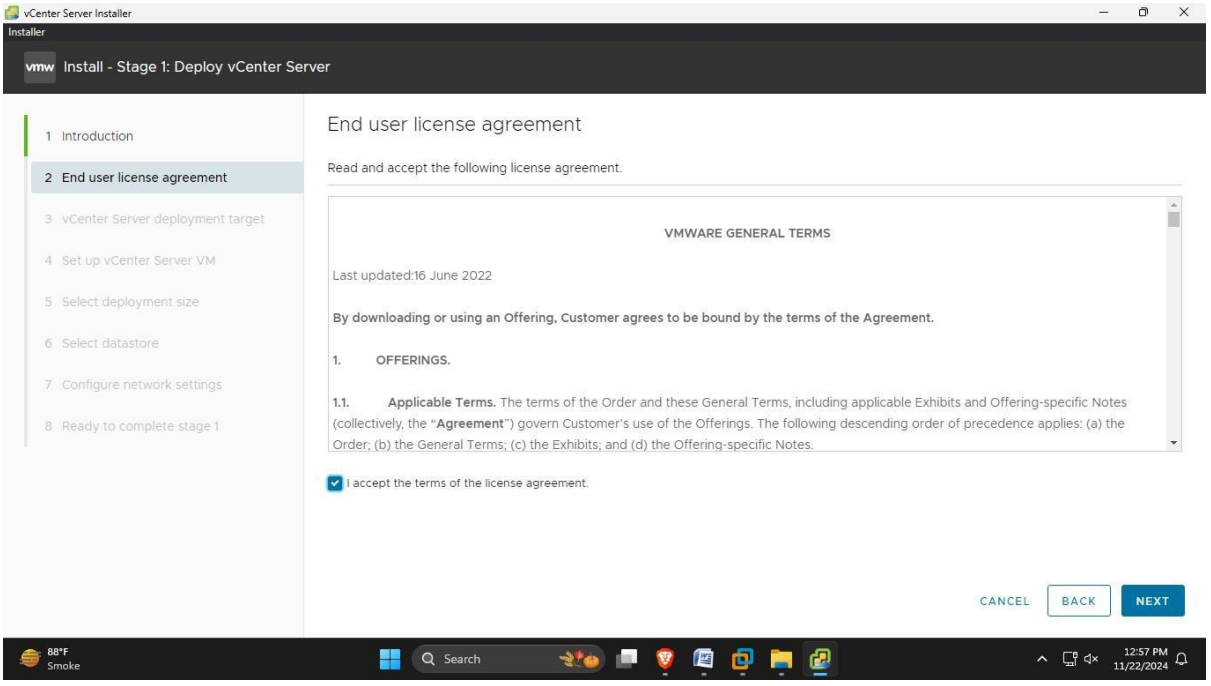
- Click on install.



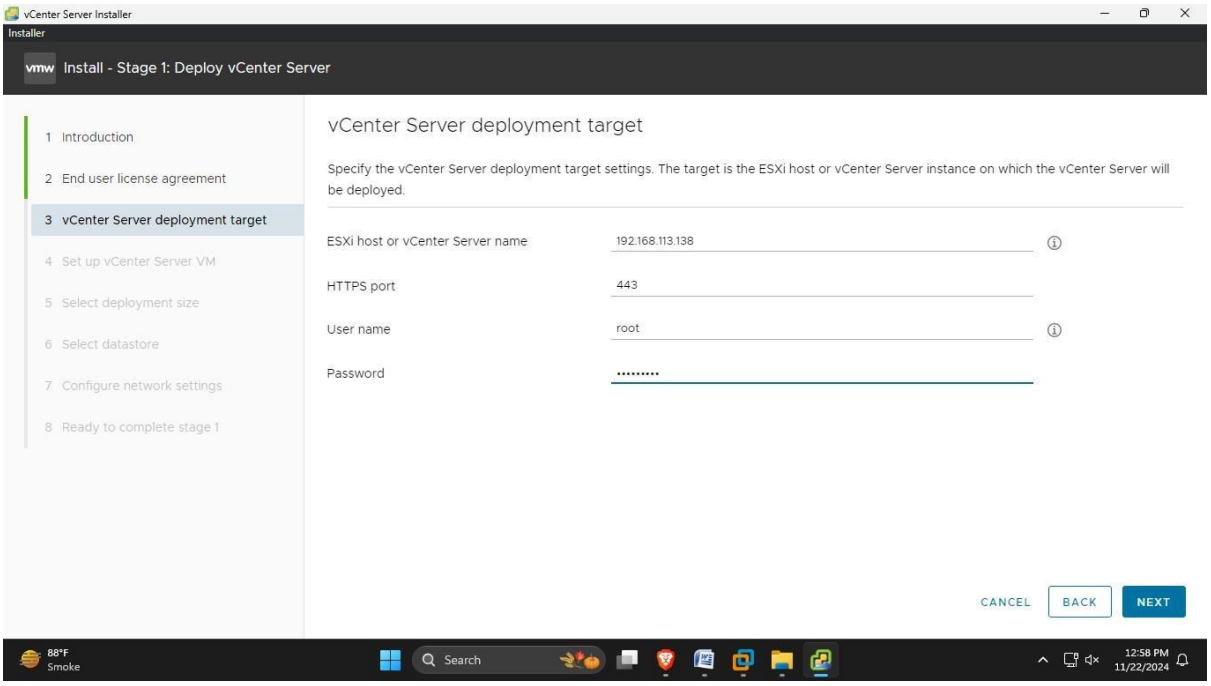
- Select Deploy vCenter and Click on next.



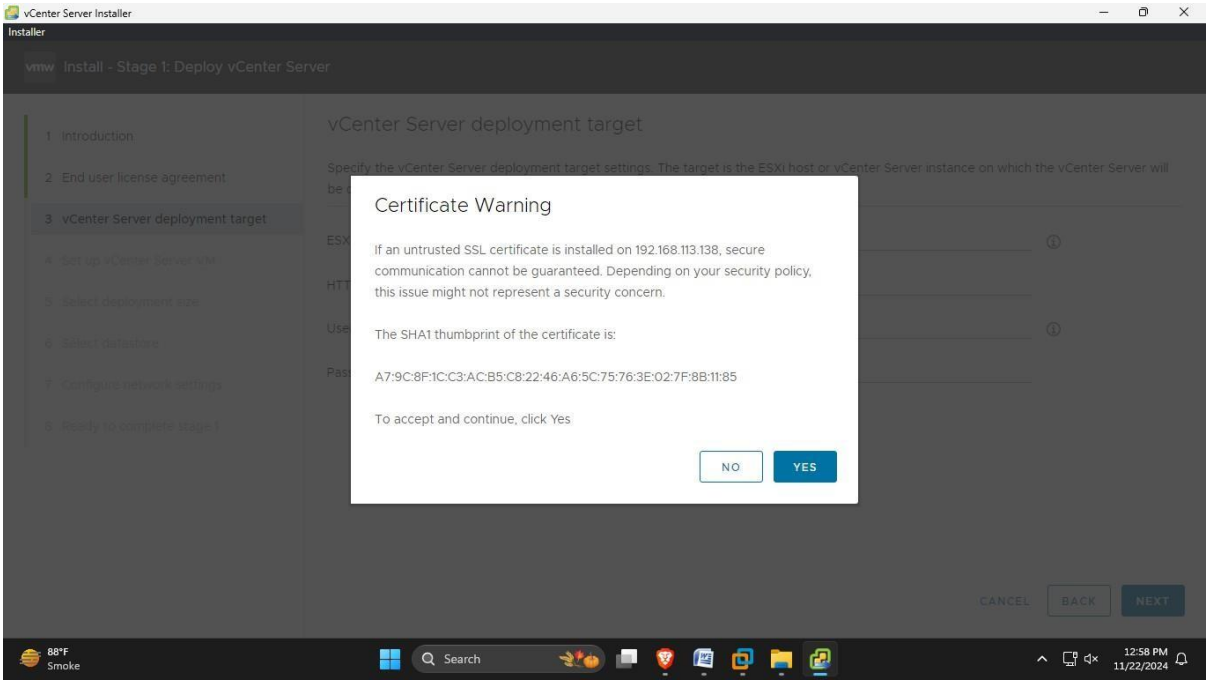
- Click on I accept……and Click on next



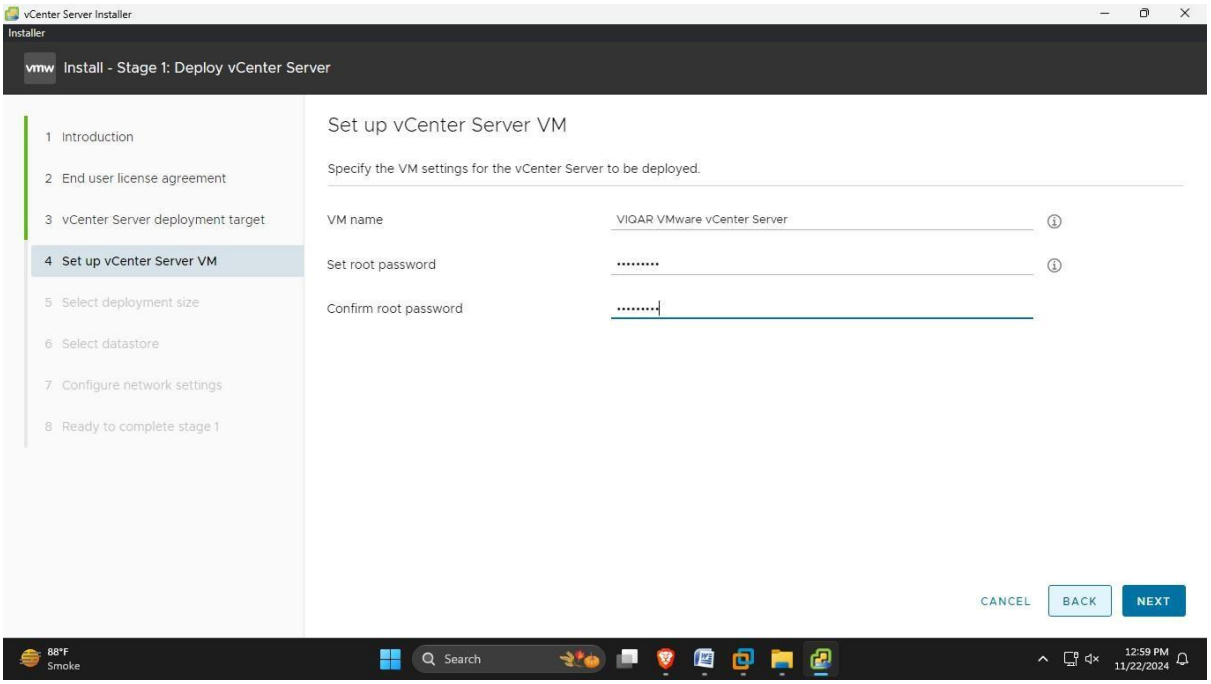
- Type your esxi's ip address and username as root and password as Name!098.



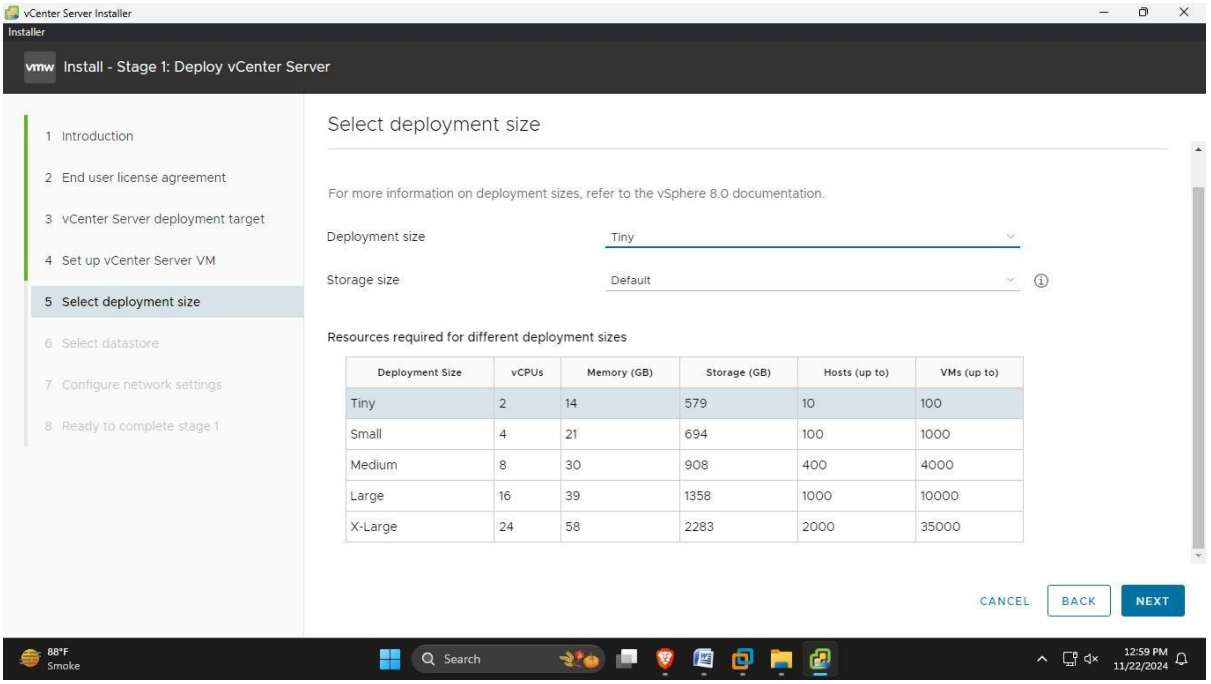
- Click on yes.



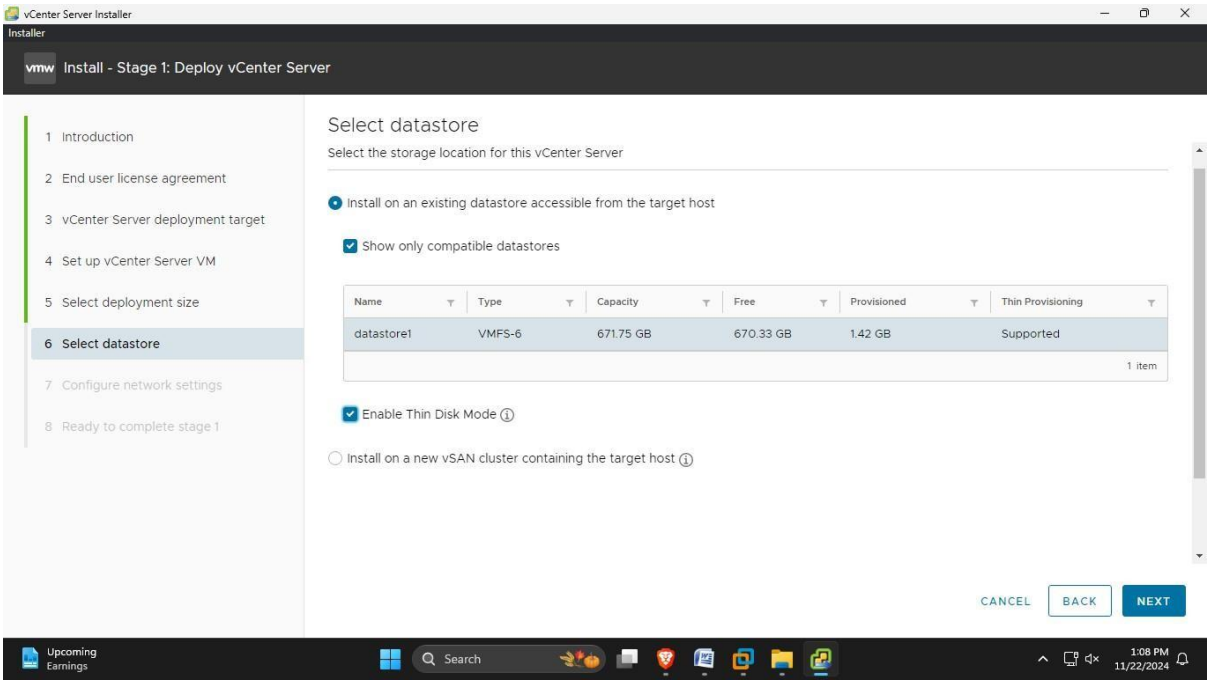
- Name it and give password.



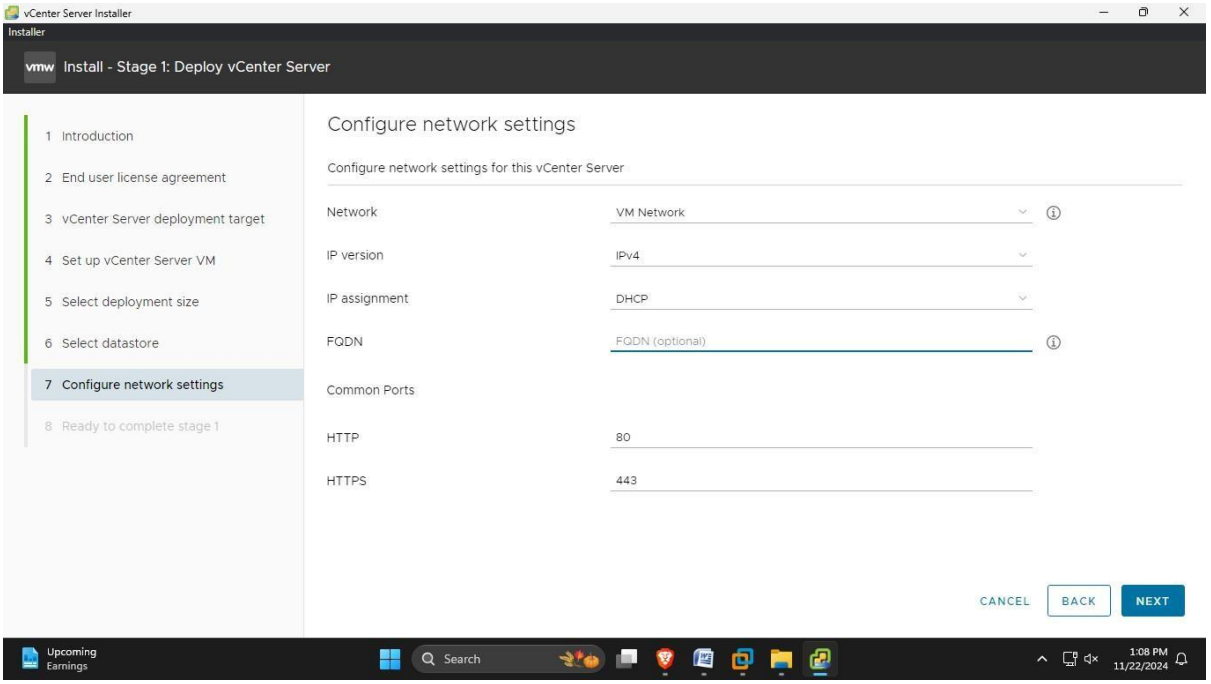
- Select ting of minimum 14 gb ram version.



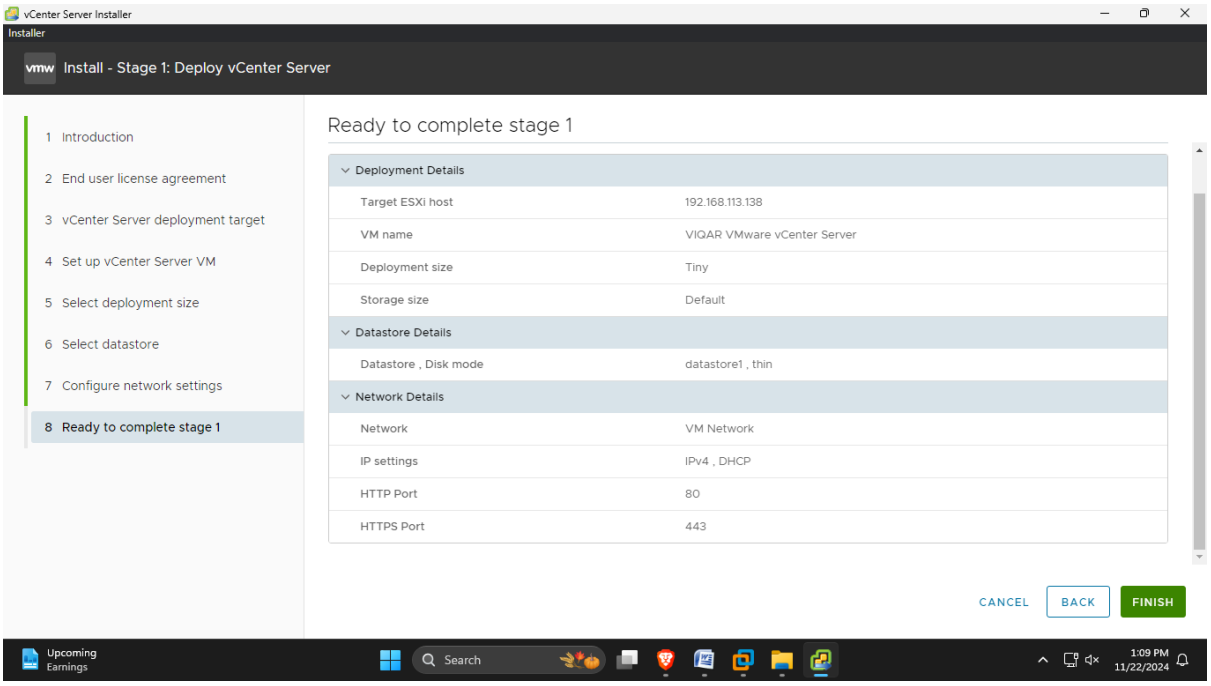
- Click on enable thin disk and Click on next.



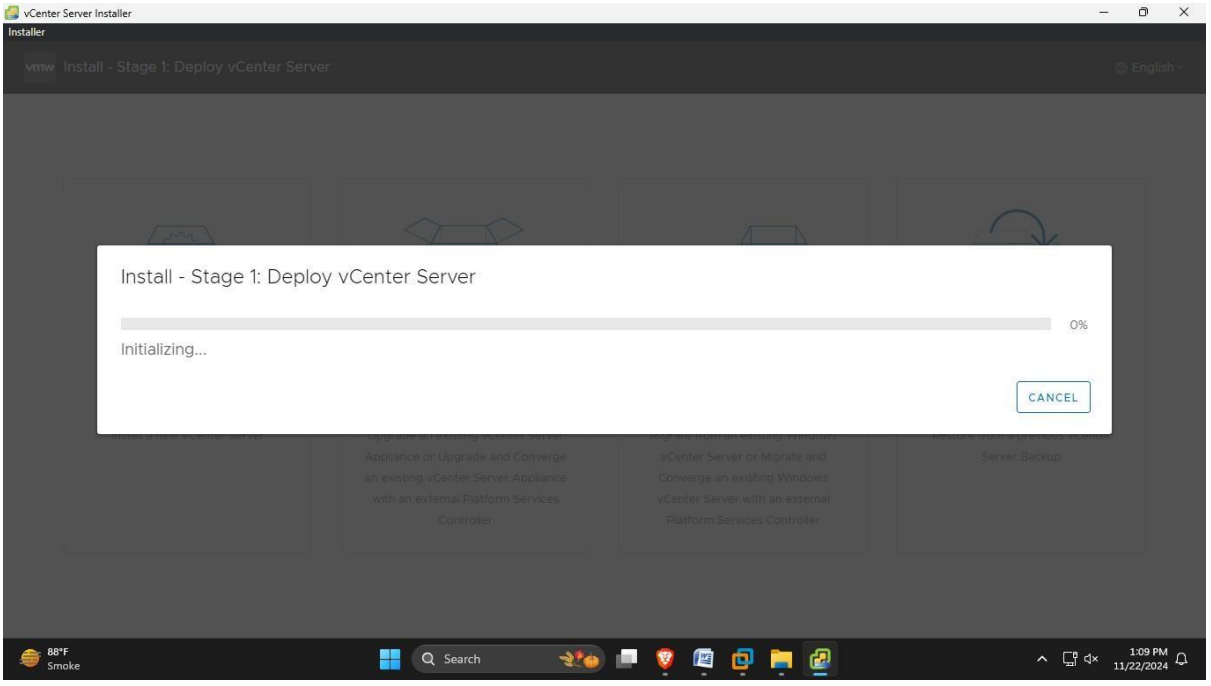
- Select DHCP and Click on next.



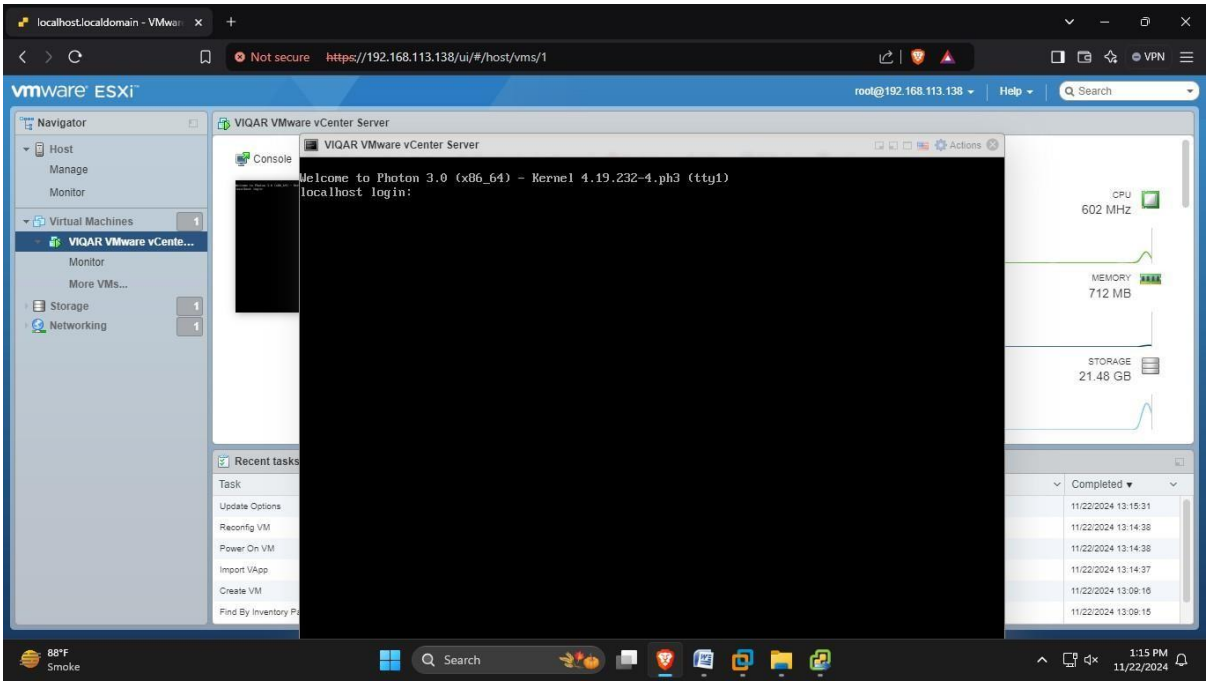
- Click on finish.



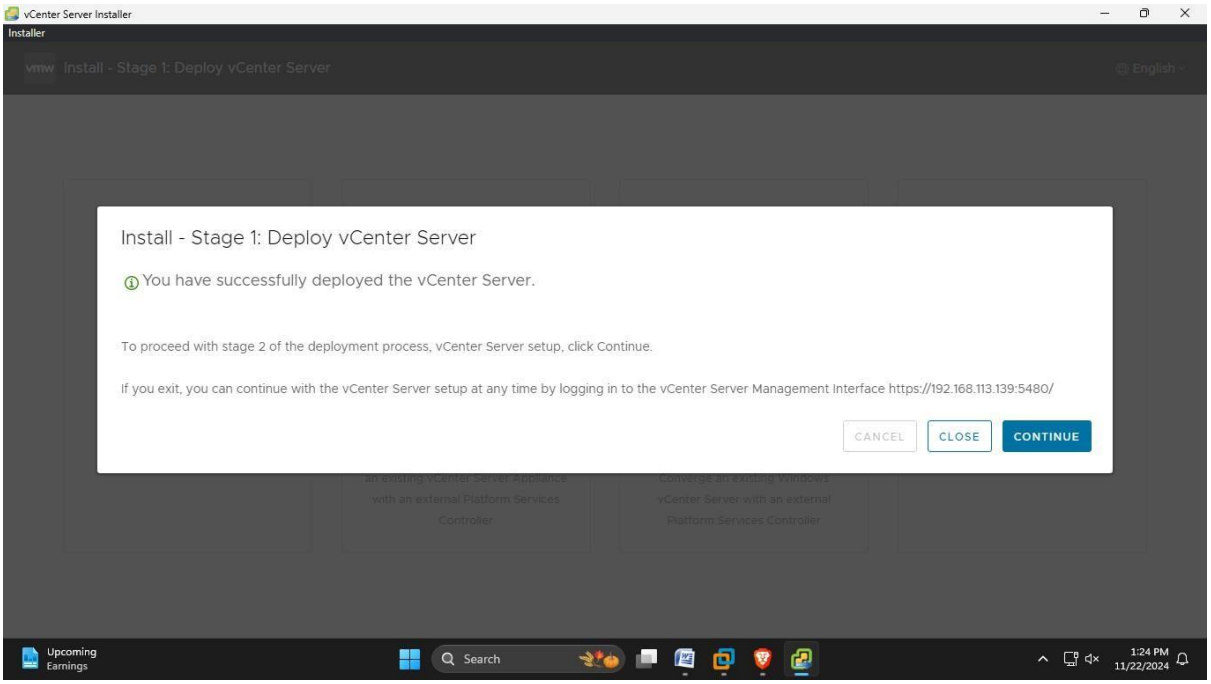
- Installation will begin.



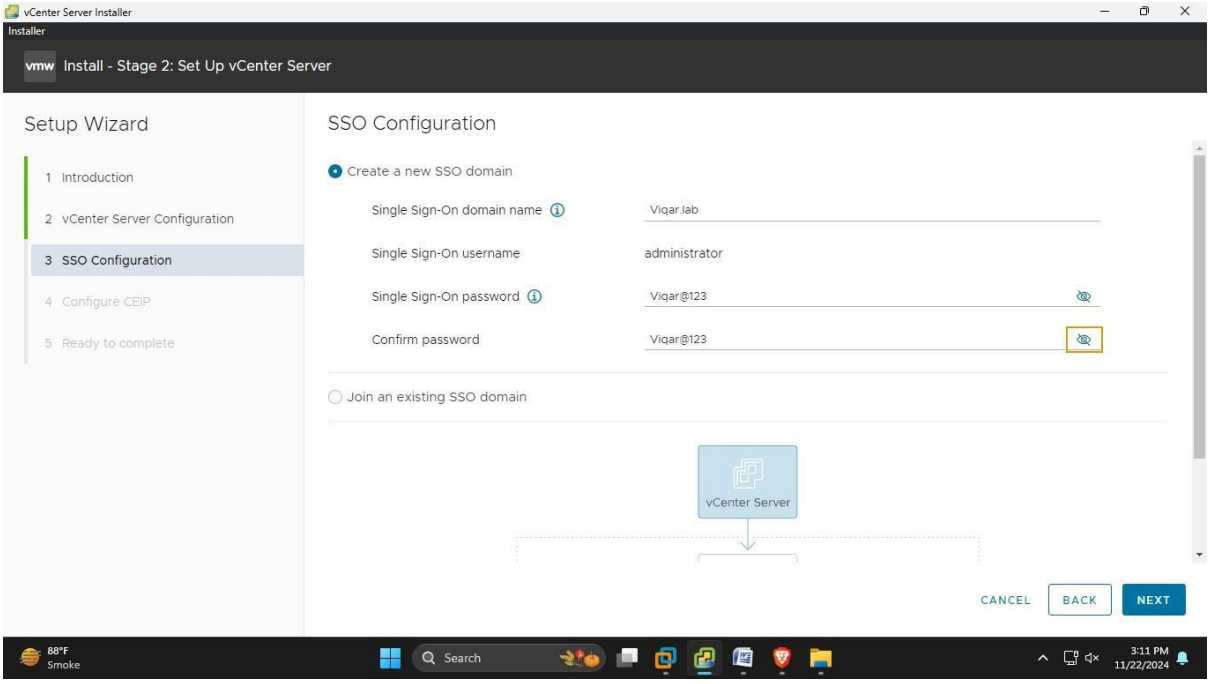
- After loging in go to vm and Run the Vcenter's VM.



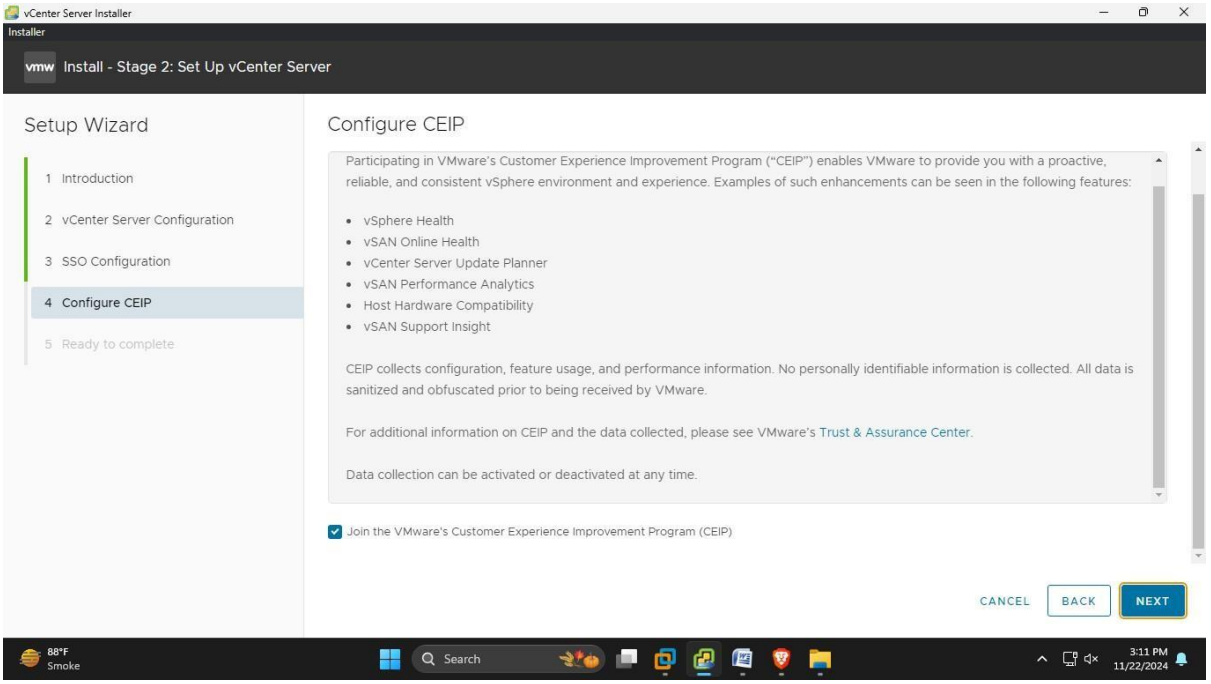
- Once the Stage 1 Installation completed click on Continue.



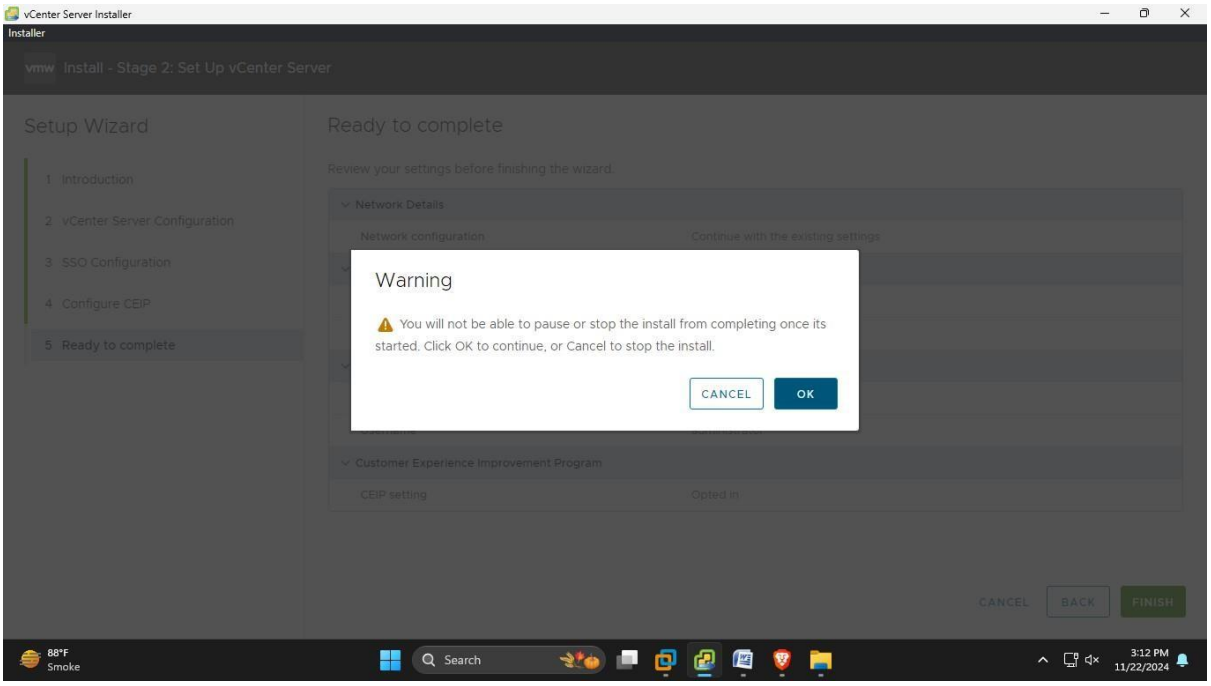
- Give your domain name and credentials.



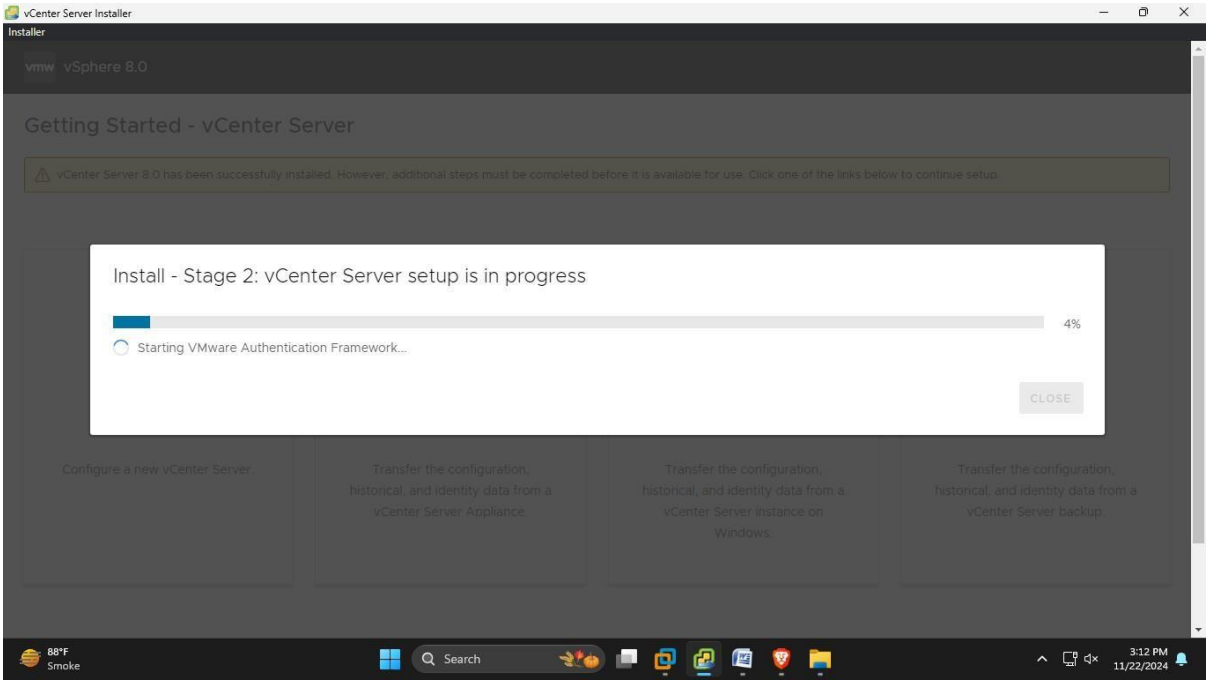
- Click on Next.



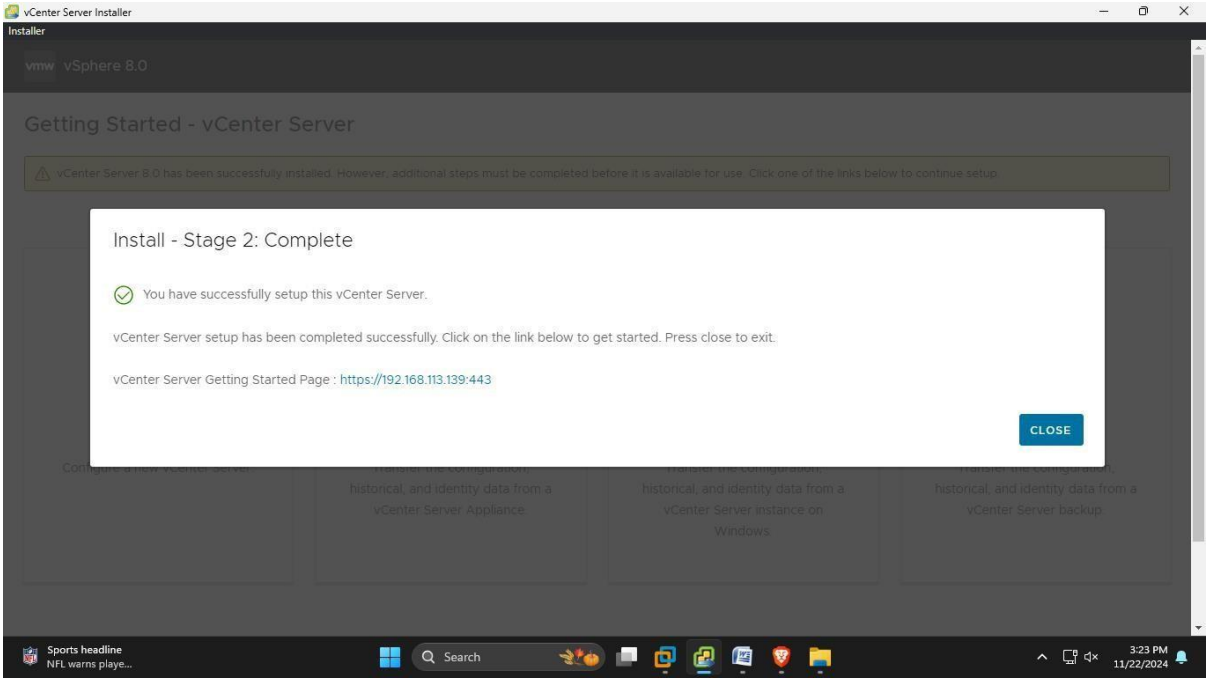
- Click on OK.



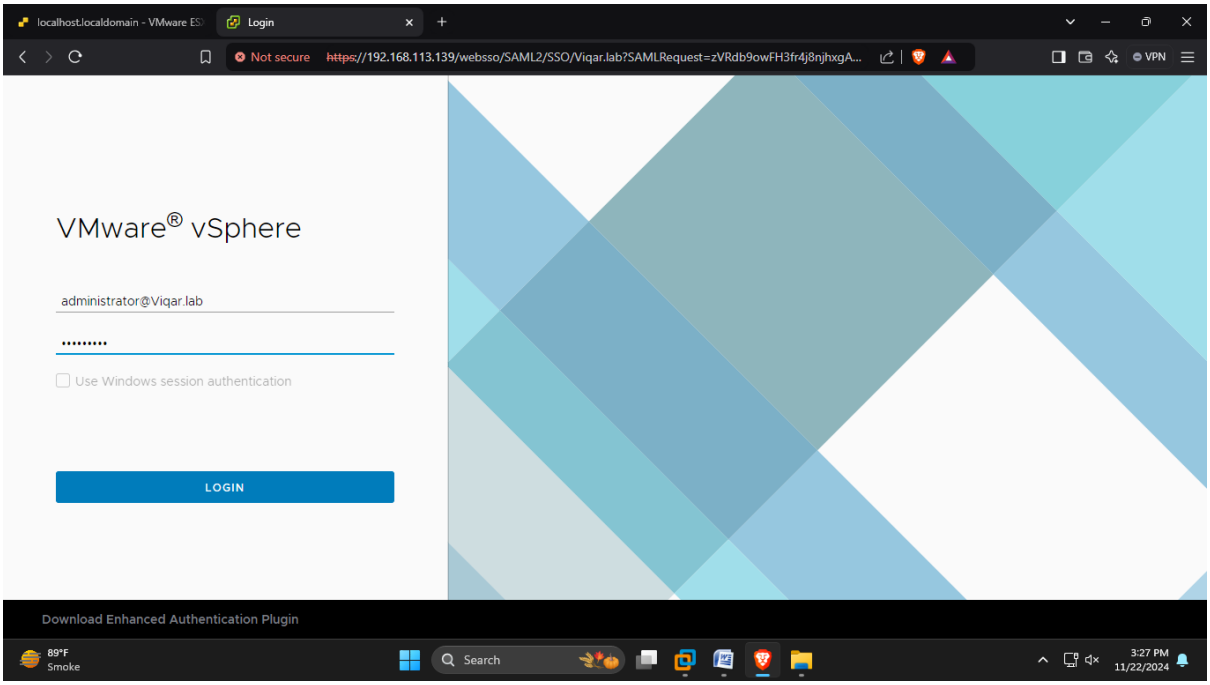
- Installation wil take around 10 minutes.



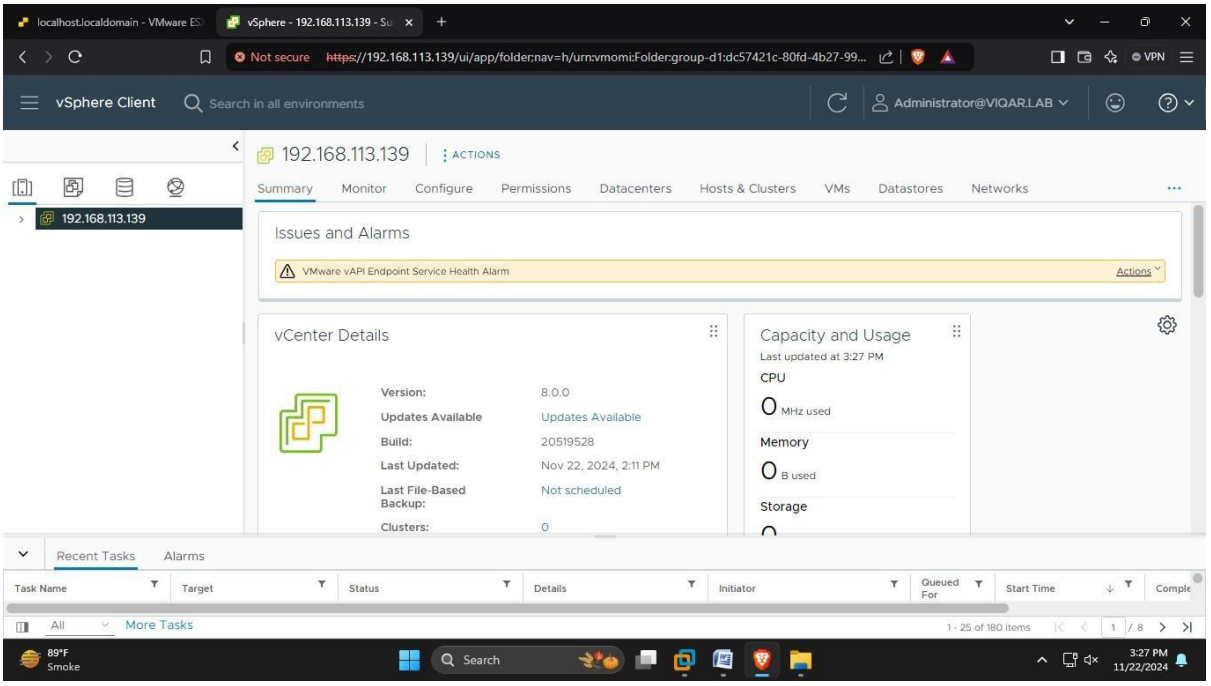
- Once installation Completed ,Click on Close.



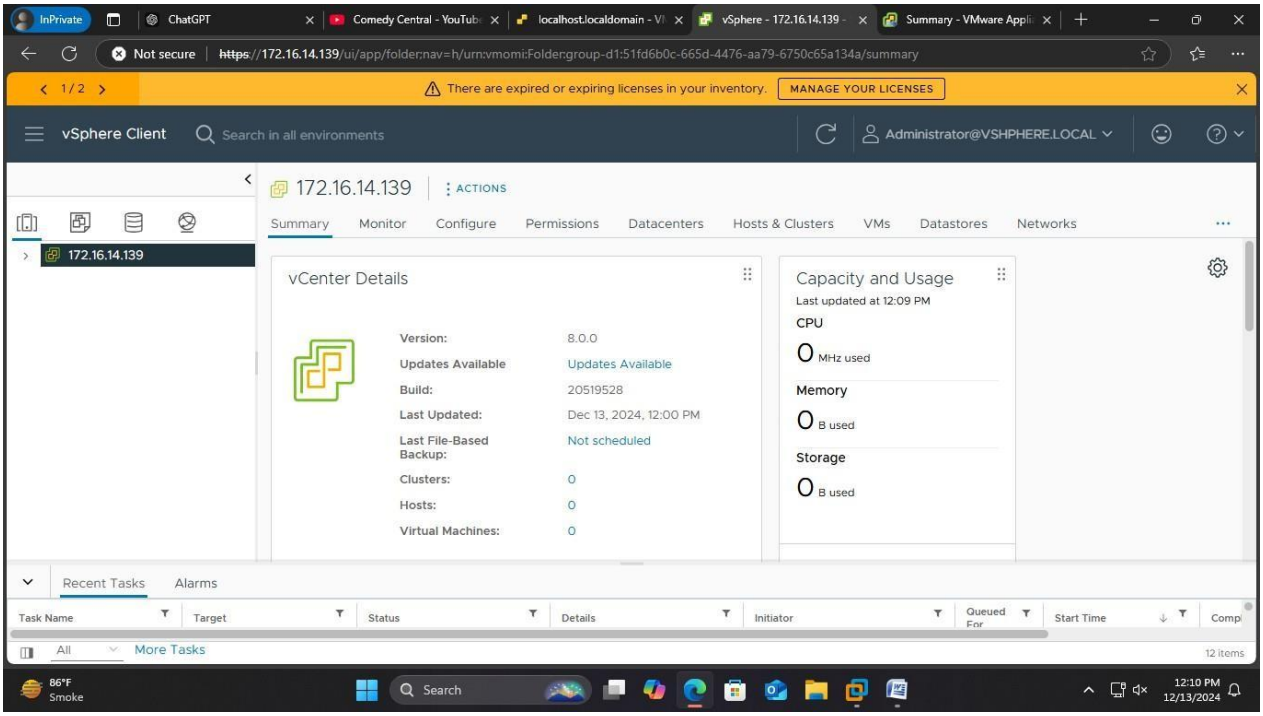
- Open a new tab in browser and type the ip which occurred on completion wizard.



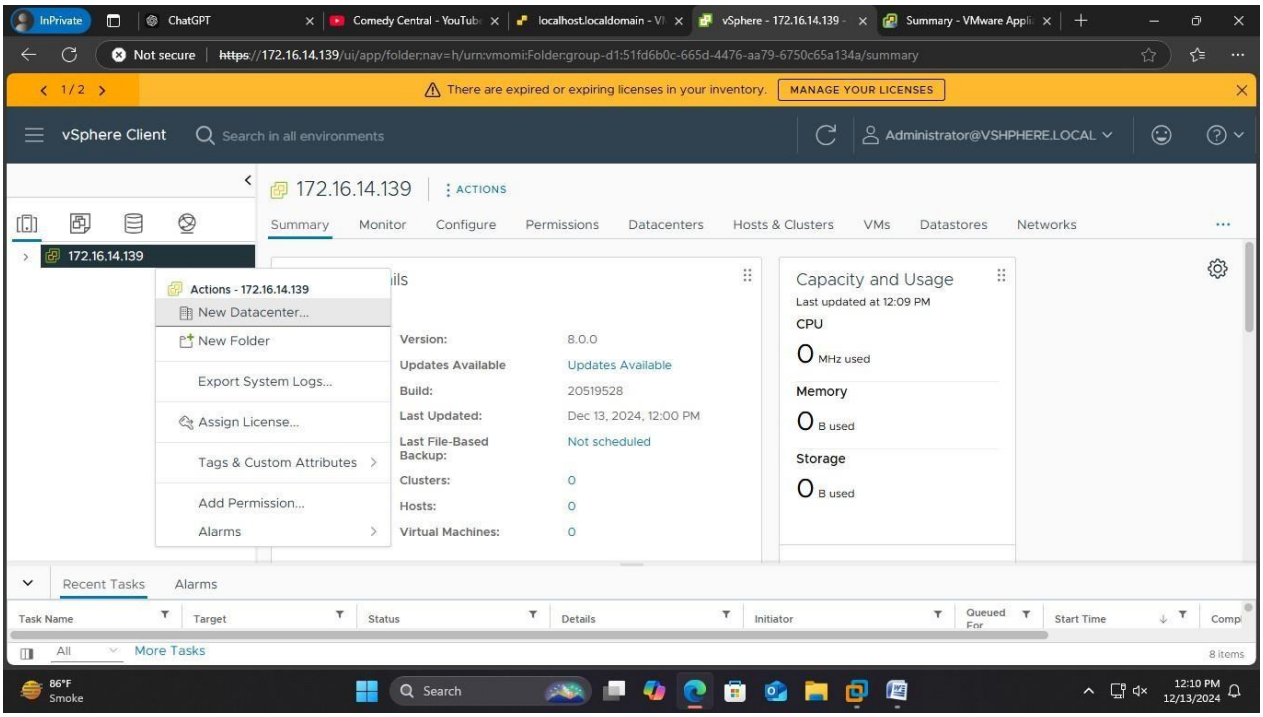
- Here! We Successfully Installed Vcenter.



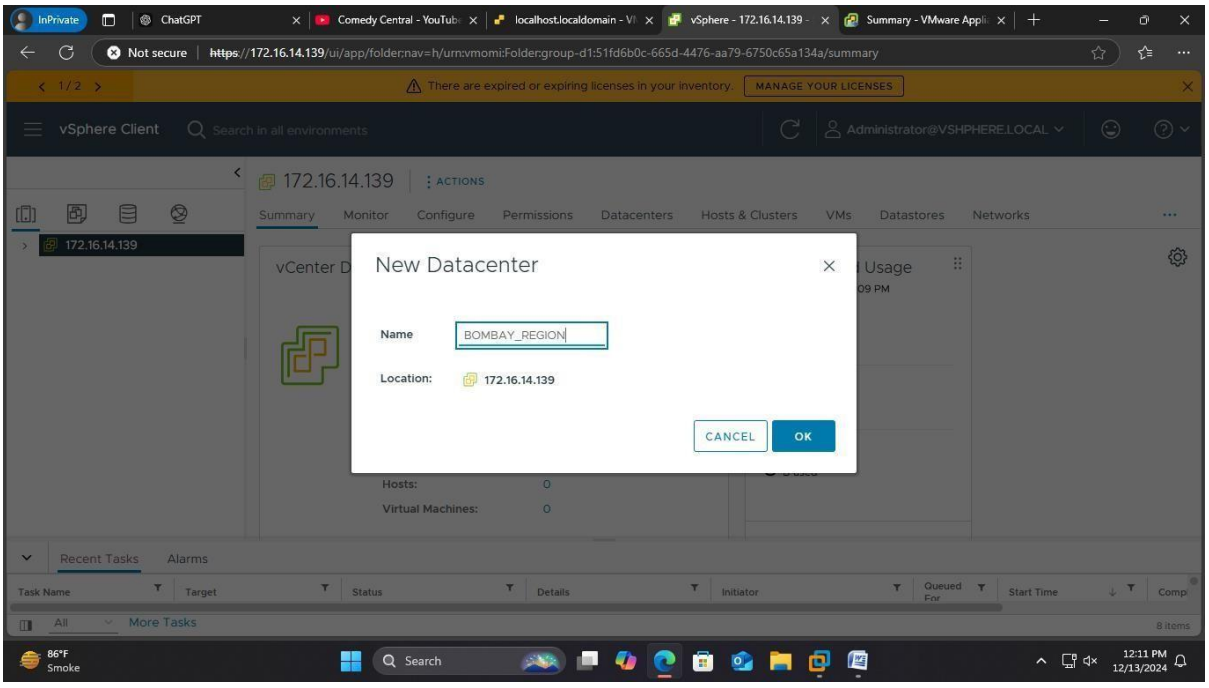
- Navigate to vcenter appliance via ip in browser



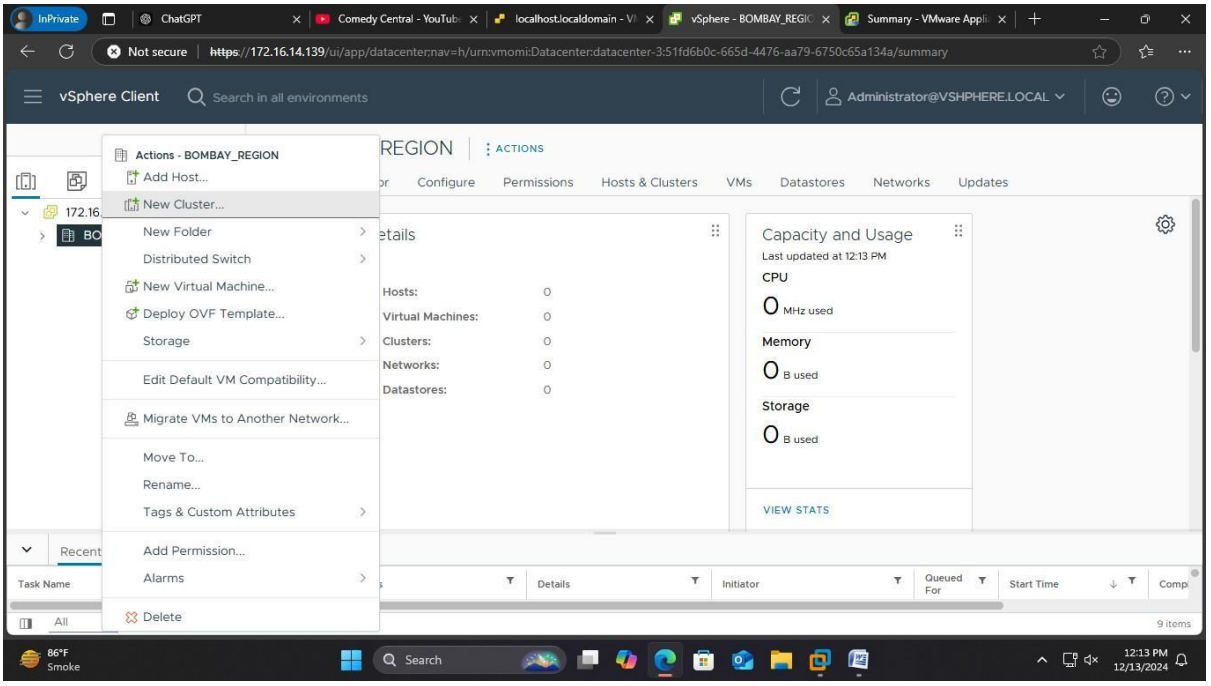
- Right click on ip address->New Datacenter



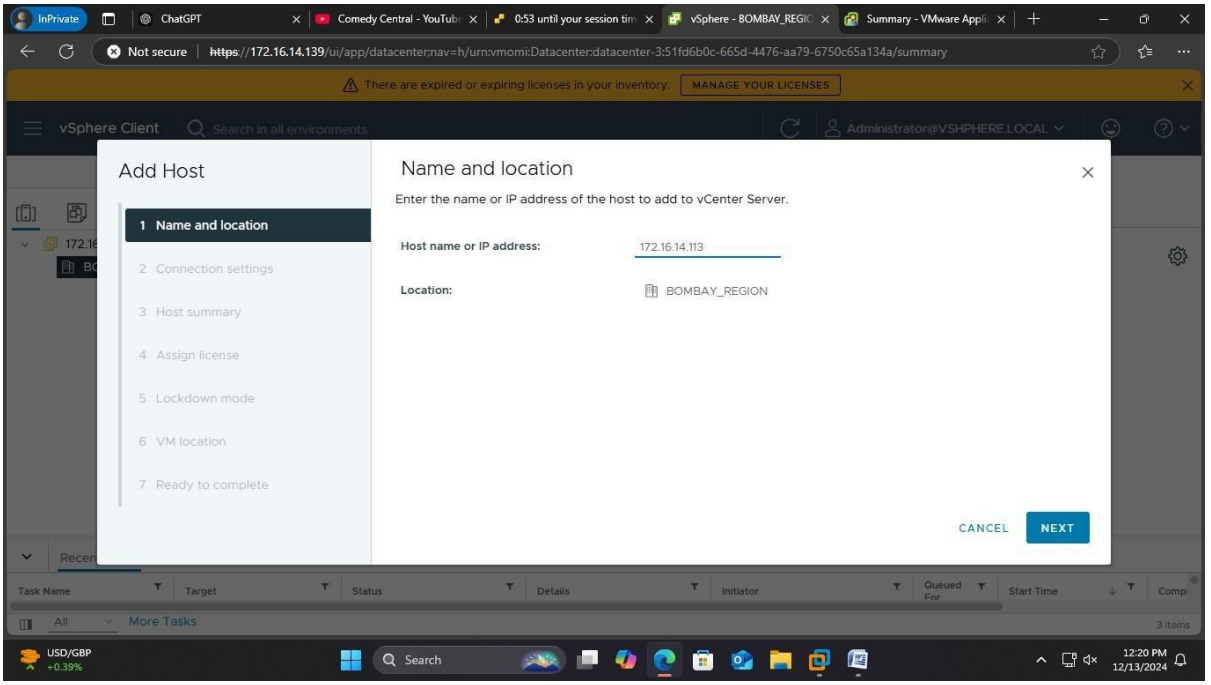
- Give your region name here it is BOMBAY_REGION



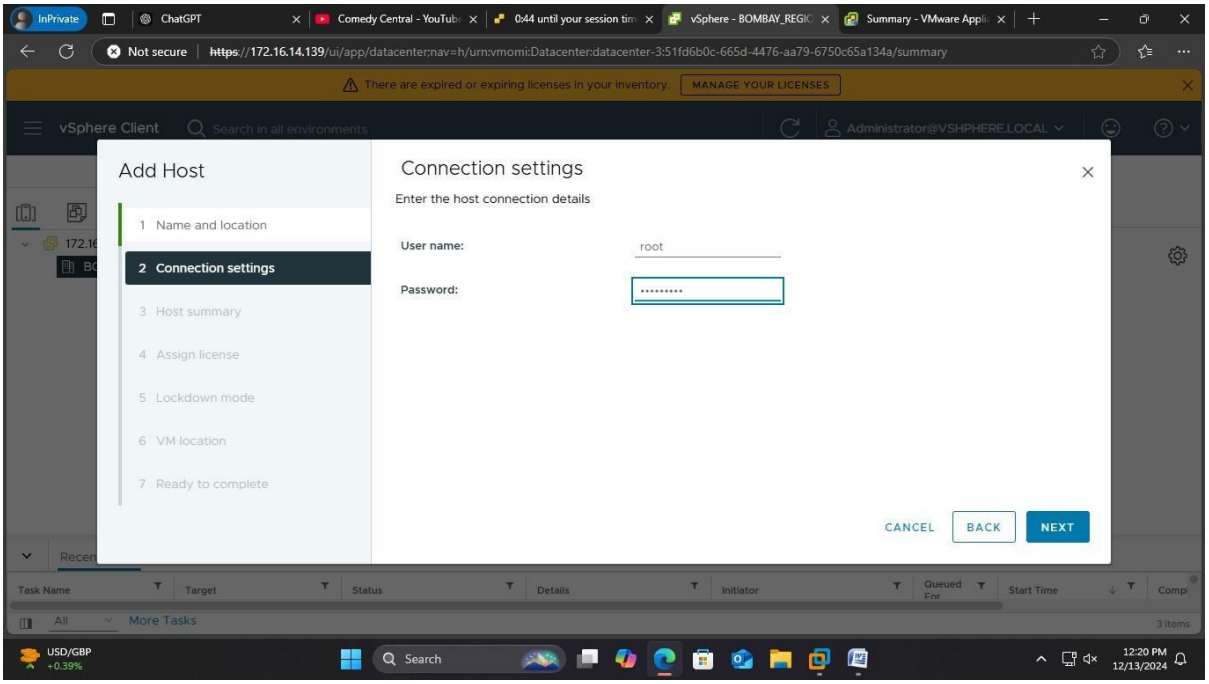
- A Region will be created right click on it select cluster



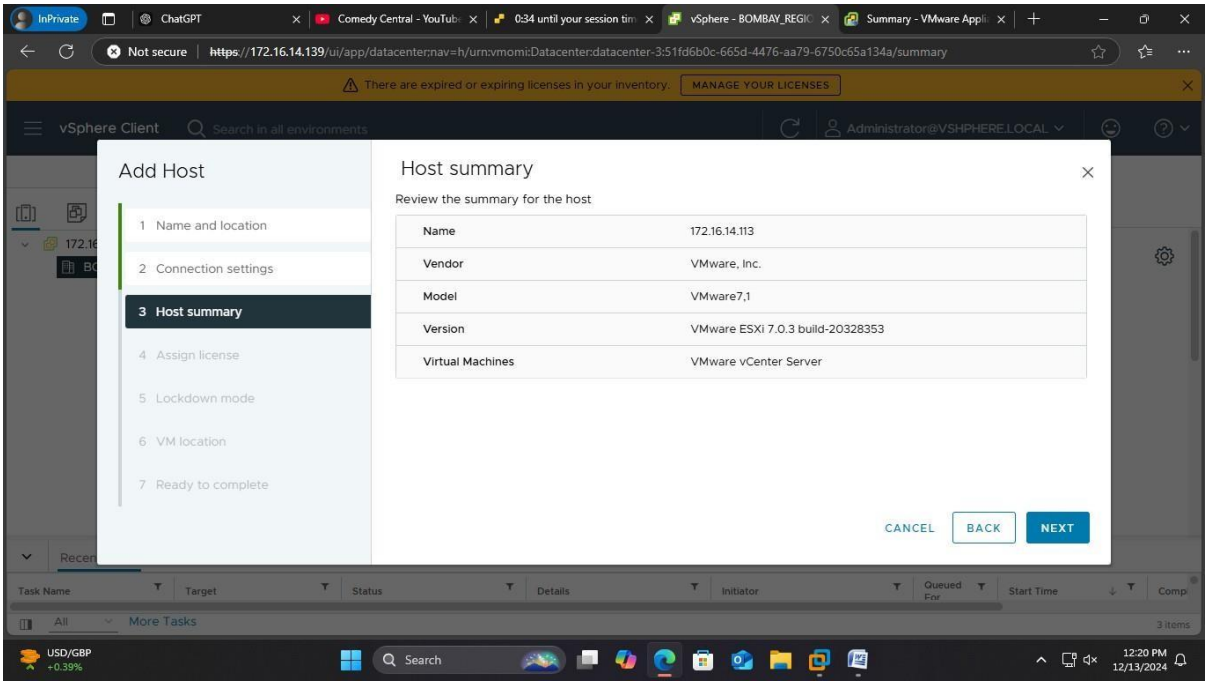
- Put ip address of your esxi server .



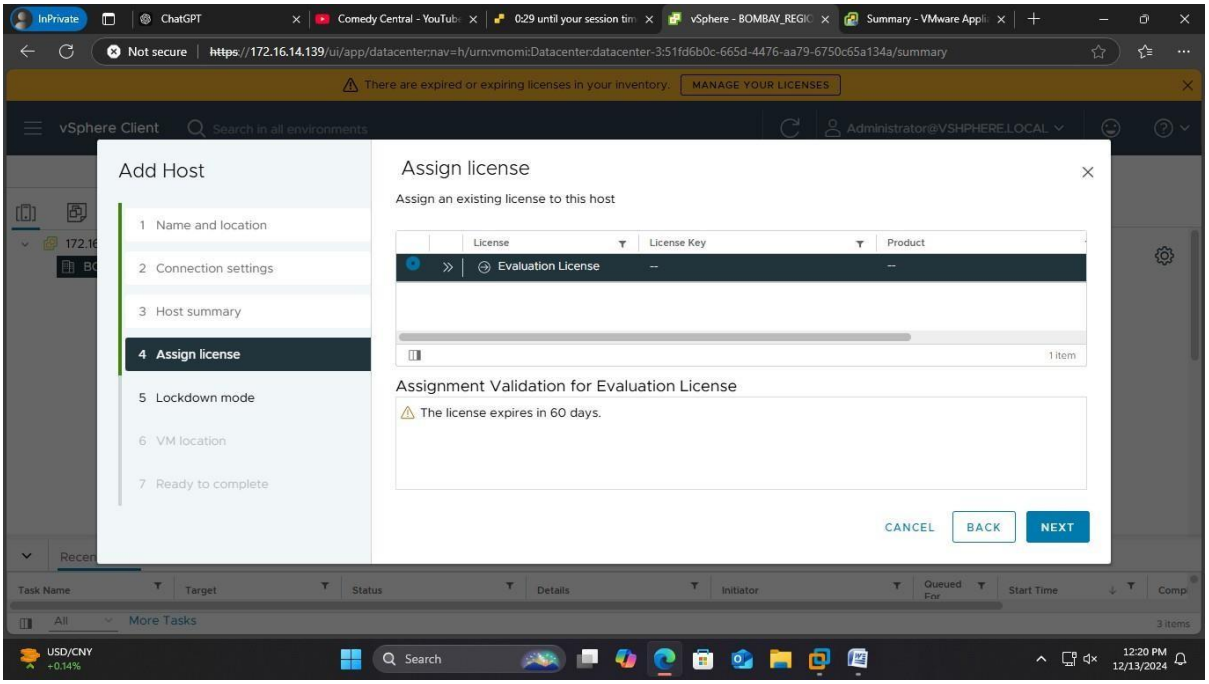
- Type root as username and give it a password



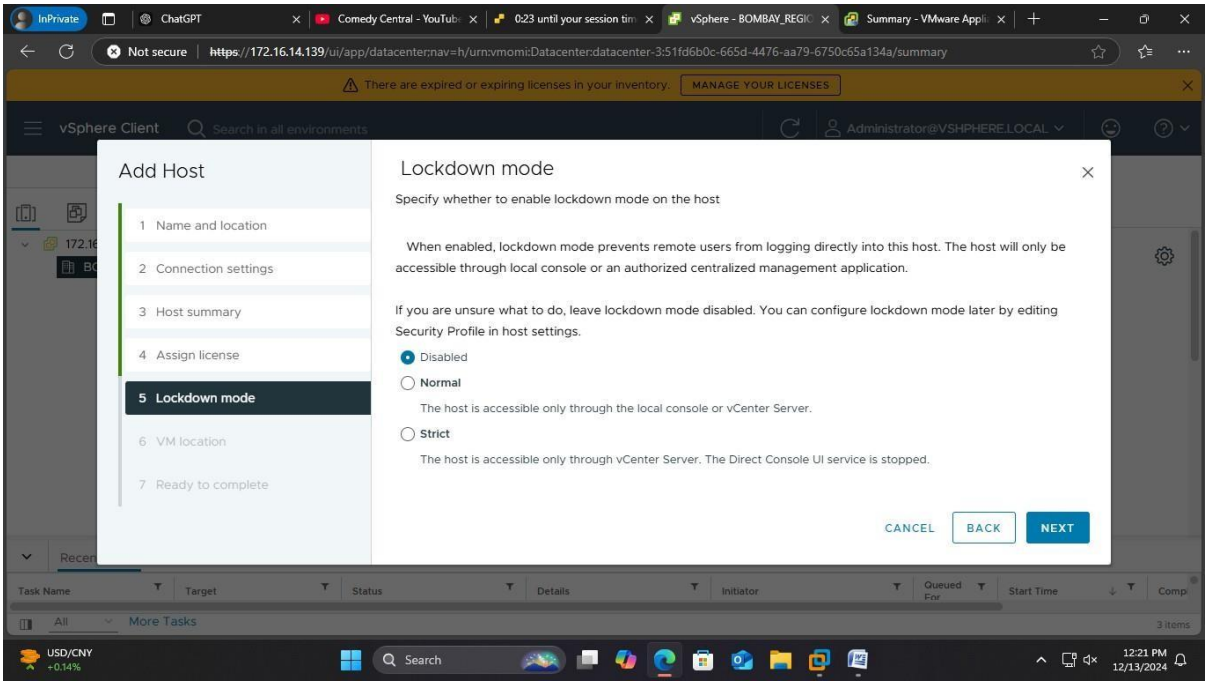
- Click on next



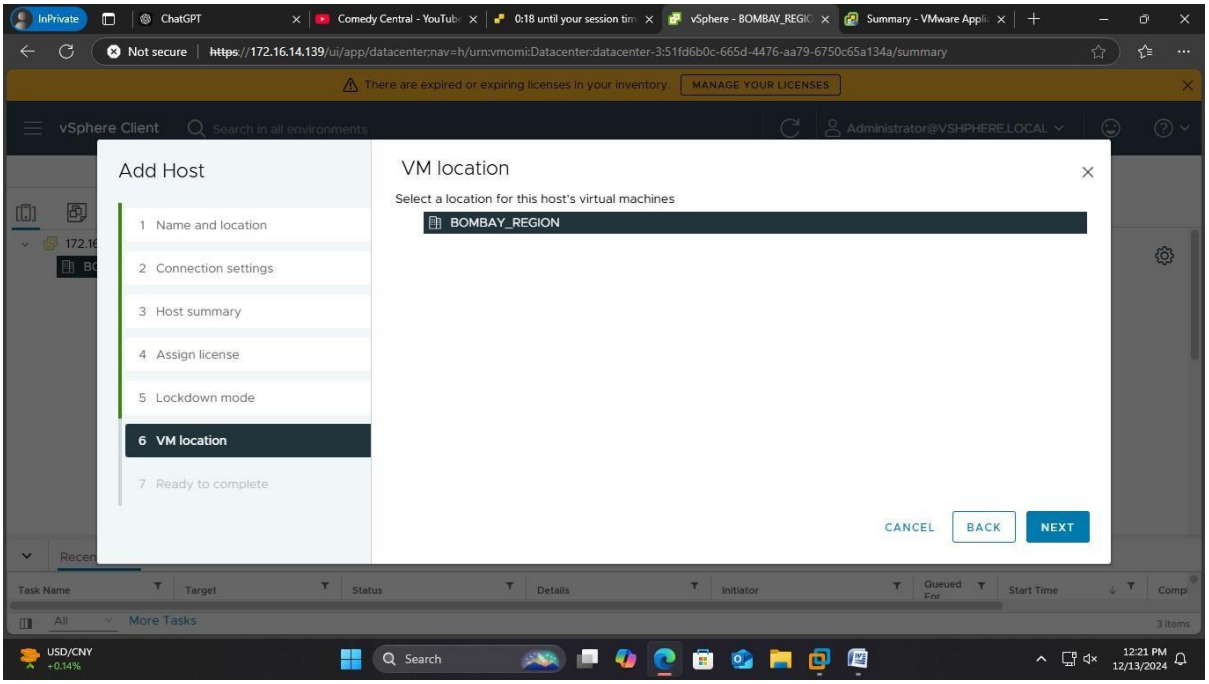
- Select Evaluation License Click on next.



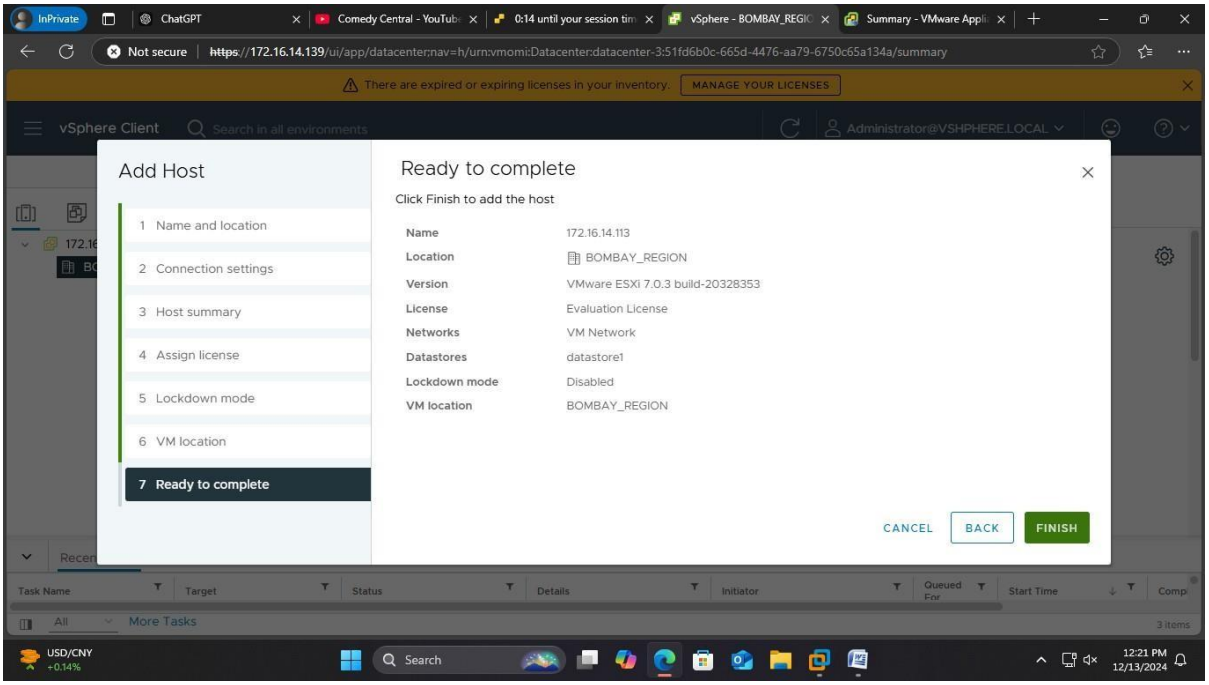
- Select Disabled and Click on next.



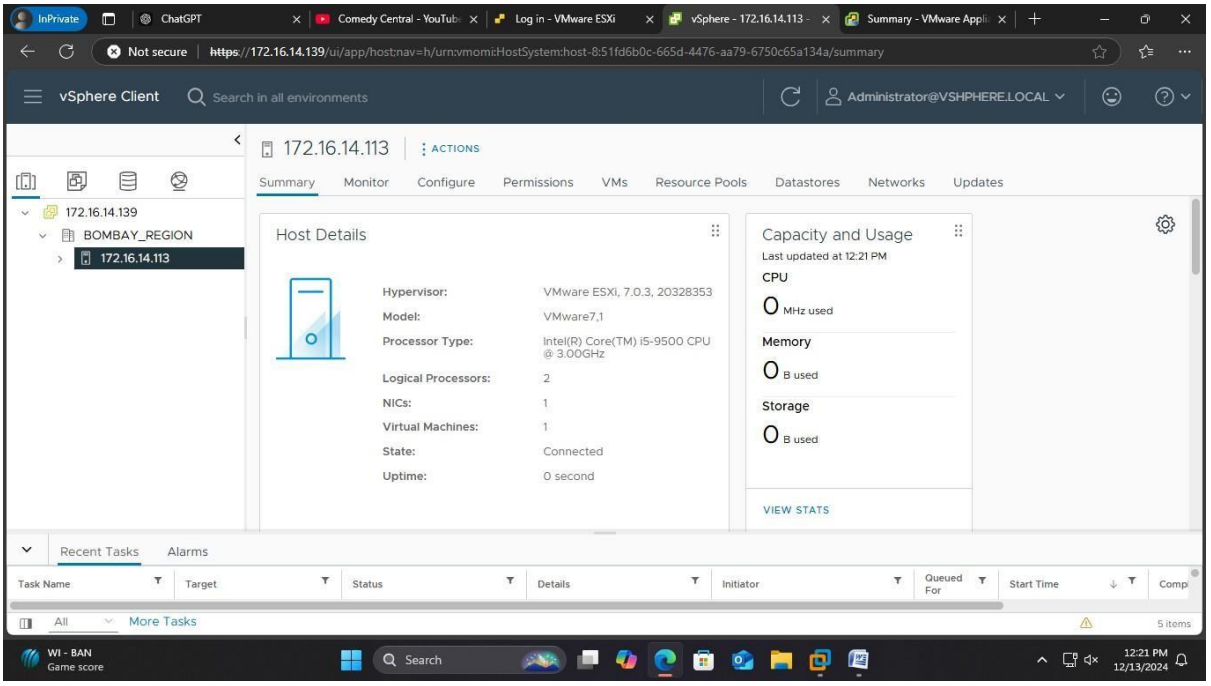
- Select Region which you created previously here it is BOMBAY_REGION.



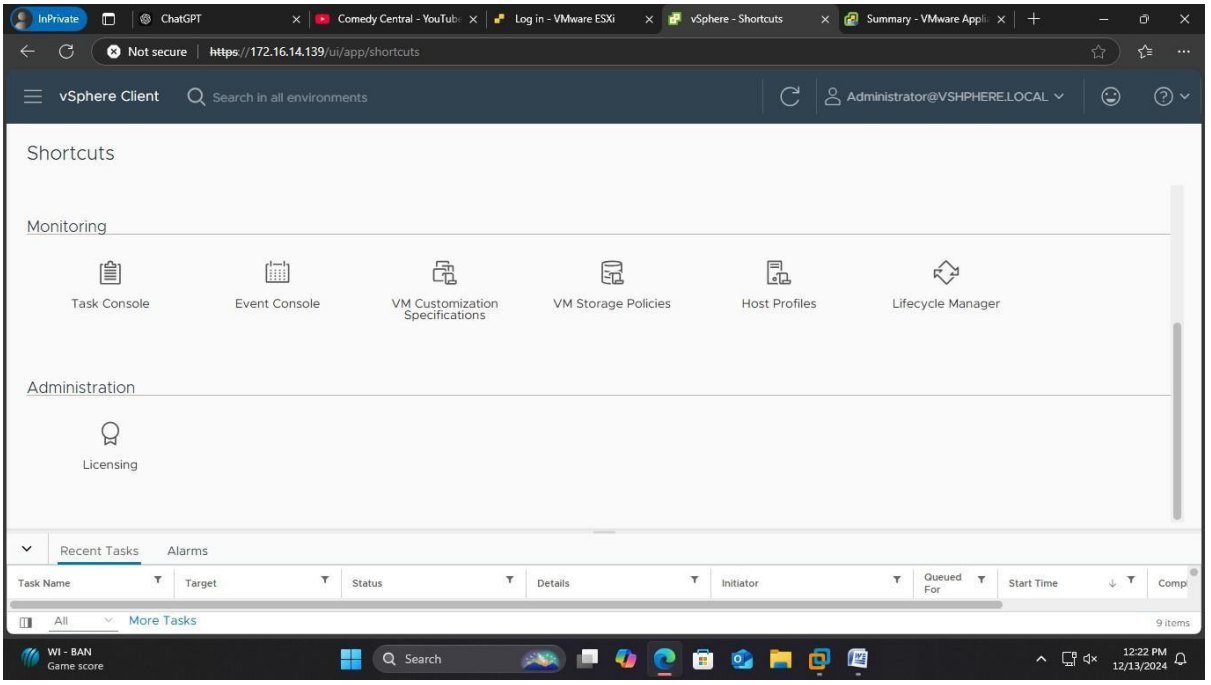
- Click on next.



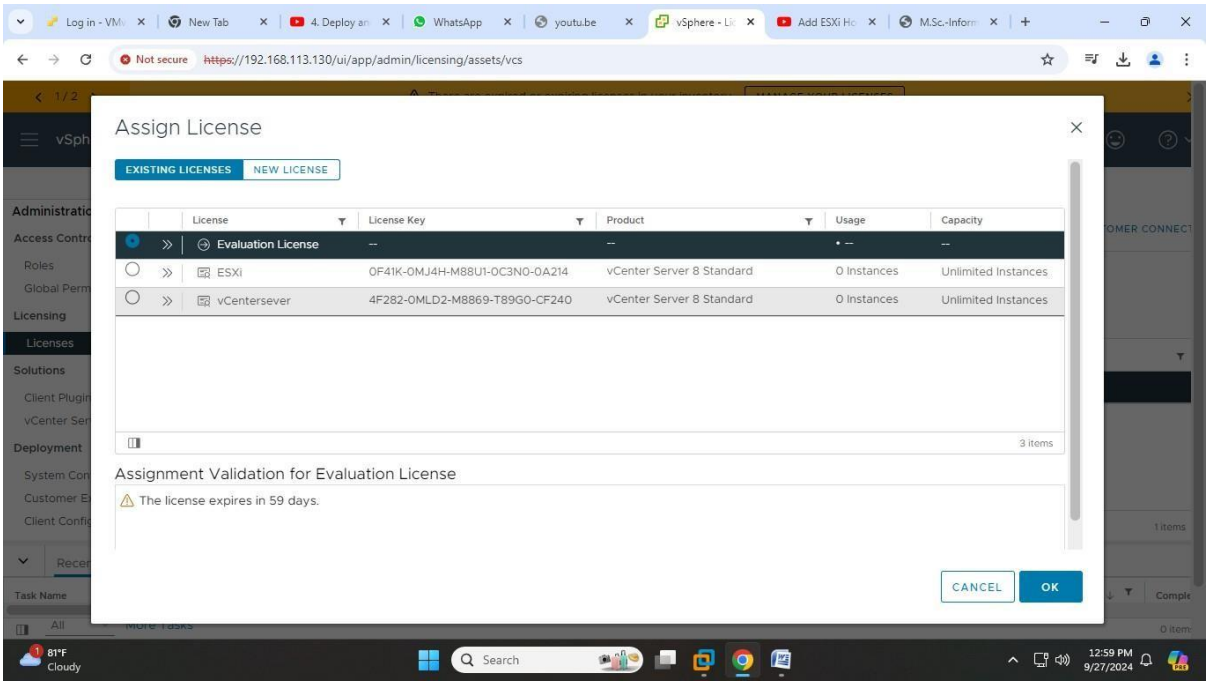
- Navigating vcenter.



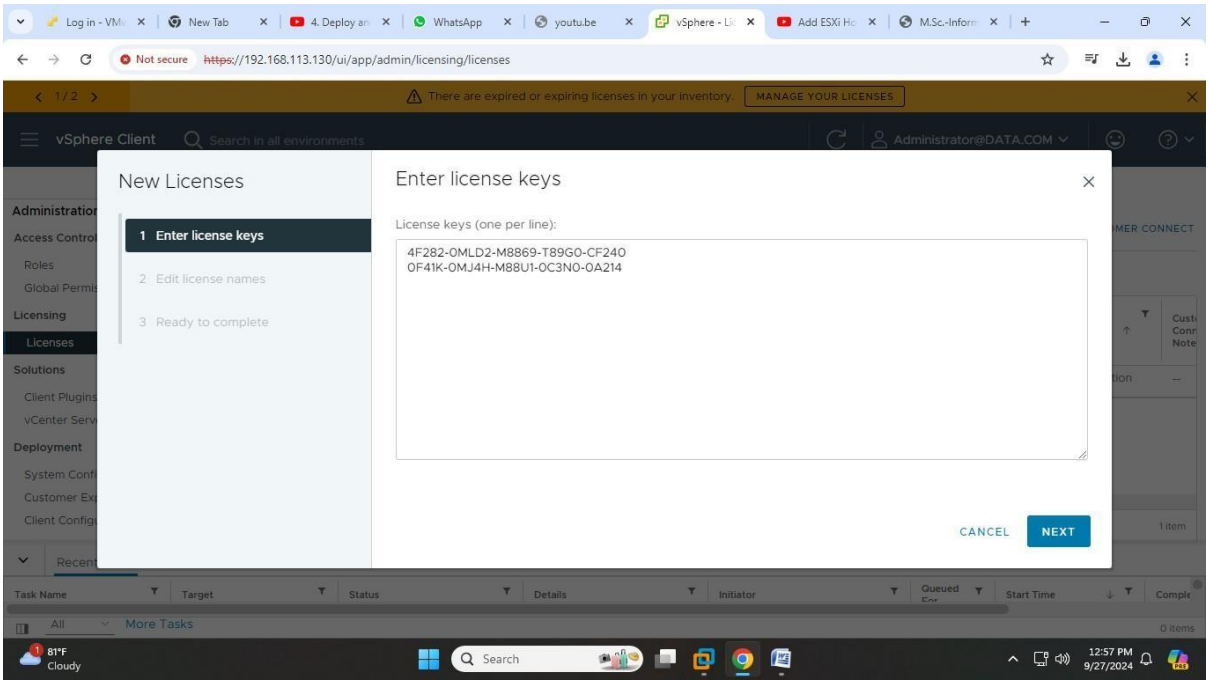
- Go to home page by clicking on vSphere Client and Select Licensing.



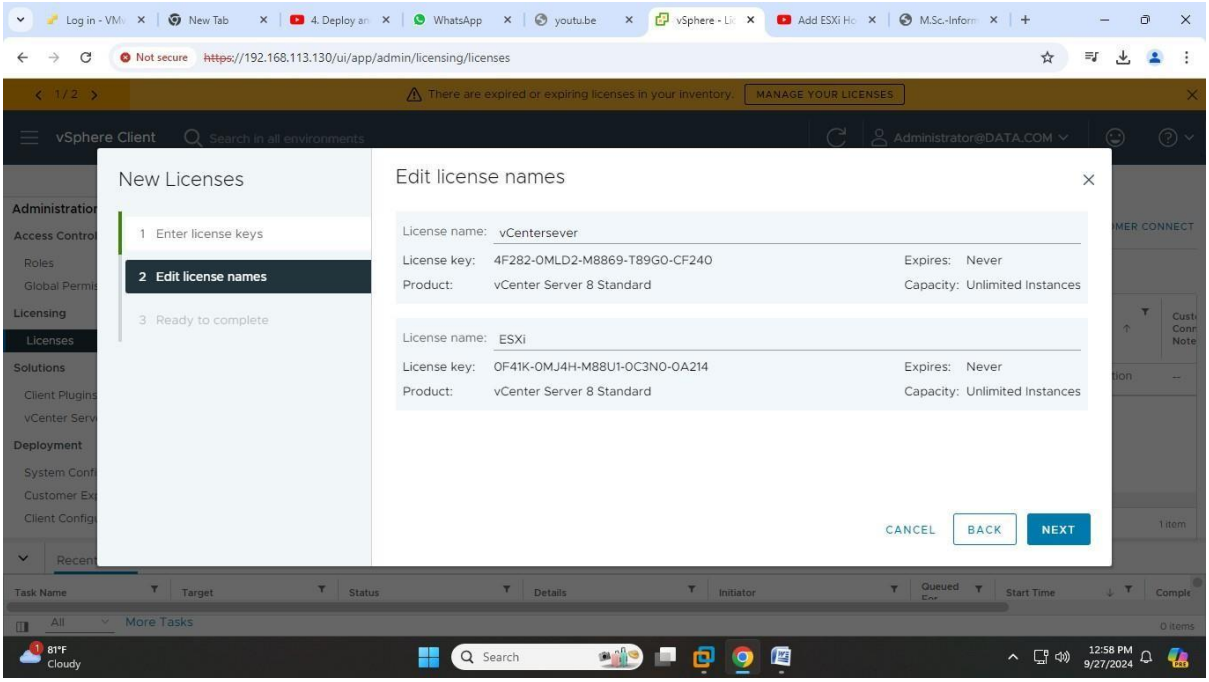
- Assign to evaluation.



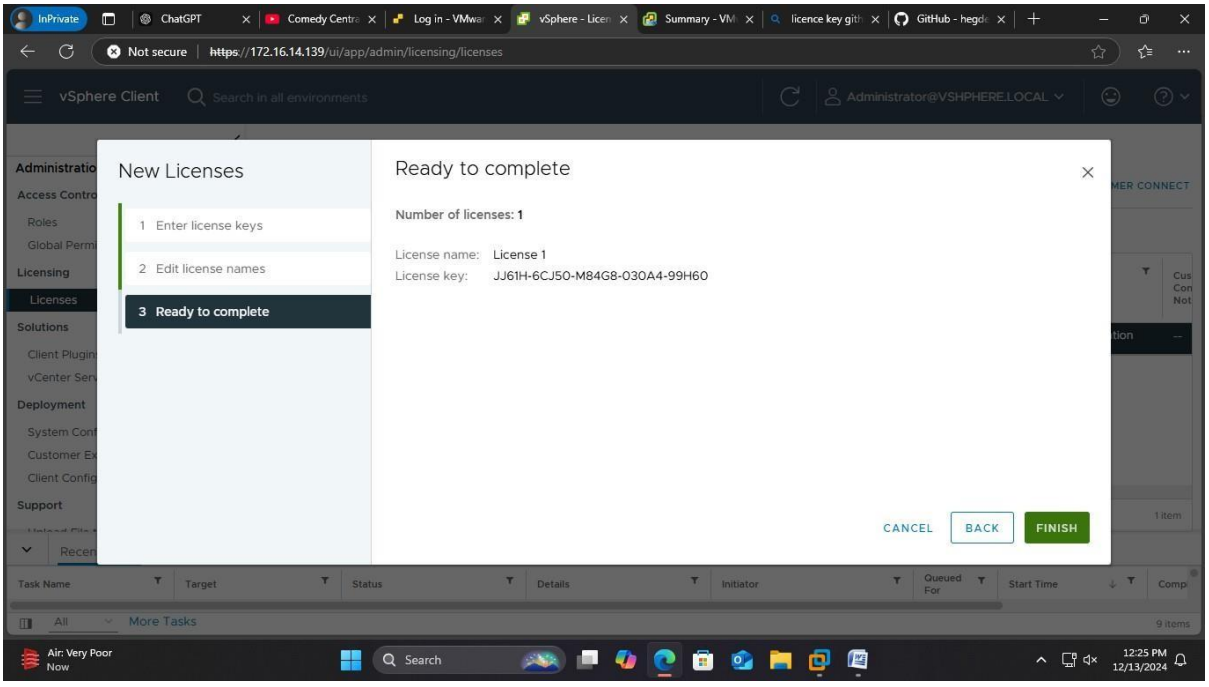
- Type Your License key.



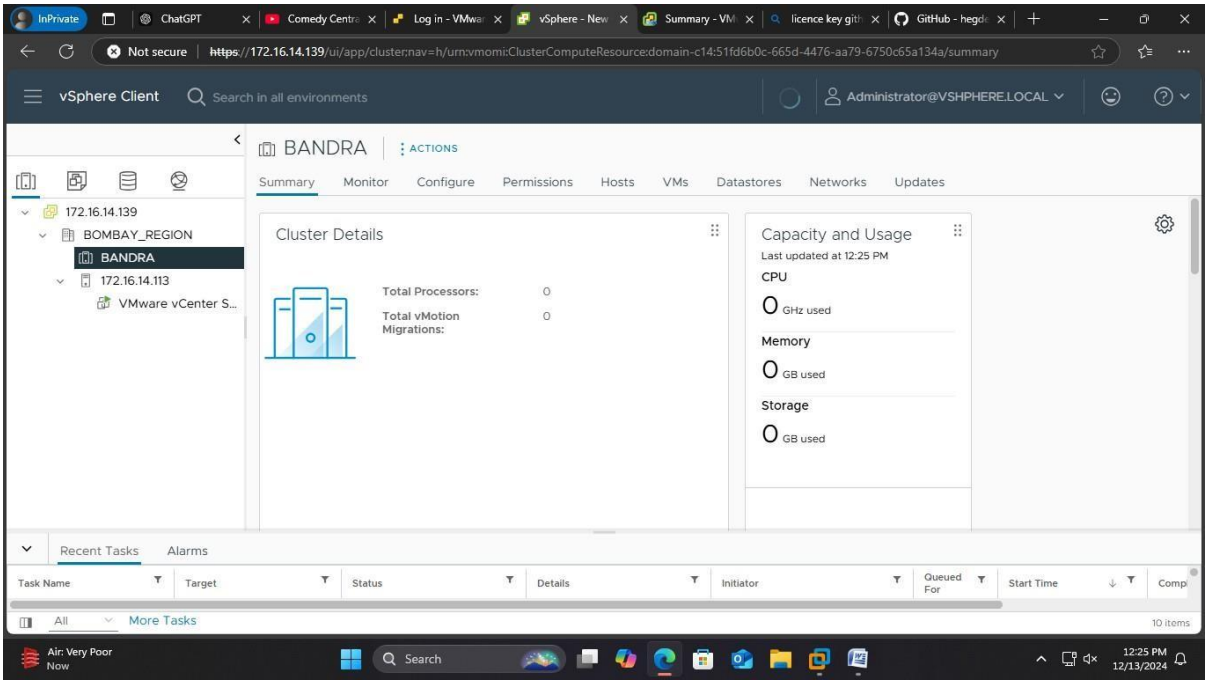
- click on next.



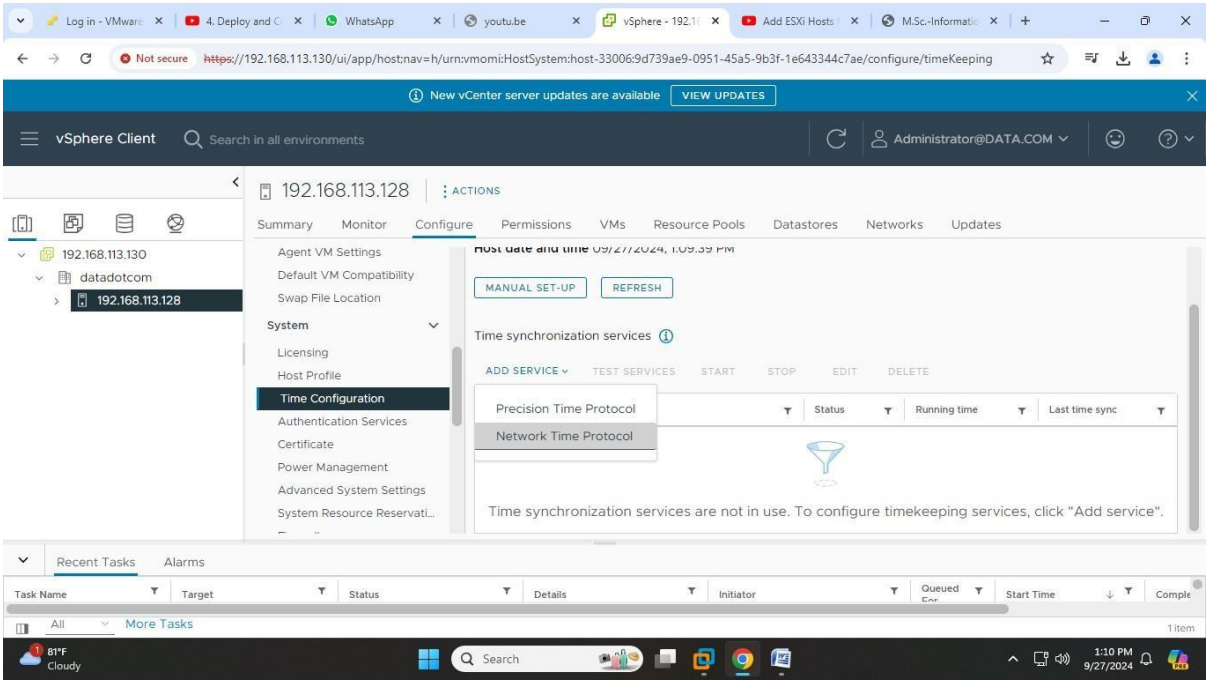
- Click on finish.



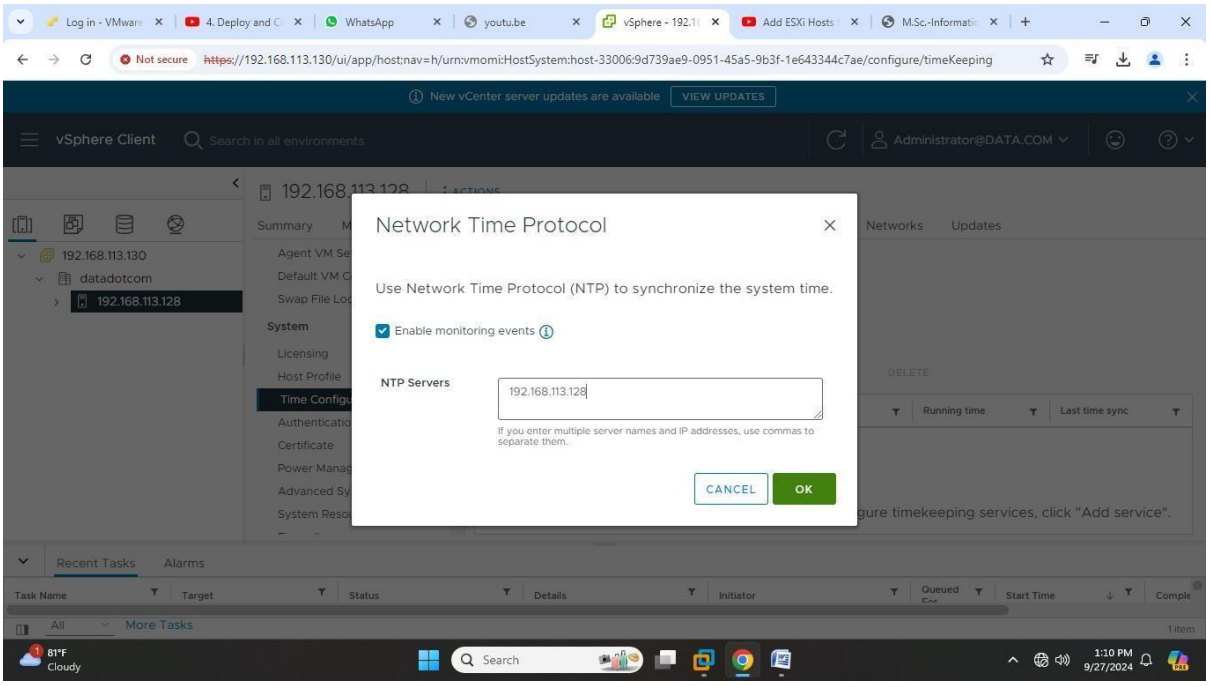
- Now we have created cluster and connected to a host and license key activated



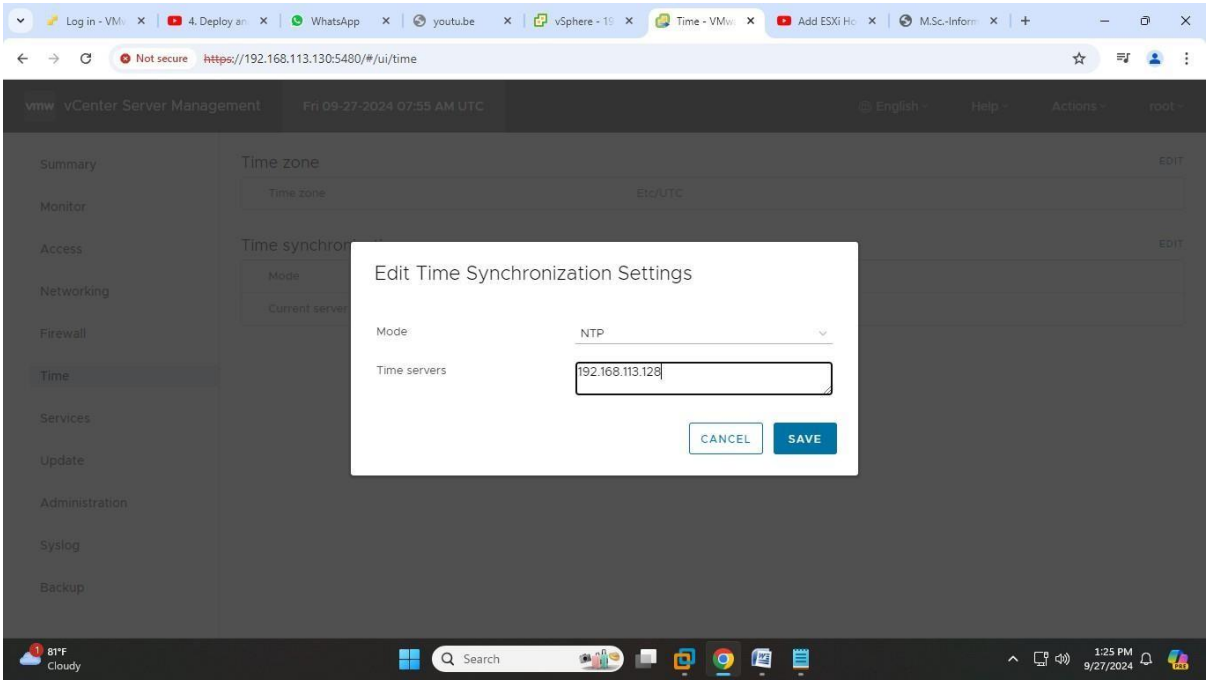
- Navigate to host configure->Time Configuration Click on ADD SERVICE



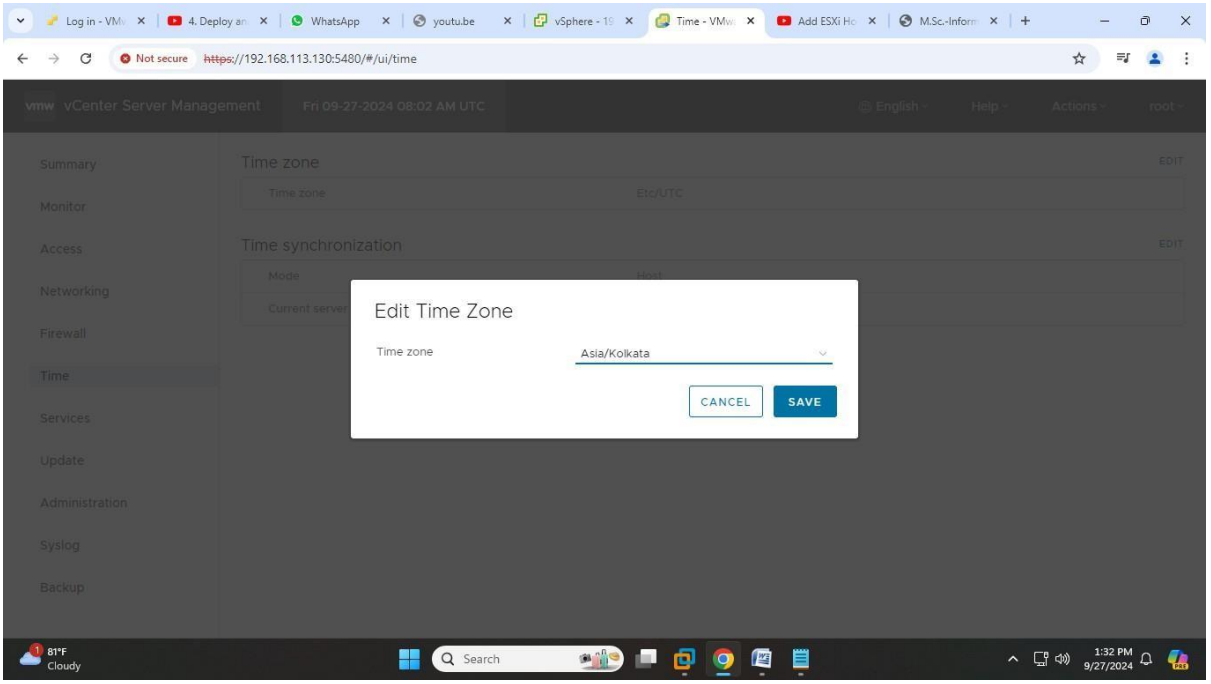
- Type esxi ip address.



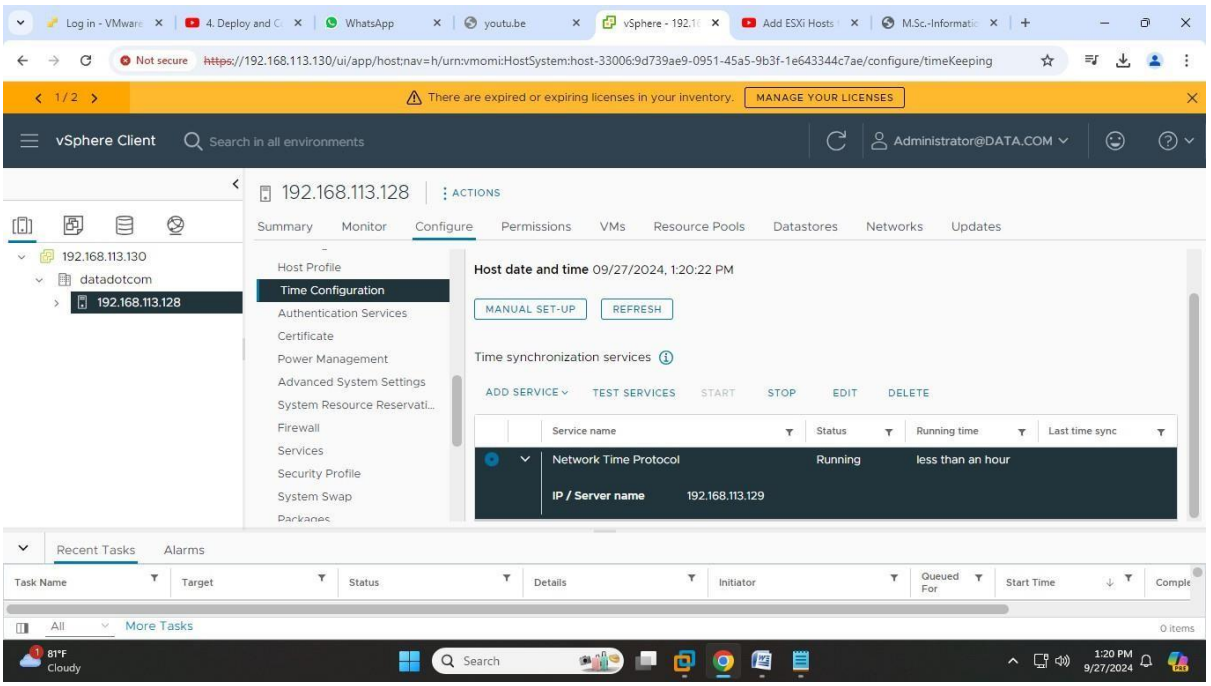
- In Vcenter management in new tab at port 5480 go to time and put ip address



- Select time zone.



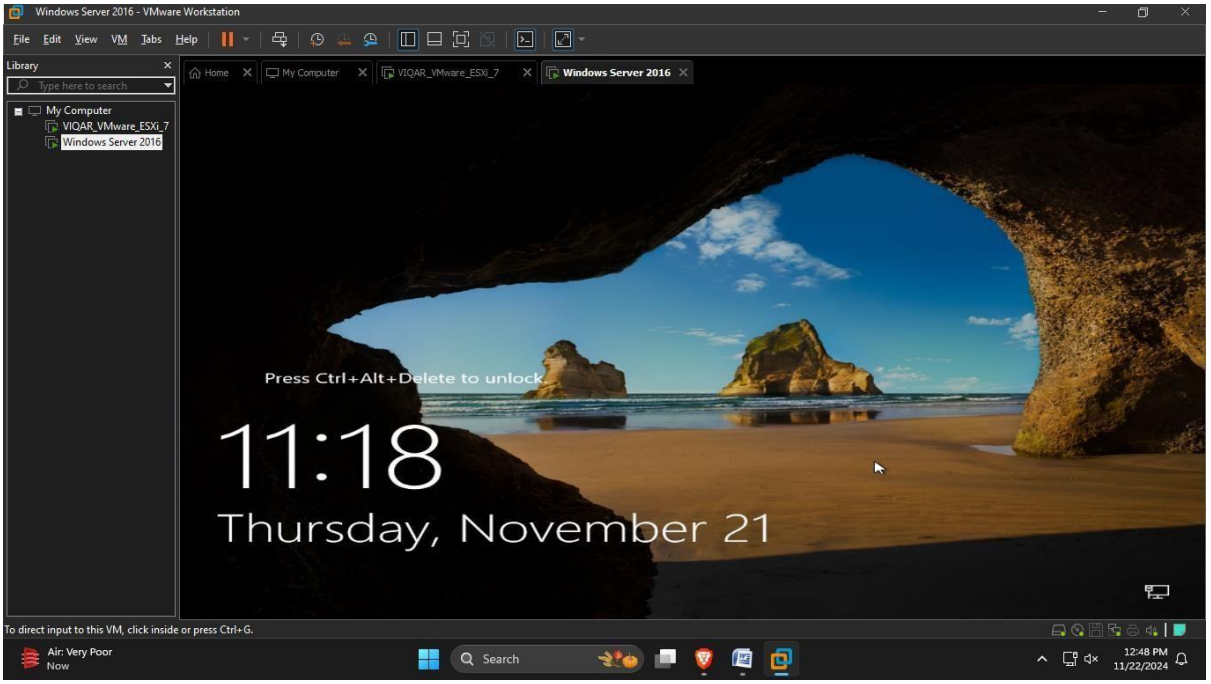
- Network Time Protocol is showing



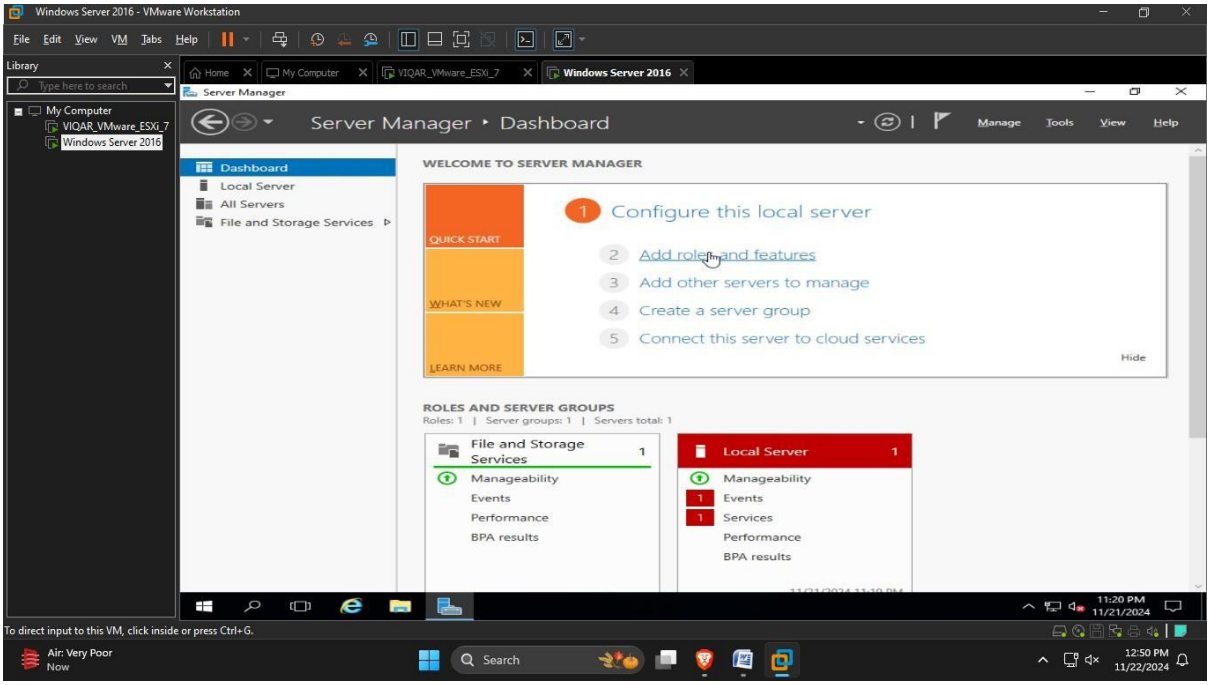
PRACTICAL 3

AIM:- Users, Groups, and Permissions.

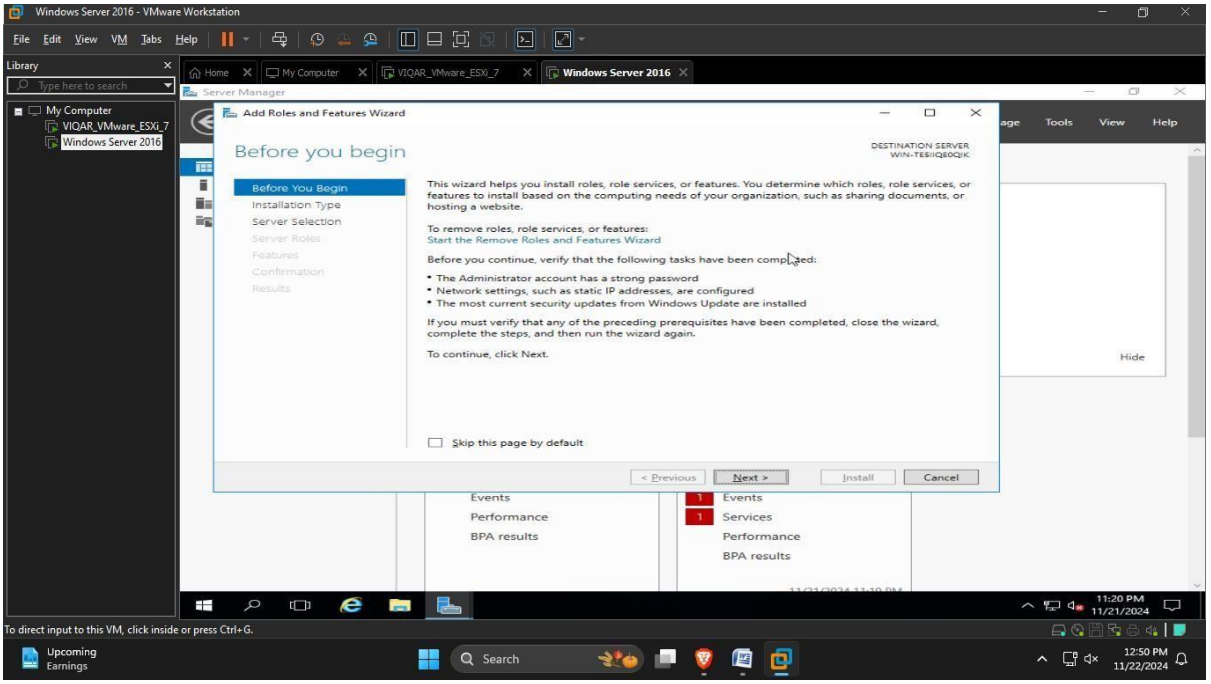
- Windows will start Type your Password.



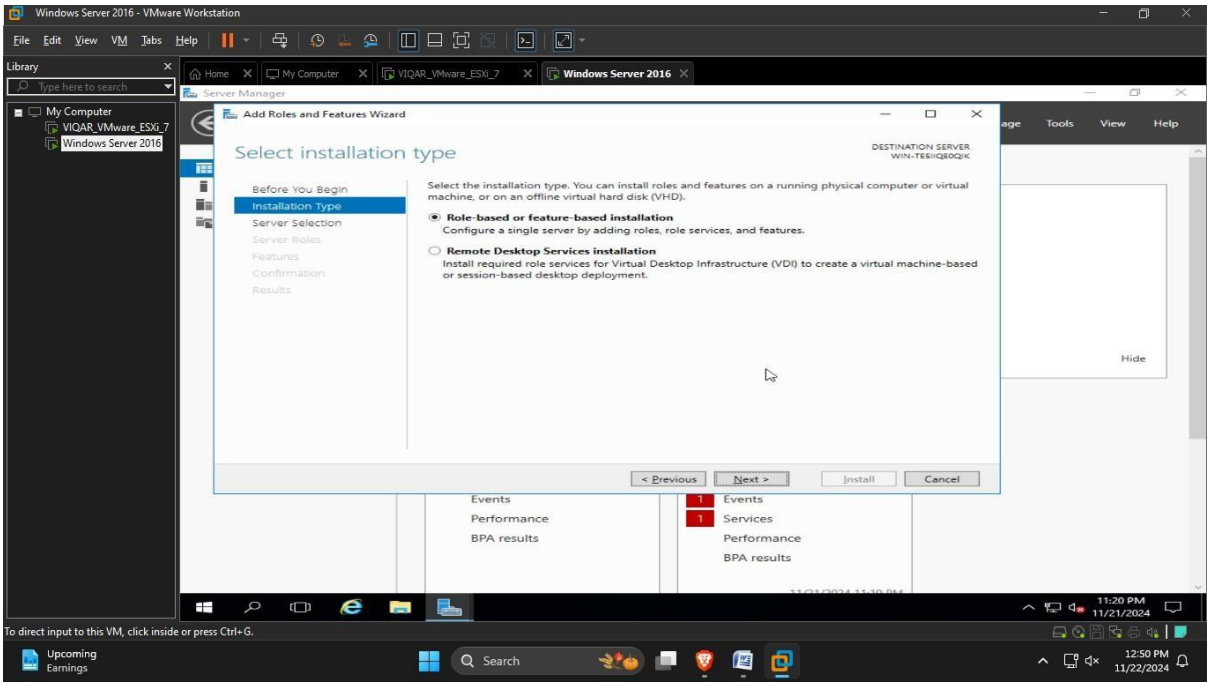
- Open server manager Click on add roles and feature.



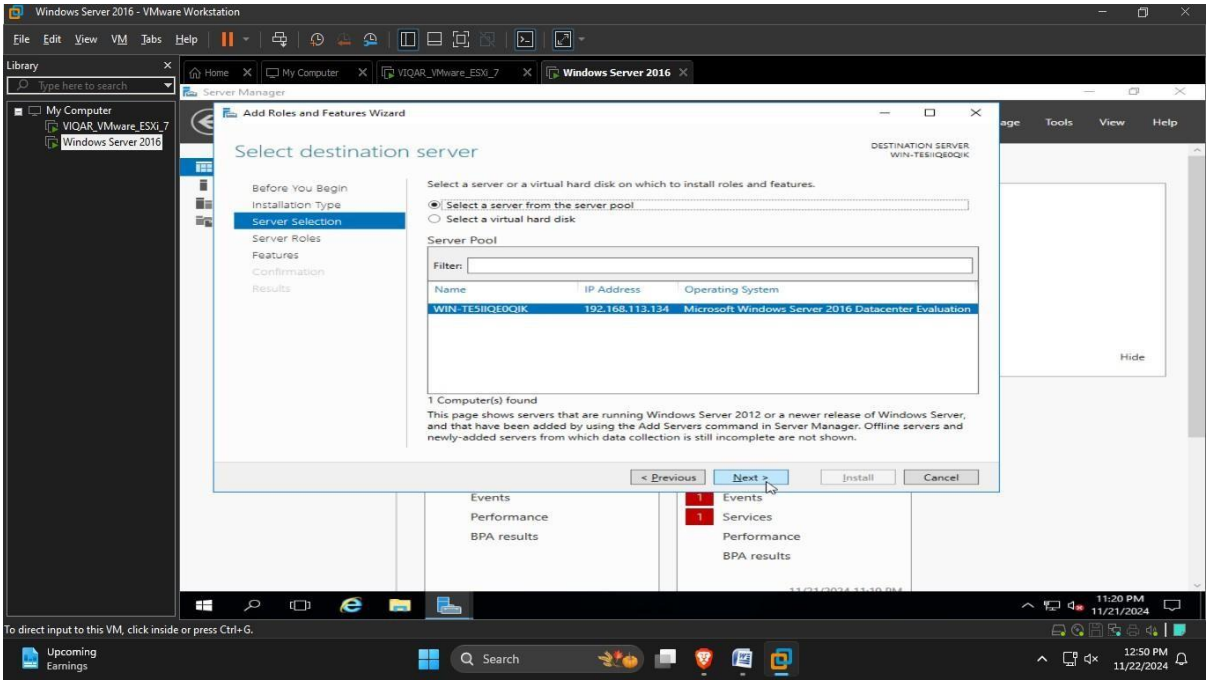
- Click on next.



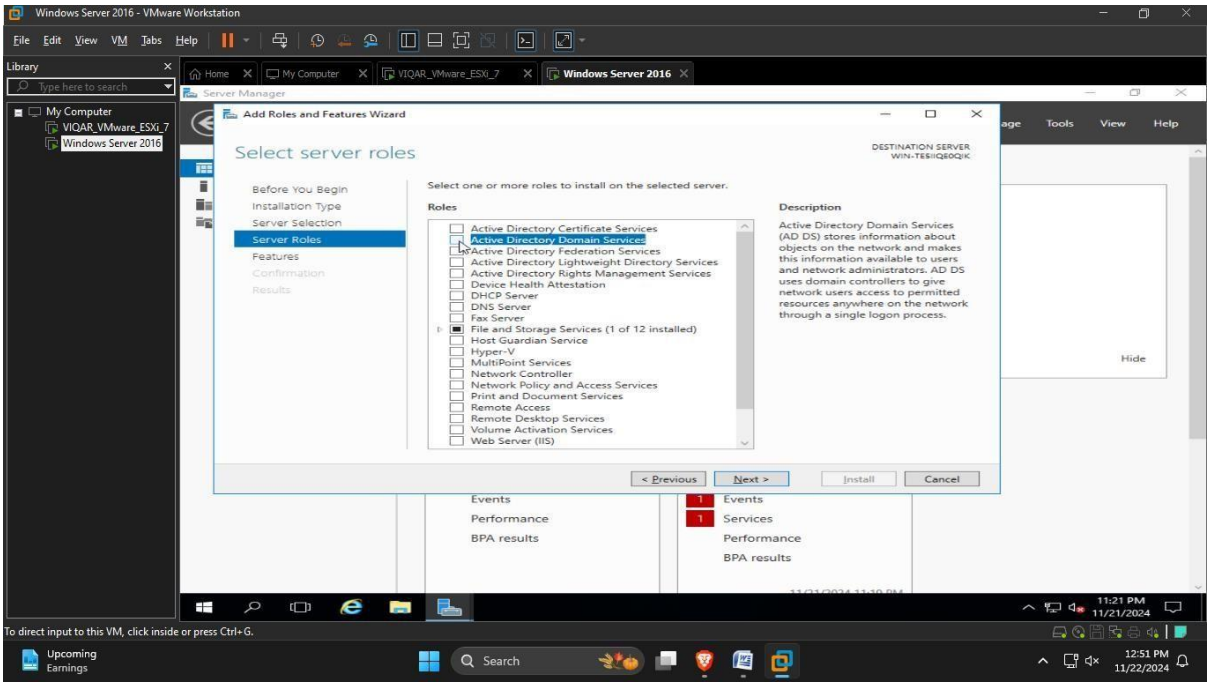
- Select role based or feature based installation.



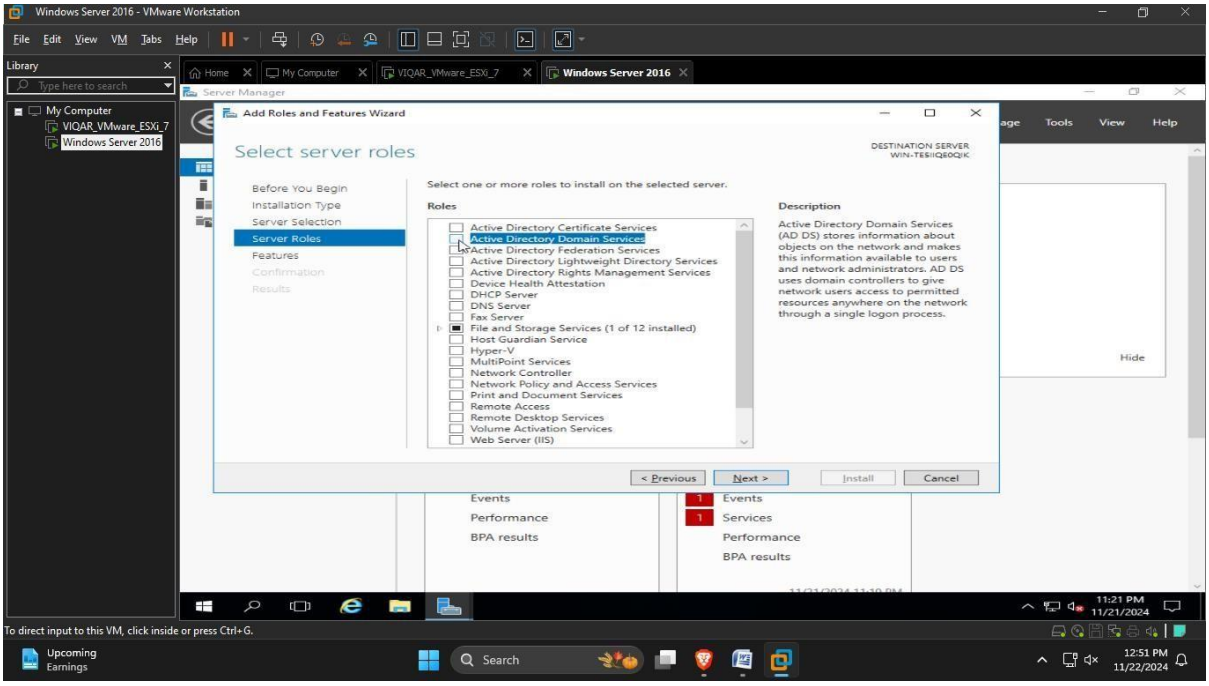
- Select server from server pool.



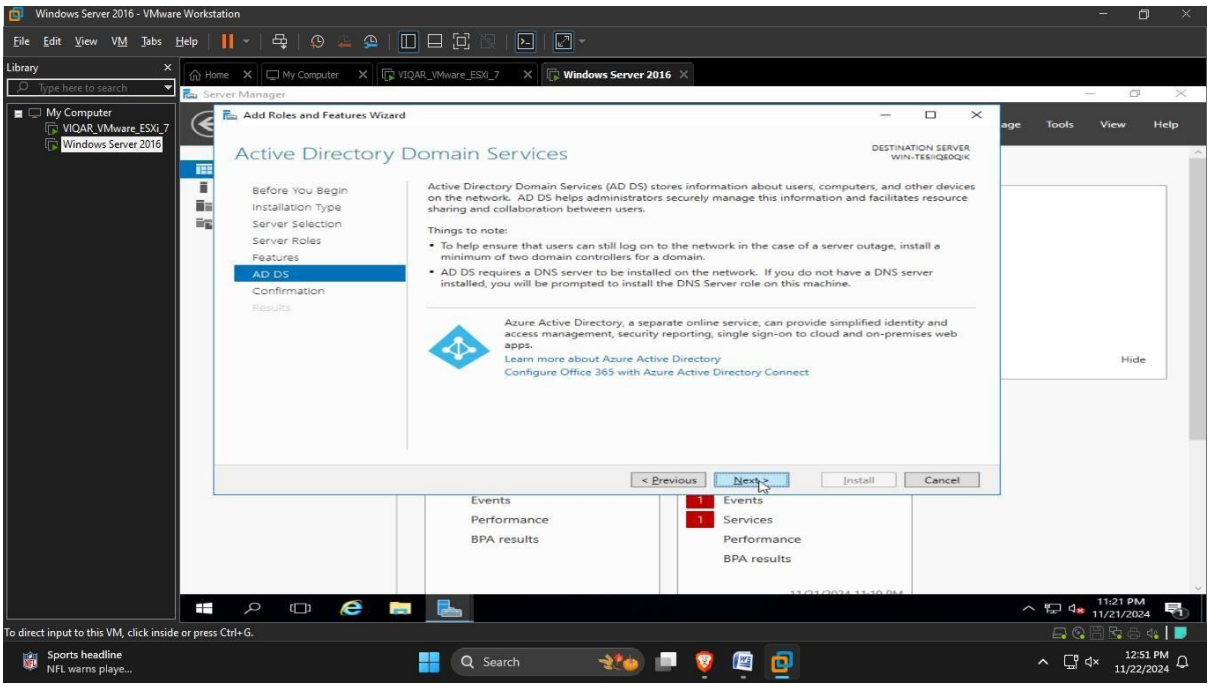
- Select active directory domain service.



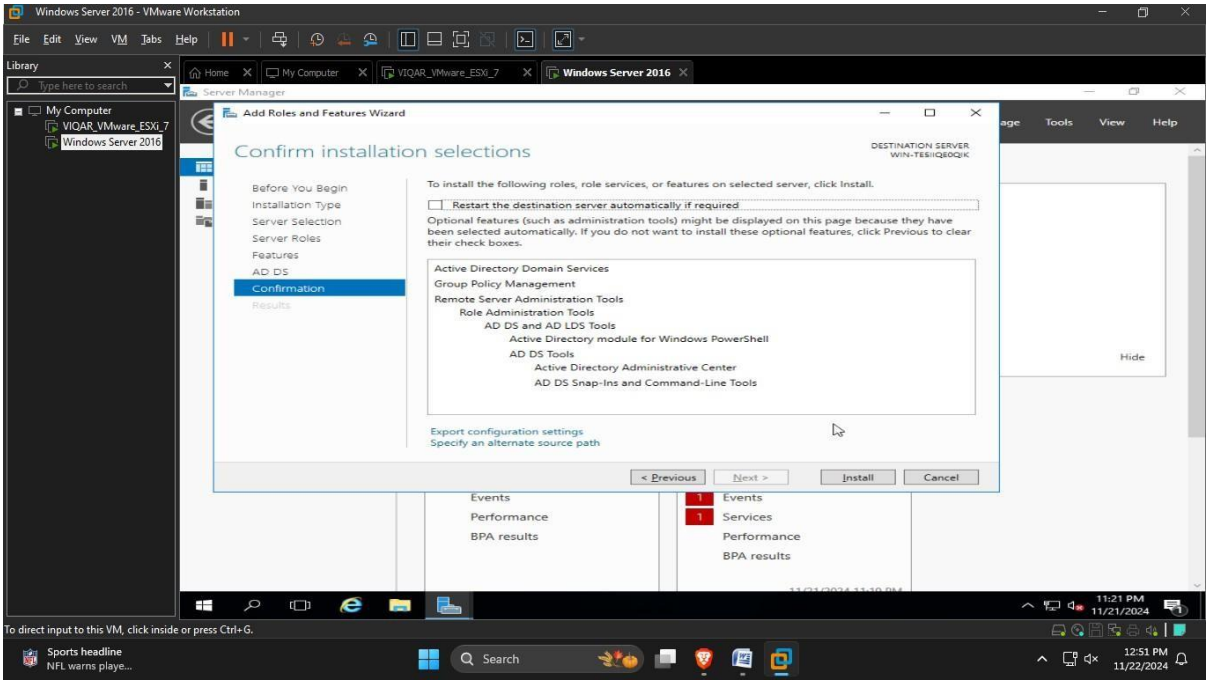
- Click on next .



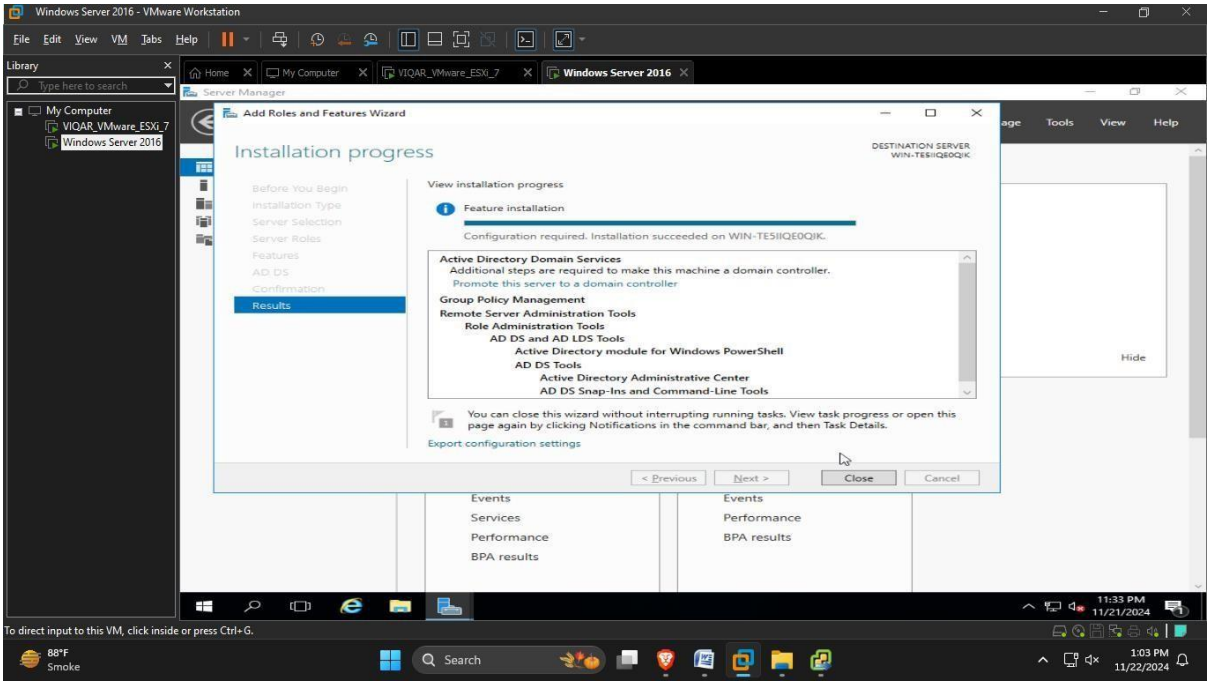
- Click on next.



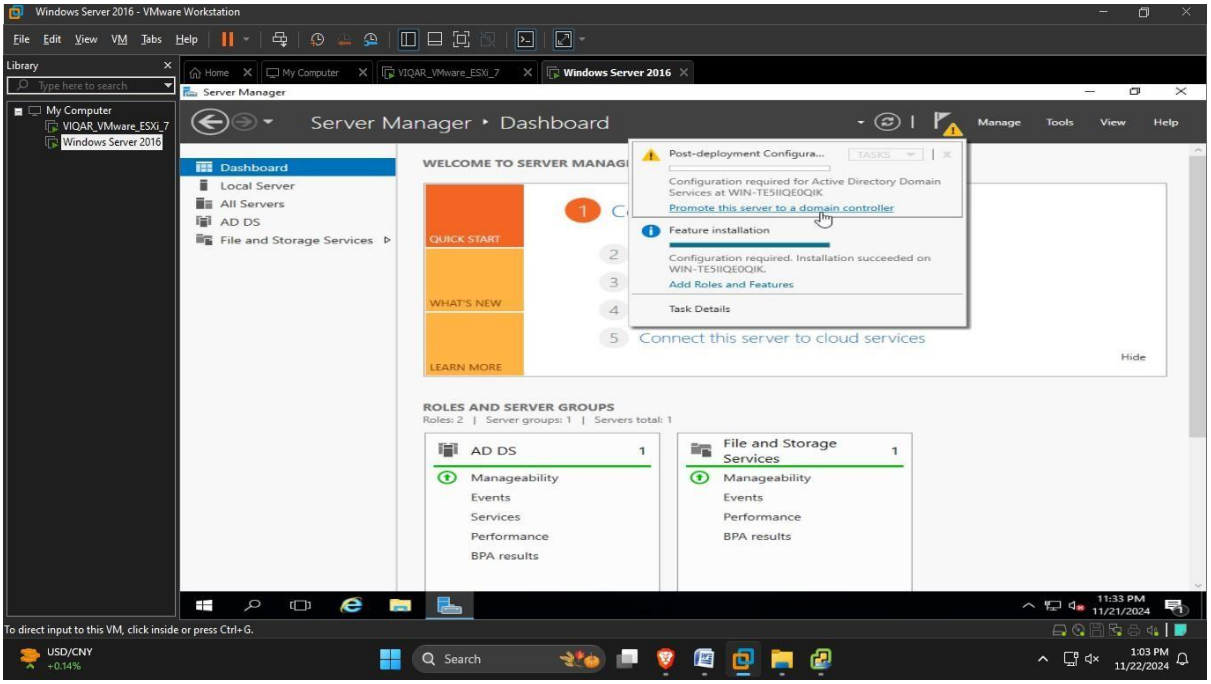
- Click on install.



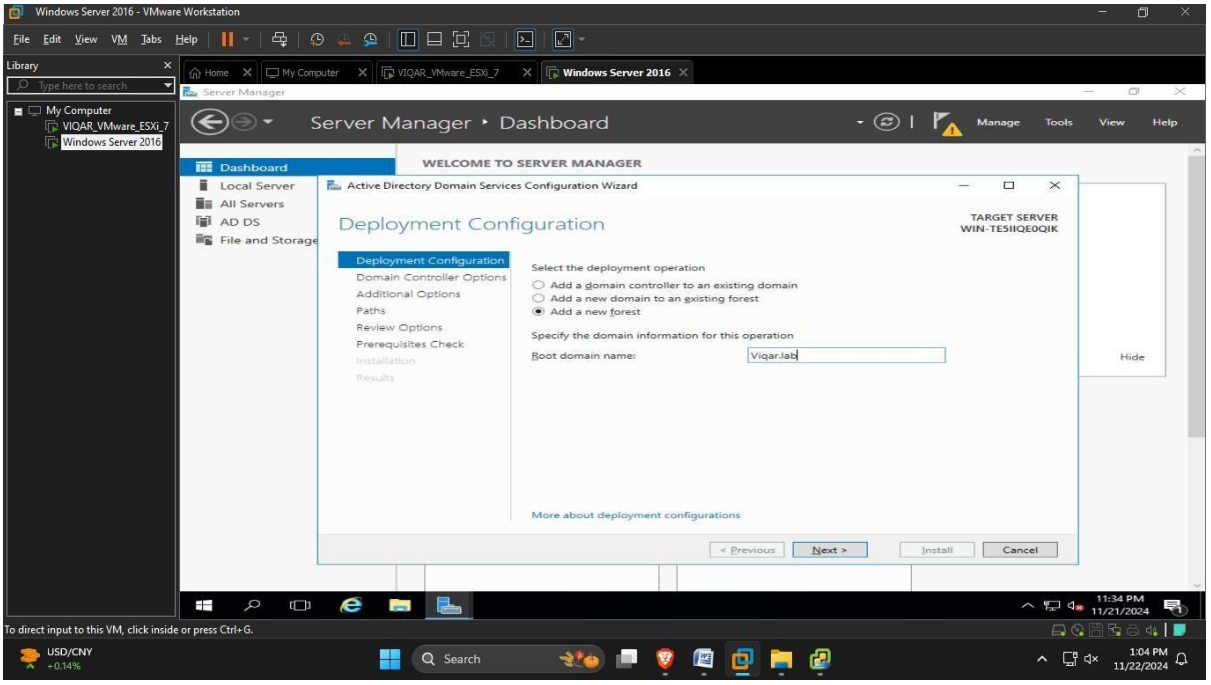
- After installation completes minimize the page.



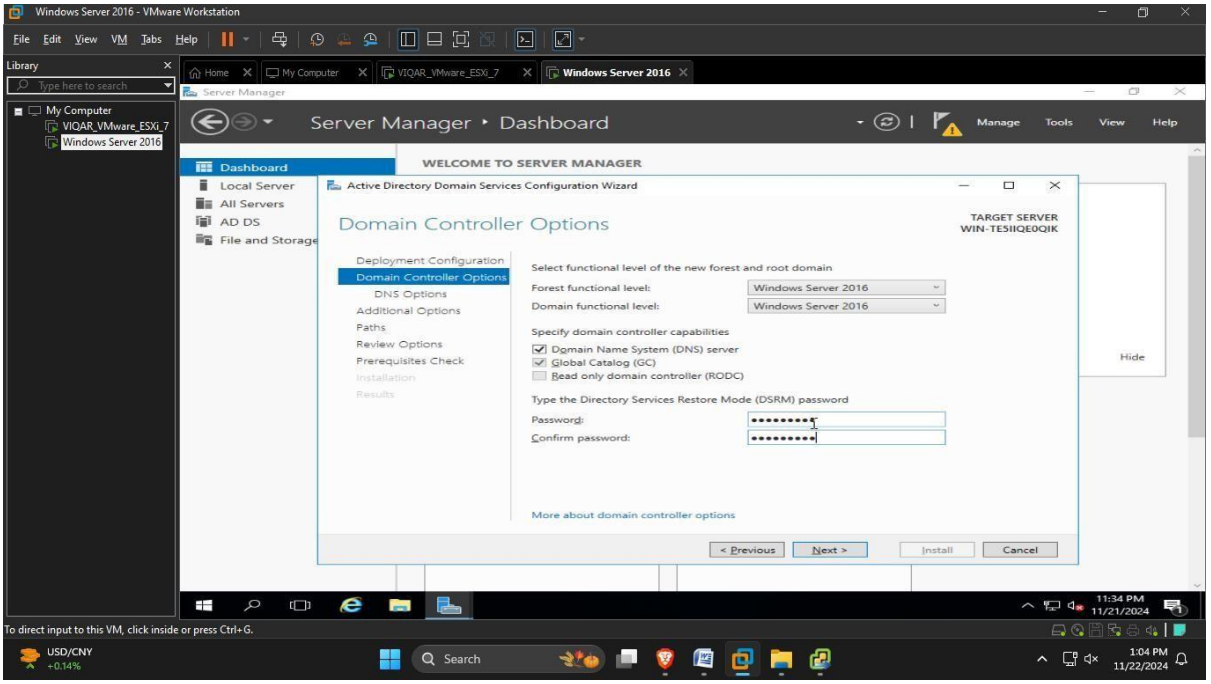
- There will be yellow flag in right corner of server manager click on it and then promote to domain controller



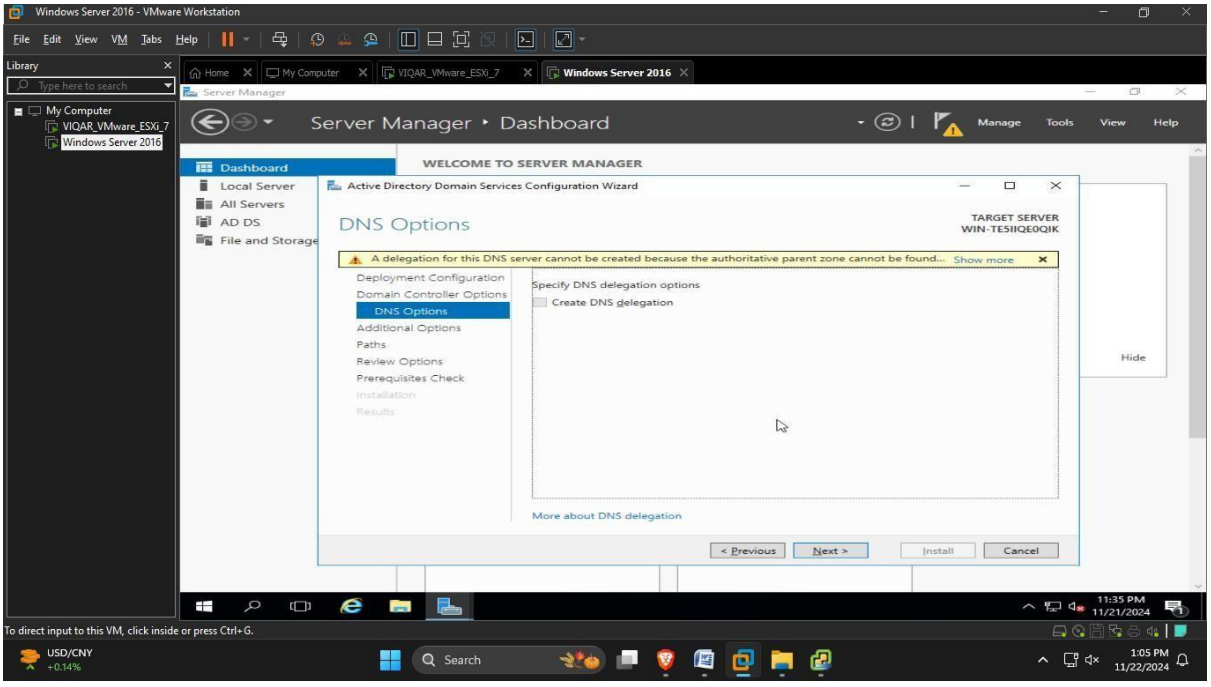
- Select add a new forest and give your domain name here it is Hannan .lab.



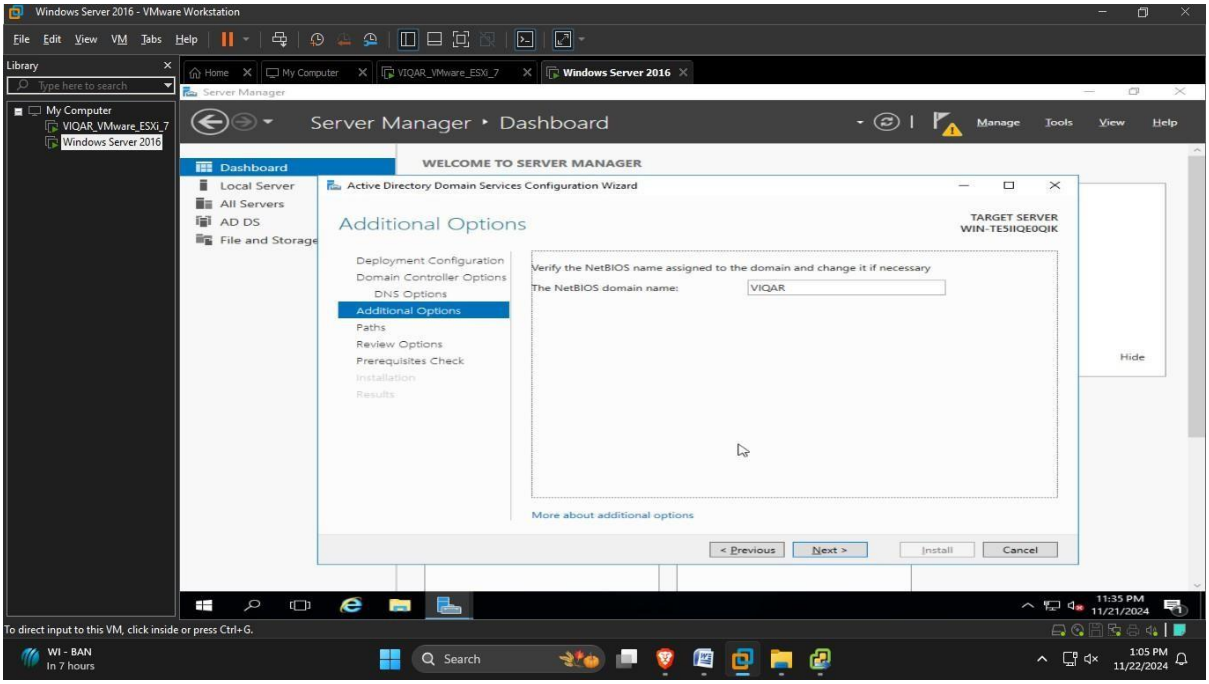
- Give a password for for your domain controller here it is Name!098.



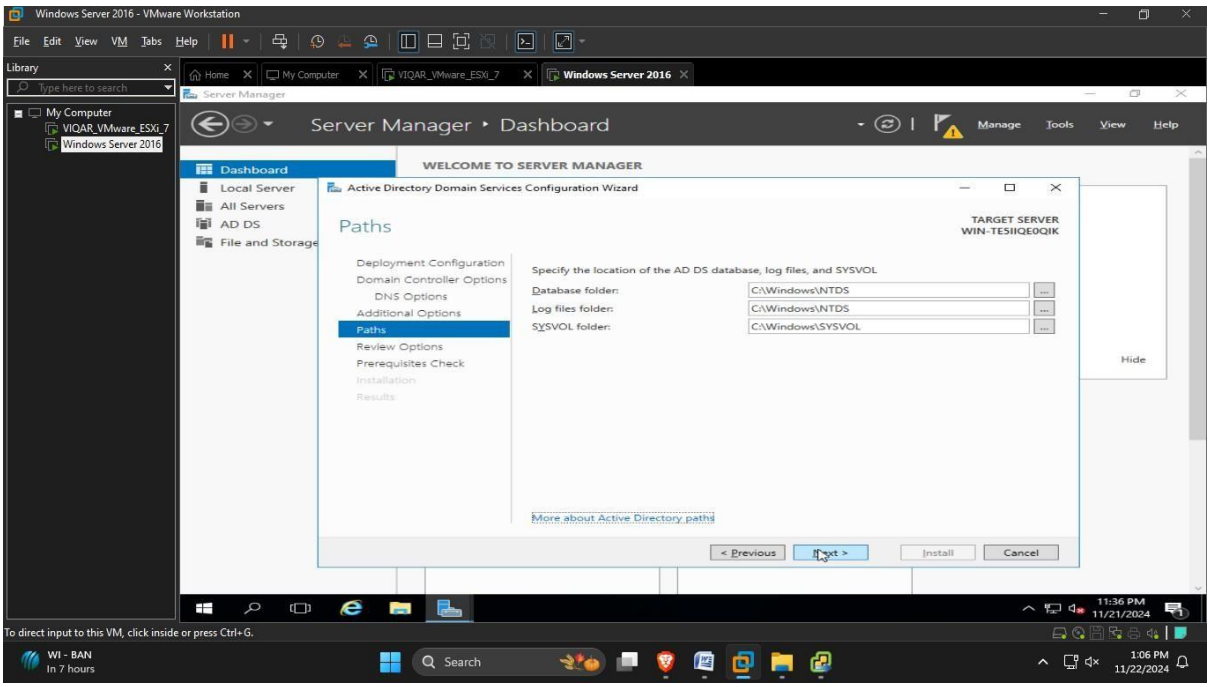
- Click on next.



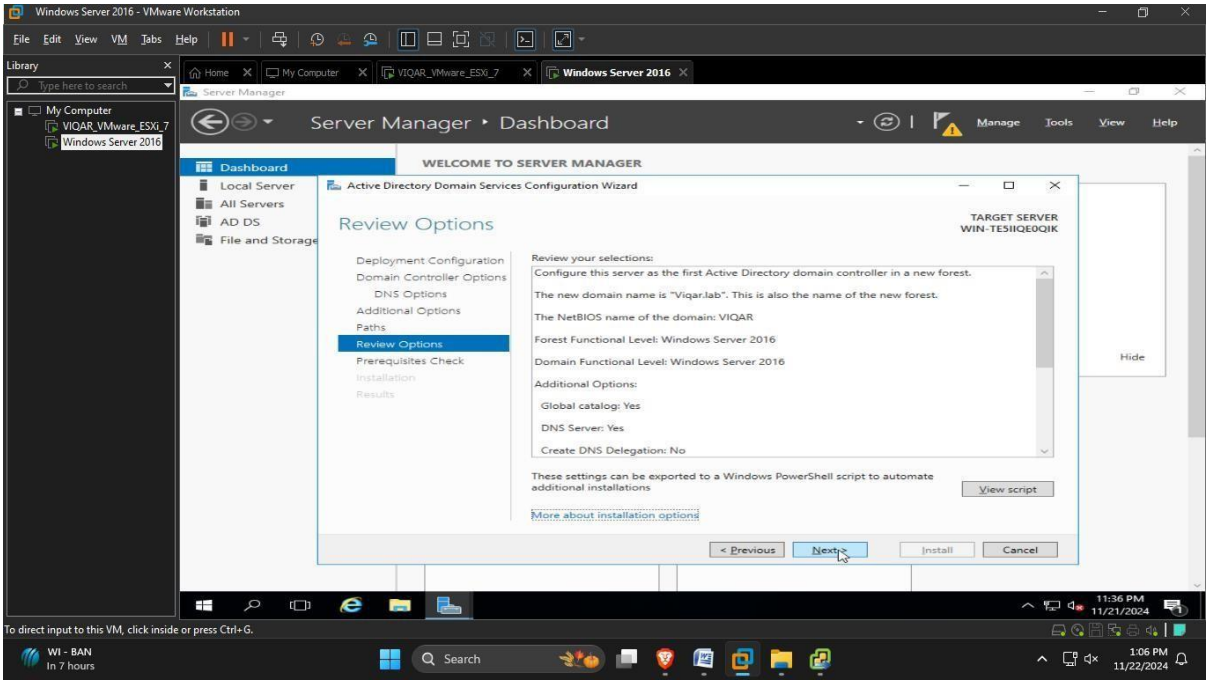
- Click on next.



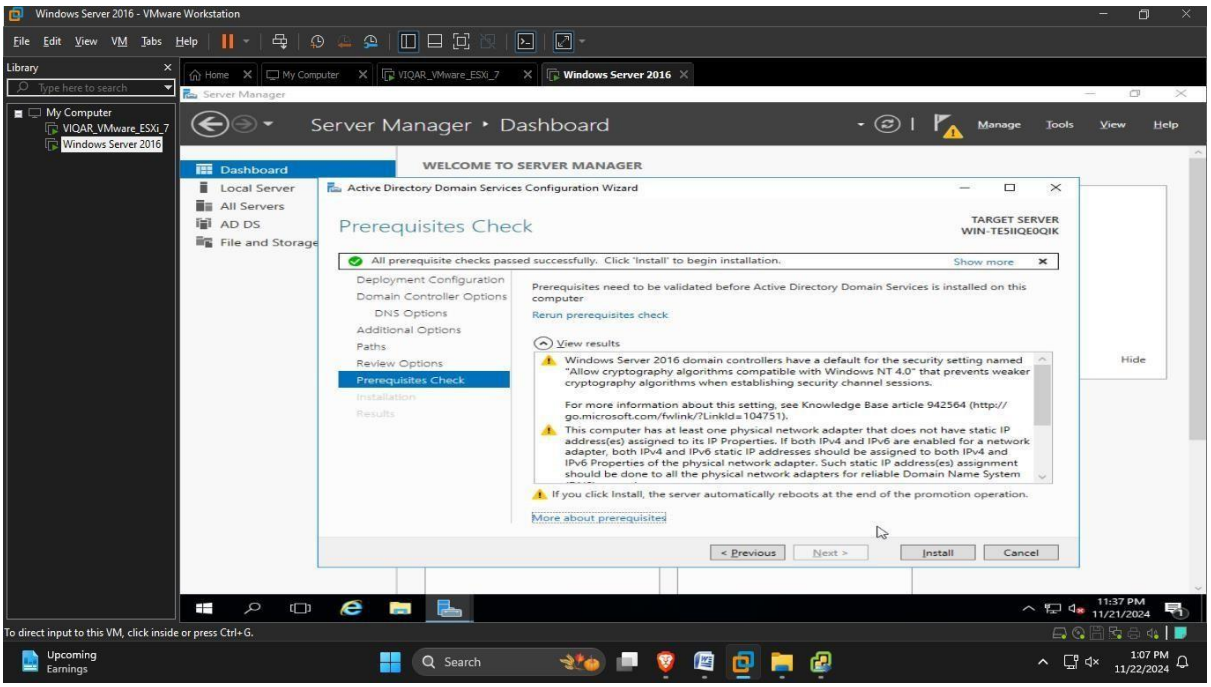
- Click on next.



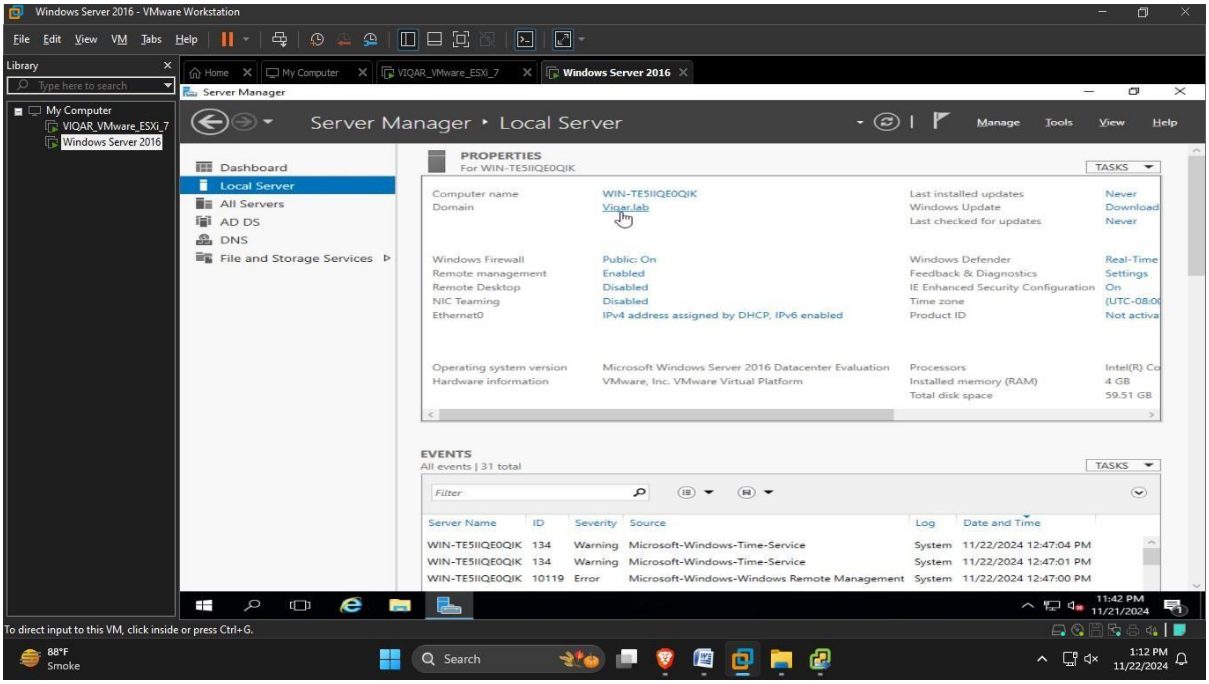
- Click on next.



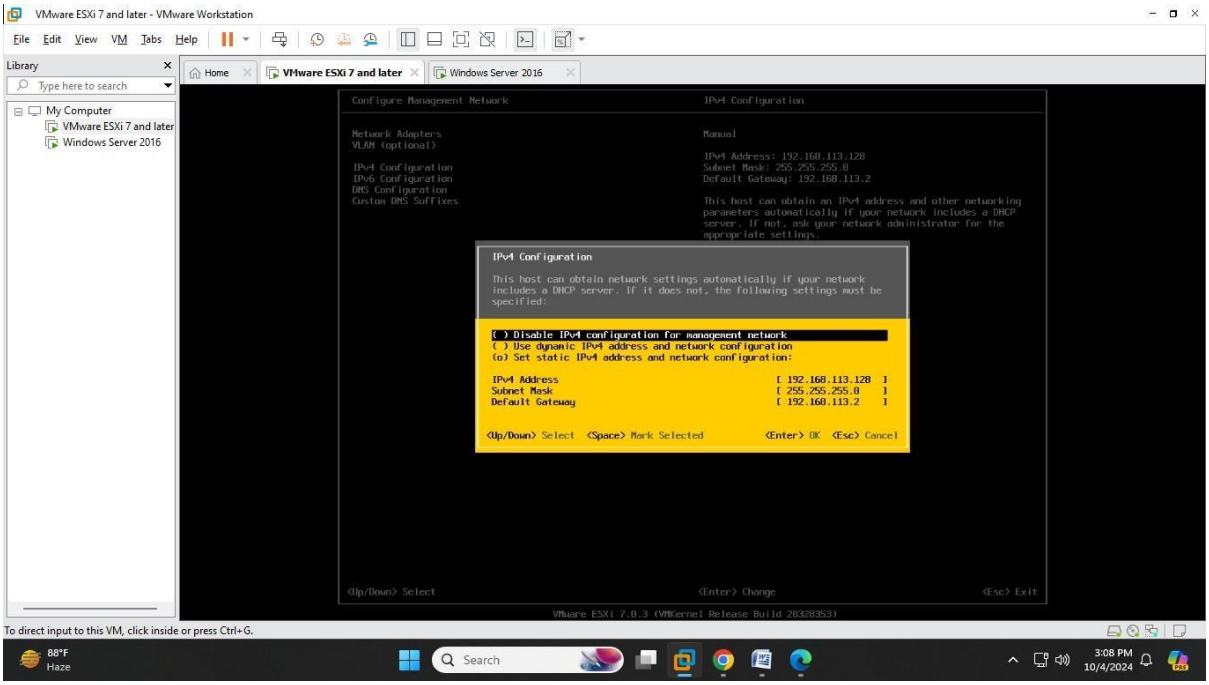
- Click on install



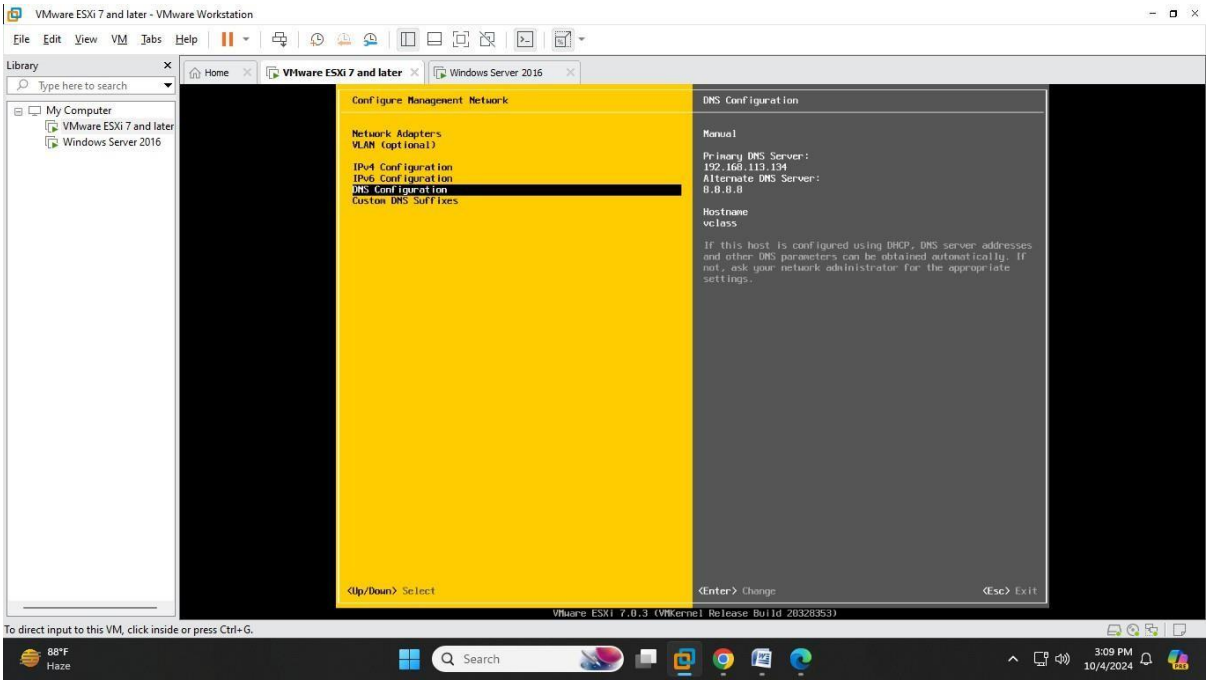
- As you can see domain controller is created.



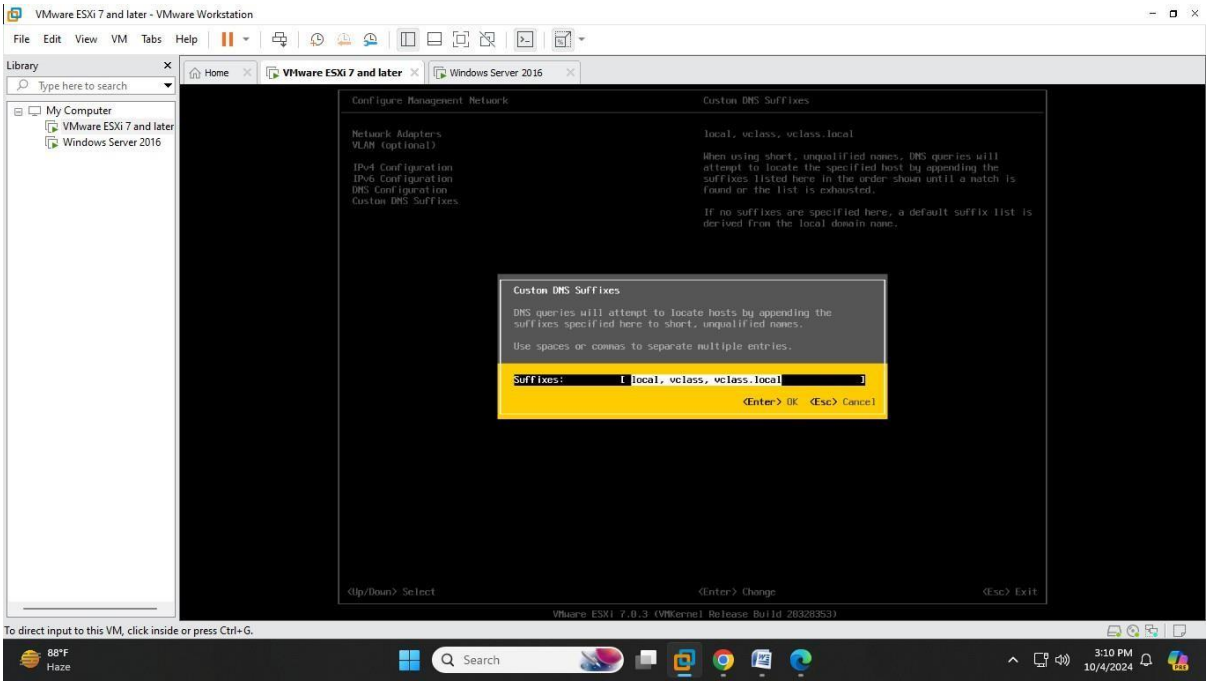
- Go to esxi in vmware workstation and set static ip addreses to connect the domain (Windows server 2016)



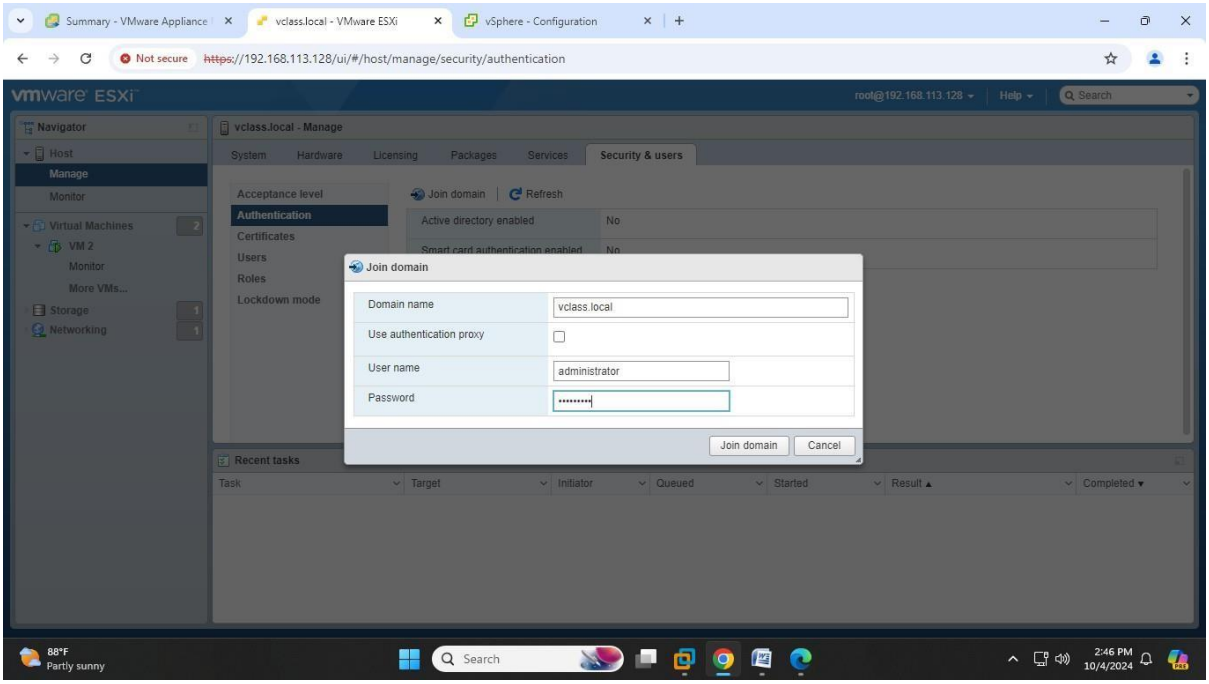
- Go to dns configuration .



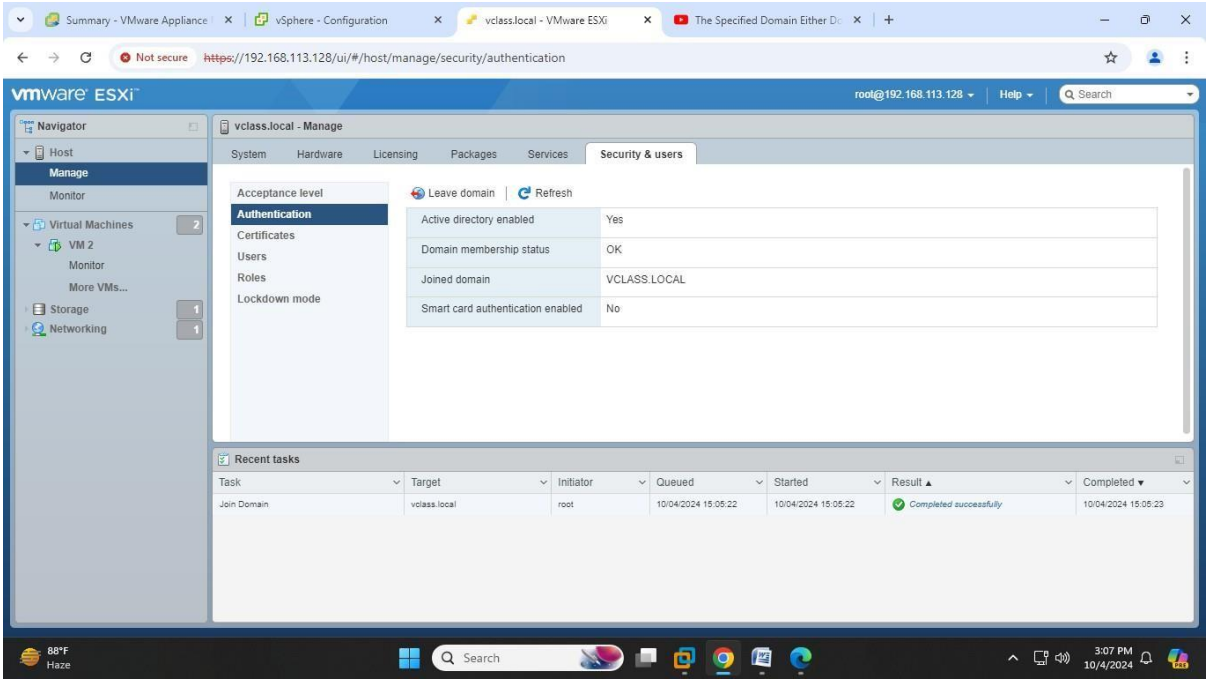
- Write your domain in suffix.



- First we have join domain in vmware esxi web accese



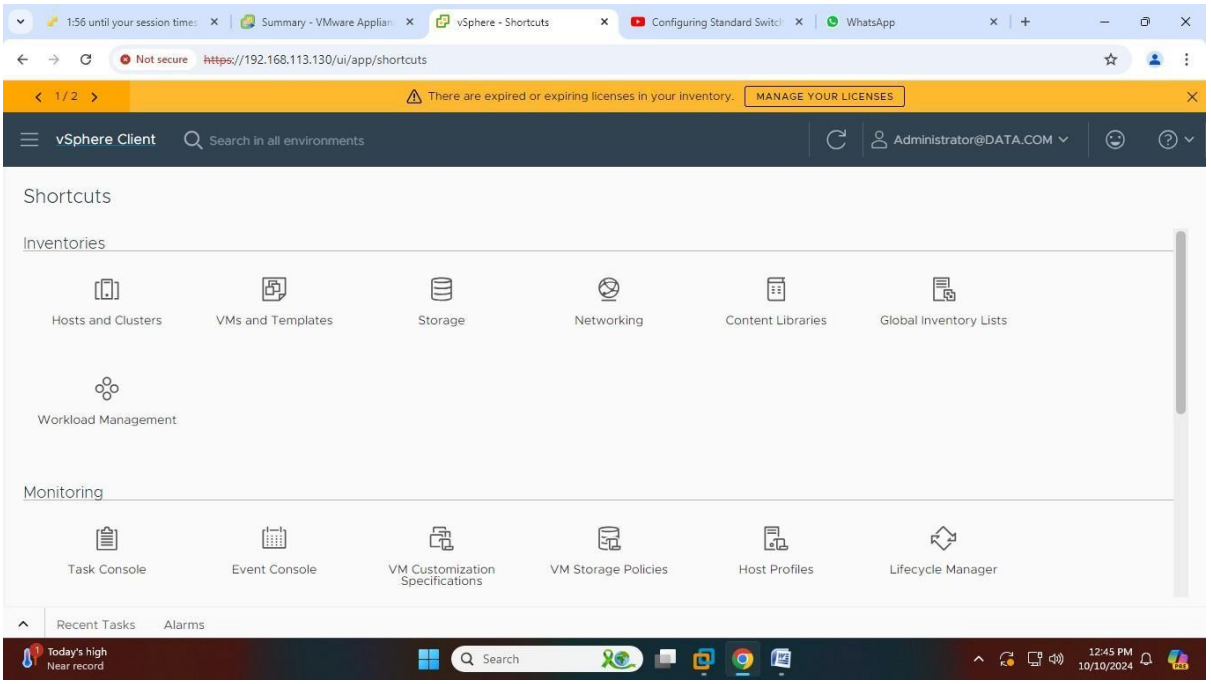
- As you can see the active directory is activated. with domain name



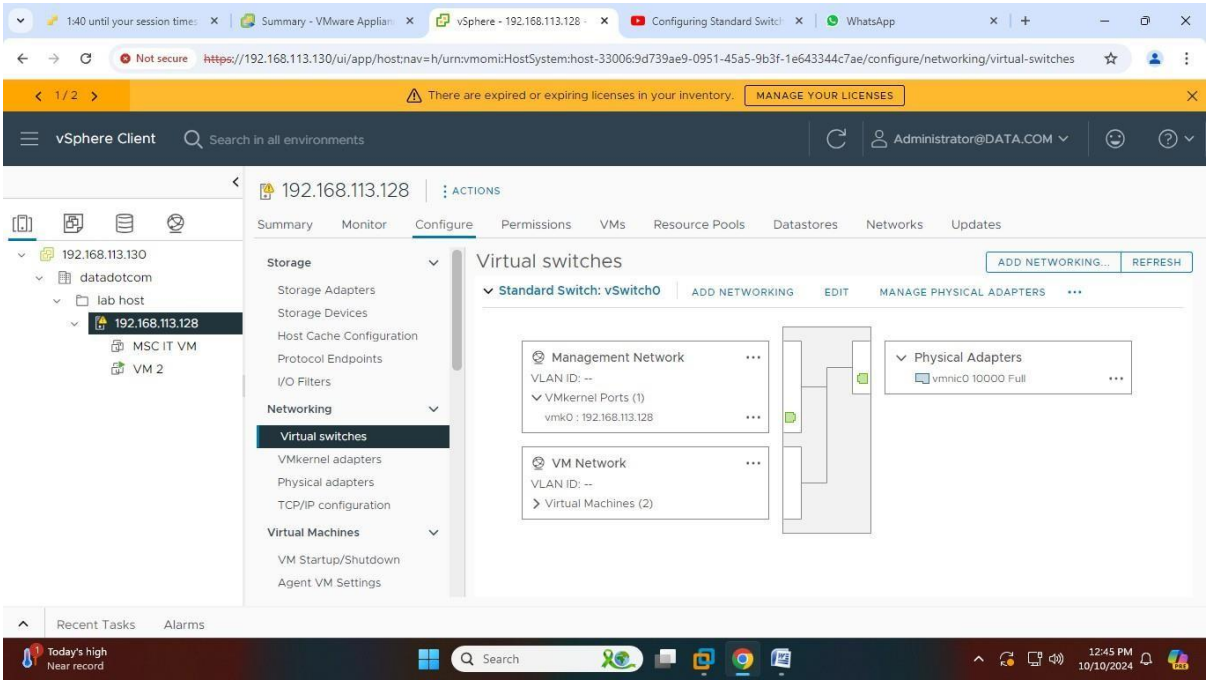
PRACTICAL 4

AIM :- Using Standard Switches.

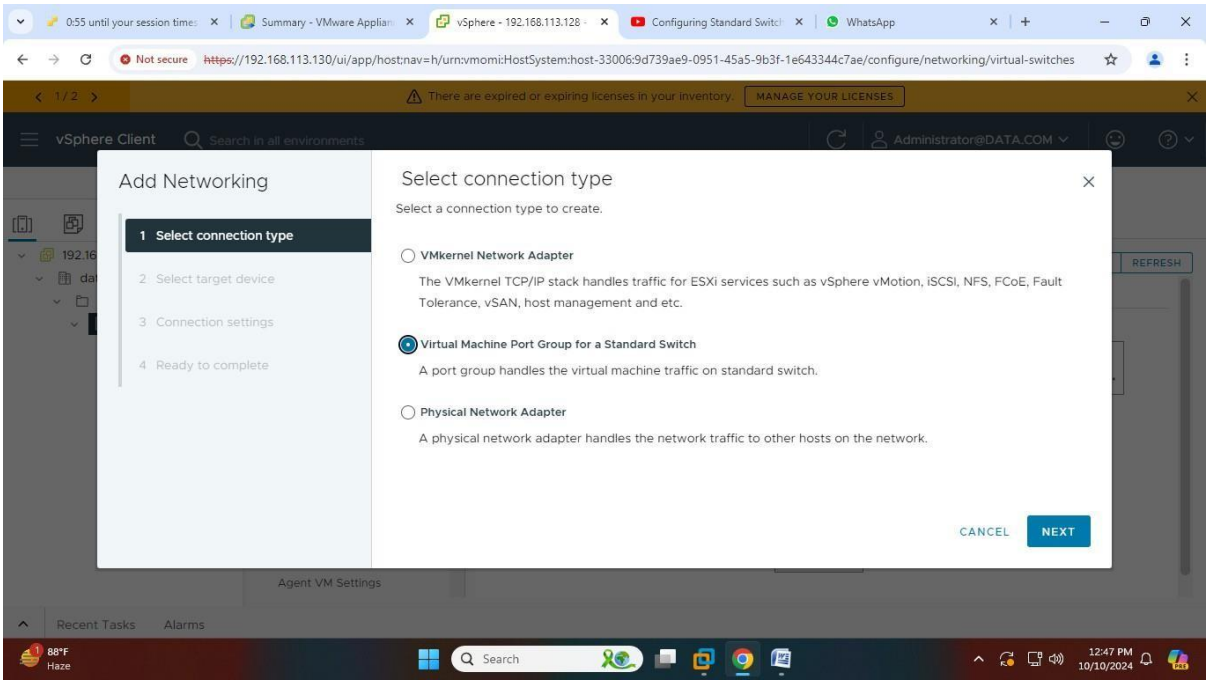
- Login to vsphere in browser.



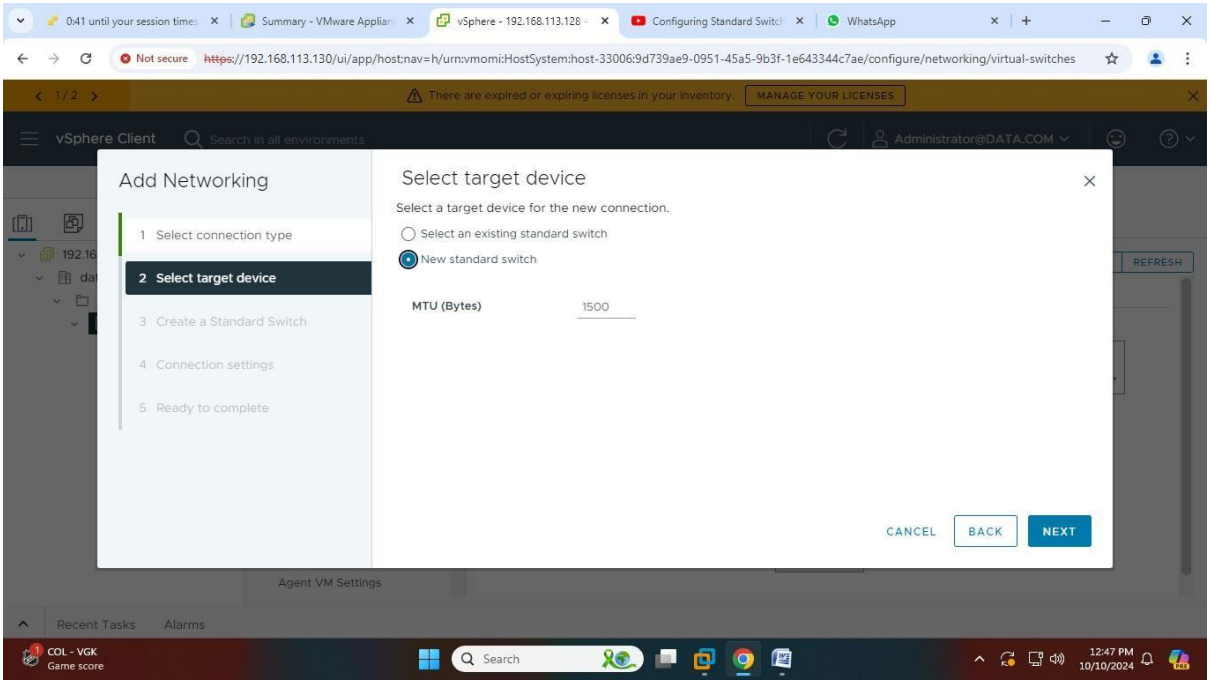
- Navigate to virtual switches.



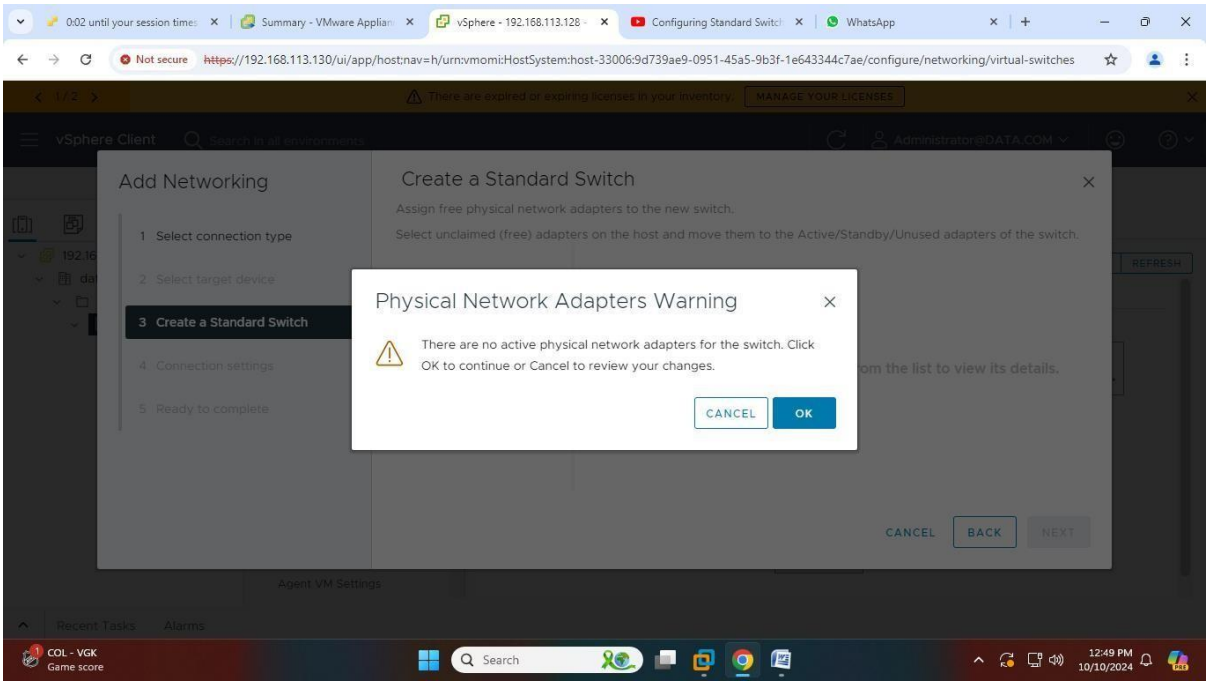
- Select Virtual Machine Port Group for a Standard Switch.



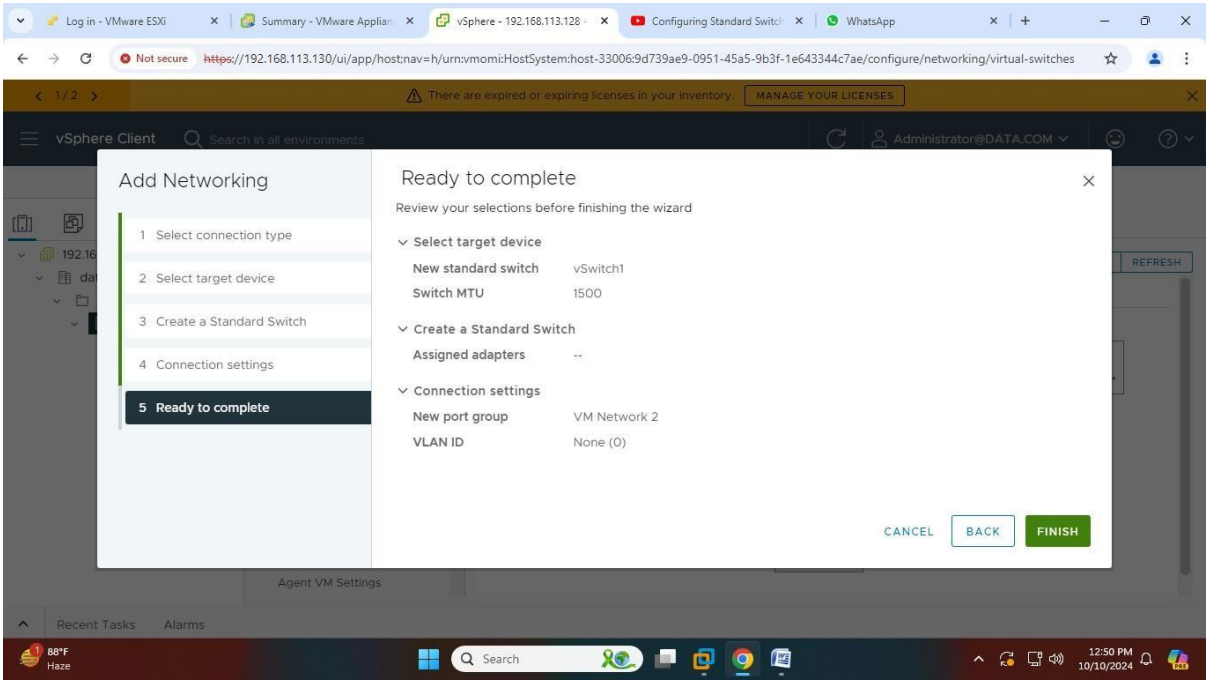
- Select New standard switch and set bytes 1500 and click on next.



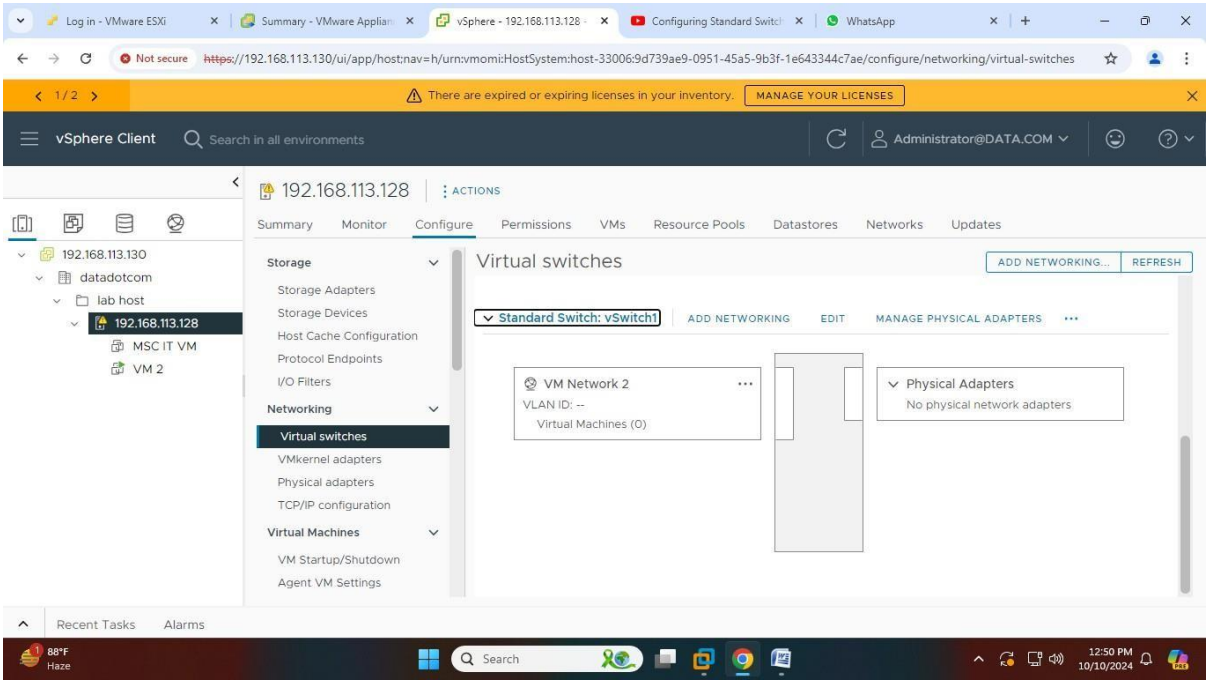
- Click on OK.



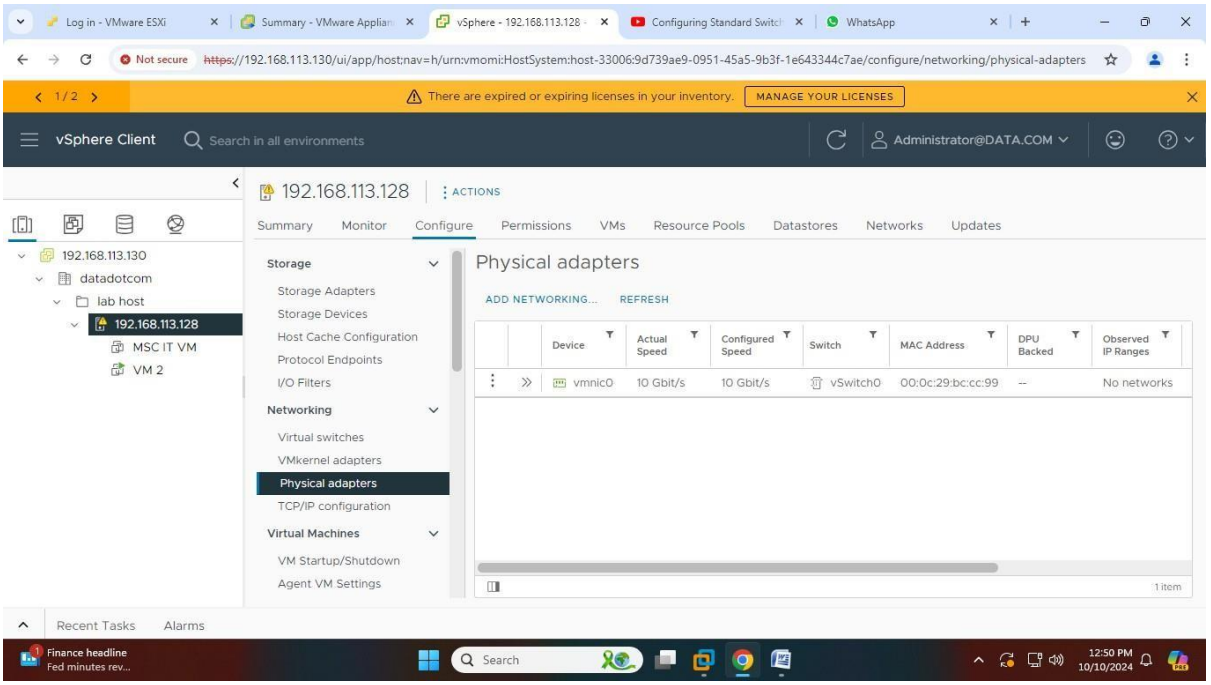
- Click on Finish.



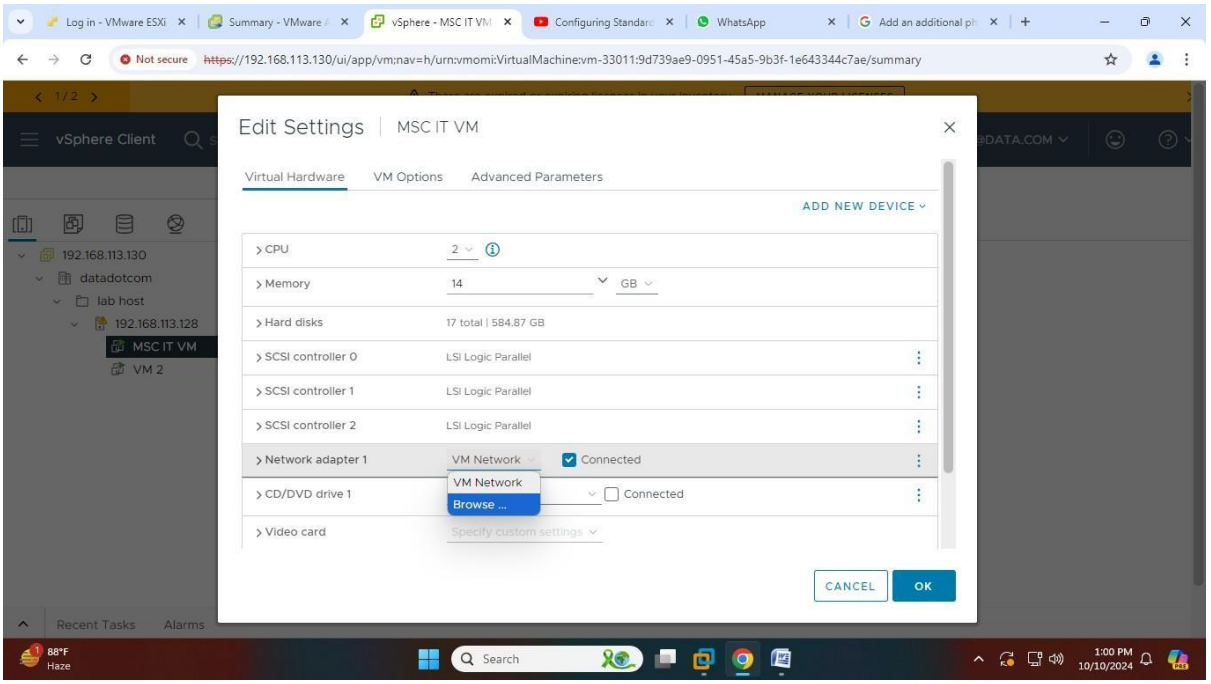
- Switch will be created but first go to vmware workstation and add network adapter .



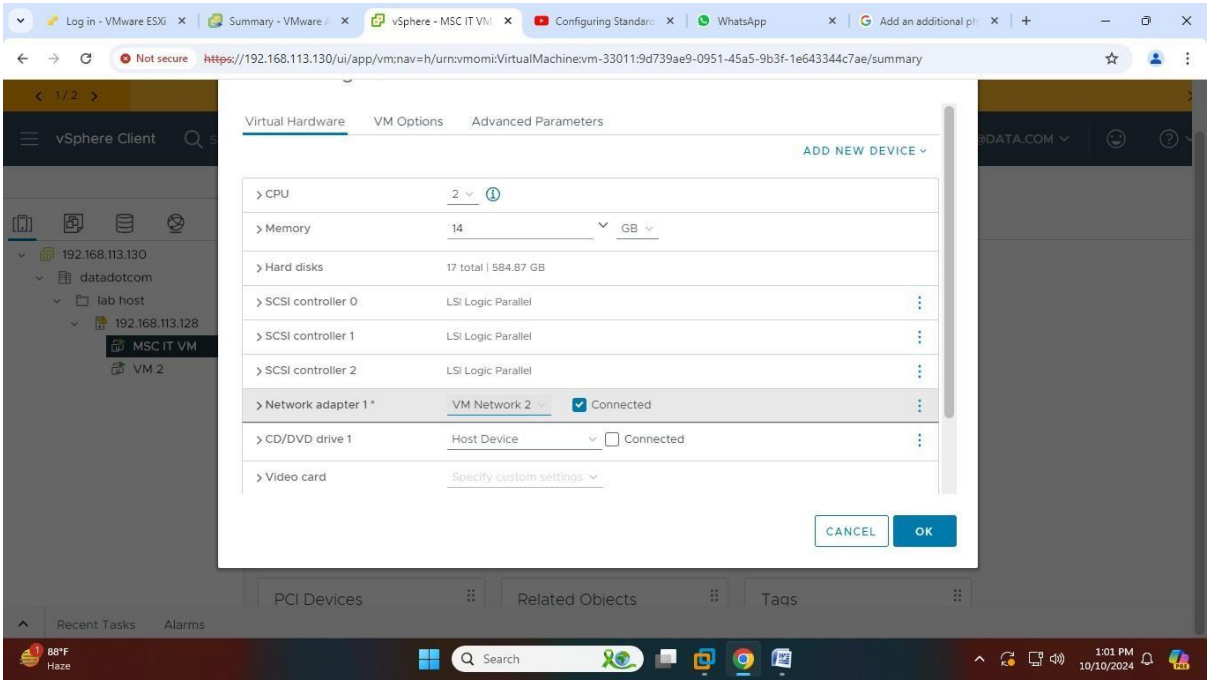
- Navigate to physical.



- **Set cpu, memory , and click on ok**



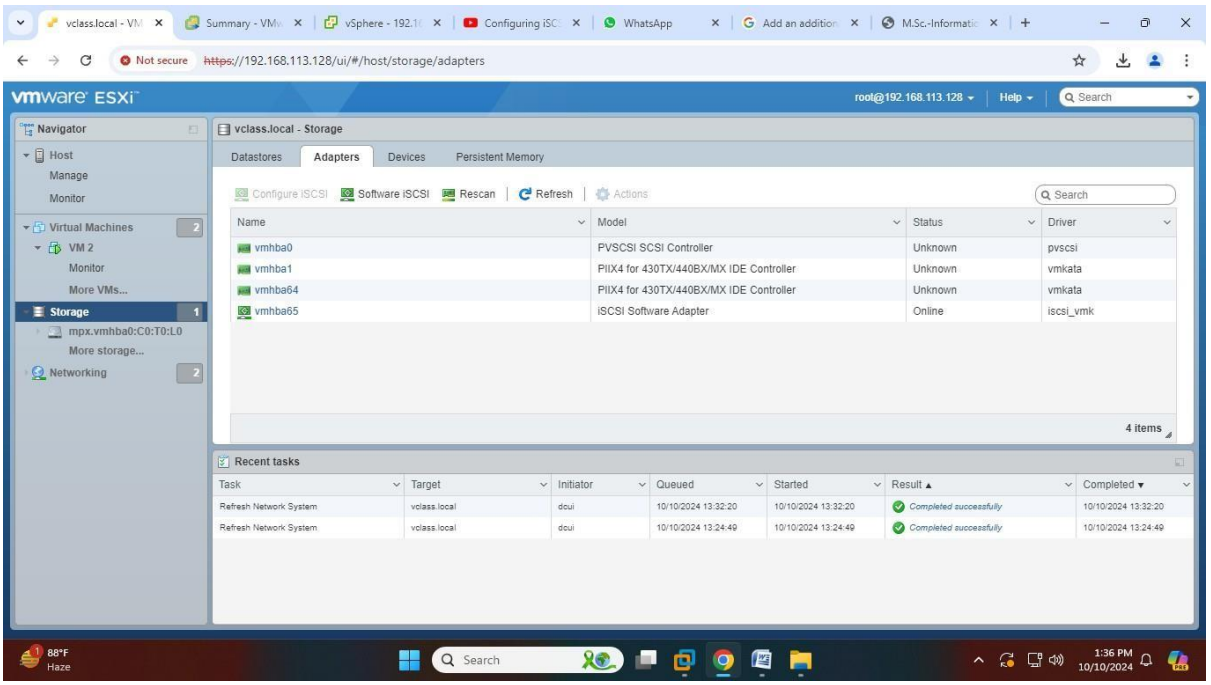
- VM Network .



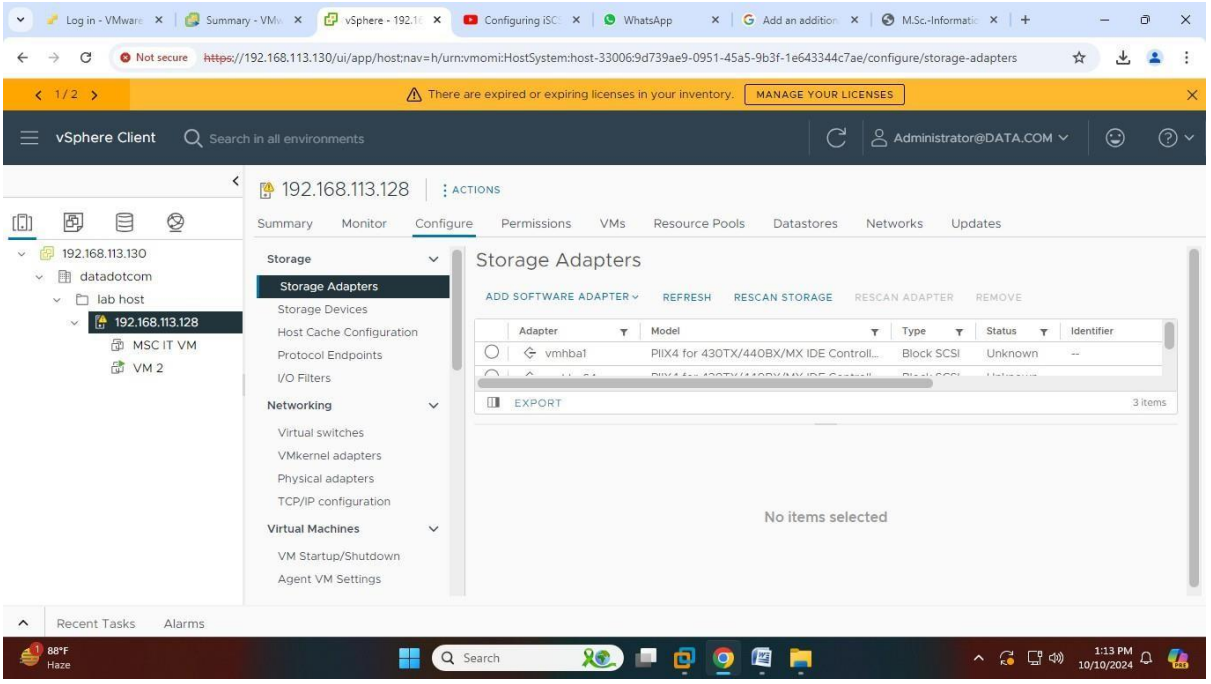
PRACTICAL 5

AIM :- Accessing iSCSI Storage.

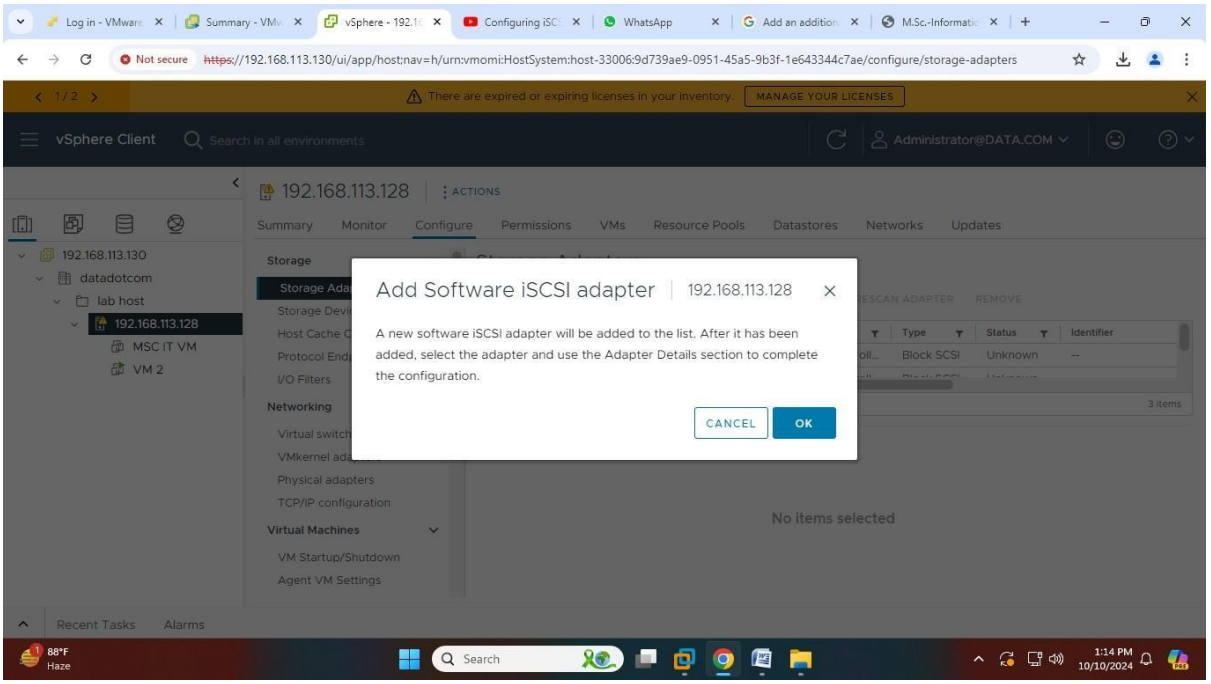
- Validating an existing esxi host icsi configuration.



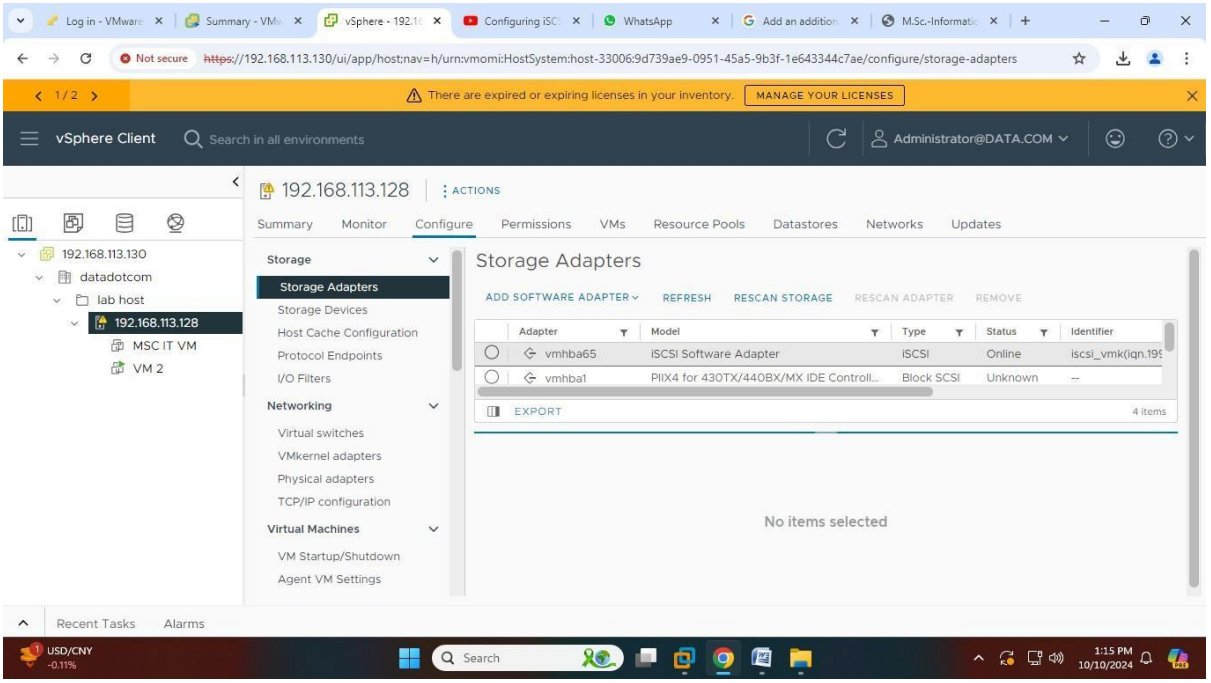
- Navigate to Storage Adapters .



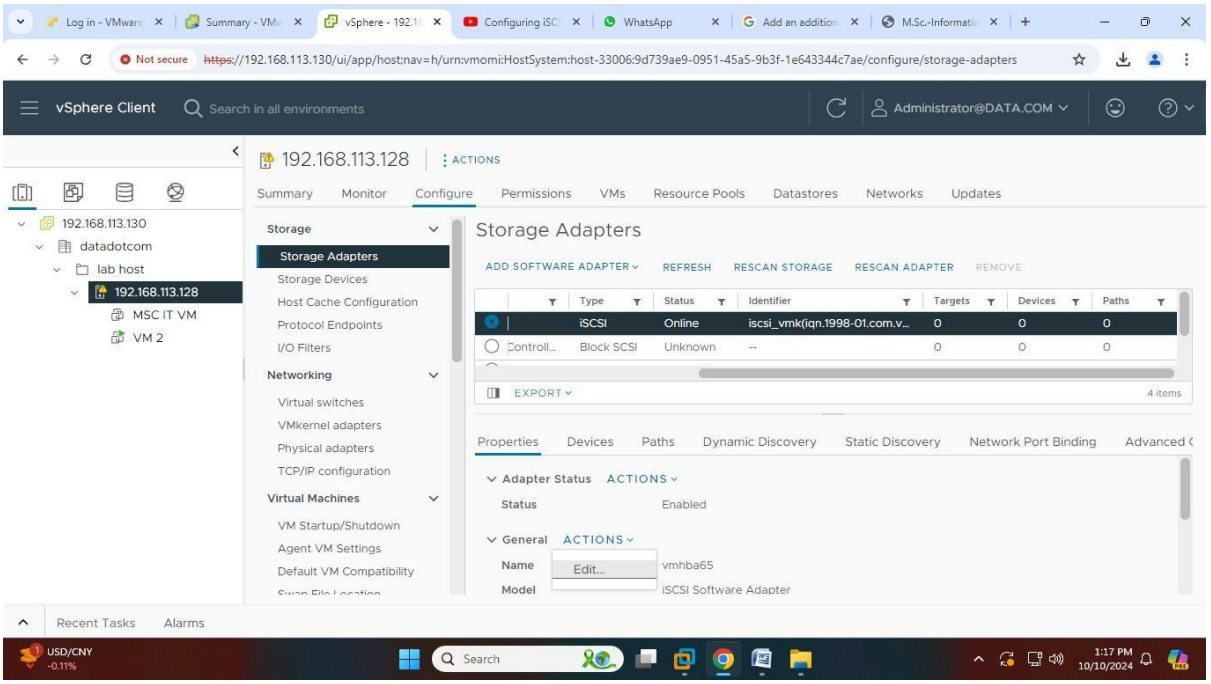
- Click oK



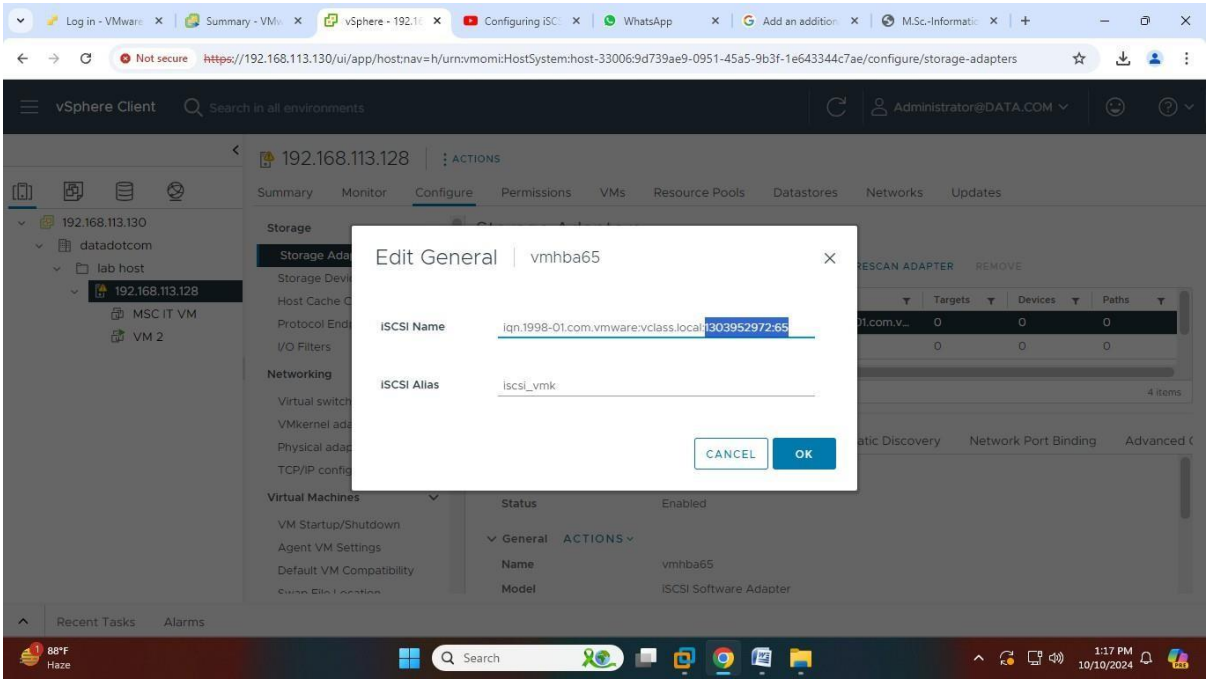
- As you can see Storage Adapters is created.



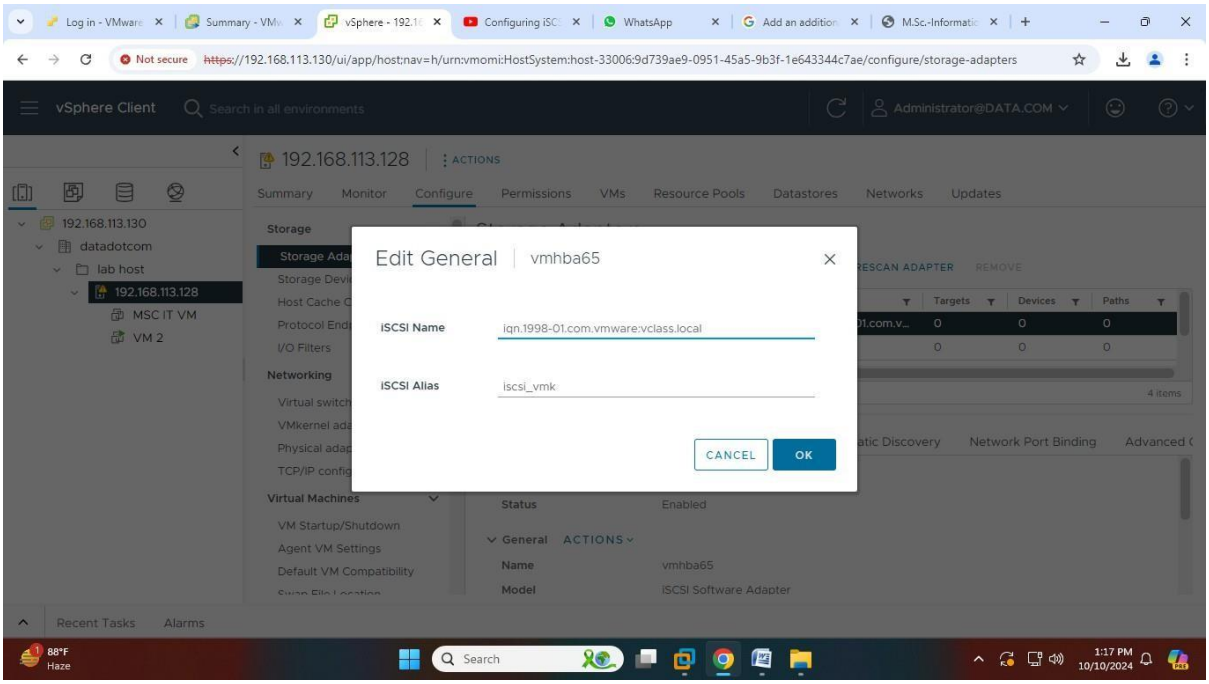
- Select iscsi.



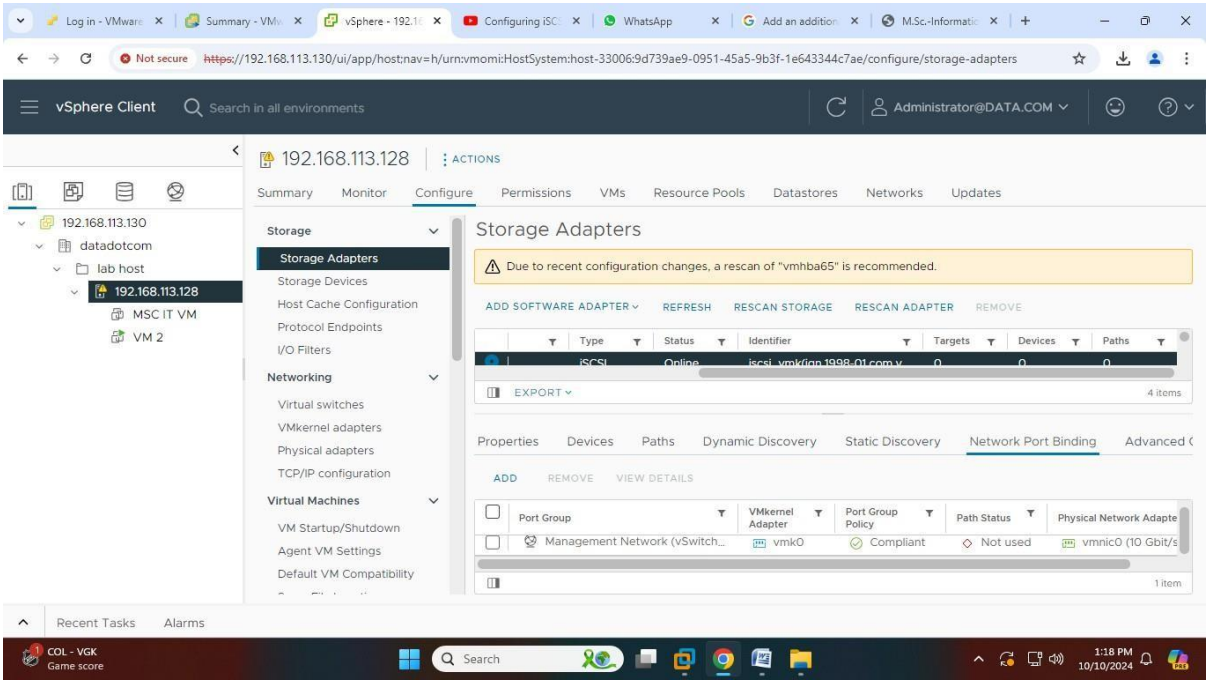
- Click on OK.



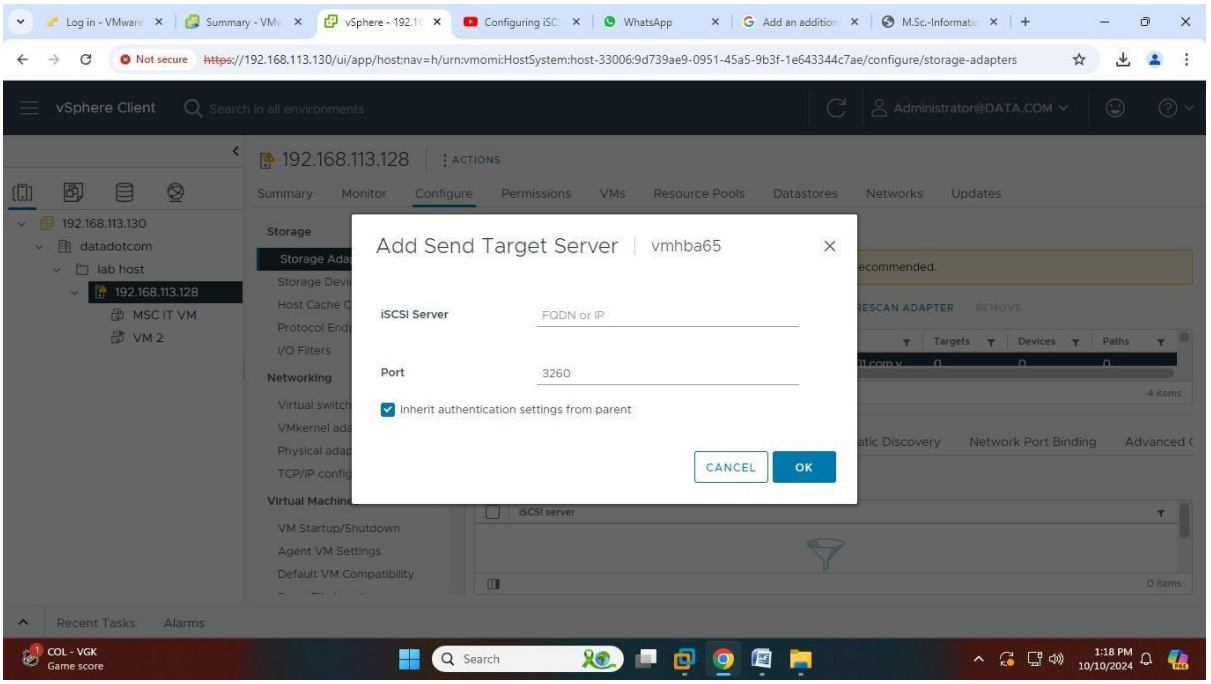
- Name icsi and Click on OK



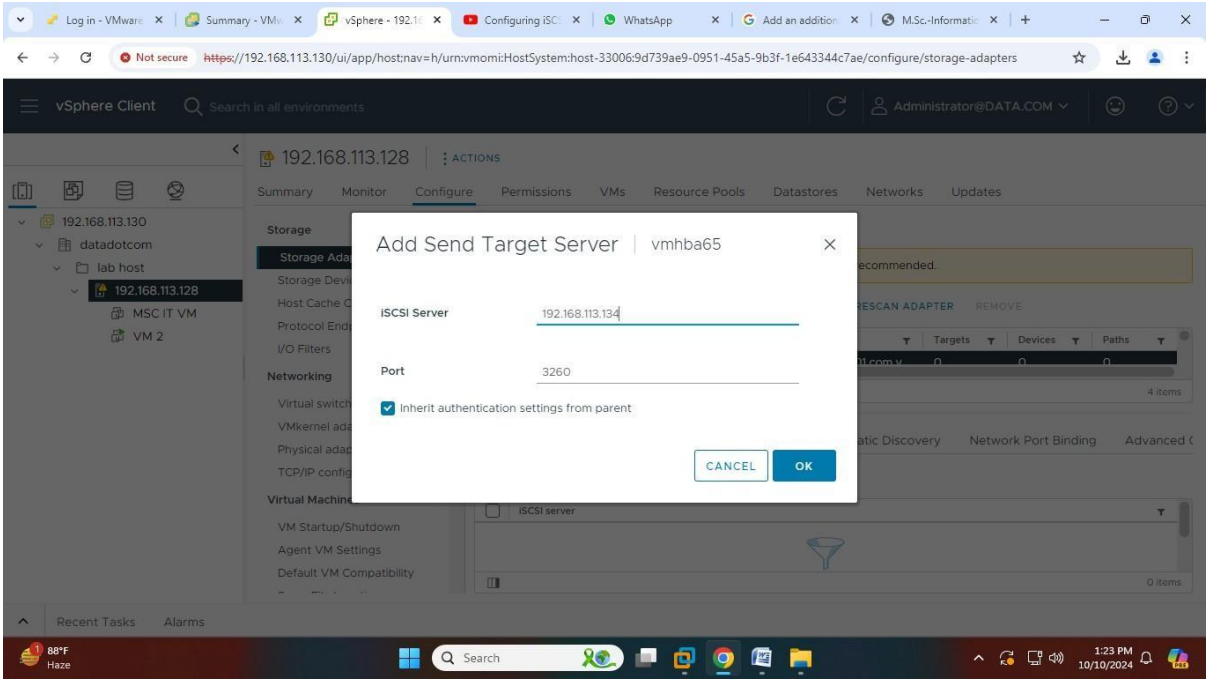
- Go on network port binding and click on add



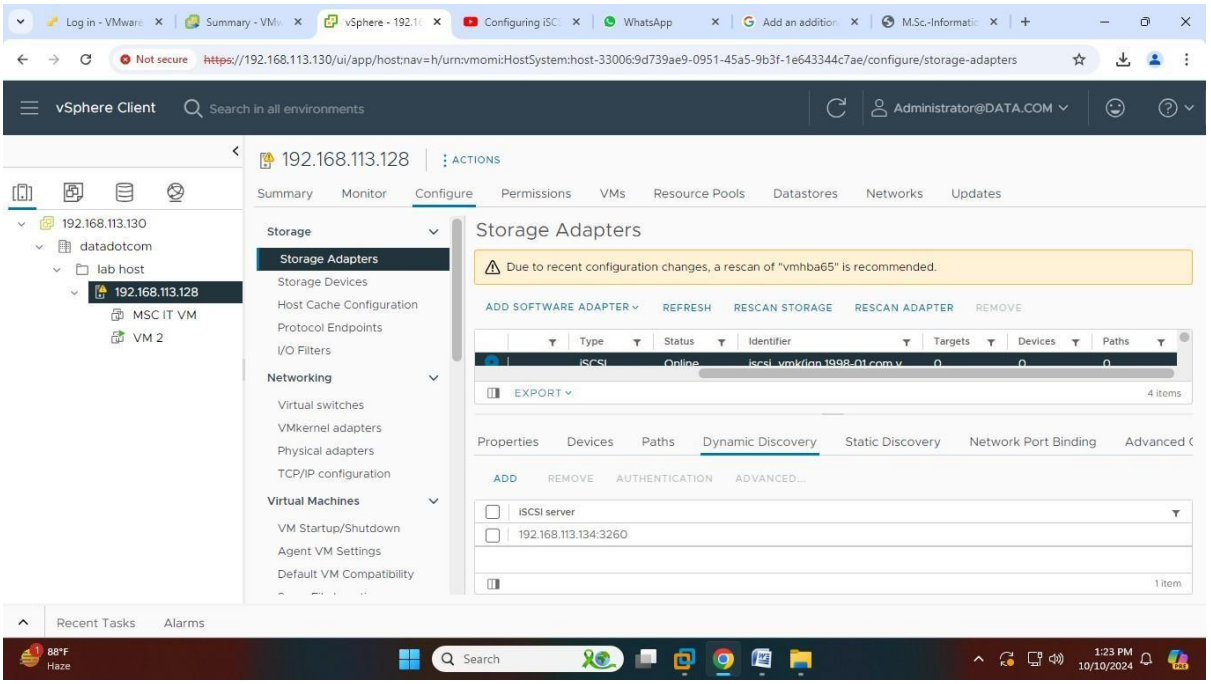
- Click on ok.



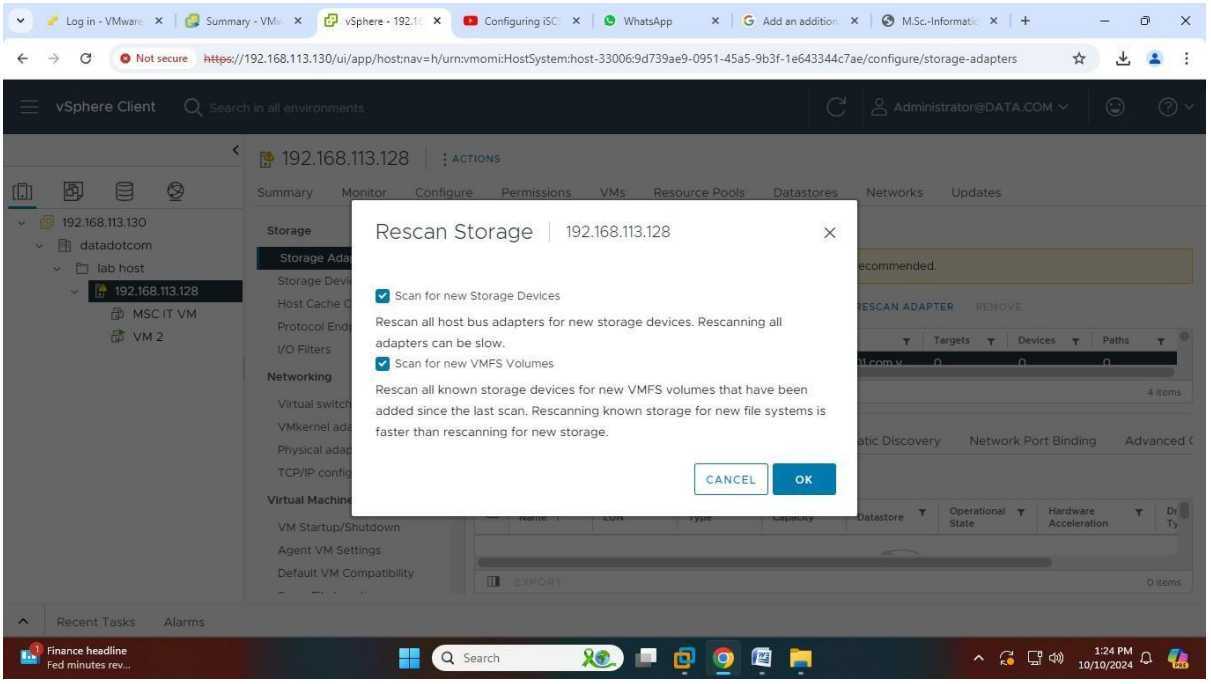
- Write ip address of host.



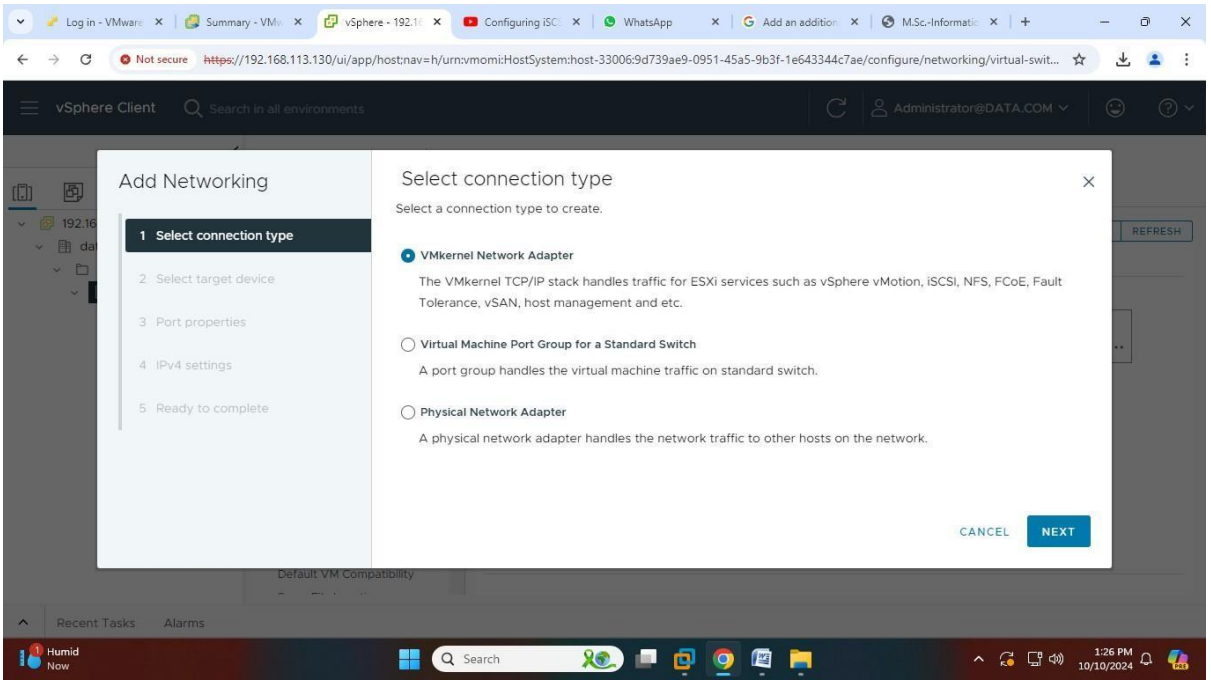
- Go to Dynamic Discovery.



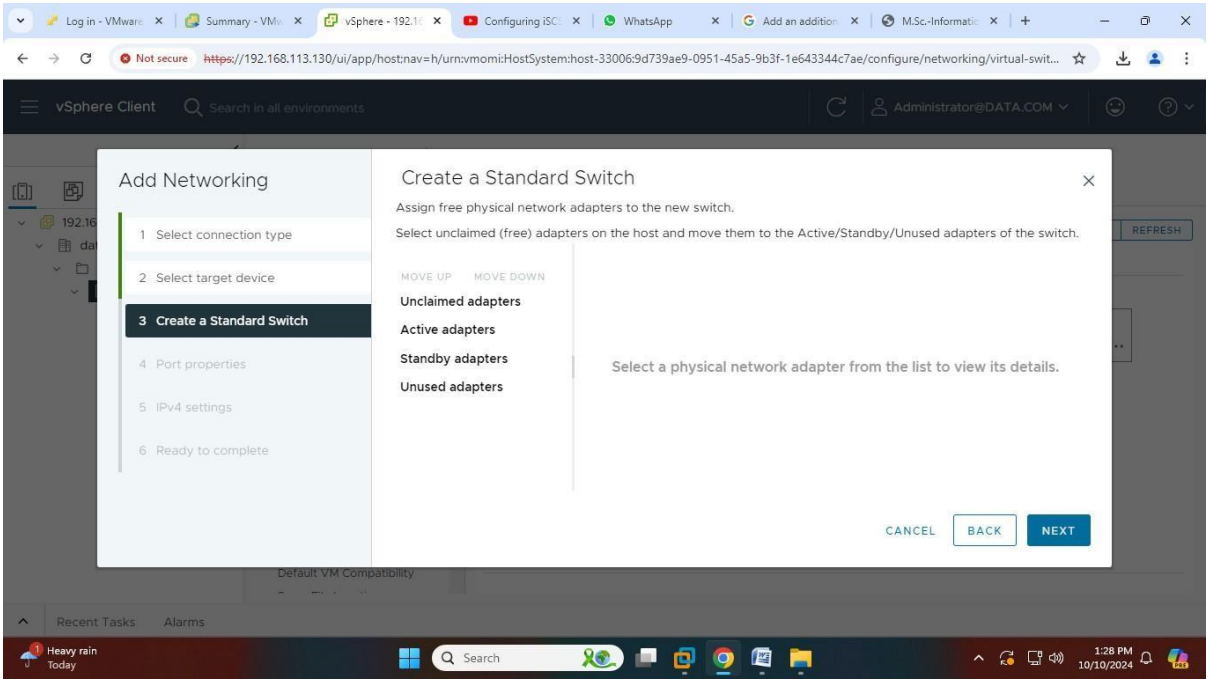
- Click on OK.



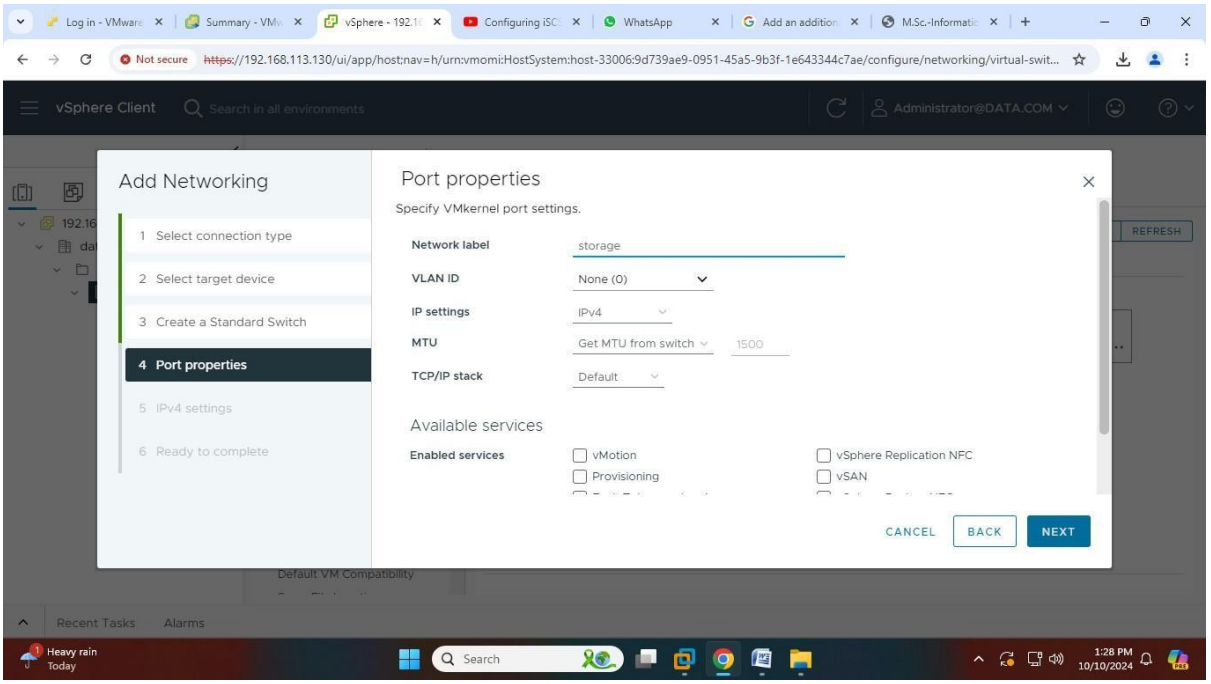
- Select VMkernel Network Adapter.



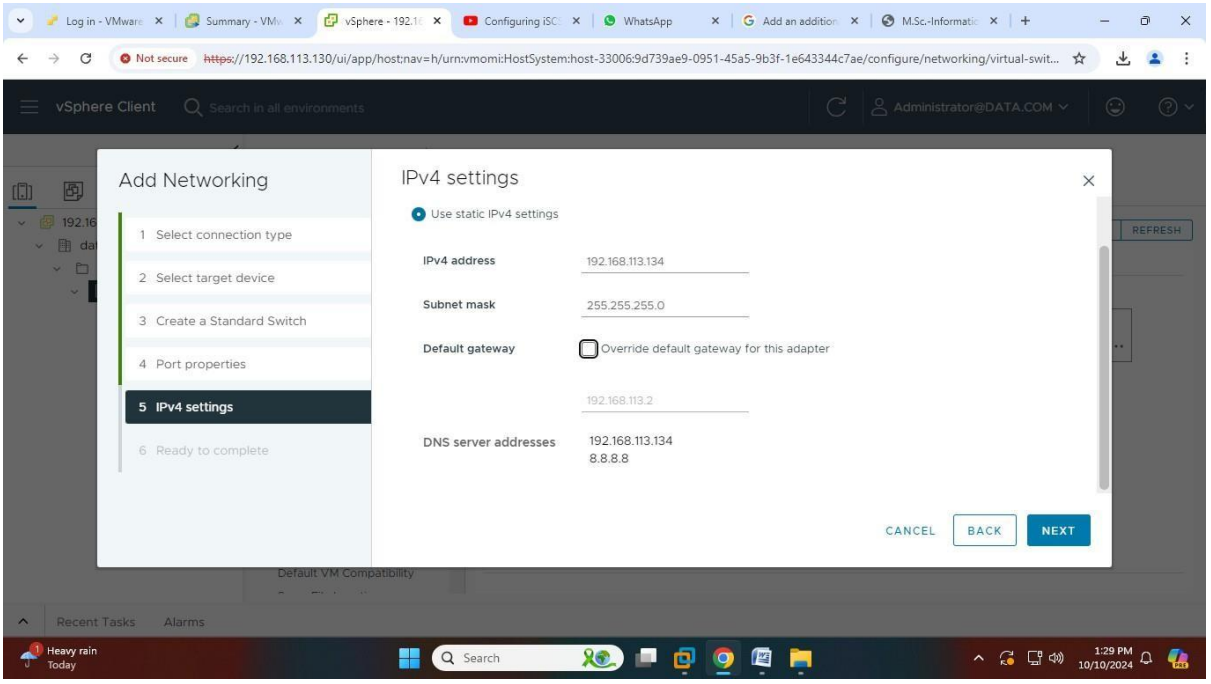
- Click on next.



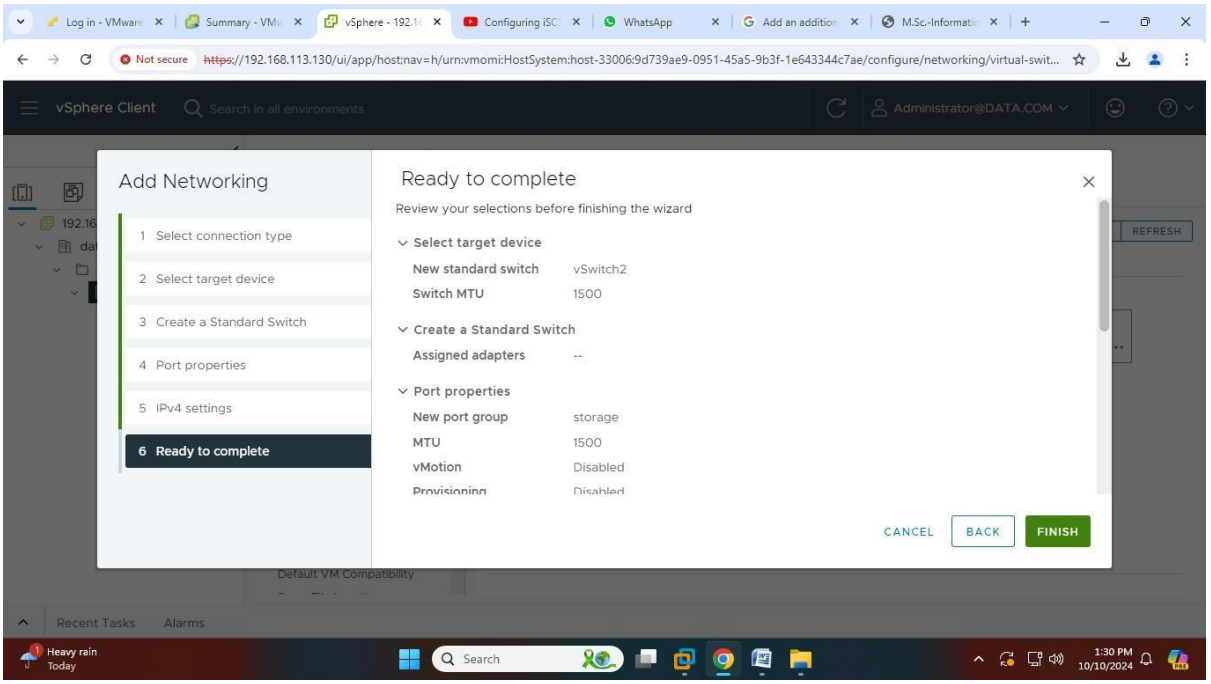
- Configuring port properties for VM kernel



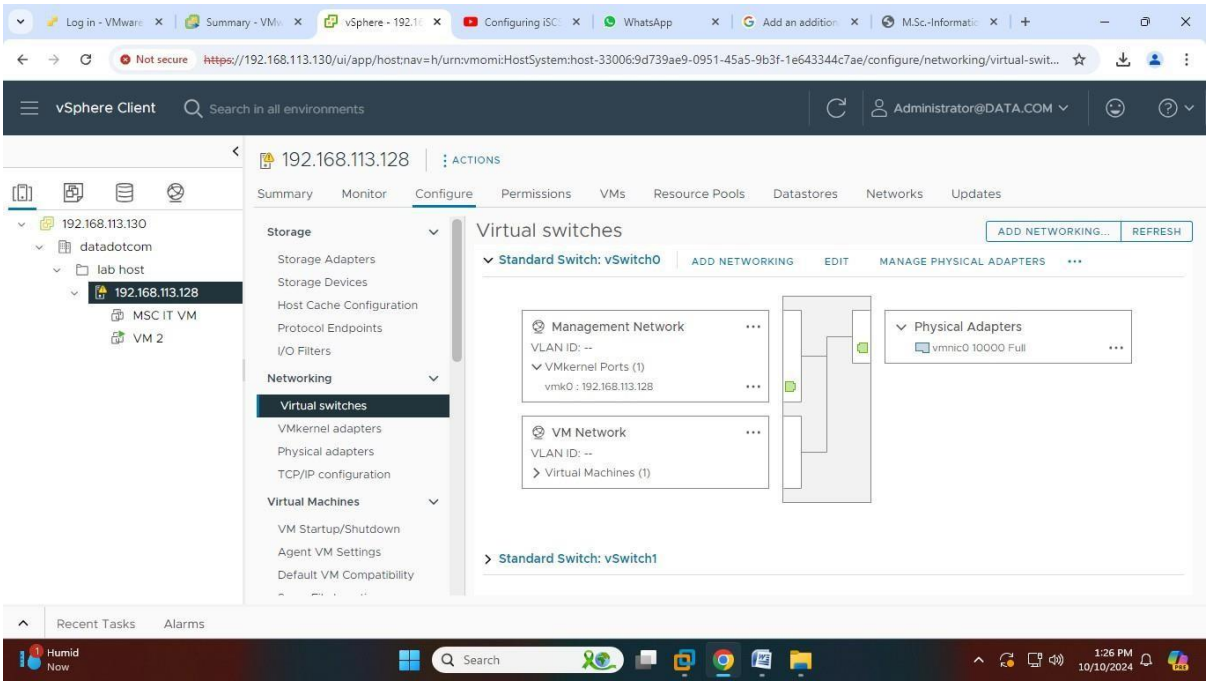
- Click on next.



- Click on Finish.



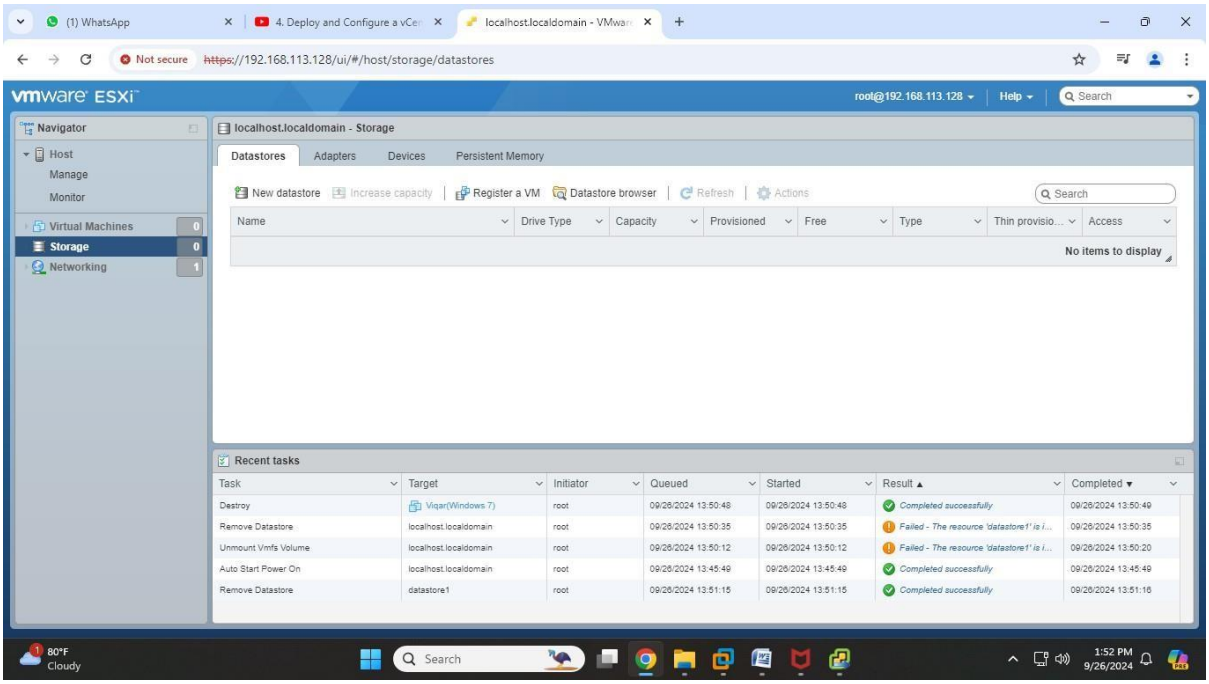
- As of now virtual network is added.



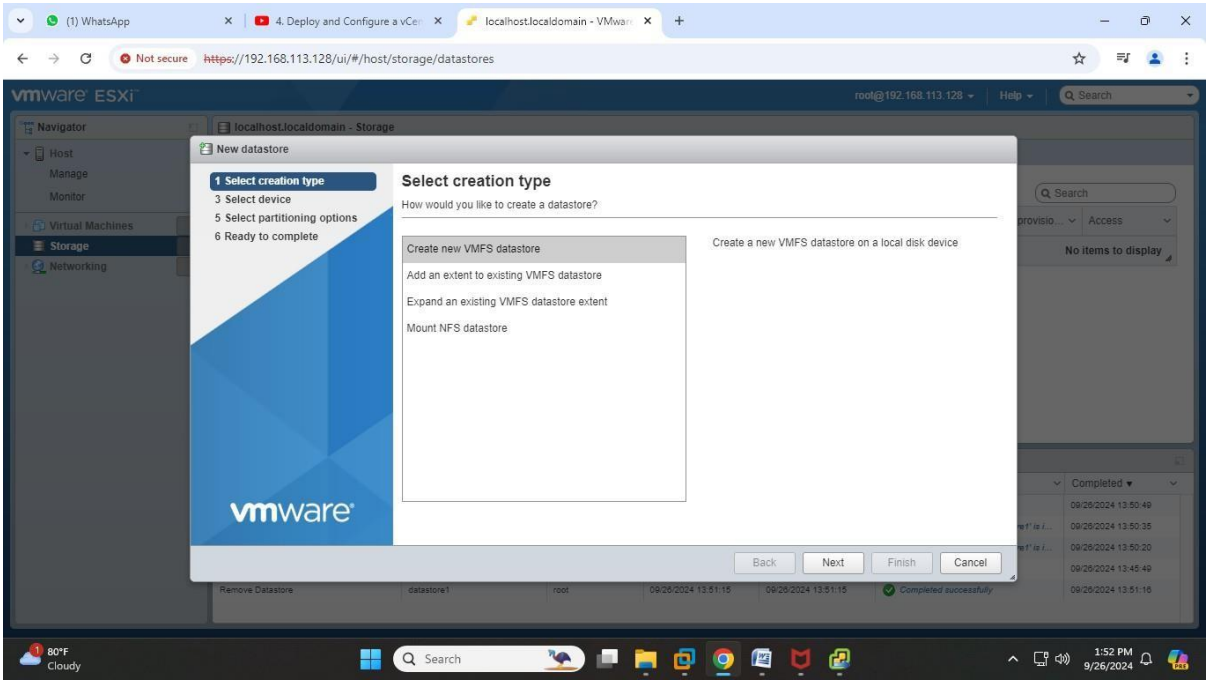
PRACTICAL 6

AIM :- Managing VMFS Datastores.

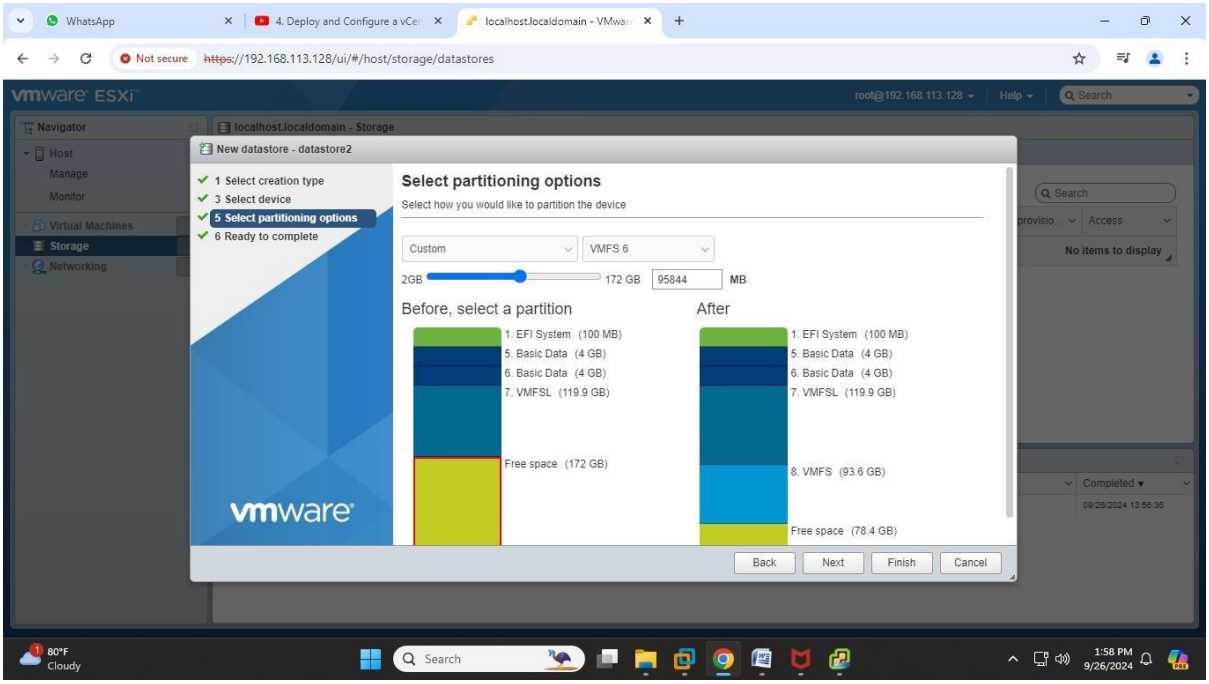
- Login to your esxi host web panel navigate to storage and click on new datastore.



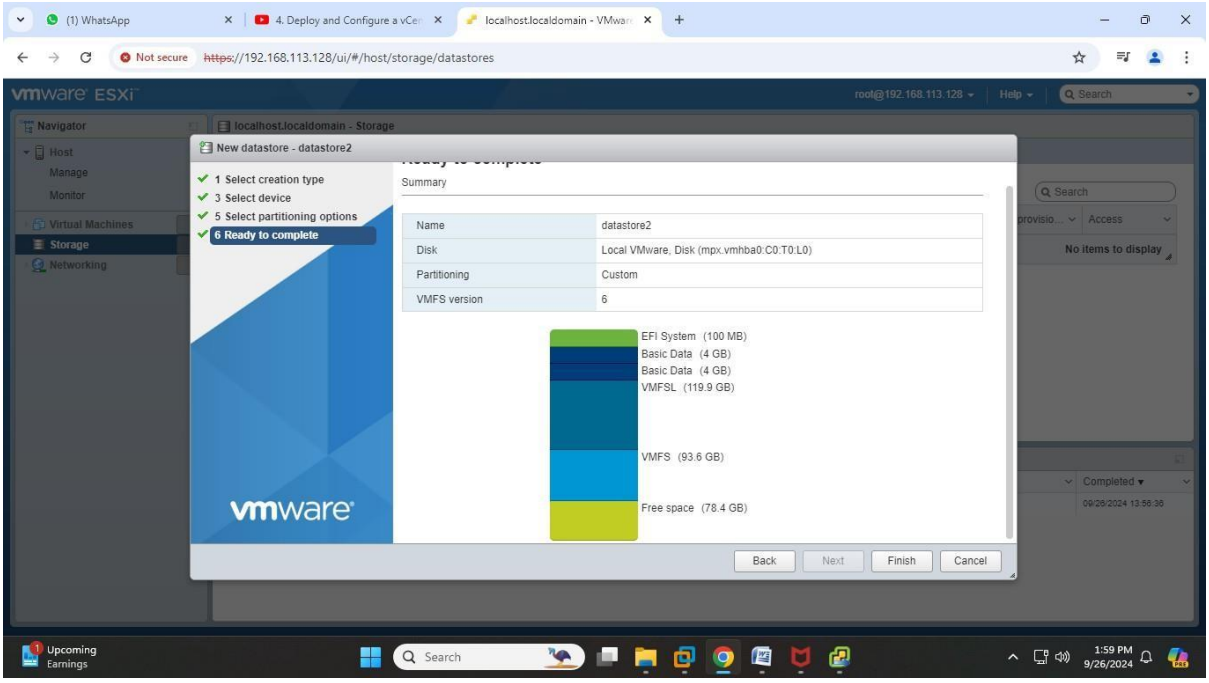
- Select Create new VMFS datastore.



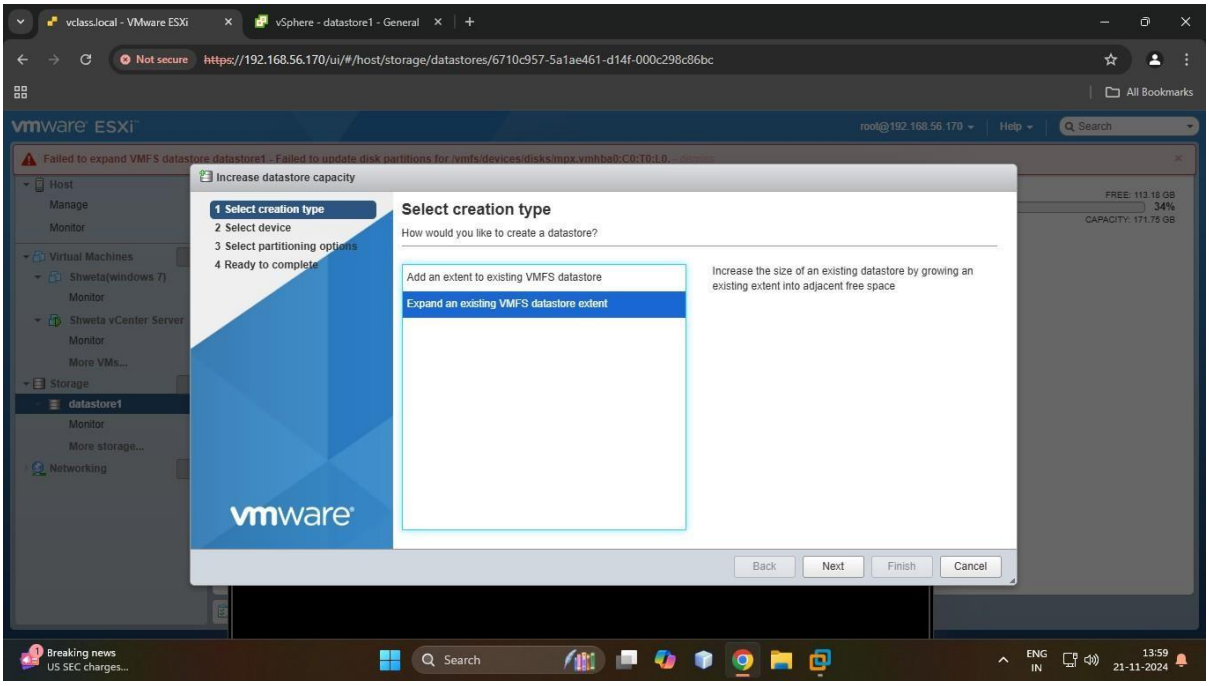
- Select the olive green division at left to set the storage for VMFS by moving the above bar.



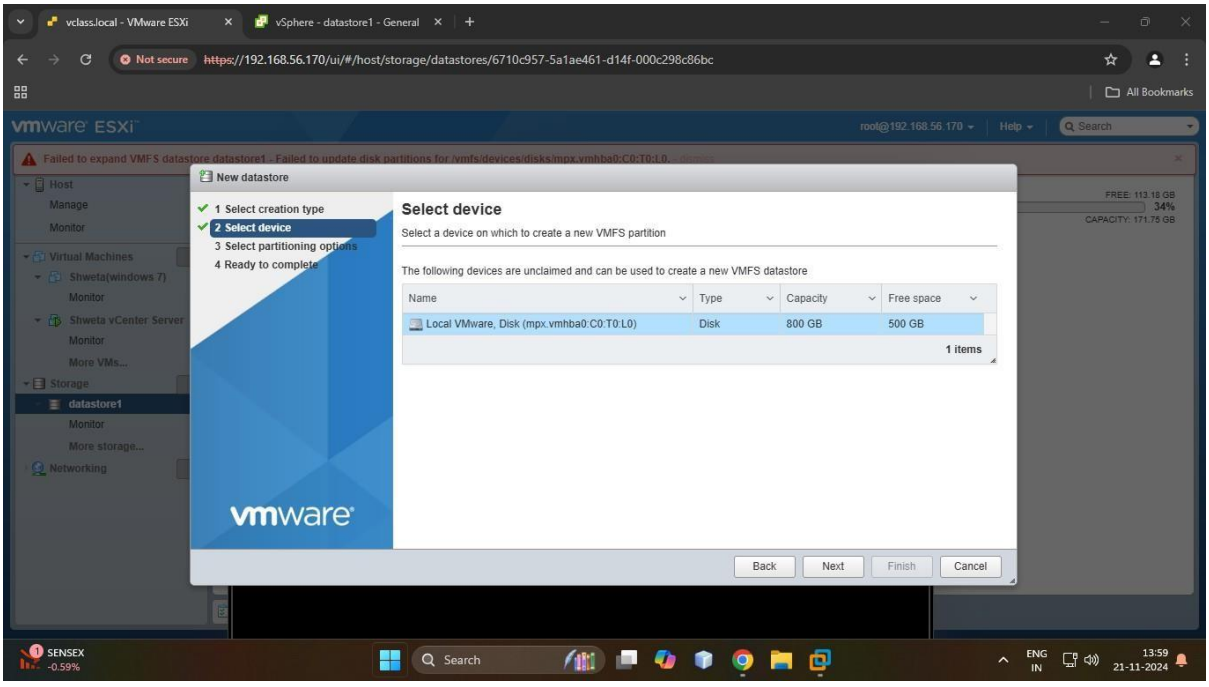
- Click on Finish.



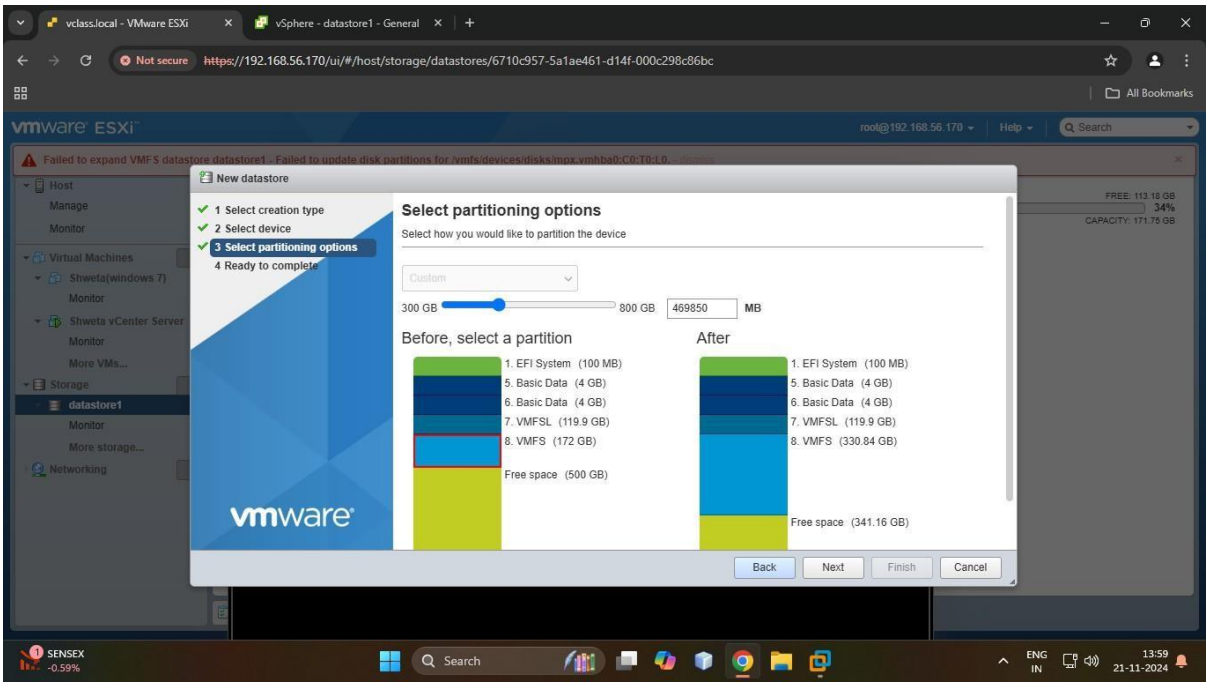
- Too expand the storage select expand and click on next.



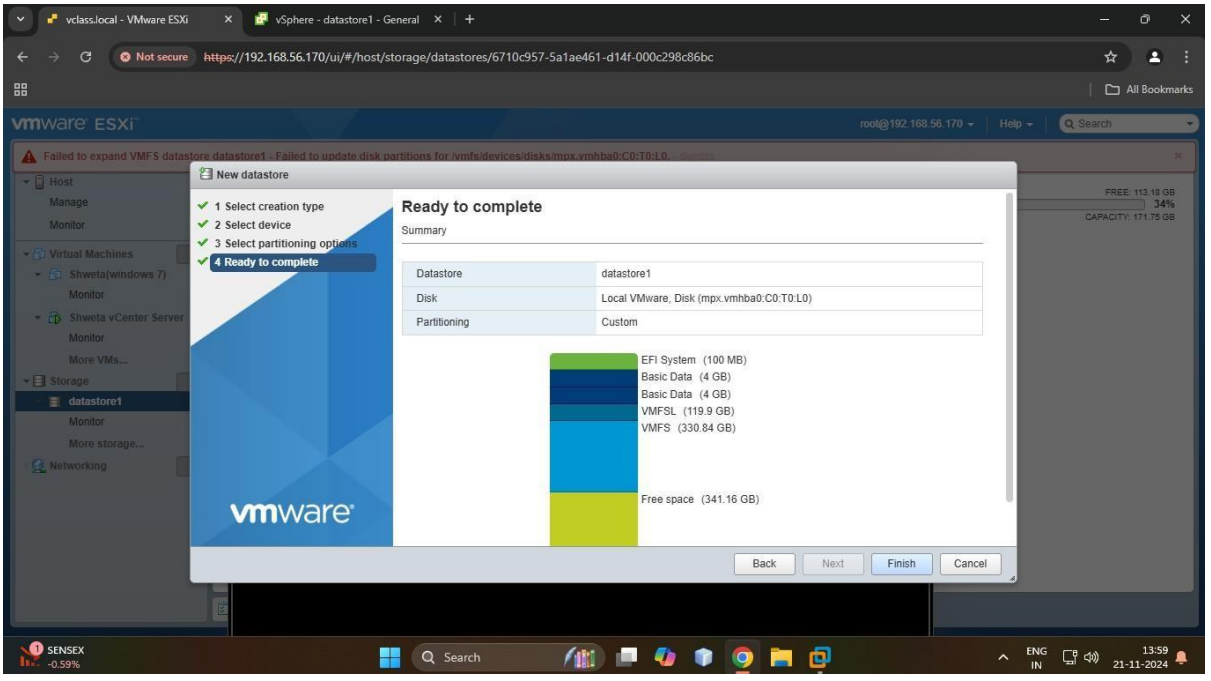
- Select the newly created datastore.



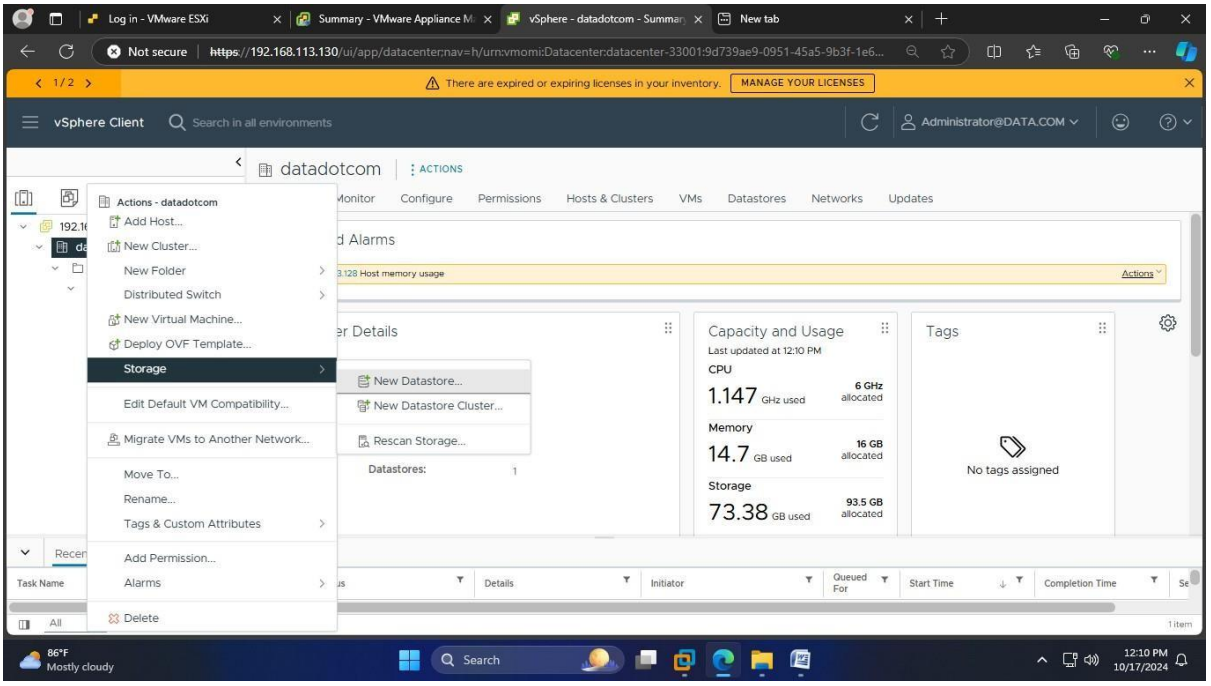
- Set the storage by scrolling bar allocate both the vmfs and free space a similar storage.



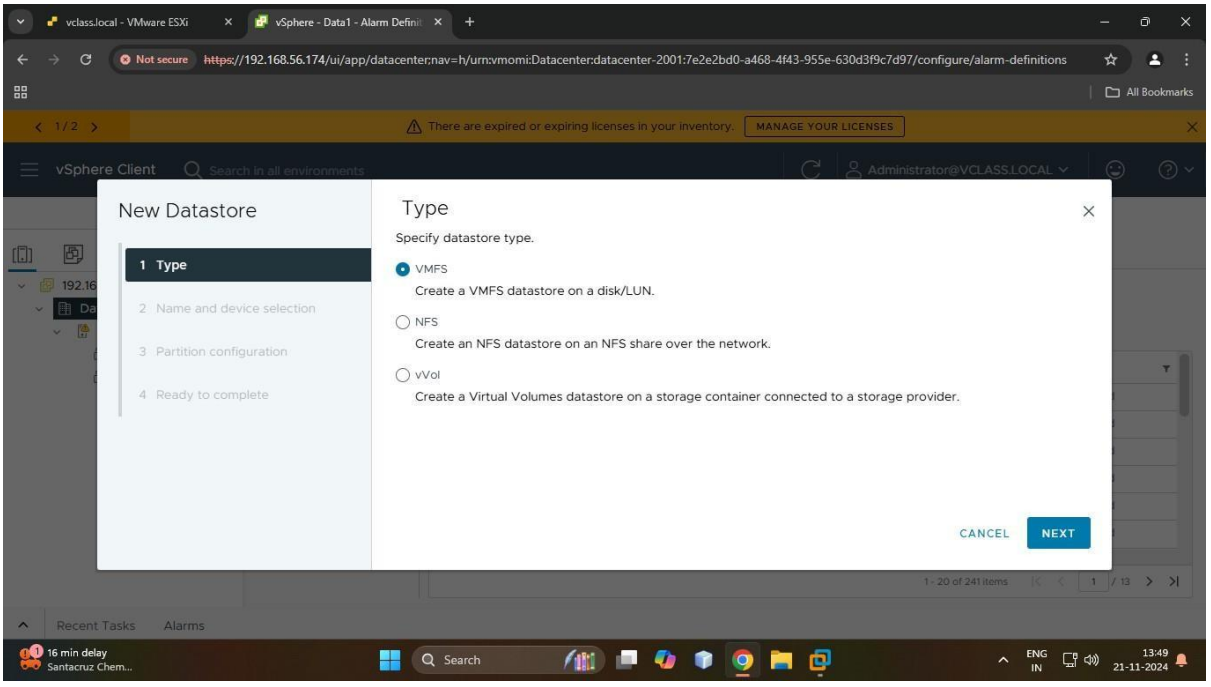
- Our VMFS expanded Click on Finish.



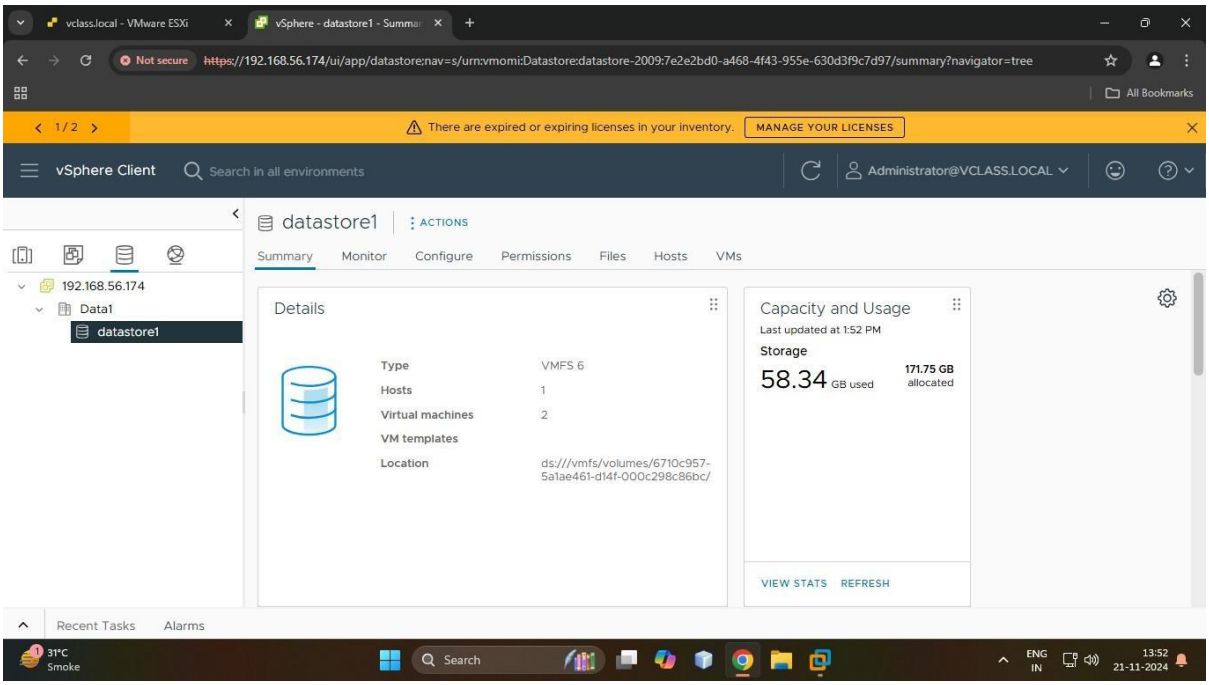
- Login vcenter and go to storage and new data store.



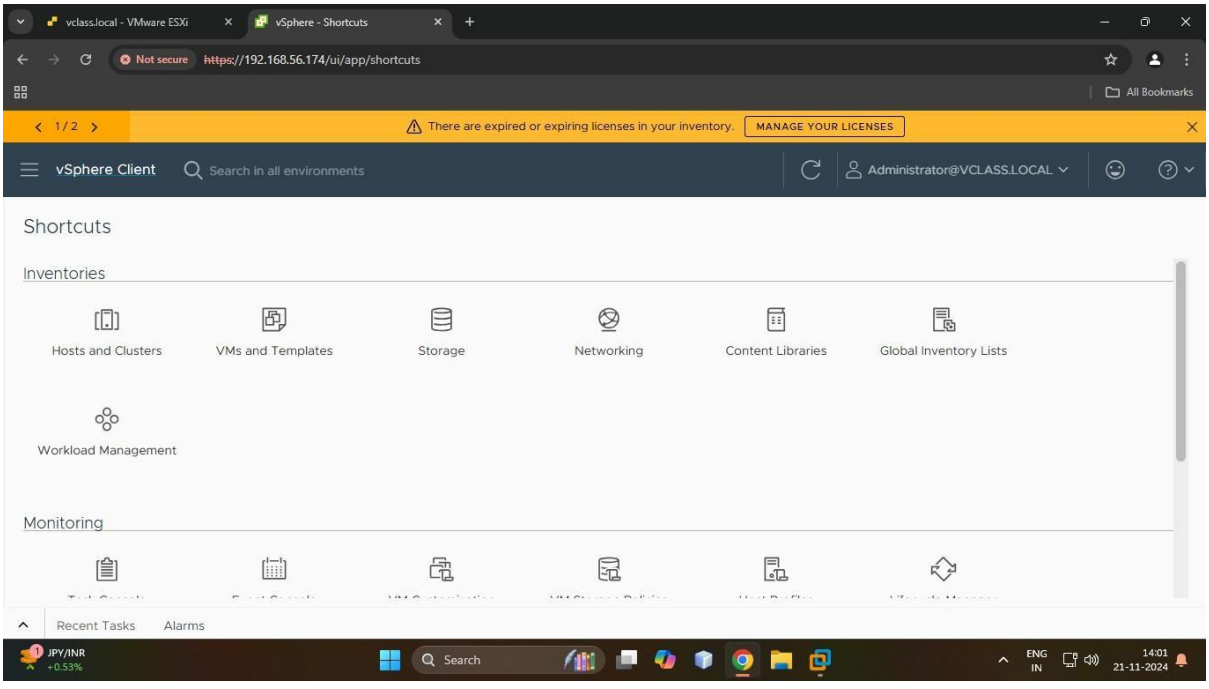
- Select VMFS and Click on next.



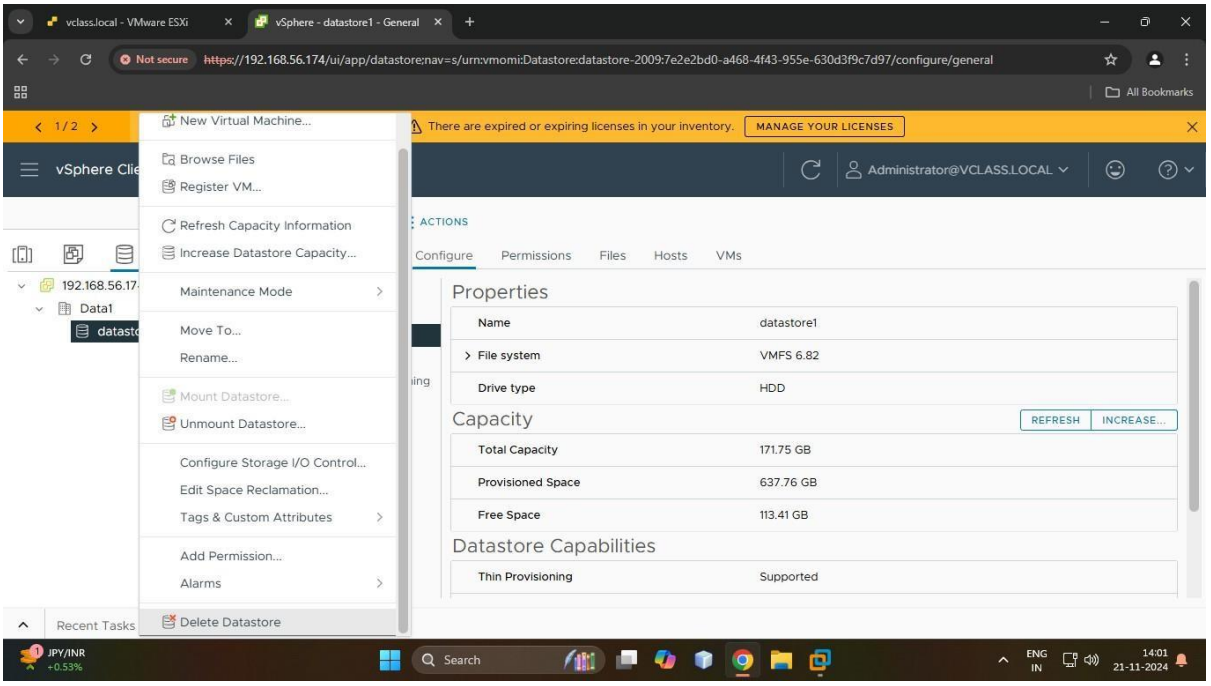
- Our datastore is created



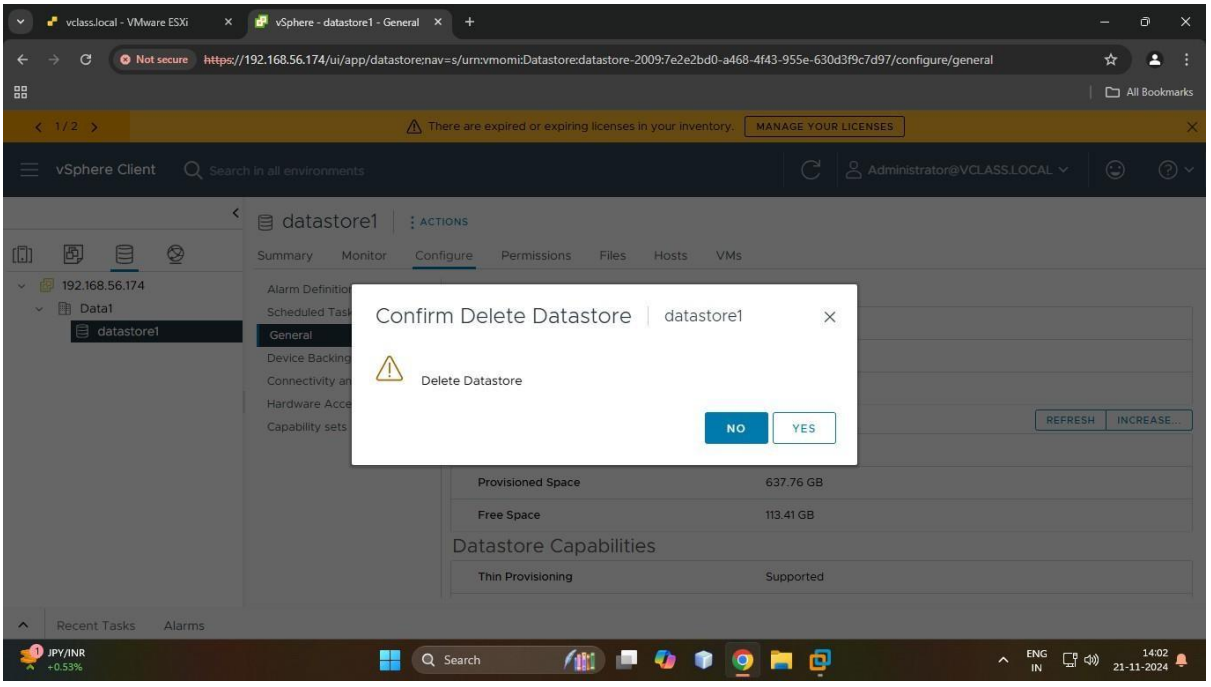
- Navigate to vcenter ->storage.



- Delete datastore.



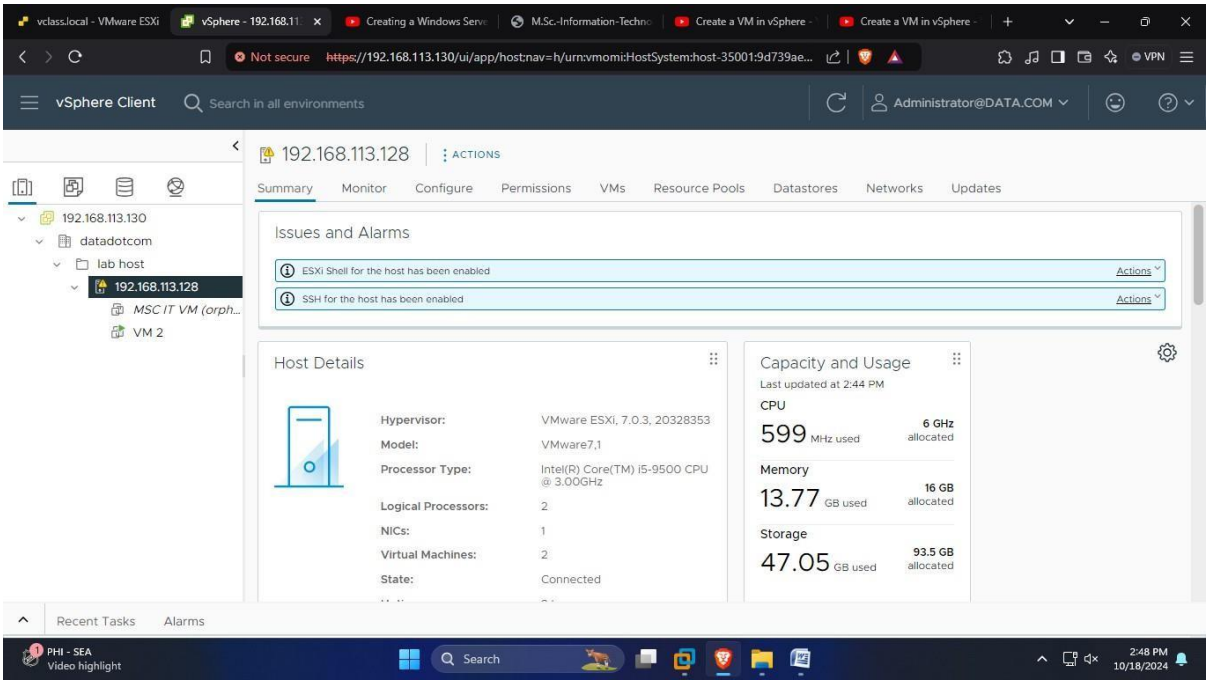
- Click on yes.



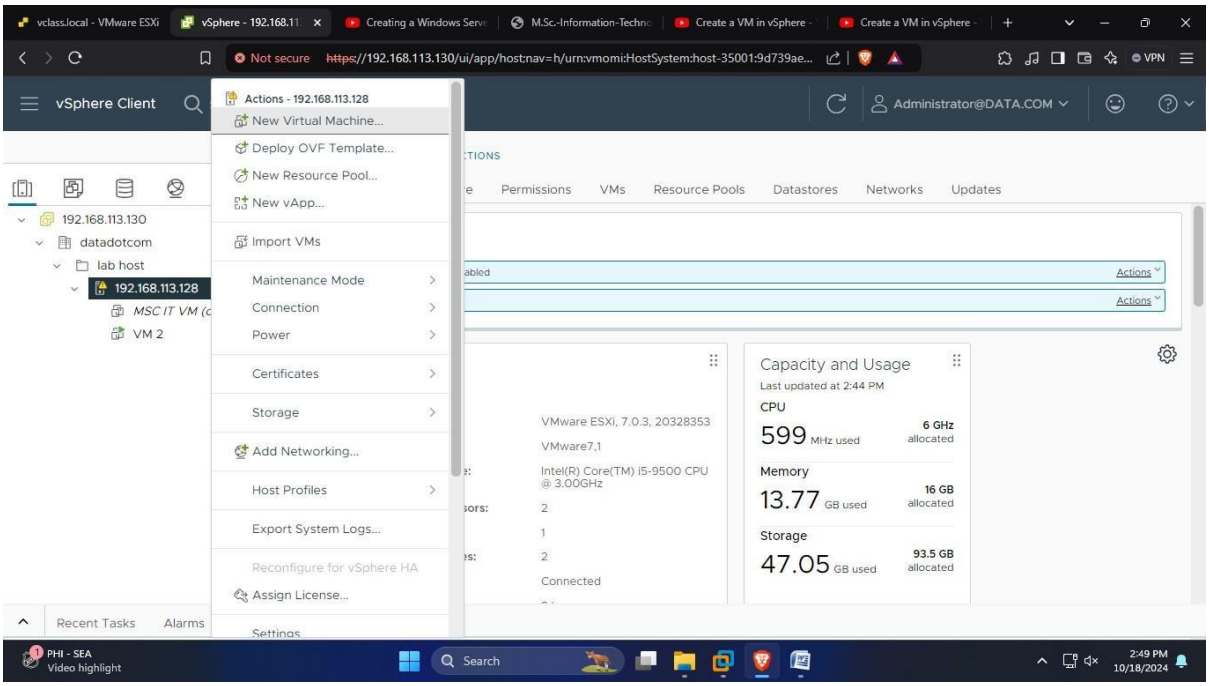
PRACTICAL 7

AIM:- Using Templates and Clones

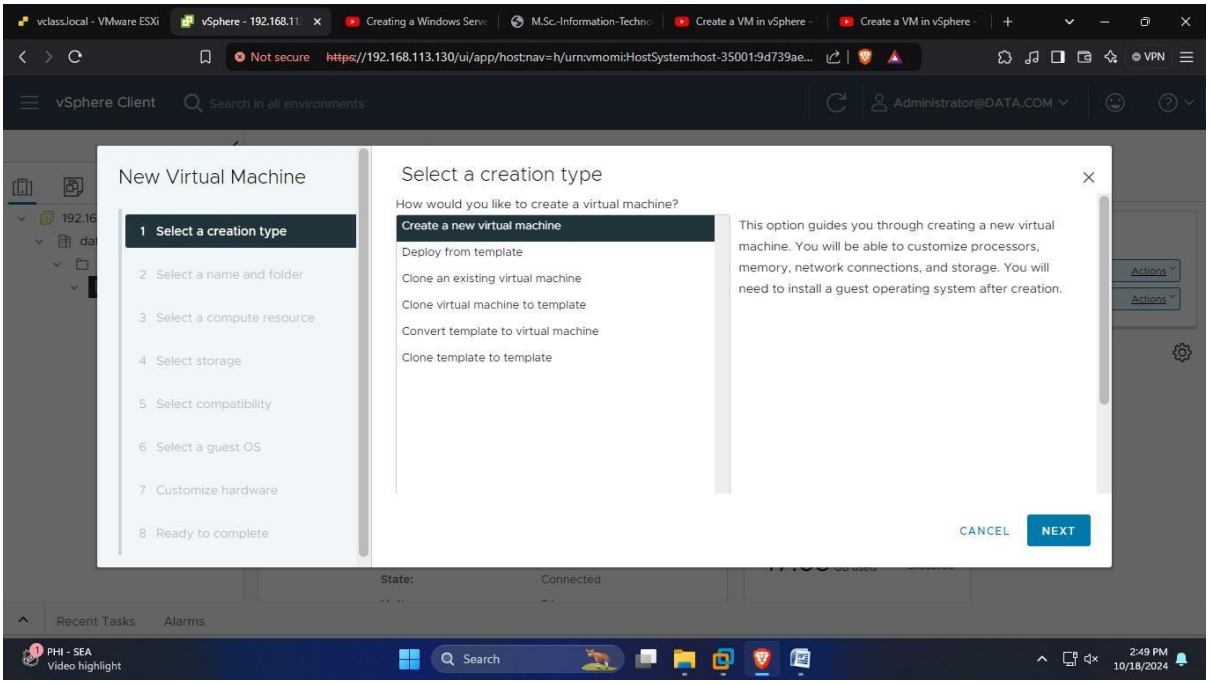
- Go to vcenter



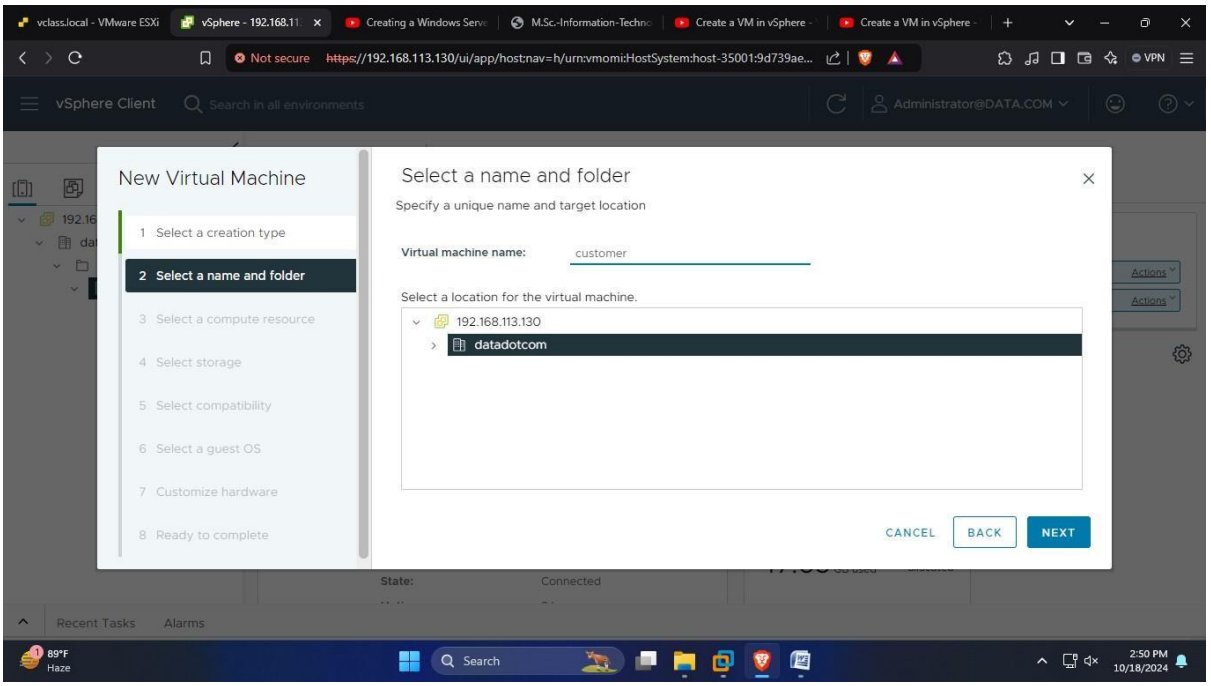
- Create a new virtual machine on your host.



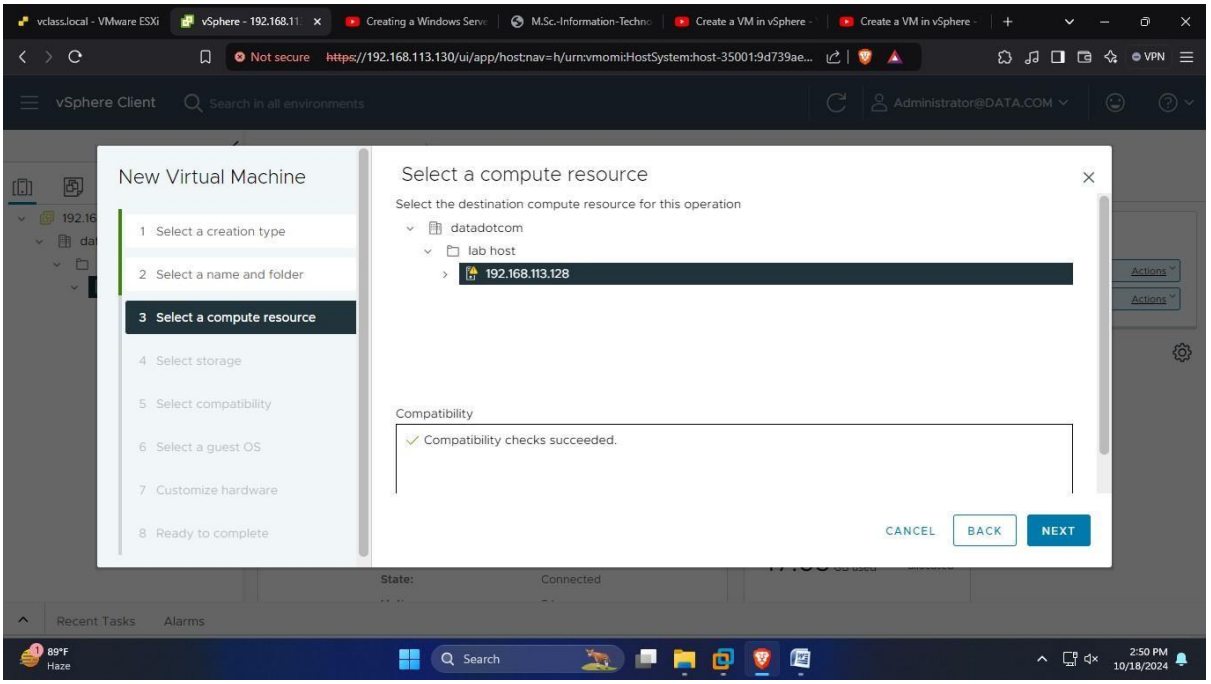
- Click on next.



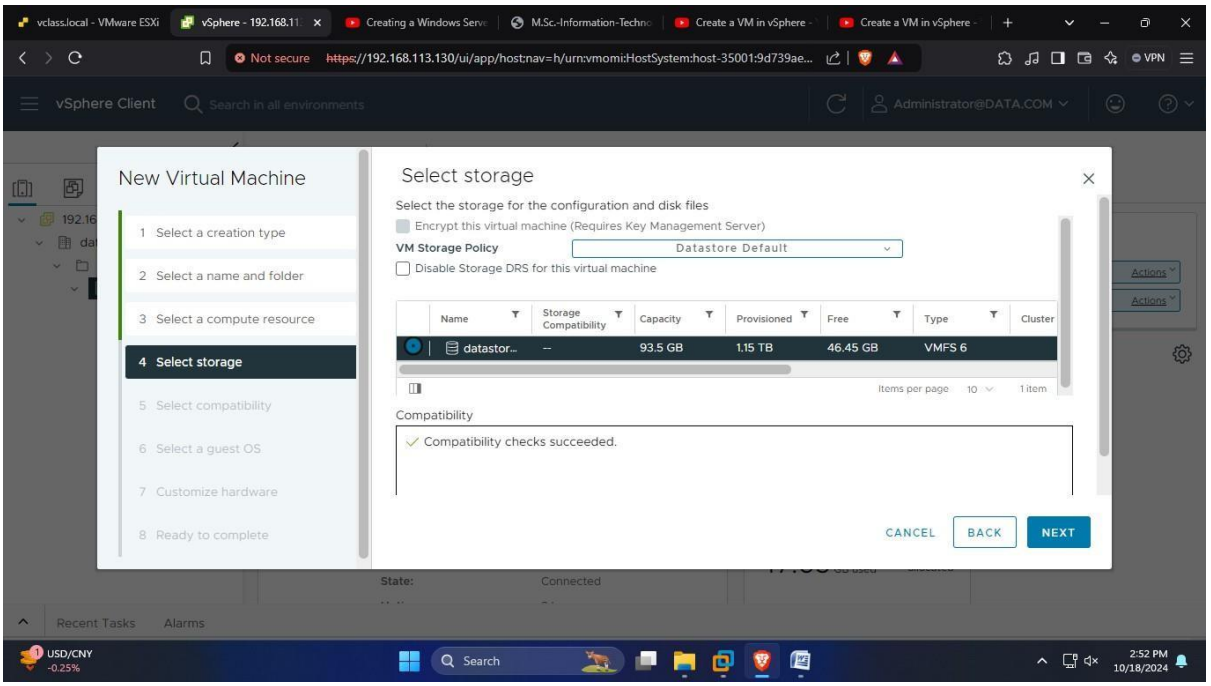
- Click on next.



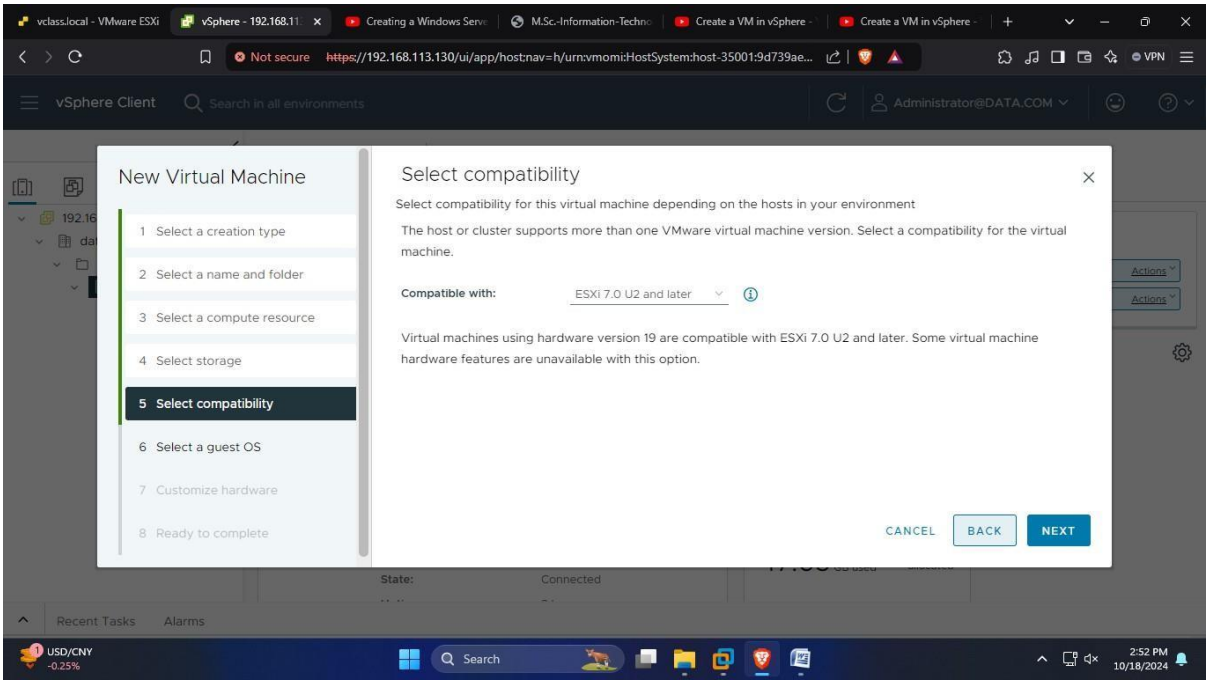
- Click on next.



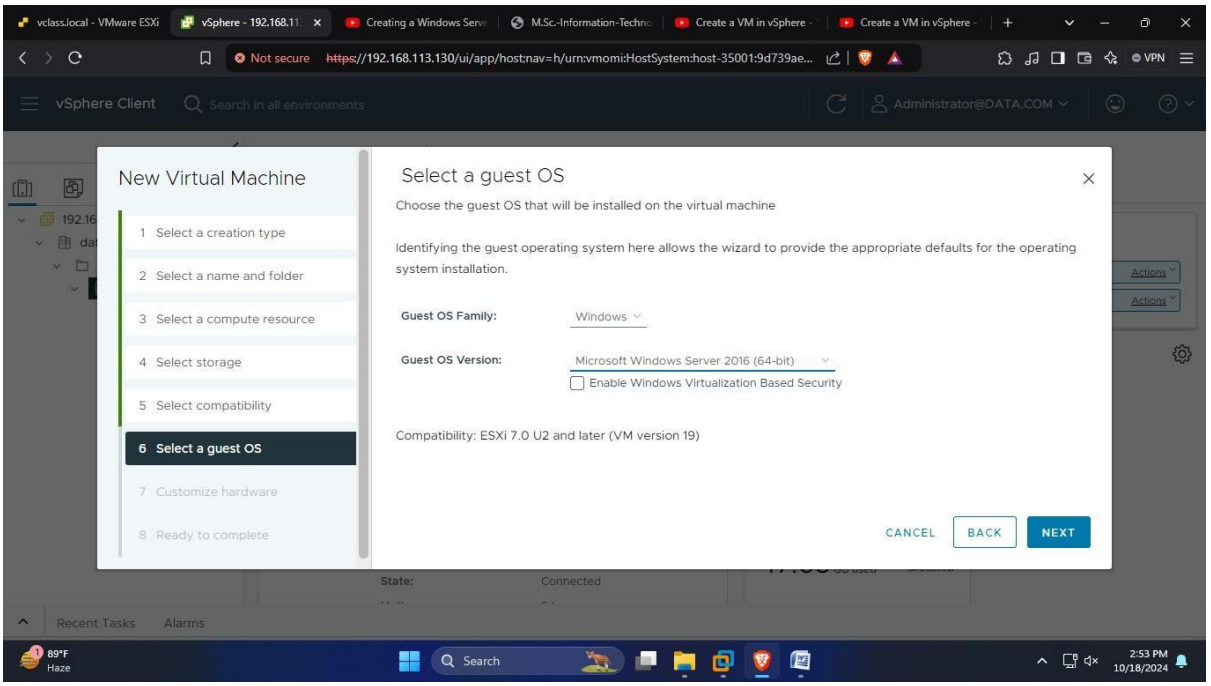
- Select datastore and Click on next.



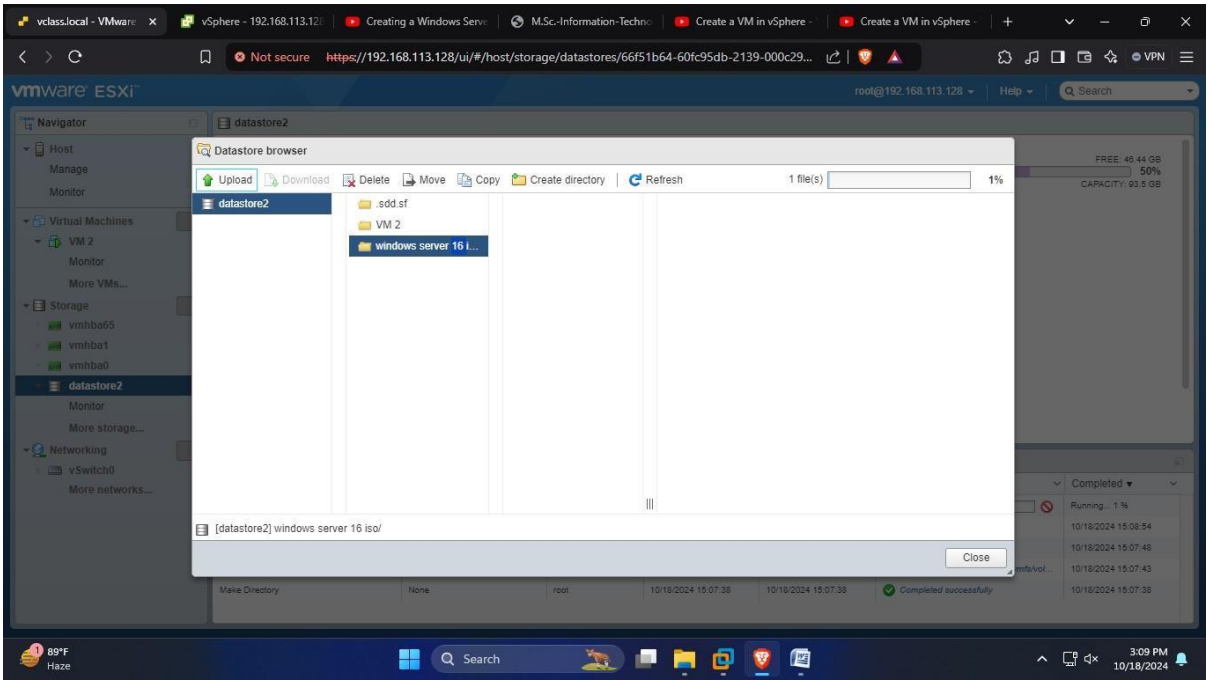
- Select esxi 7 and Click on next.



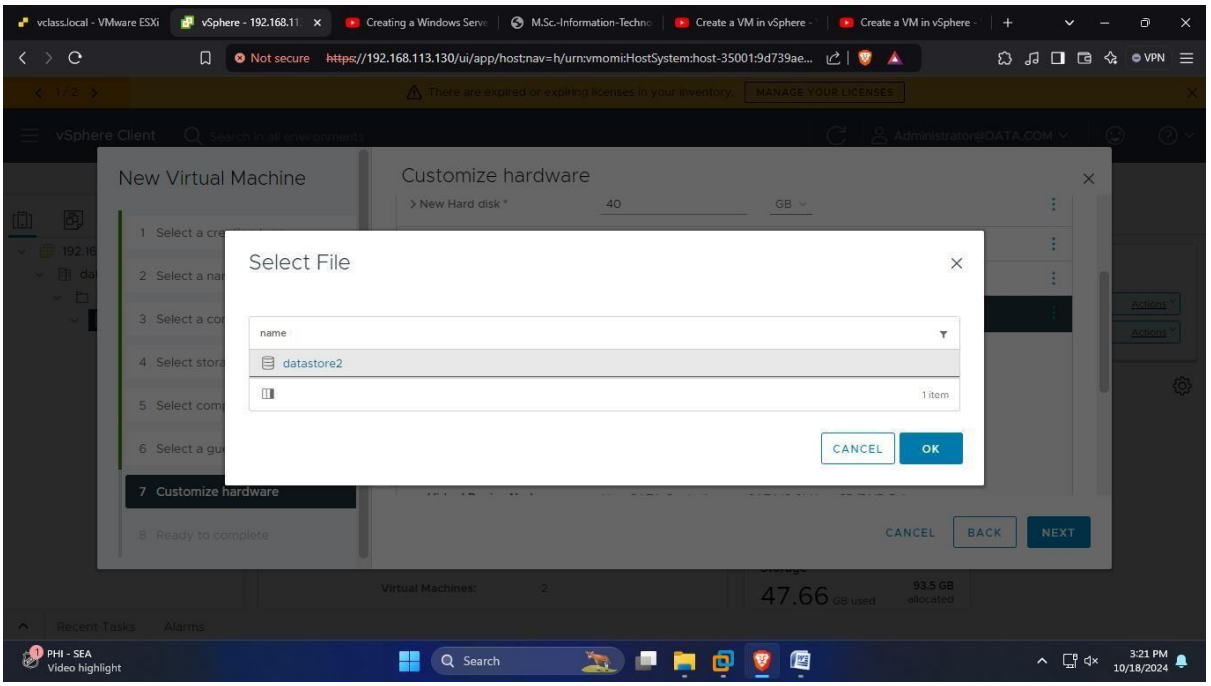
- Guest OS Select Cent os because it is open source.



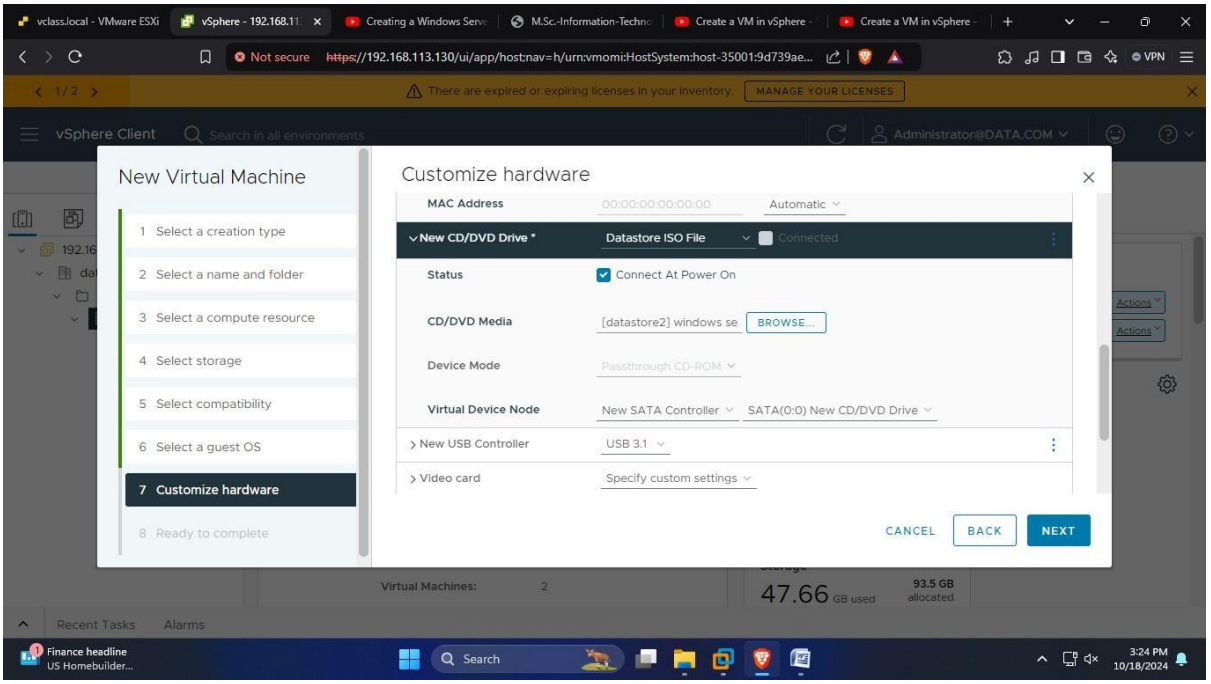
- Upload iso file of OS.



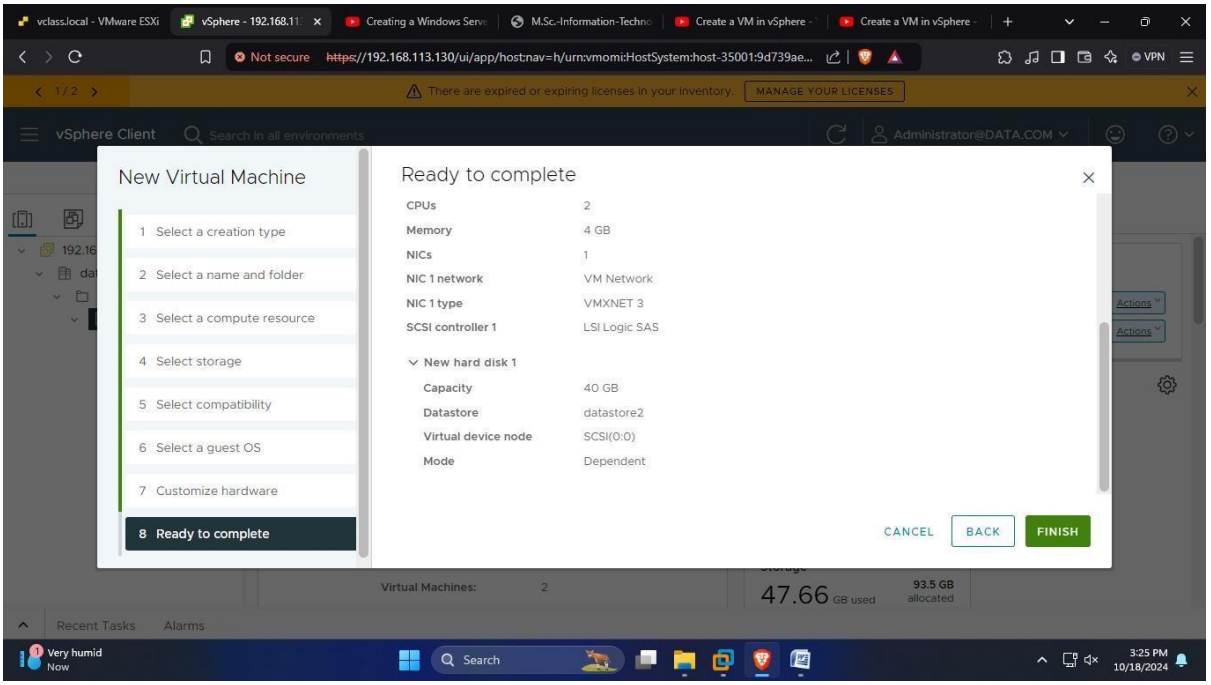
- Select data store.



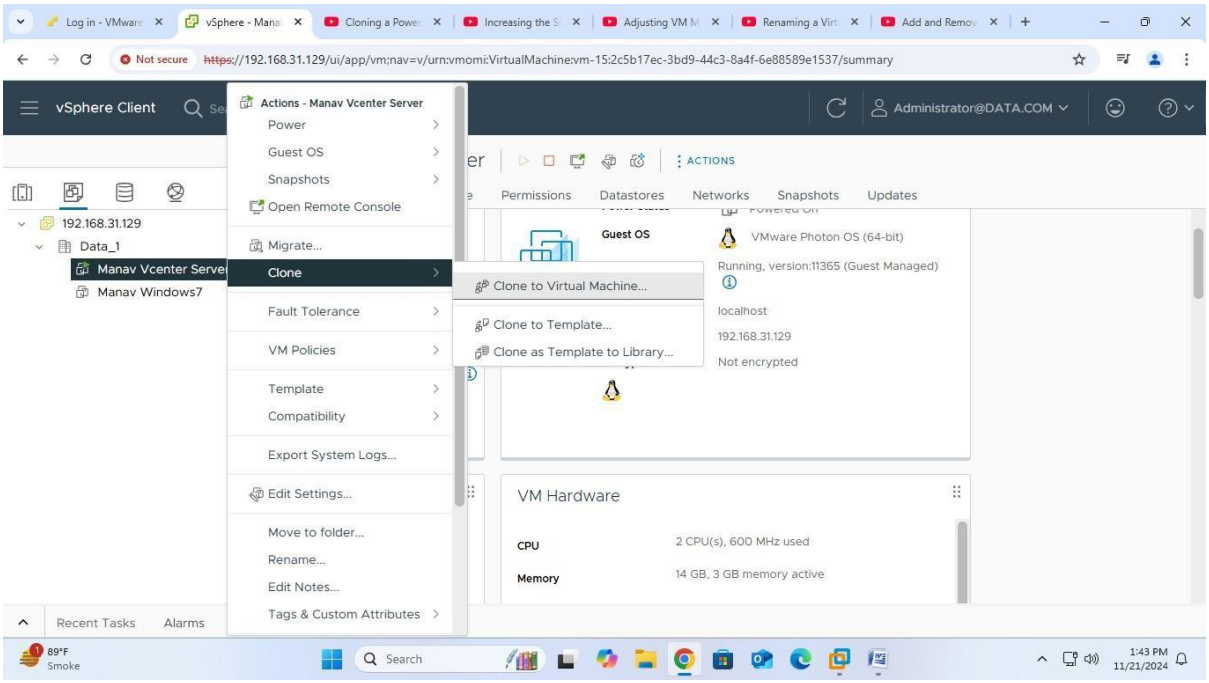
- Connect at power on . and click on next.



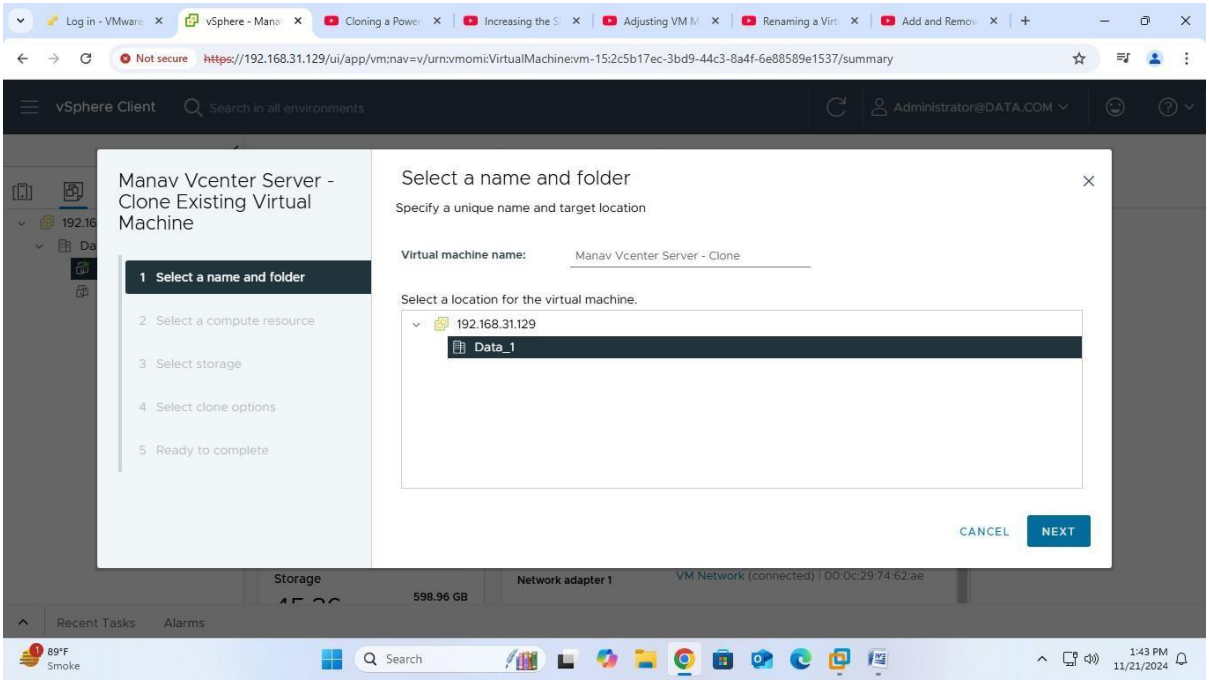
- Click on Finish .



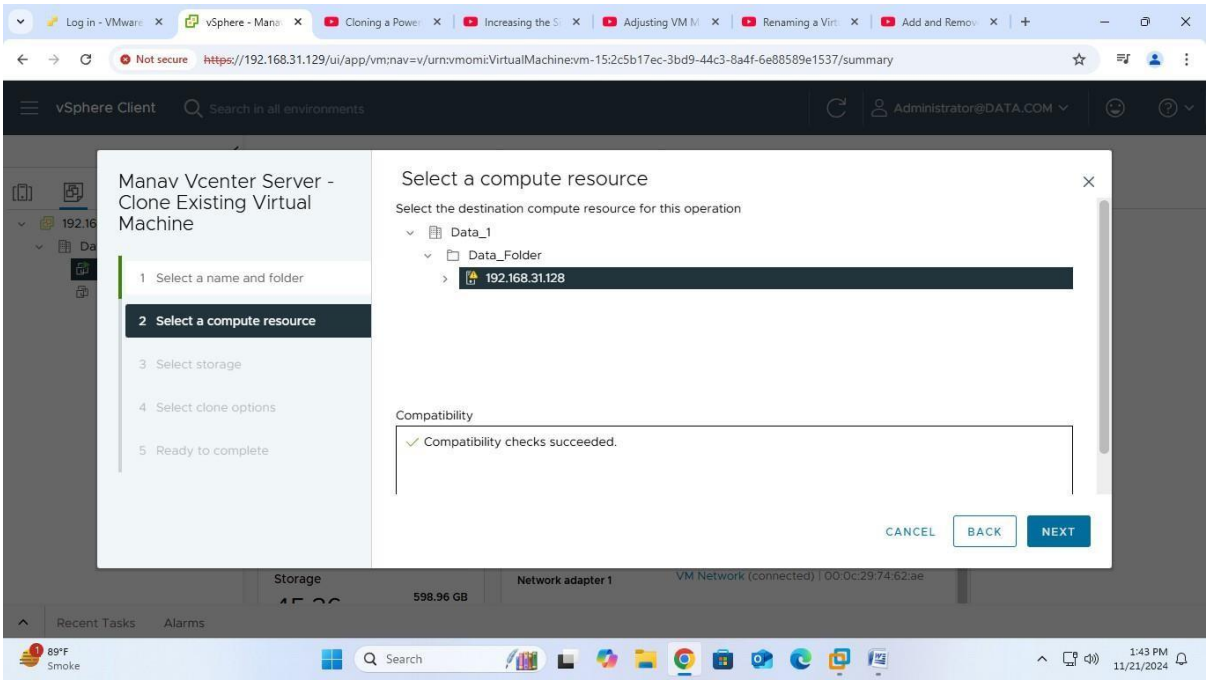
- Right Click on The Virtual Machine You Just Created.



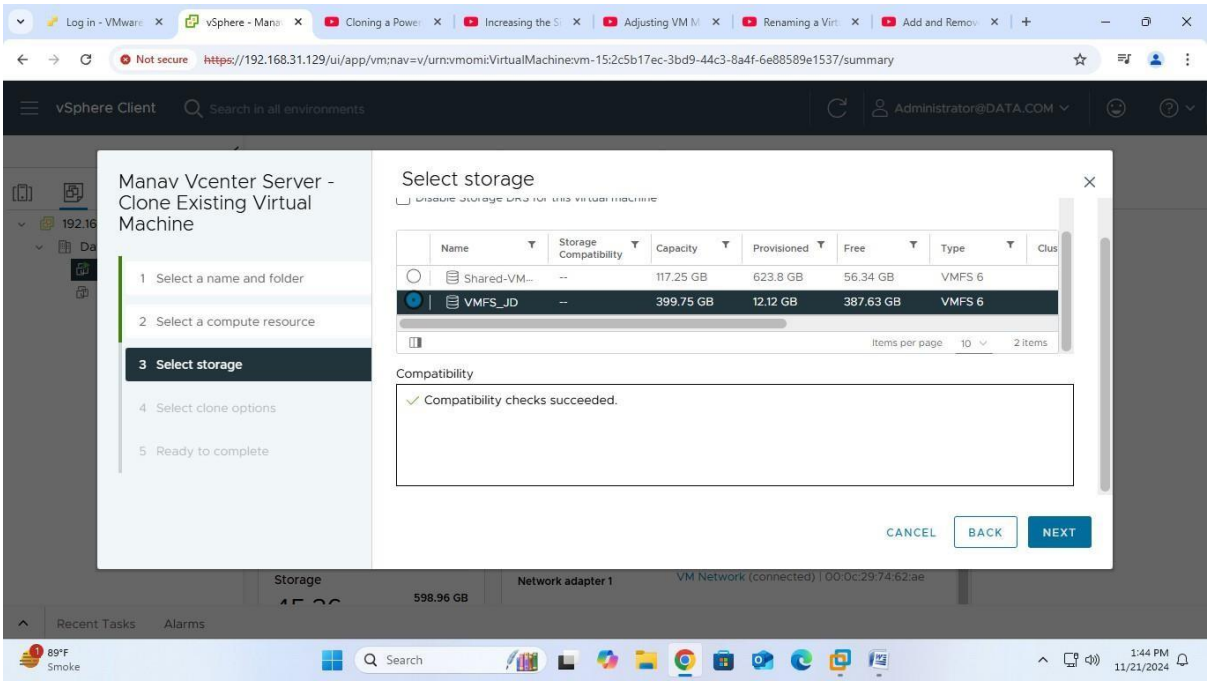
- Give name it a name and select host and then Click next.



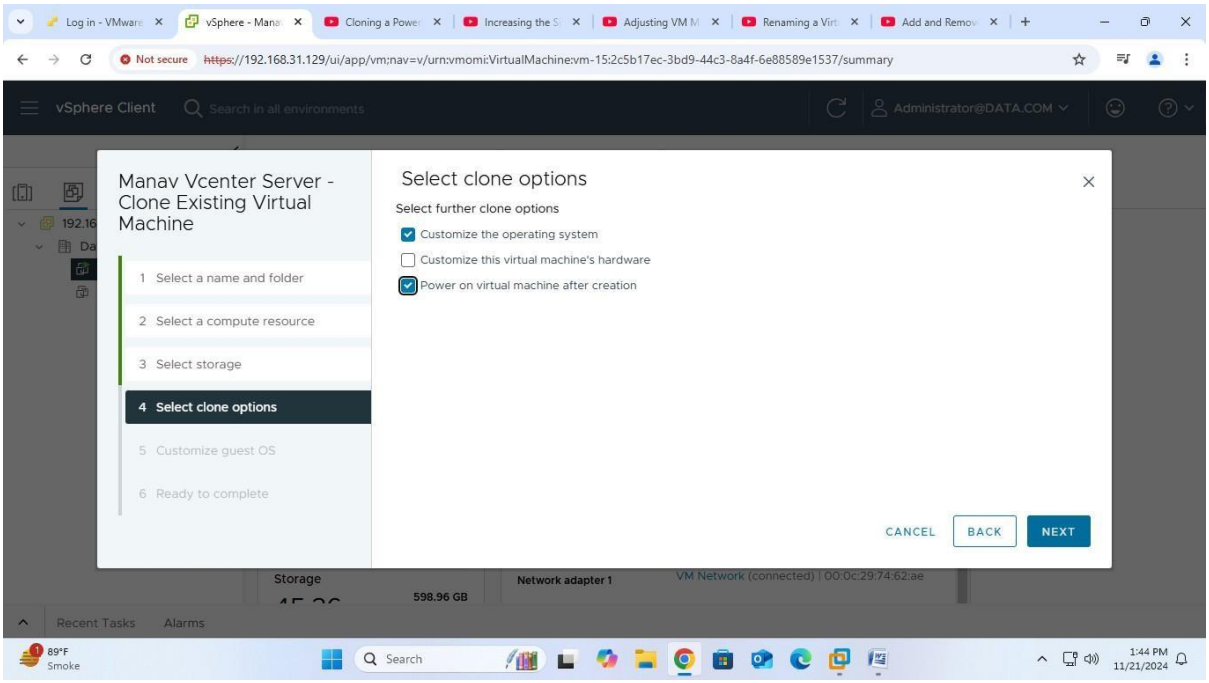
- Click on next .



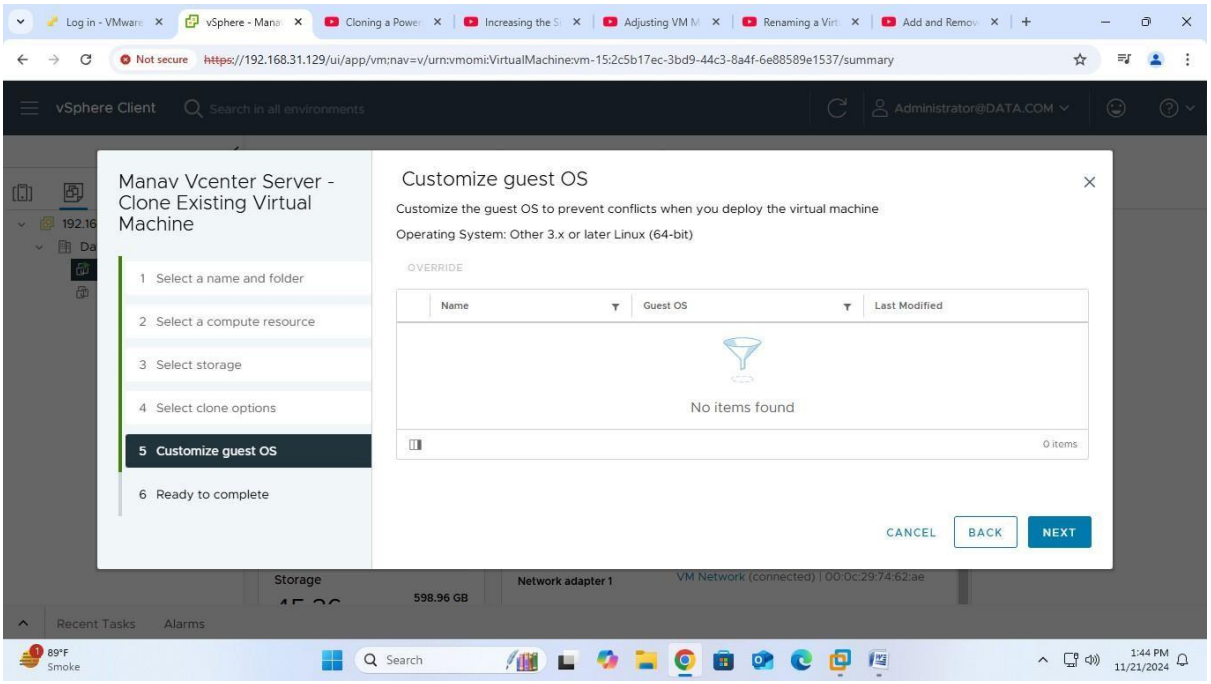
- Select VMFS.



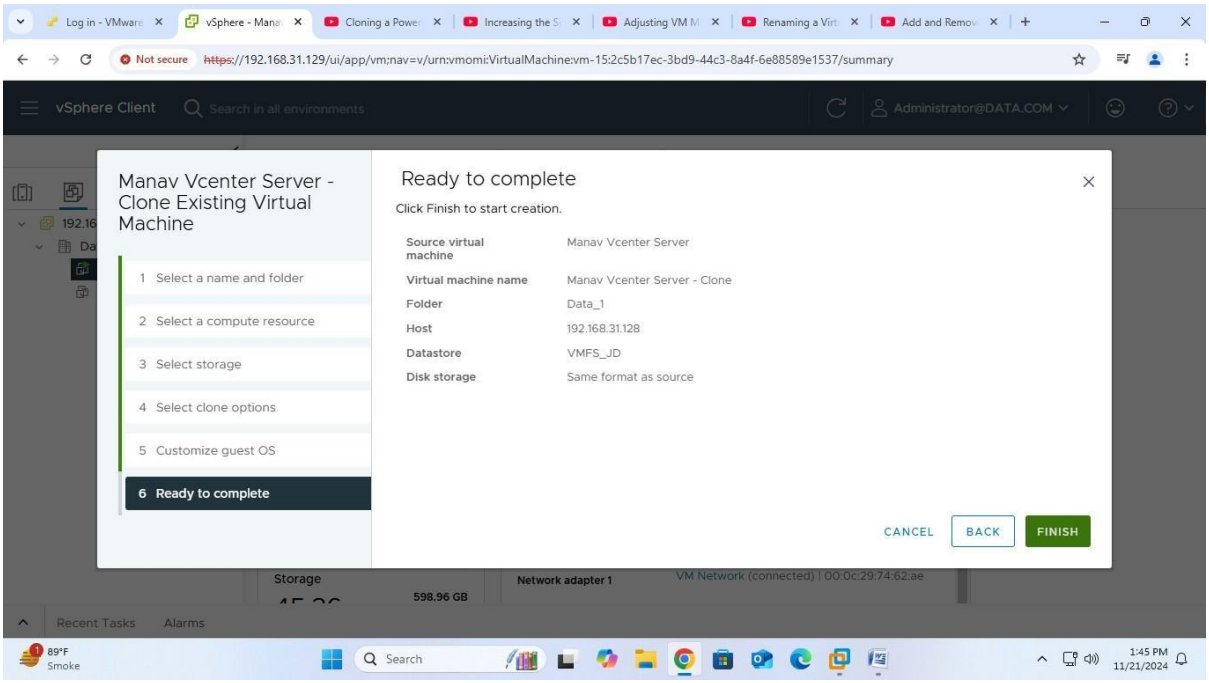
- Select customize the os and power on the virtual machine and Click on next.



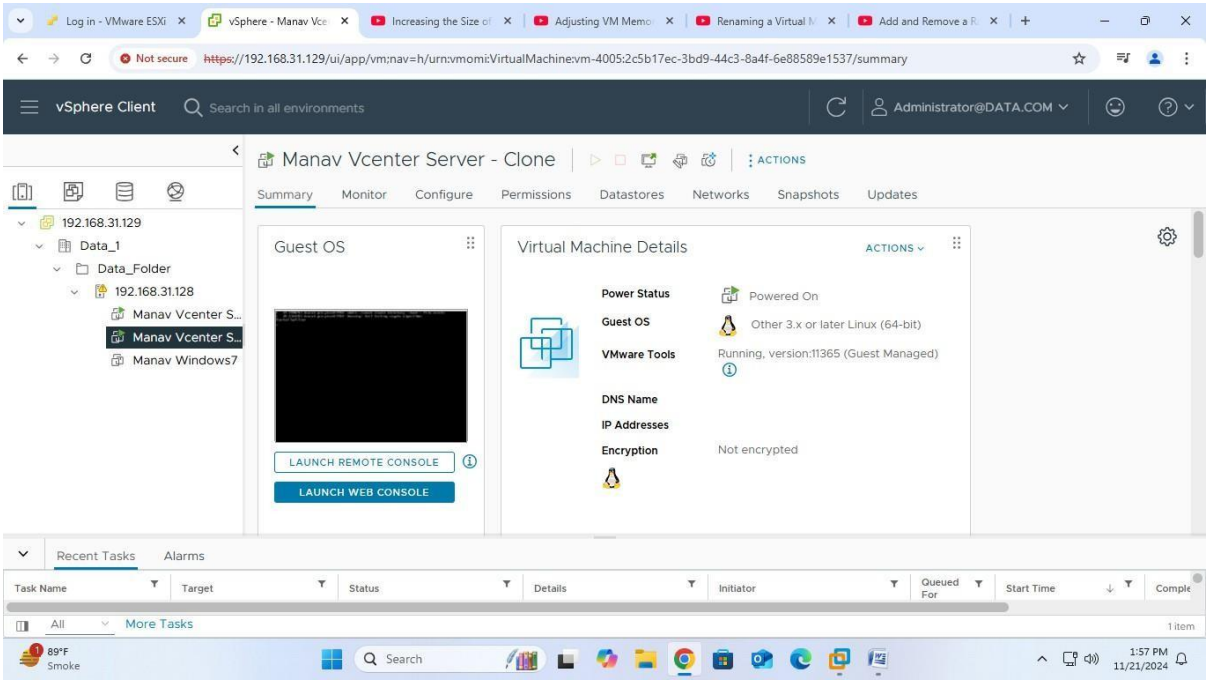
- Click on next.



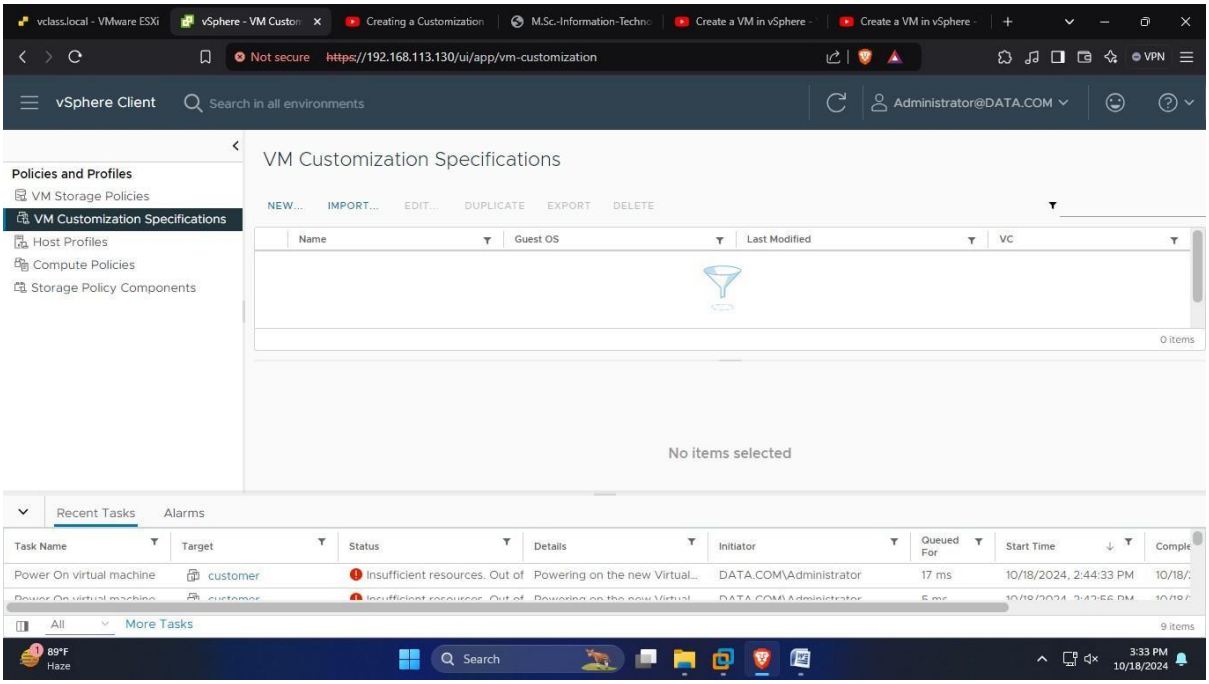
- Click on Finish.



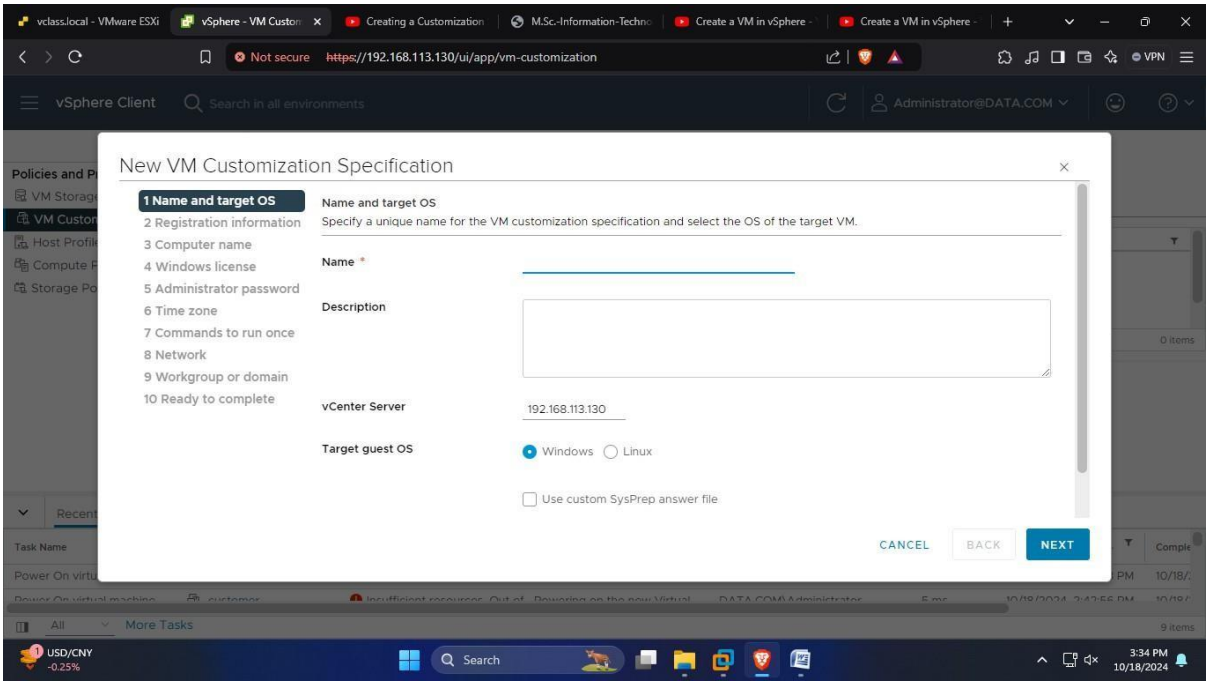
- Here we created clone of our virtual machine.



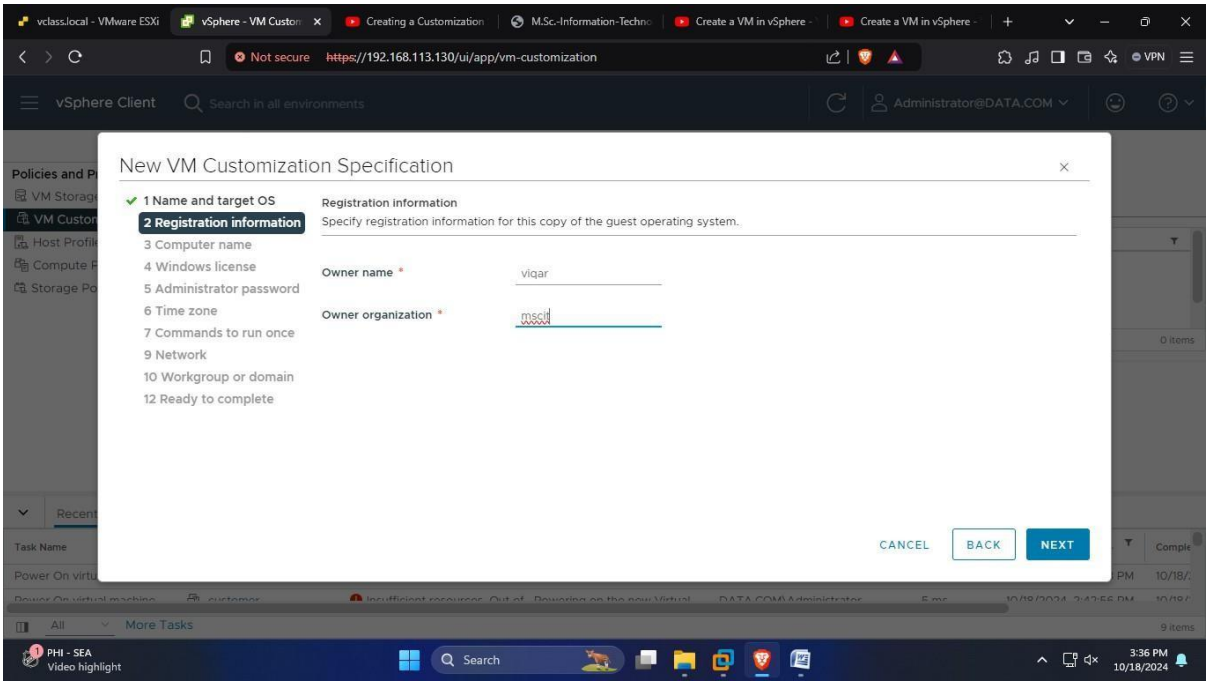
- Navigate to policies and profiles ->VM Customization Specfications.



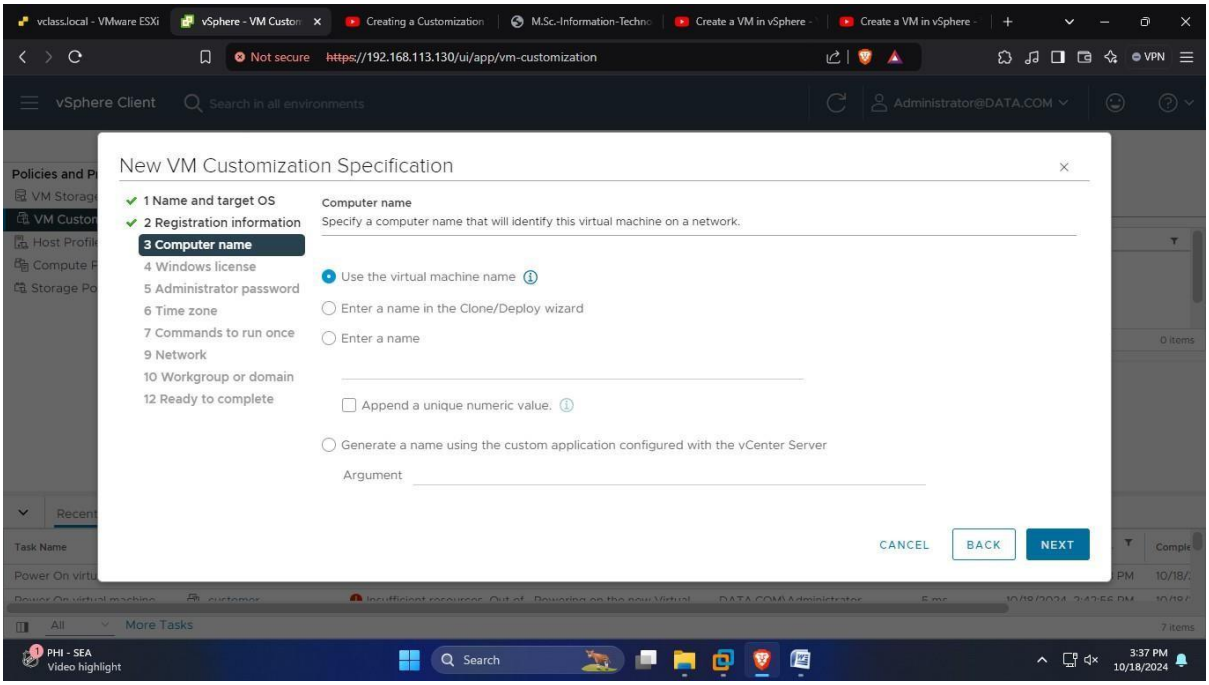
- Name it and Select Linux and Click on next.



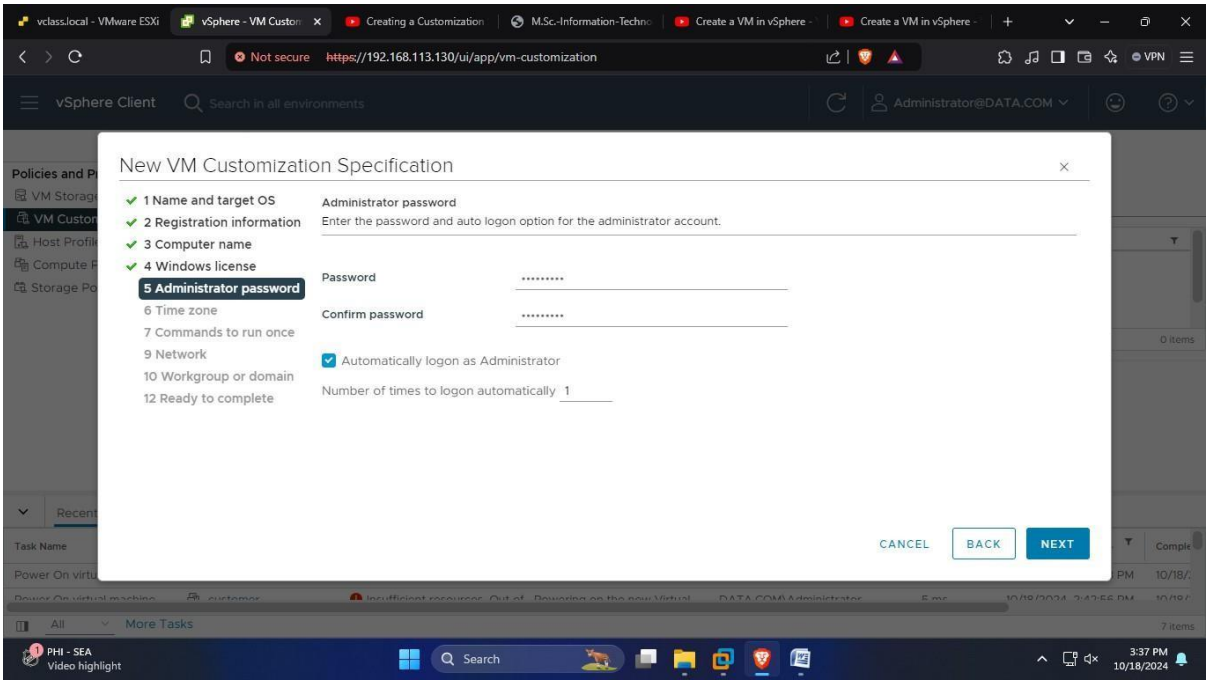
- Fill the Required credentials.



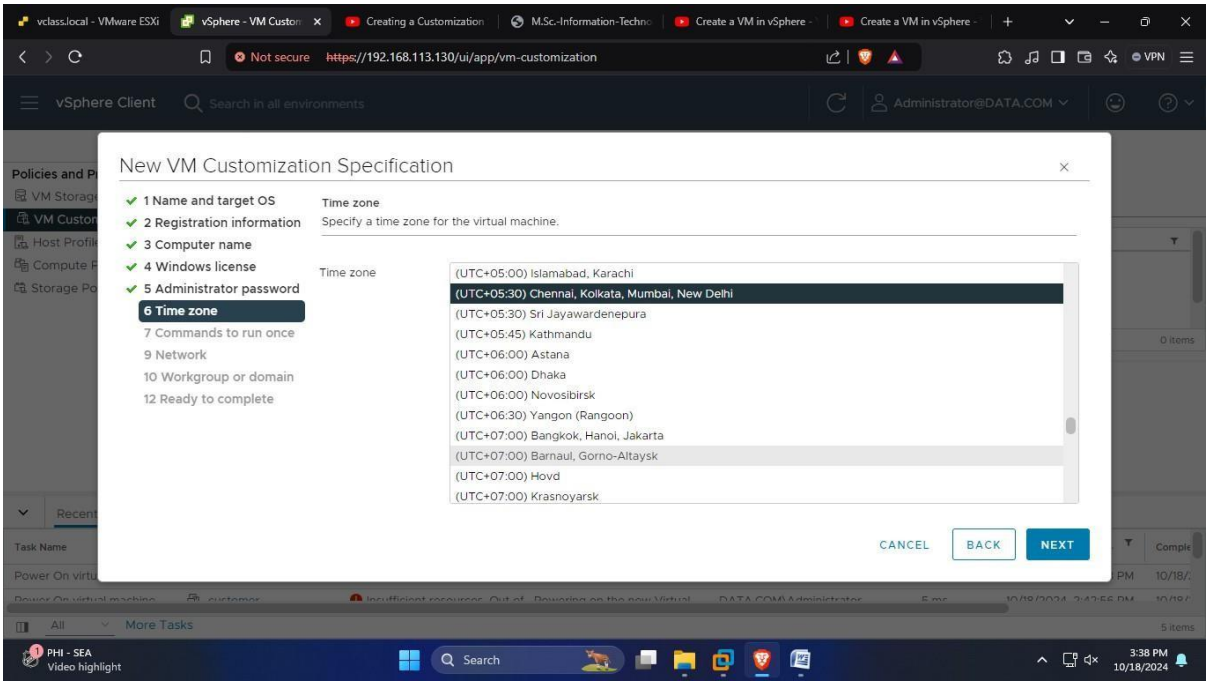
- Select use the virtual machine .



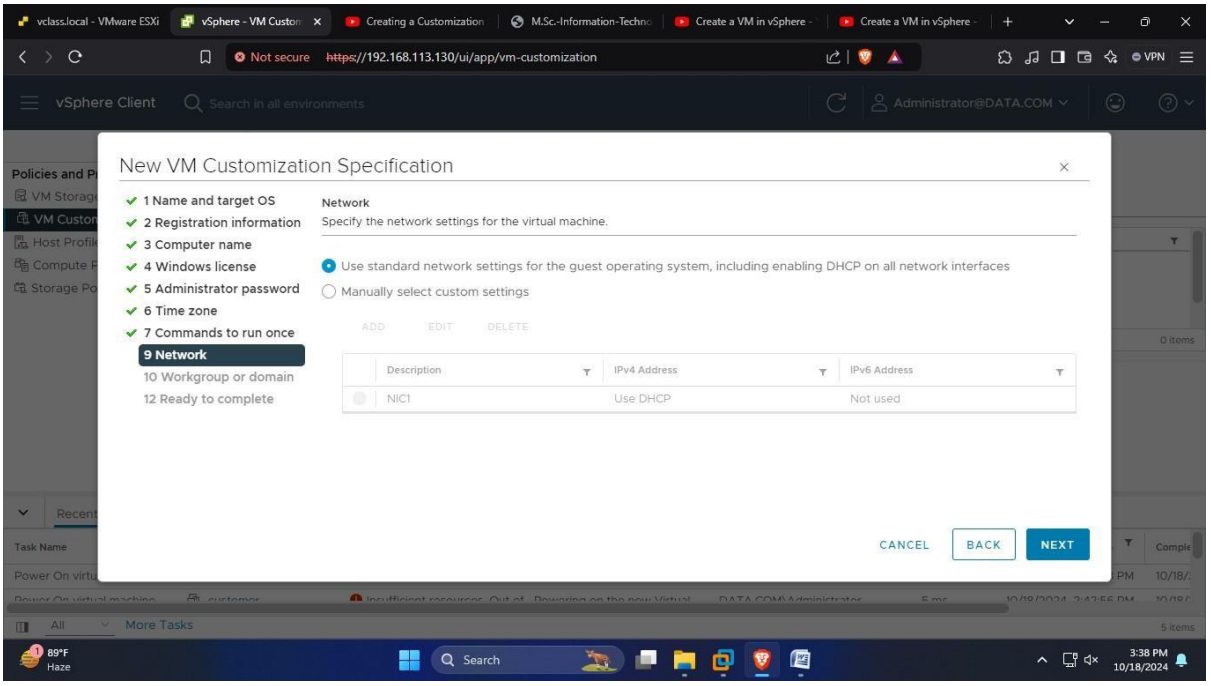
- Give it a password here it is Name!098.



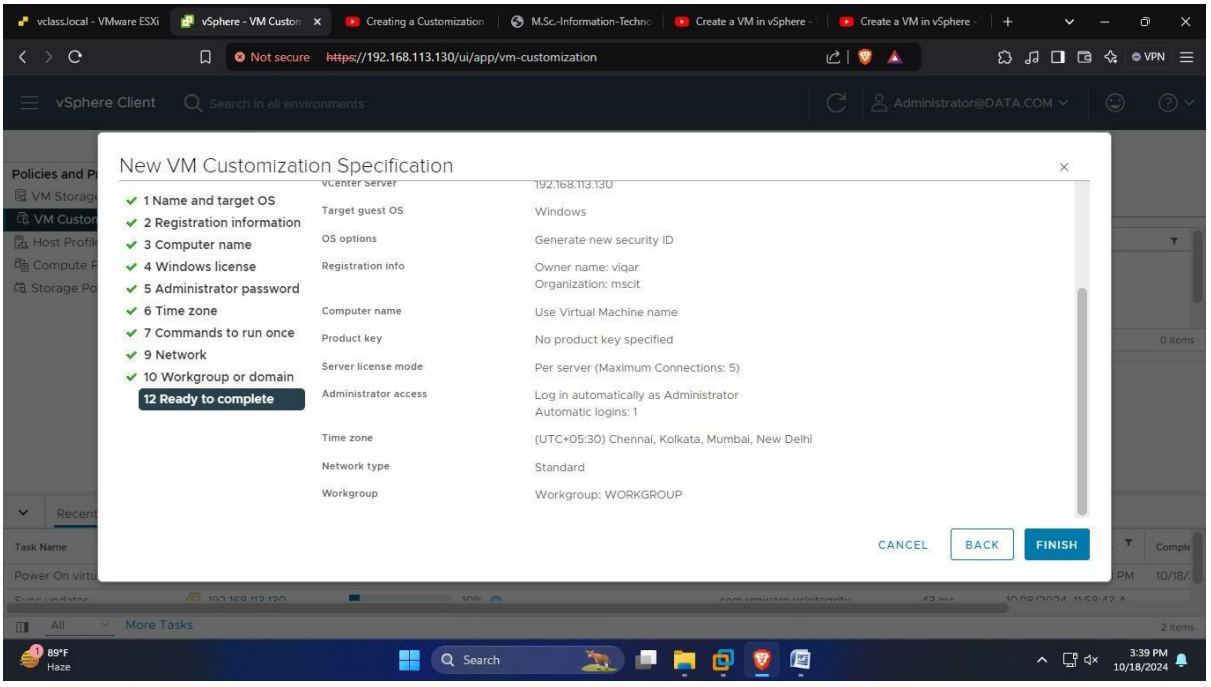
- Select According to your time zone.



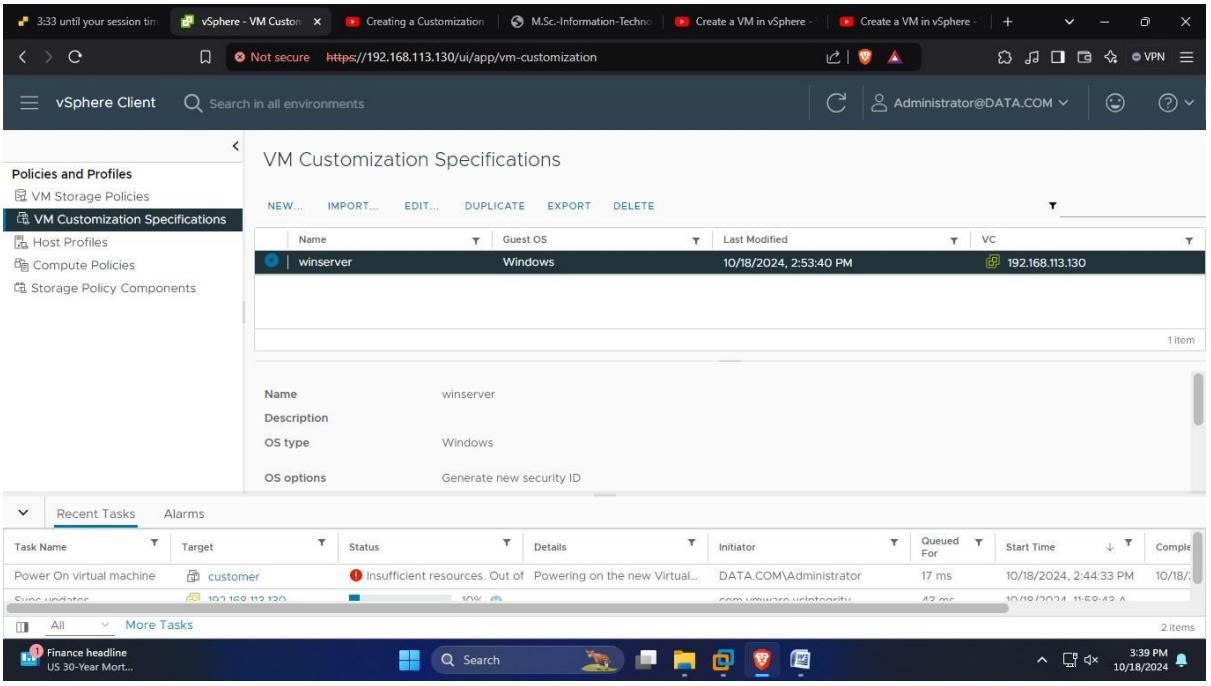
- Select use Standard network Settings.



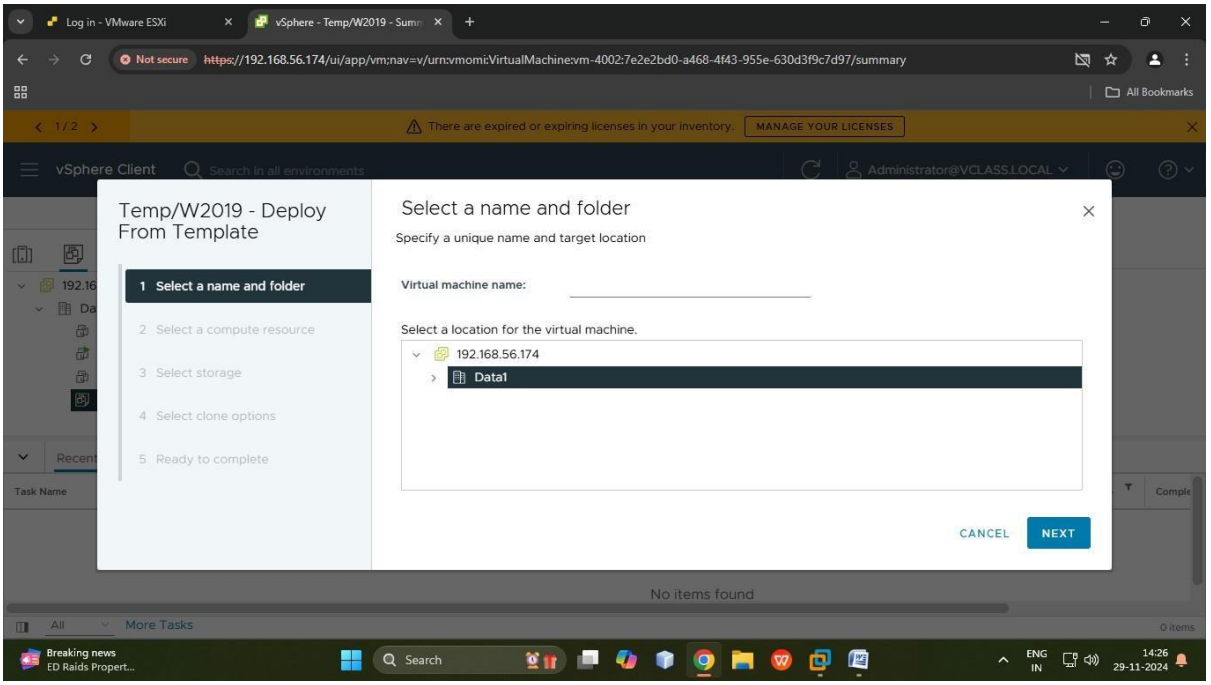
- Click on Finish.



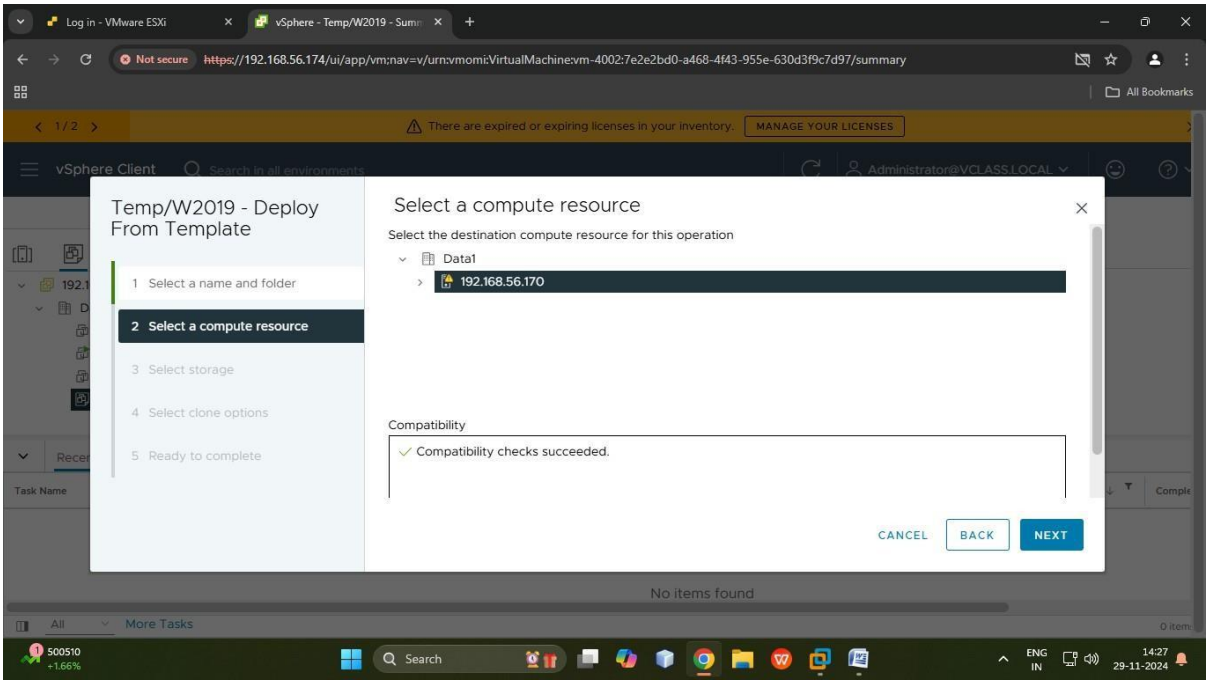
- VM customization and specification Done.



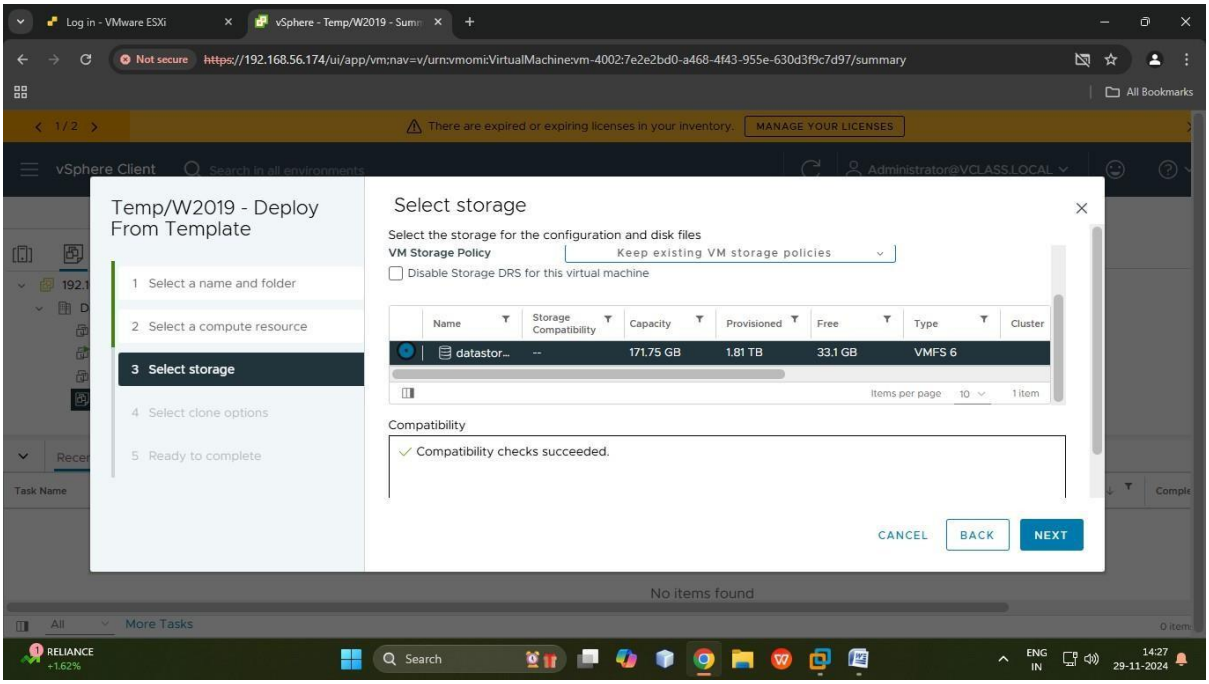
- Right Click on Virtual Machine and Click on Deploy from template and name it and Click on next.



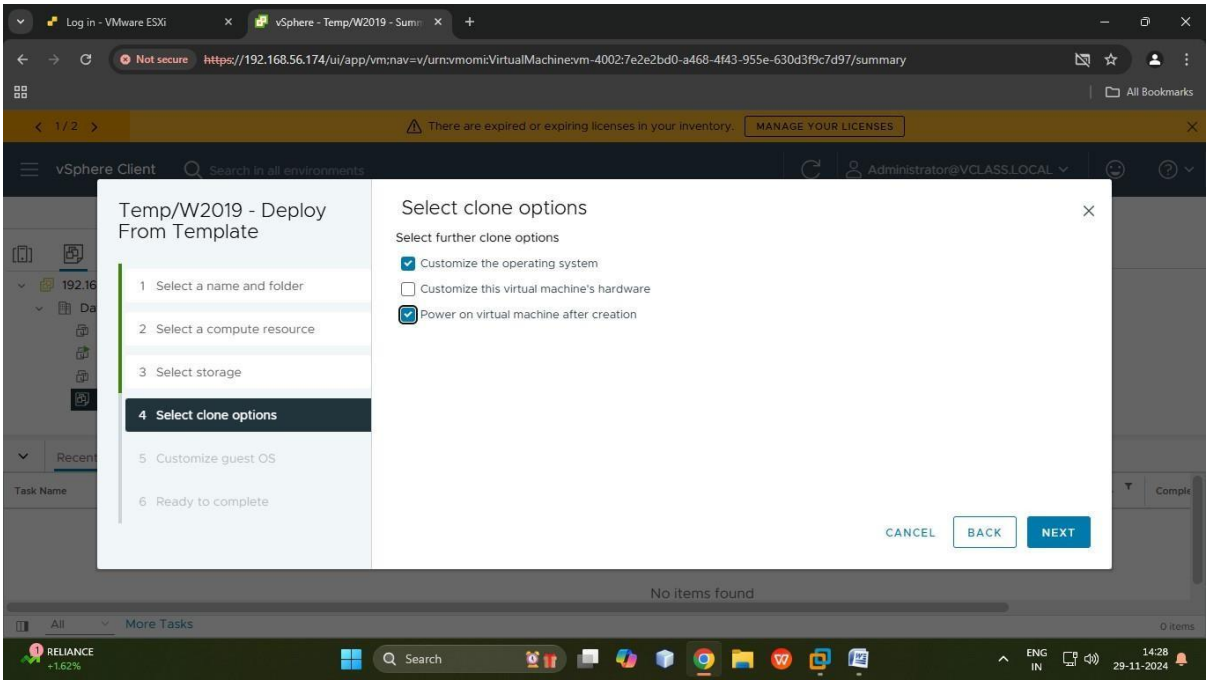
- Select Your Host.



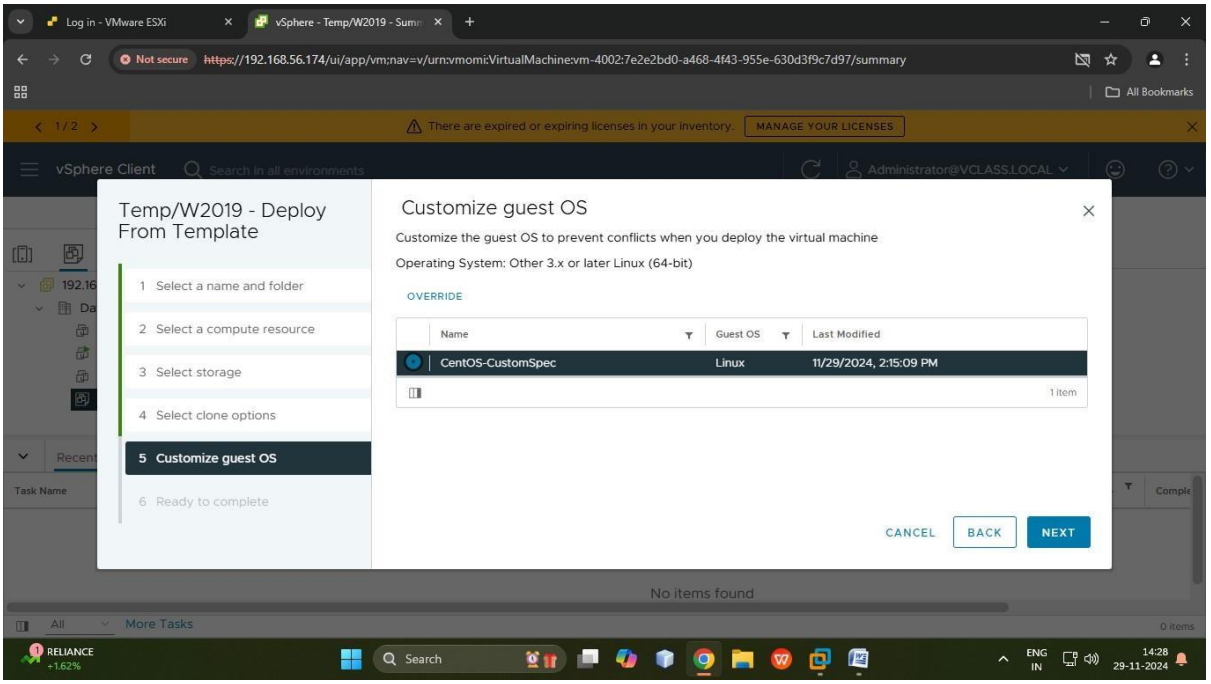
- Select Your Storage and Click on Next.



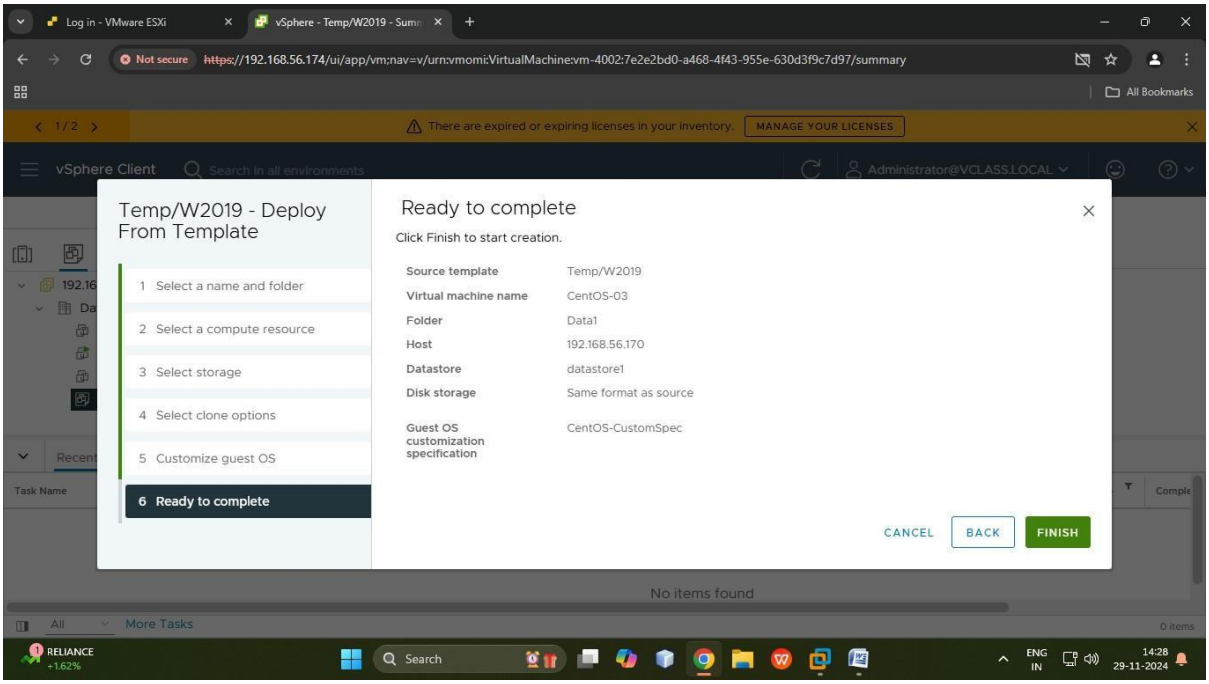
- Slect Customize the os and Select Power on the virtual machine after creation.



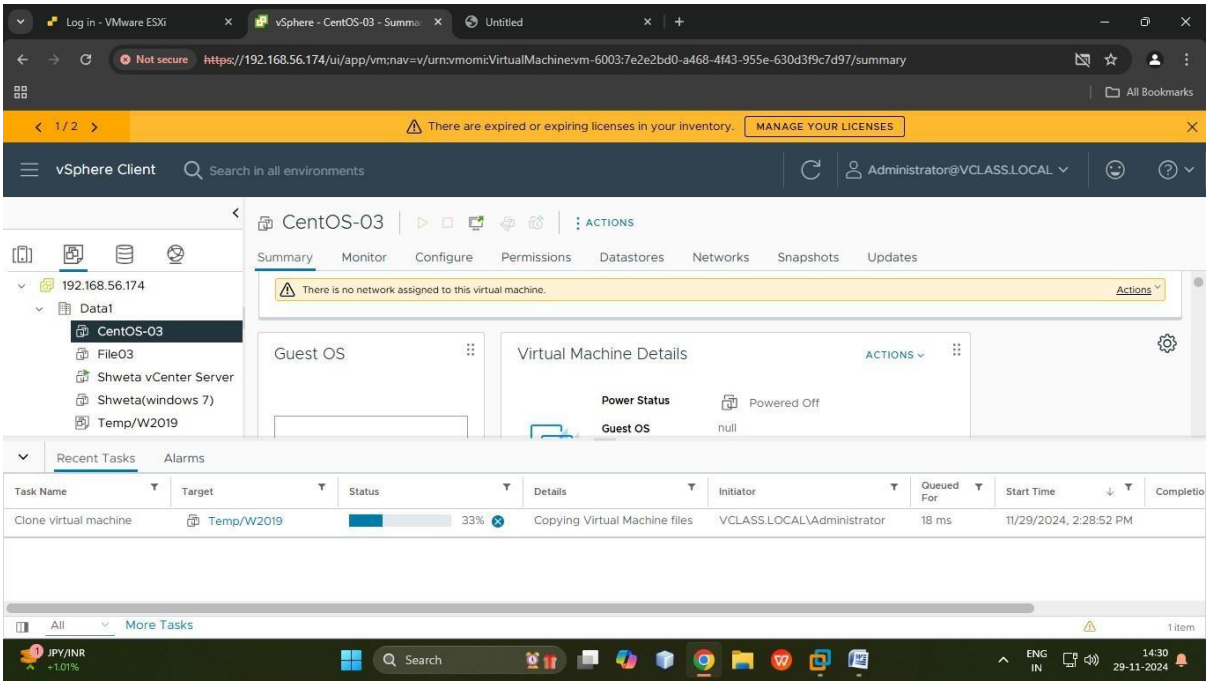
- Click On Next.



- Click on Finish.



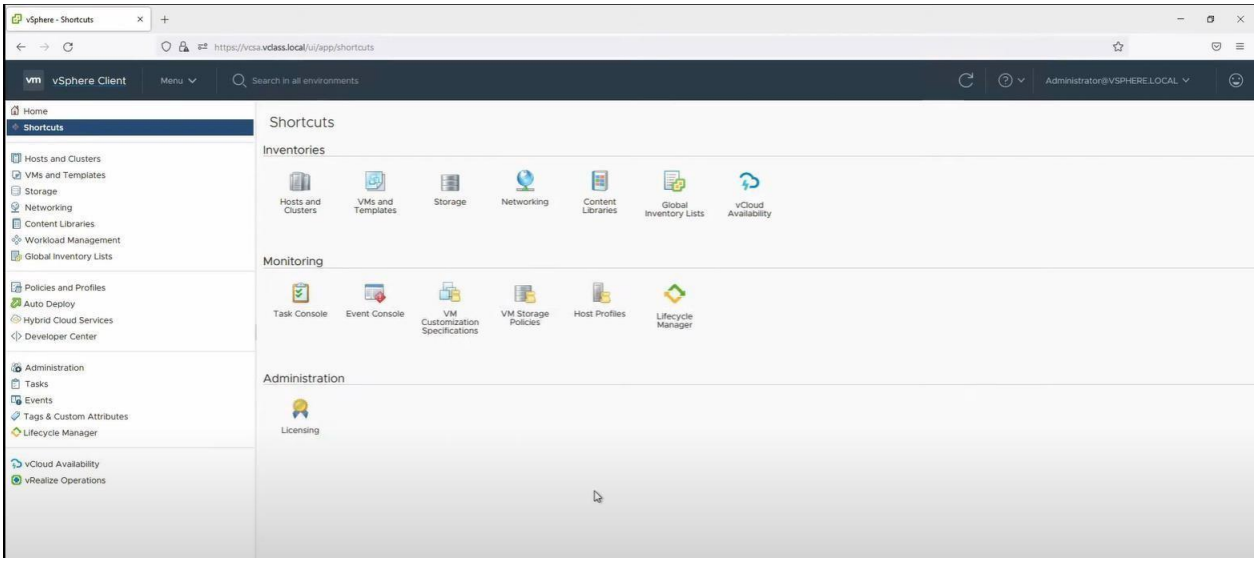
- Here we created rom template.



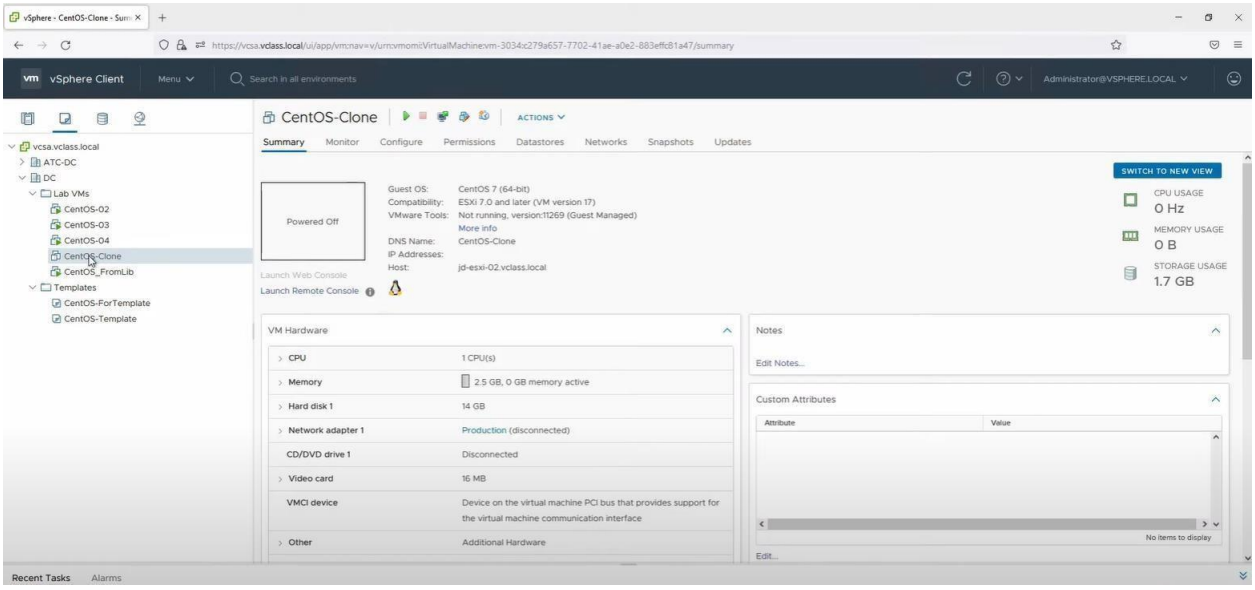
PRACTICAL 8

AIM:- Modifying Virtual Machines.

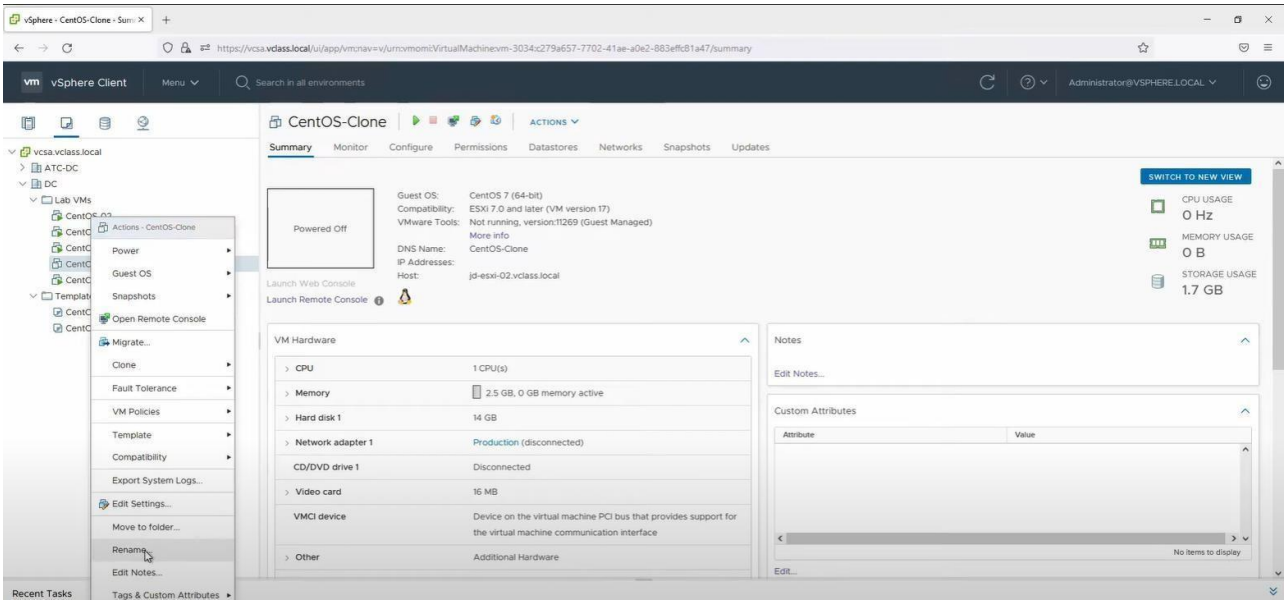
- Navigate to Vsphere home page.



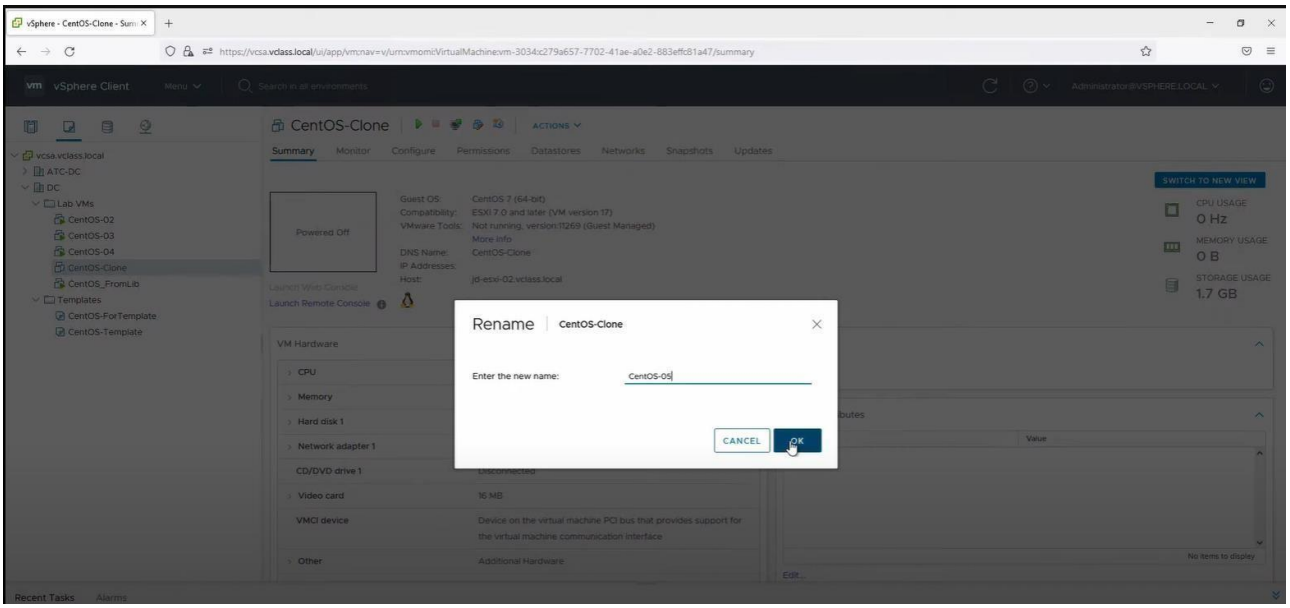
- Click on VM & Template.



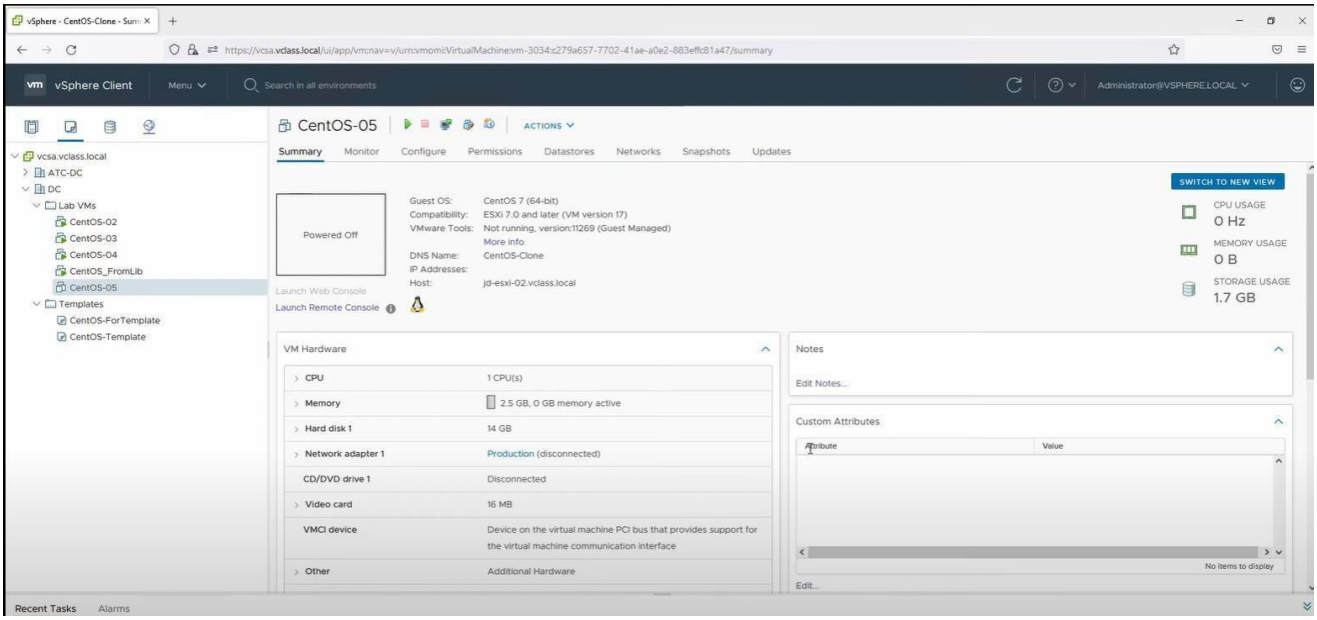
- Select a VM and Right Click Go to Rename.



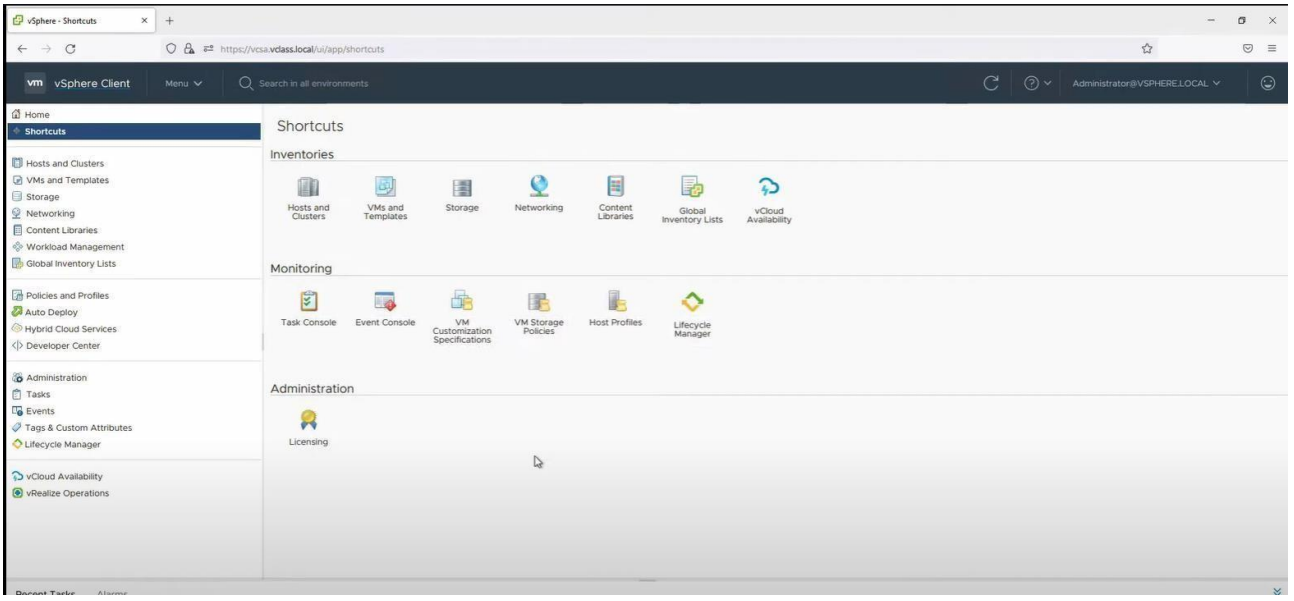
- Rename your VM and Click on Ok.



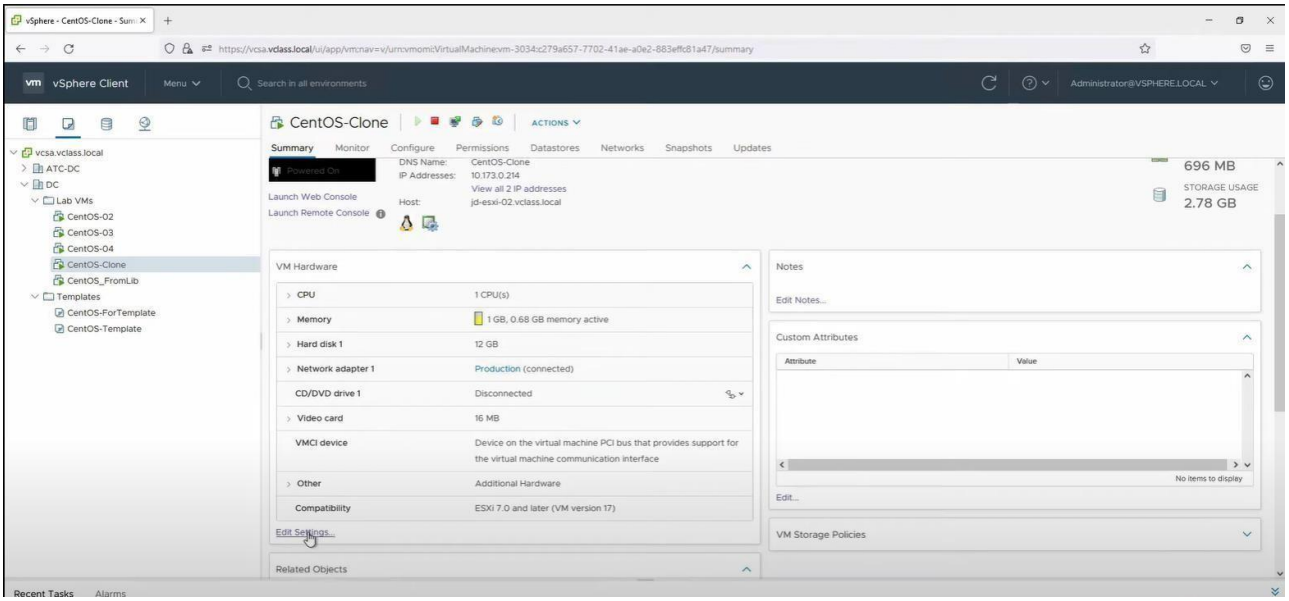
- As you can see Virtual Machine is Successfully Renamed.



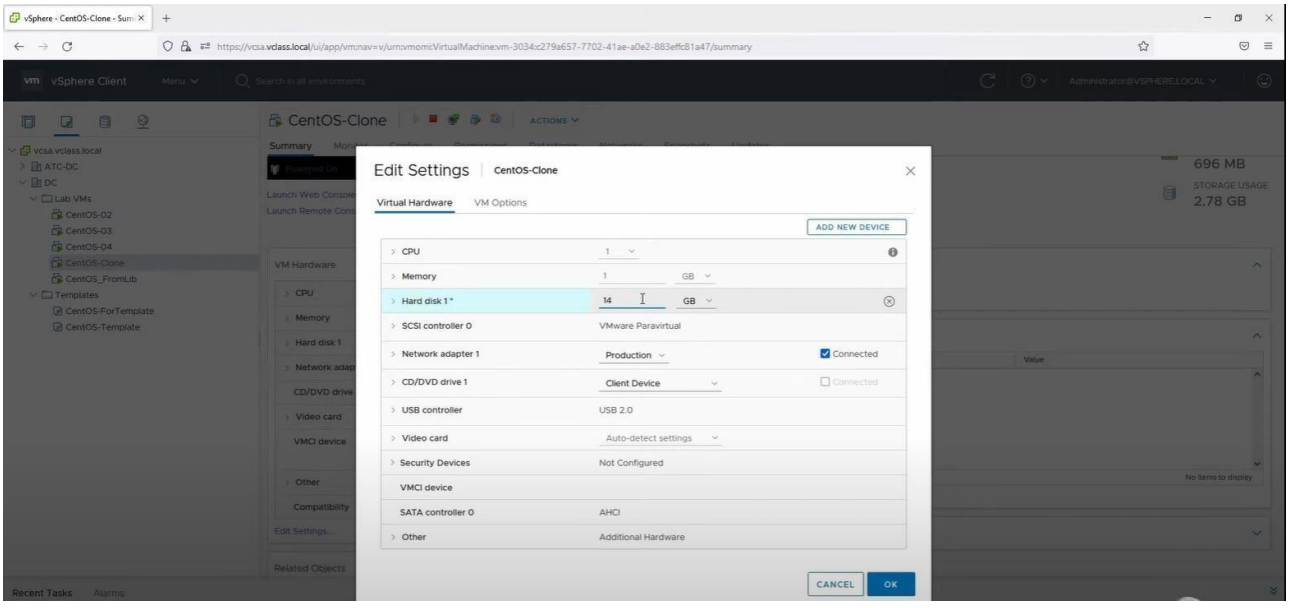
- Now go back to home page of Vsphere.



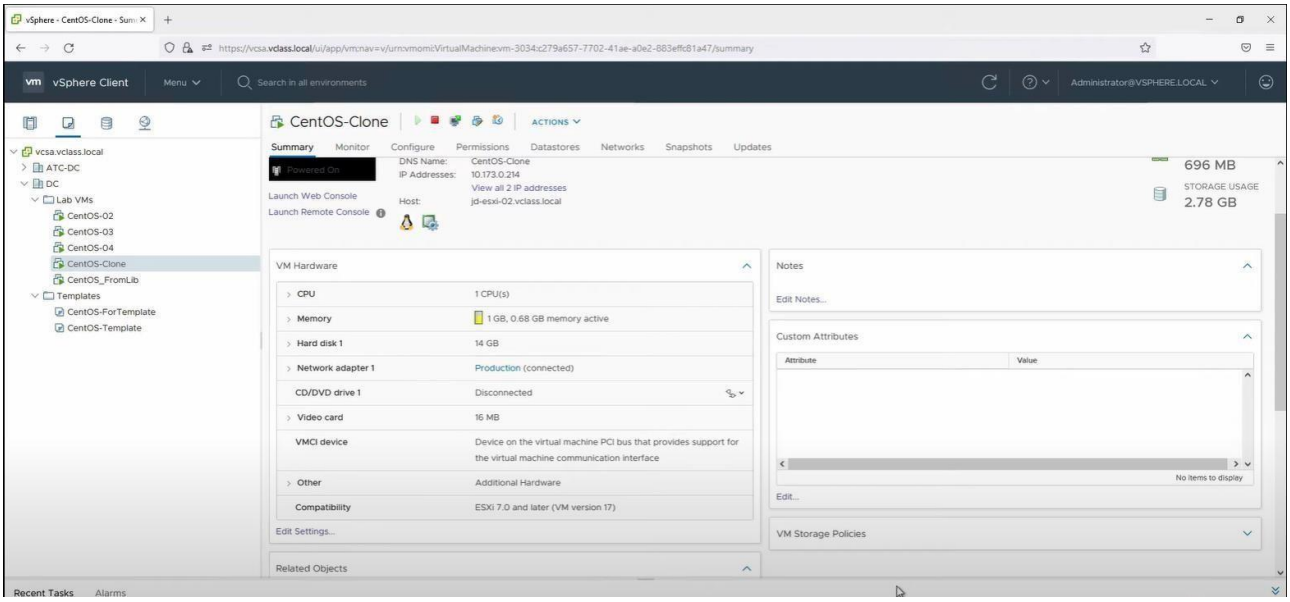
- Go to VM & Template and Select and Scroll down and Click on Edit Settings.



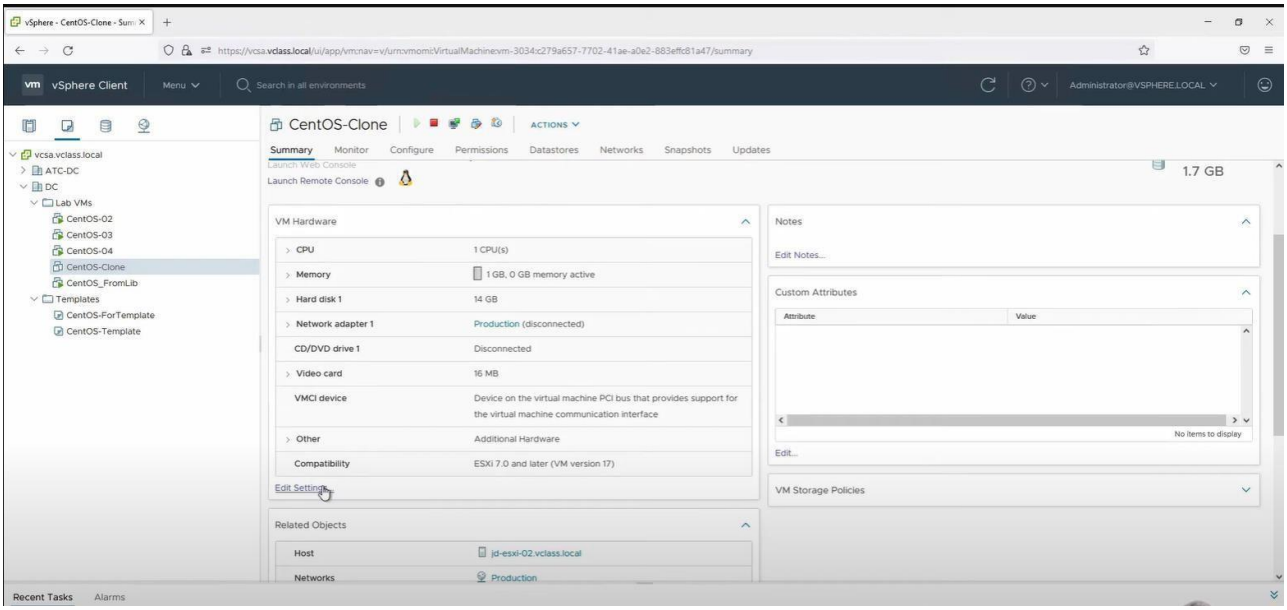
- Modify Your Hard Disk According to requirement and Click on Ok.



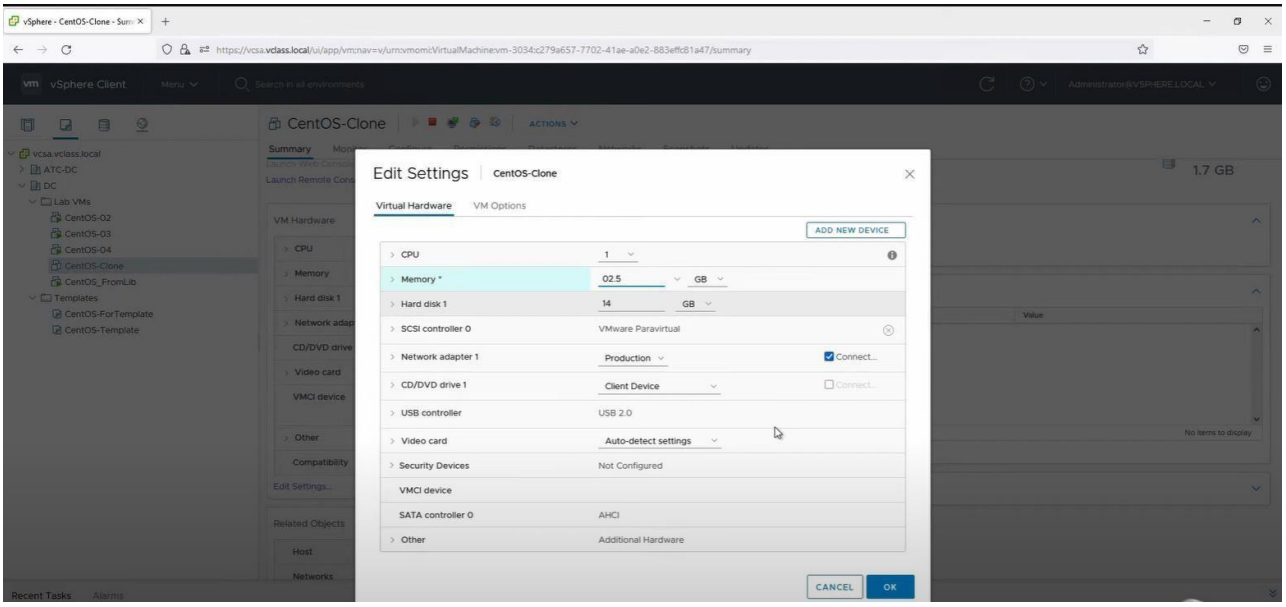
- As you can see here we Successfully Increased Storage Space.



- Again Click on Edit Settings to Edit Memory.



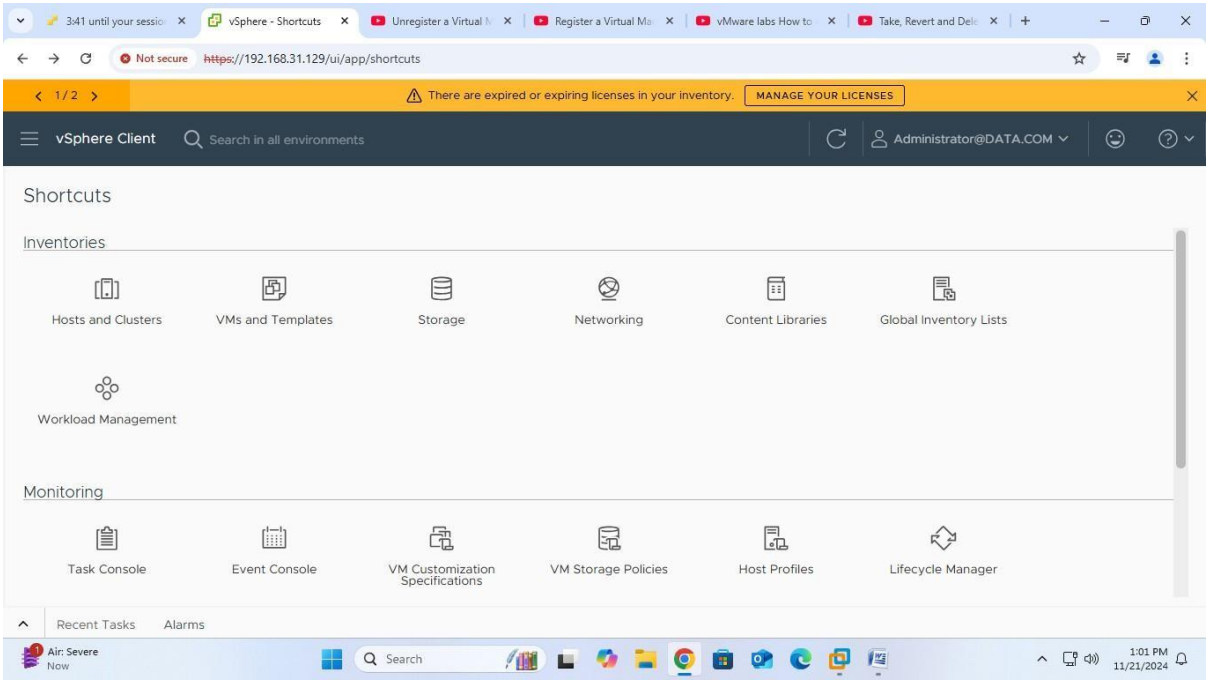
- Now Modify Your Memory According to Requirement.



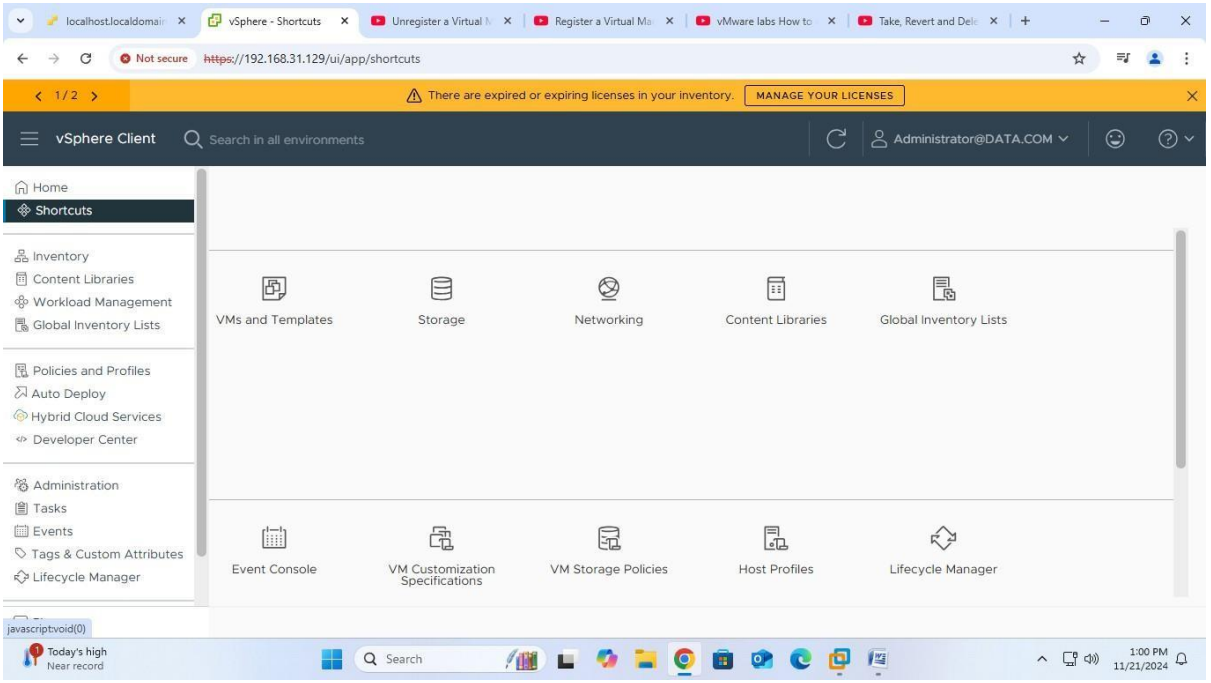
PRACTICAL 9

AIM:-Managing virtual machines.

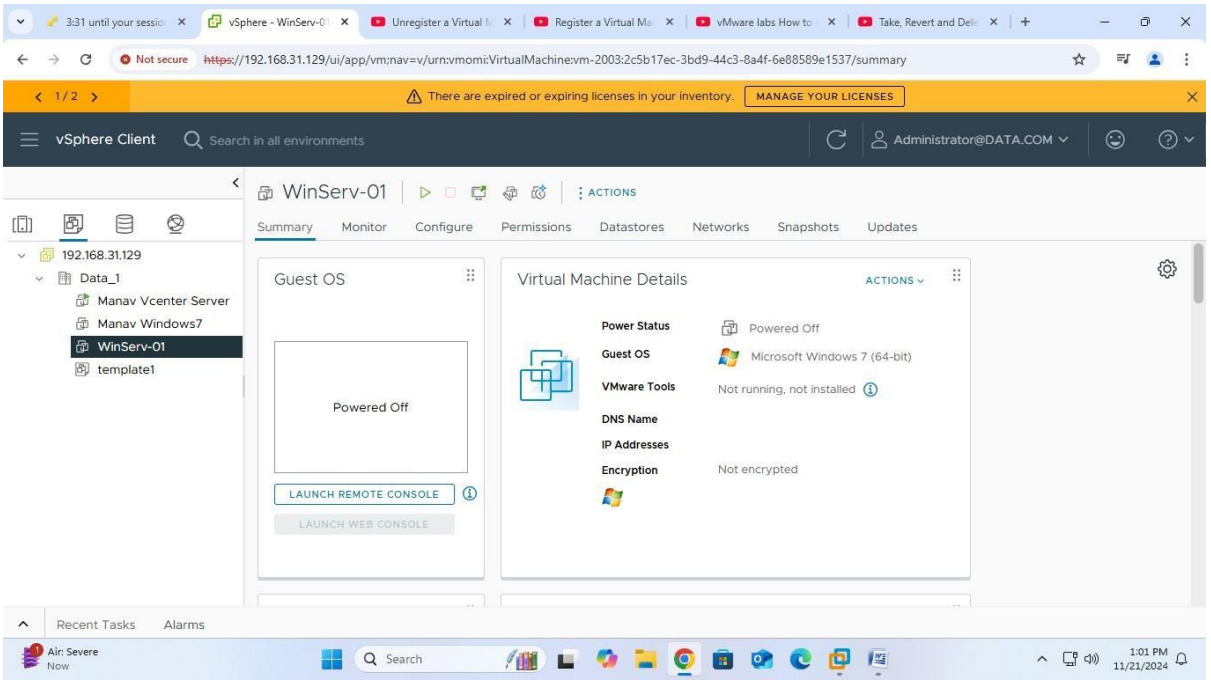
- Navigate to vcenter.



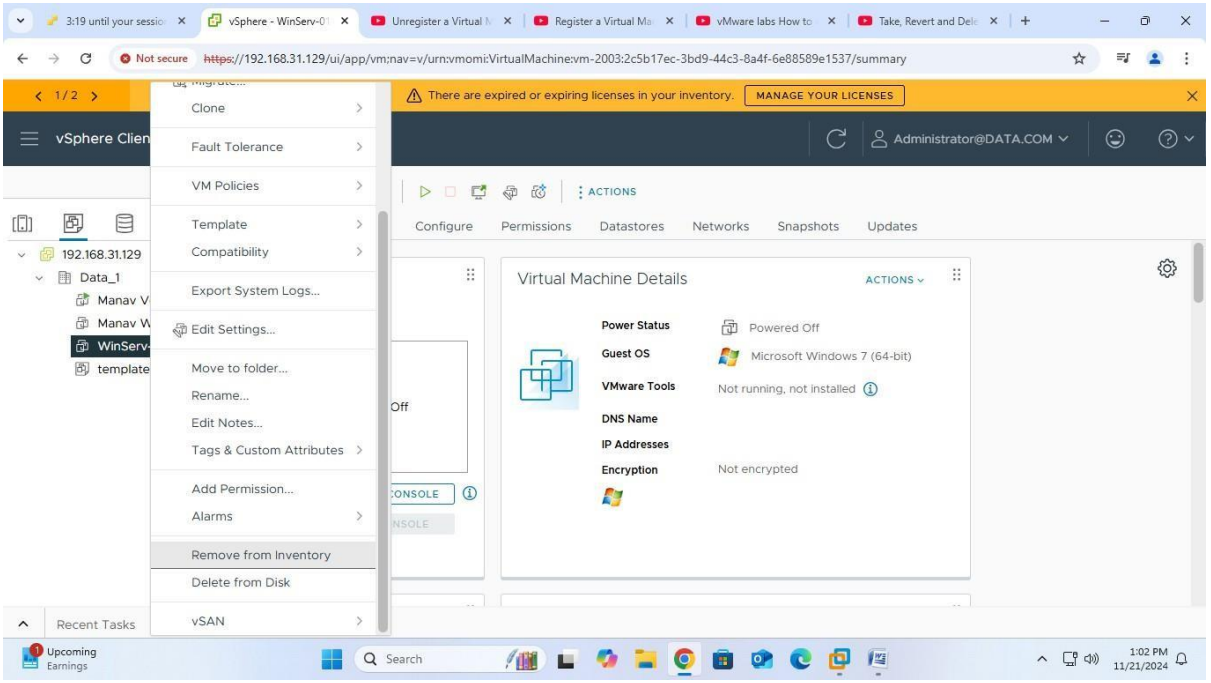
- Select VM and Template.



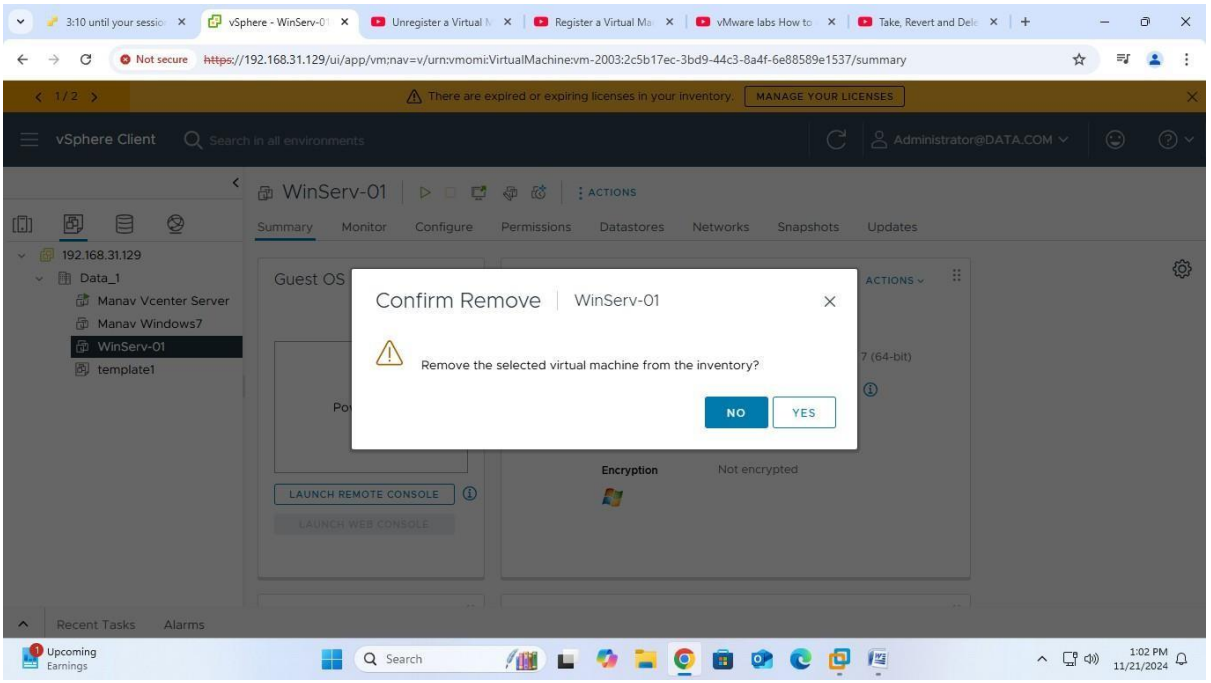
- Select VM.



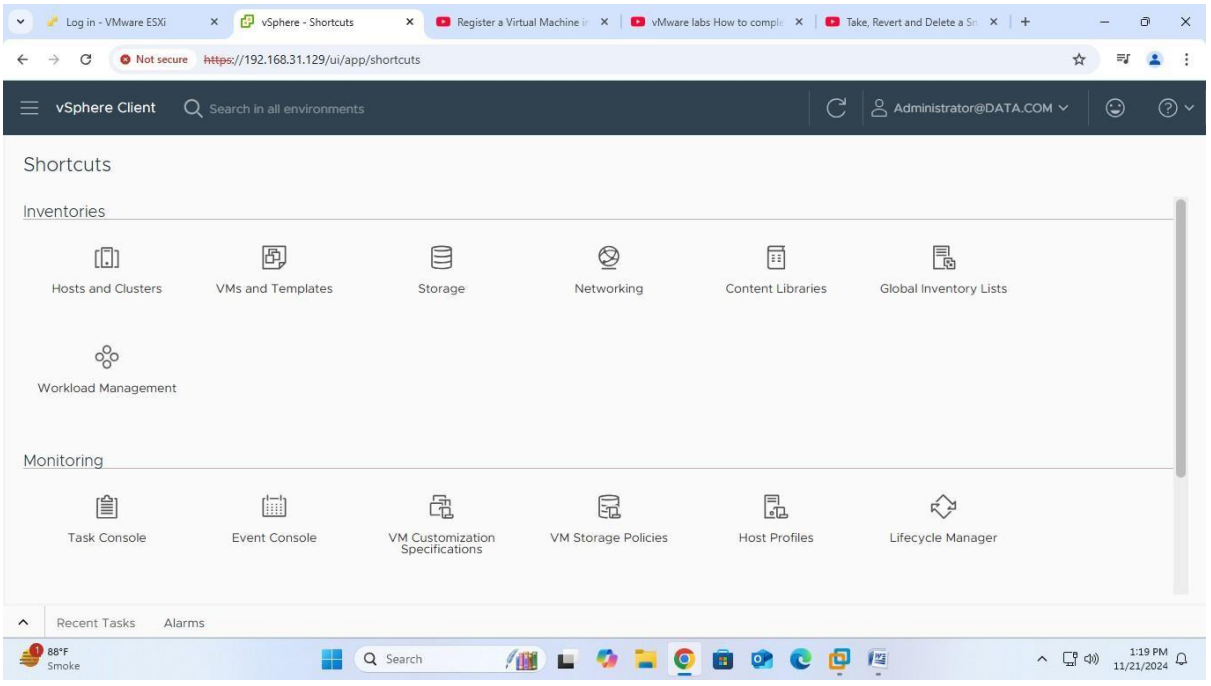
- Right Click on VM and then Select Remove Inventory.



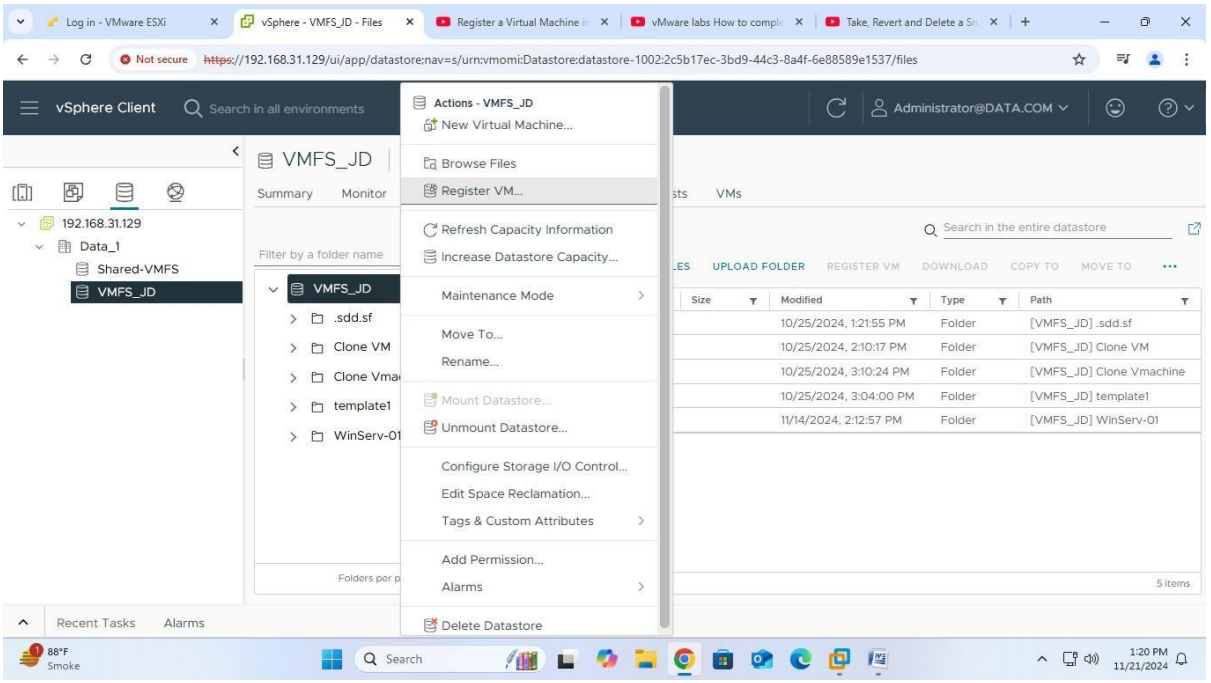
- Click on Yes.



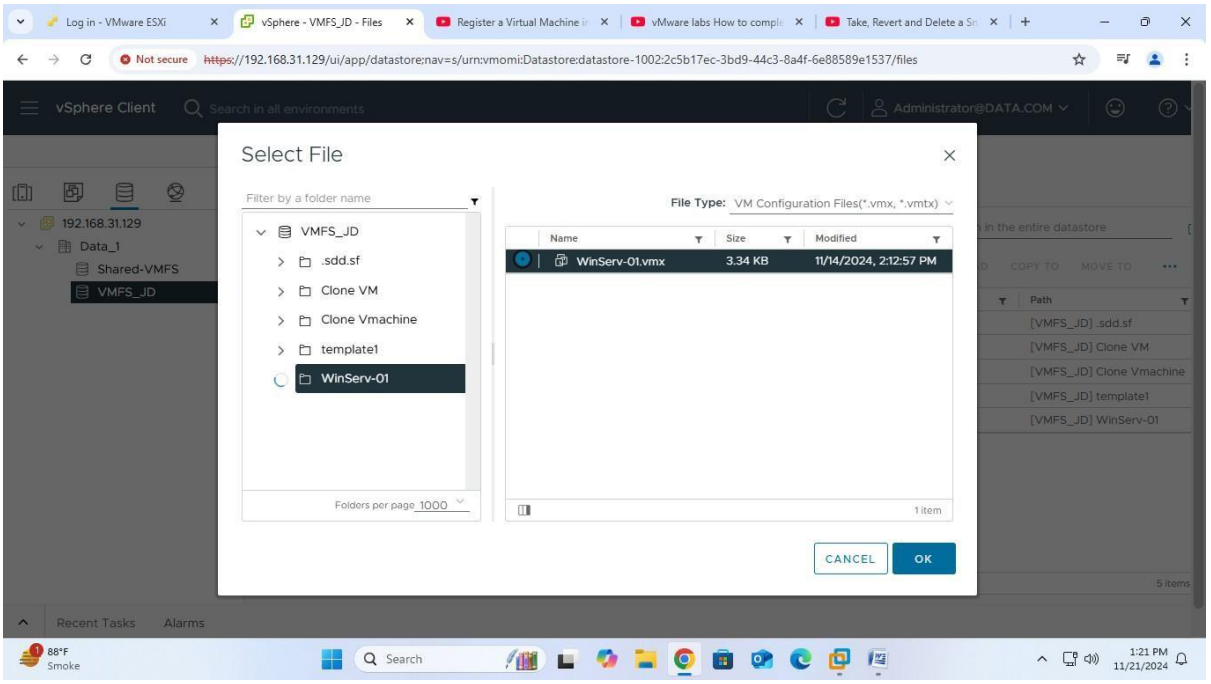
- Select Storage.



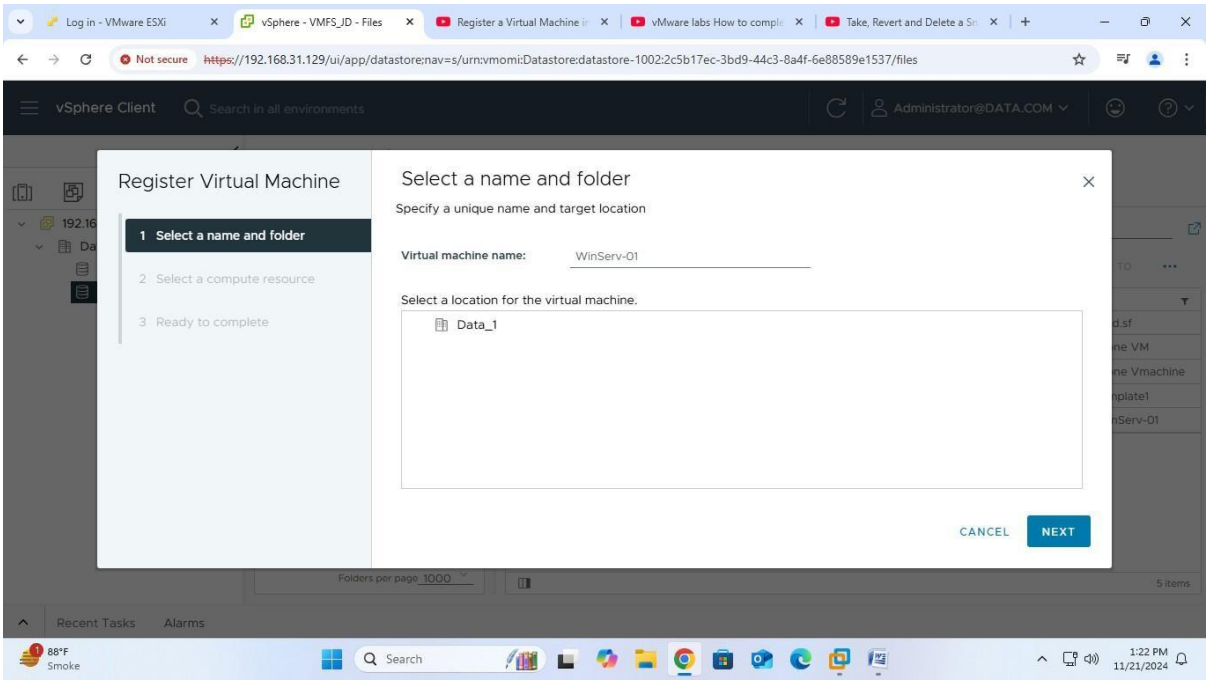
- Right Click on Sufficient Datastore and Click on Register VM.



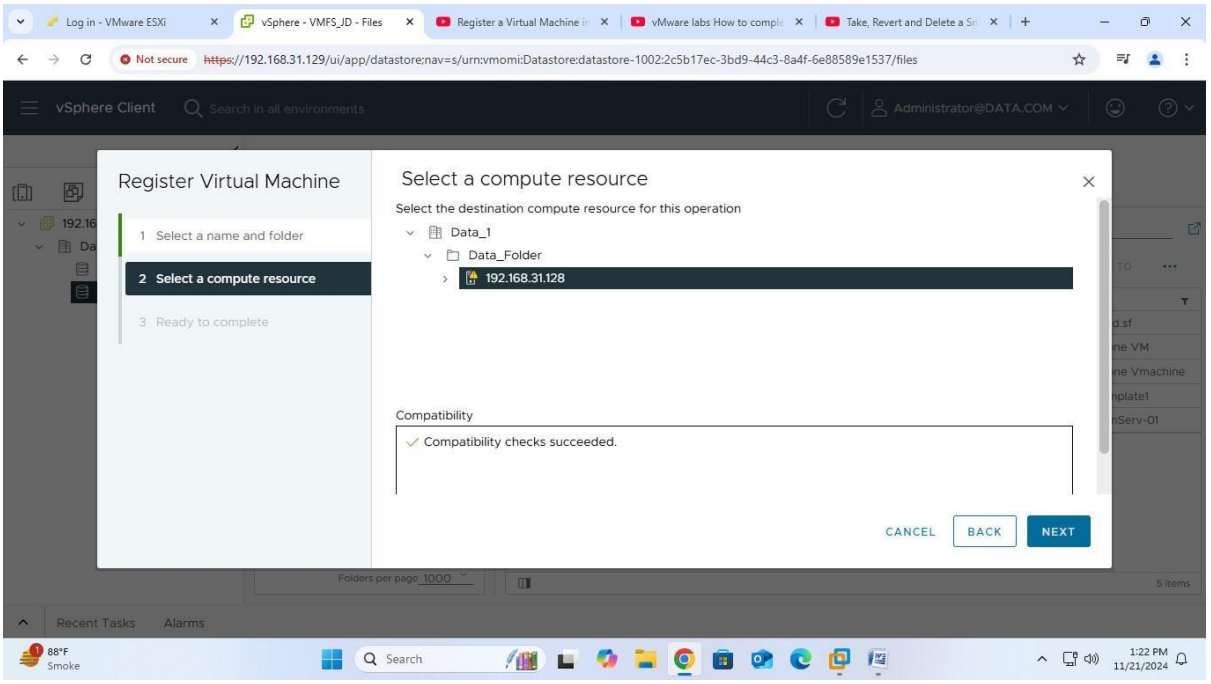
- Select os File.



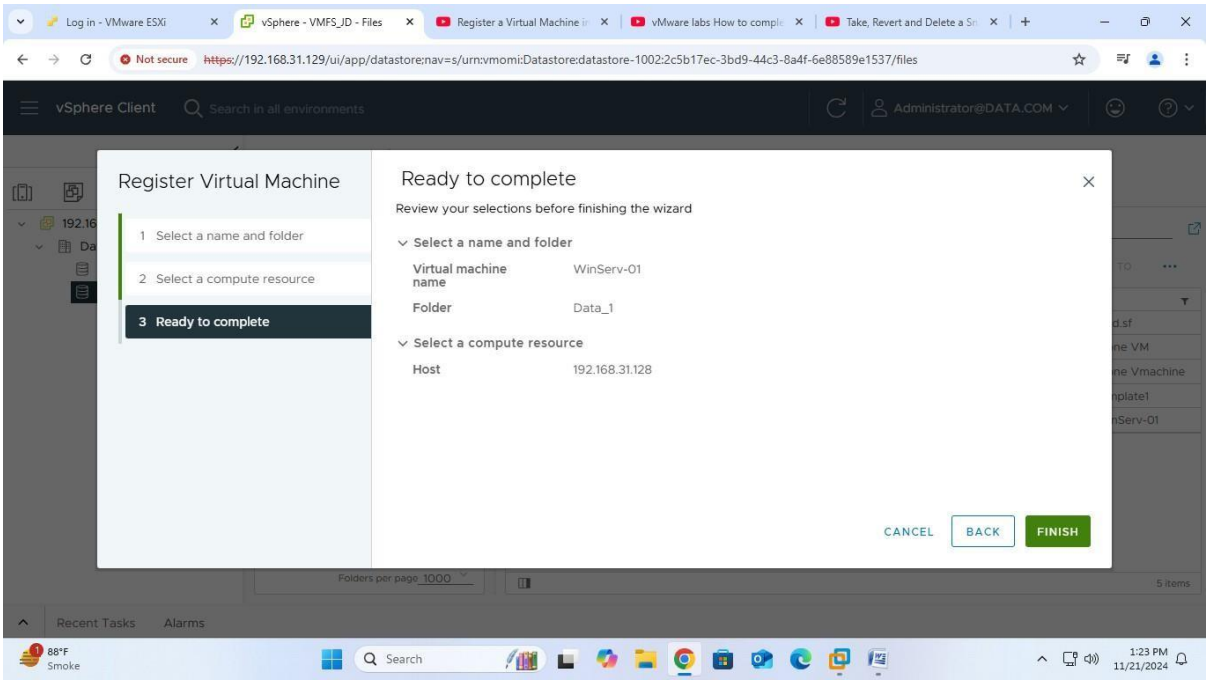
- Give it a name.



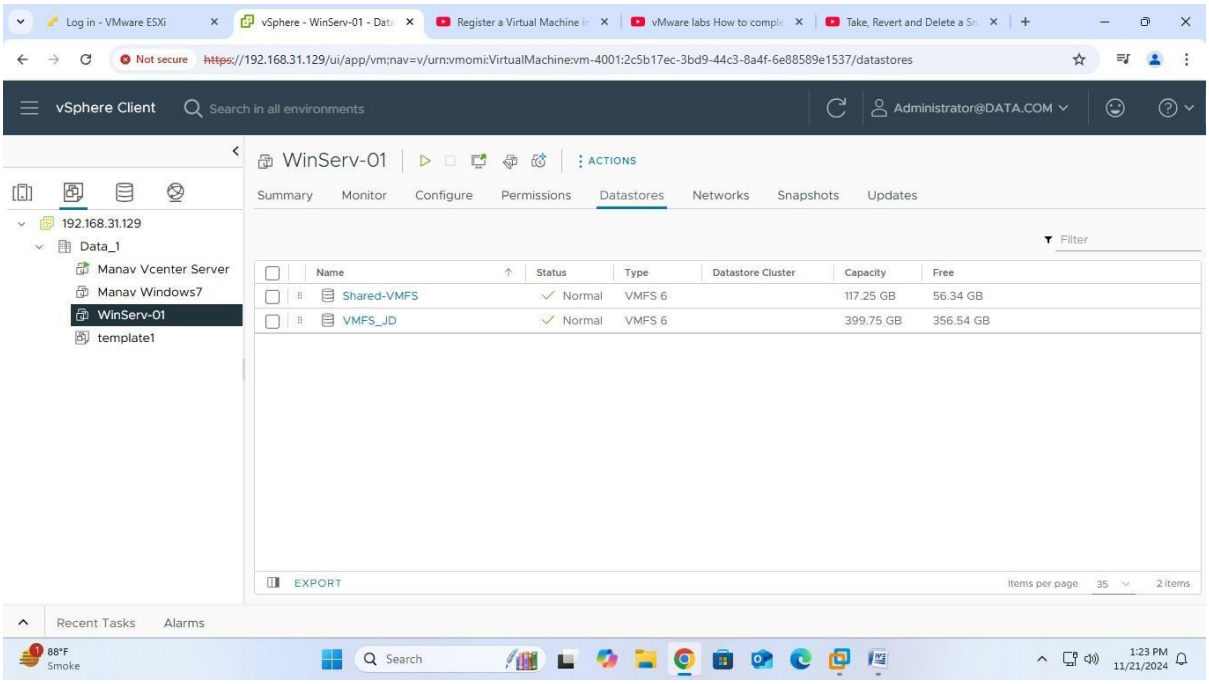
- Select your host and Click on next.



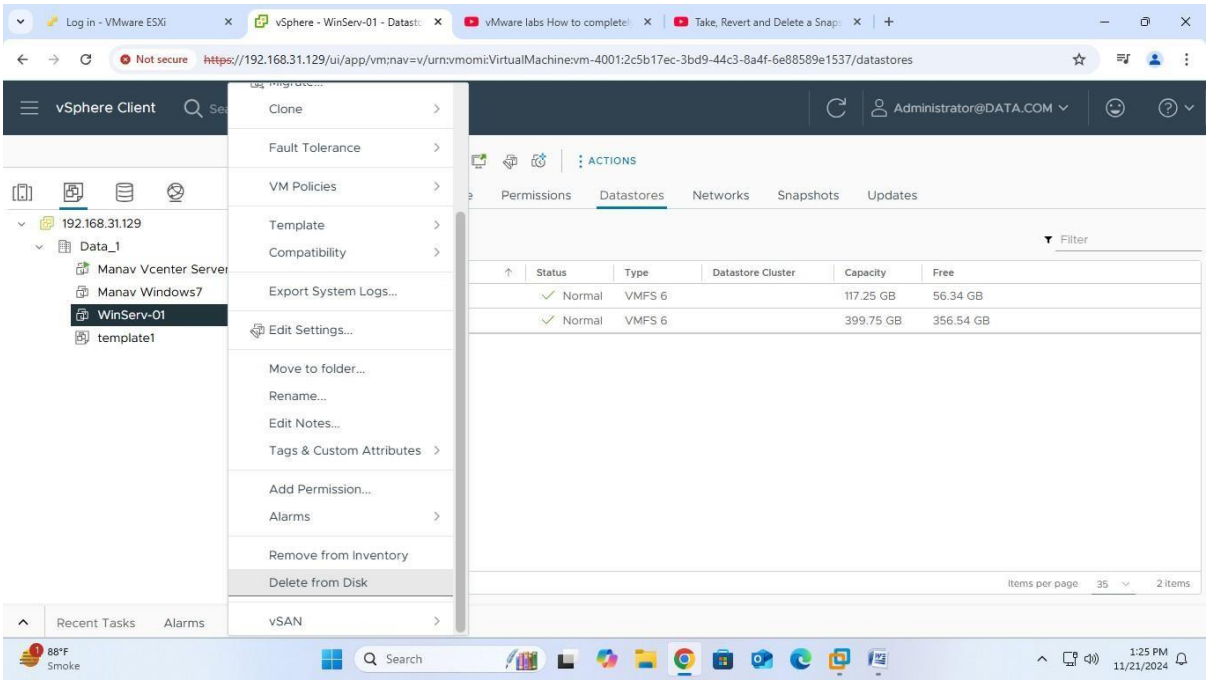
- Click on Finish.



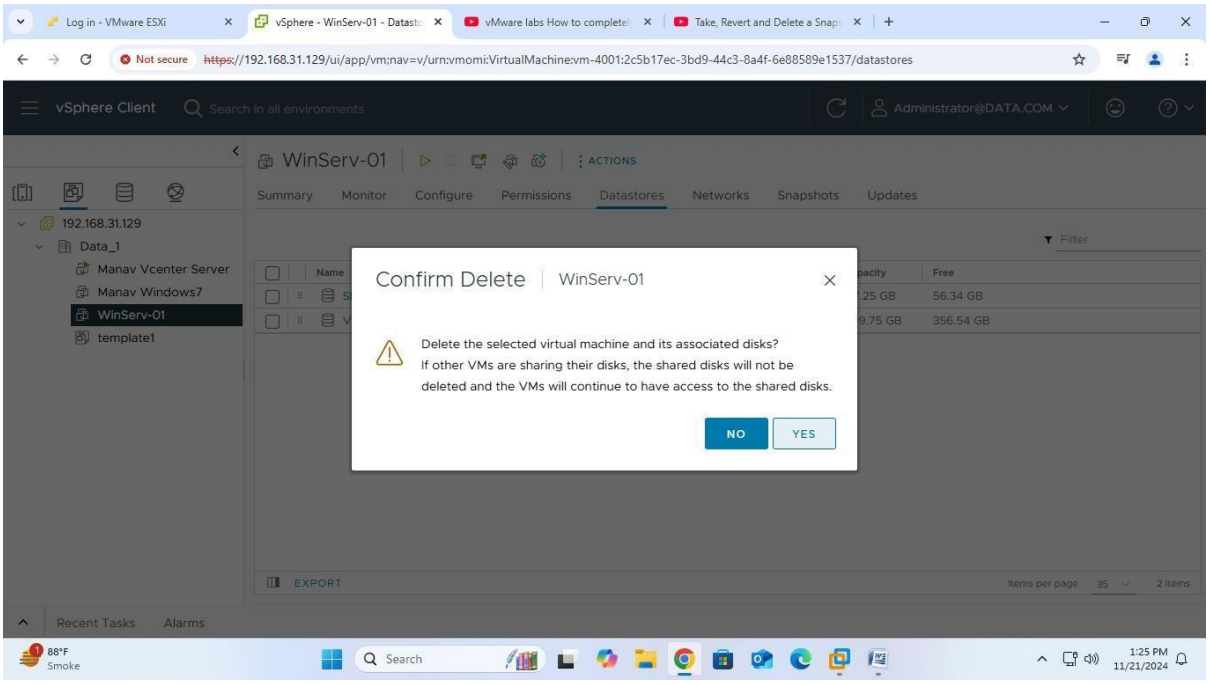
- Here Successfully registered a VM.



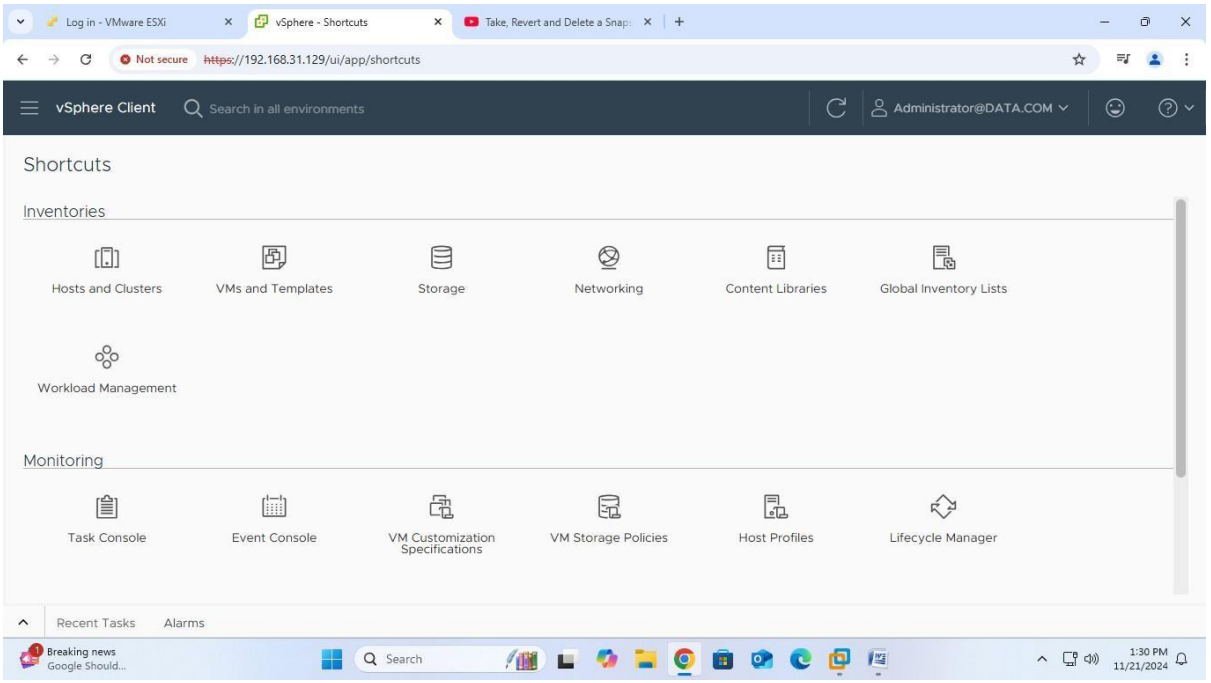
- Right Click on VM and Select Delete from disk.



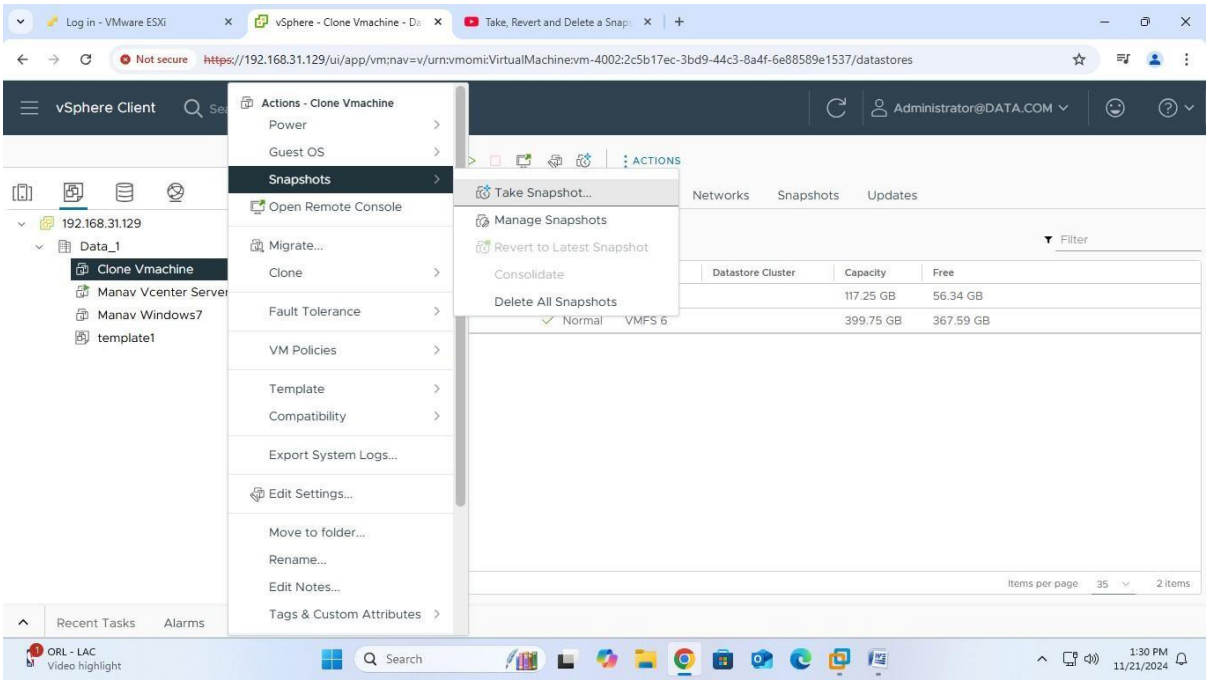
- Click on yes.



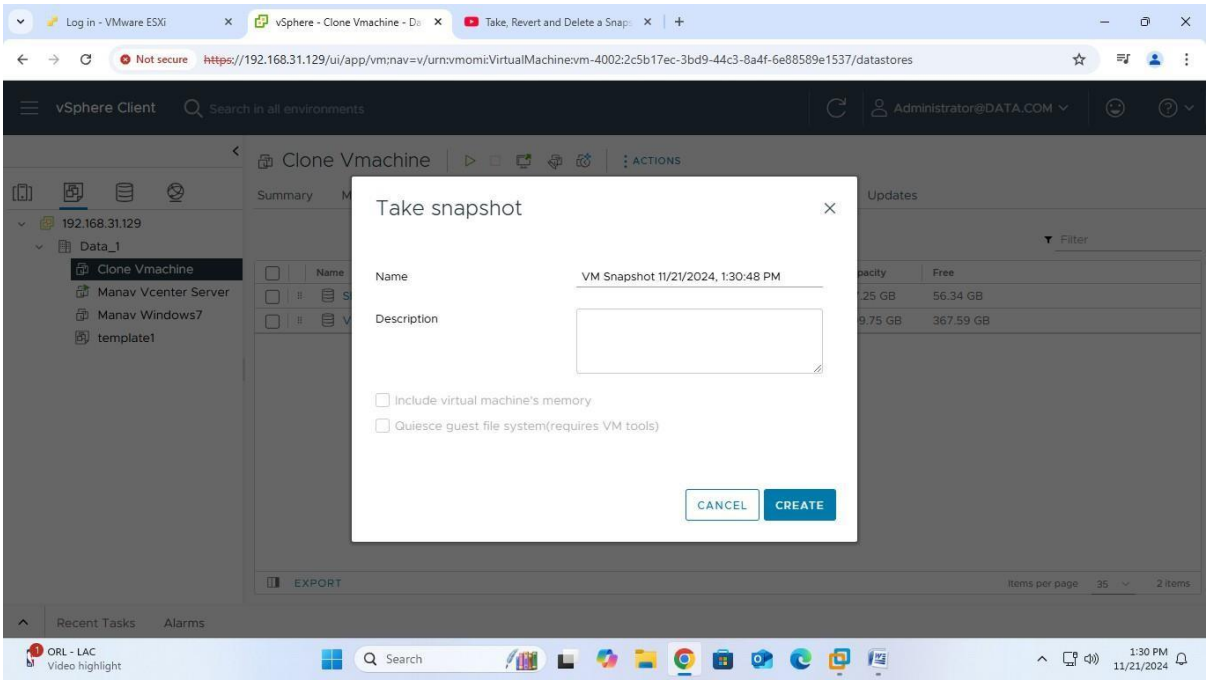
- Navigate to home page and Select VM and Template.



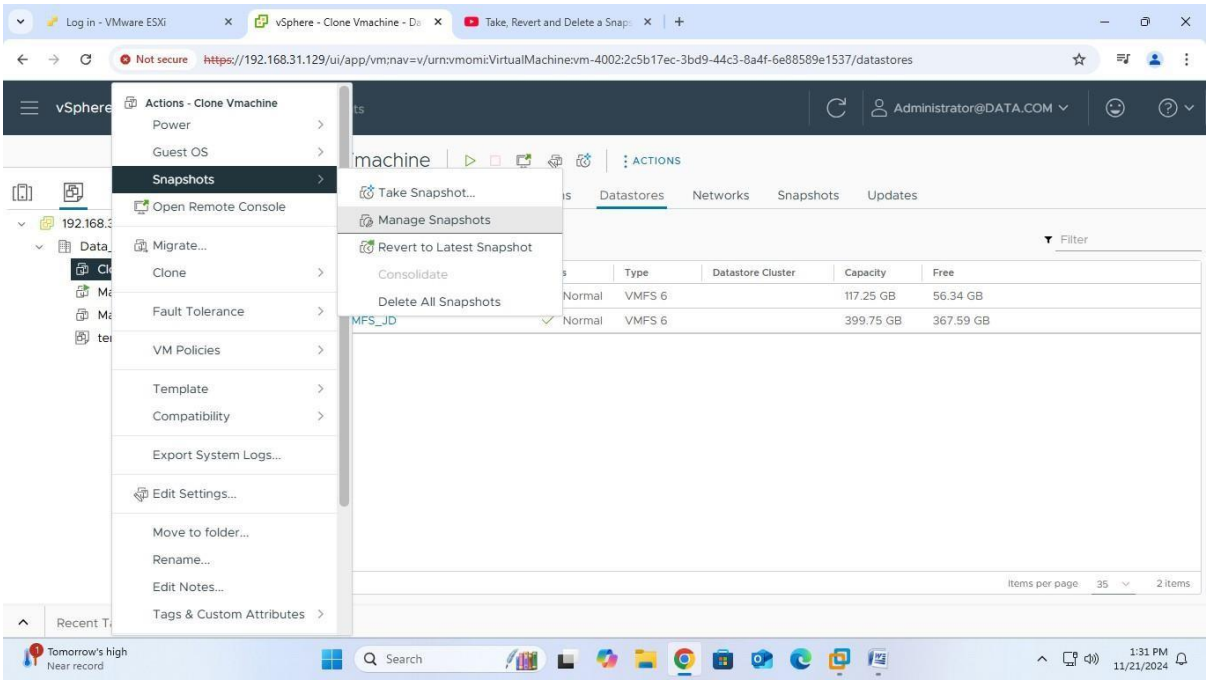
- Select a VM and Right Click And Click on Snapshots.



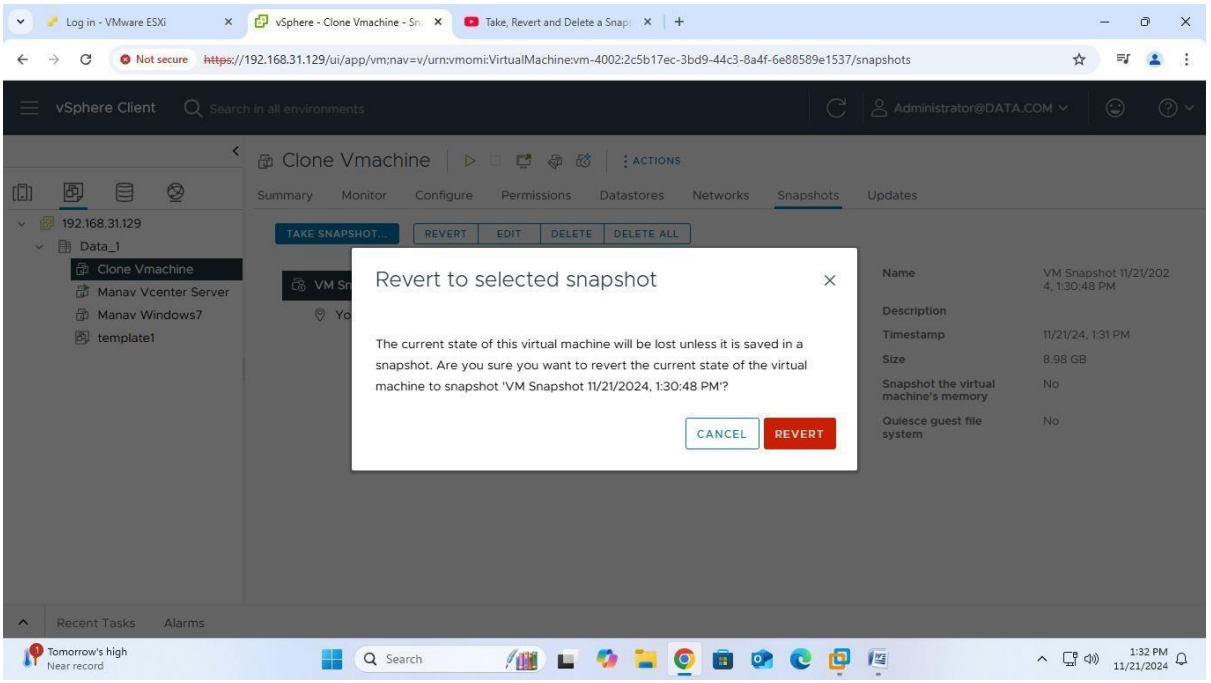
- Click on Create.



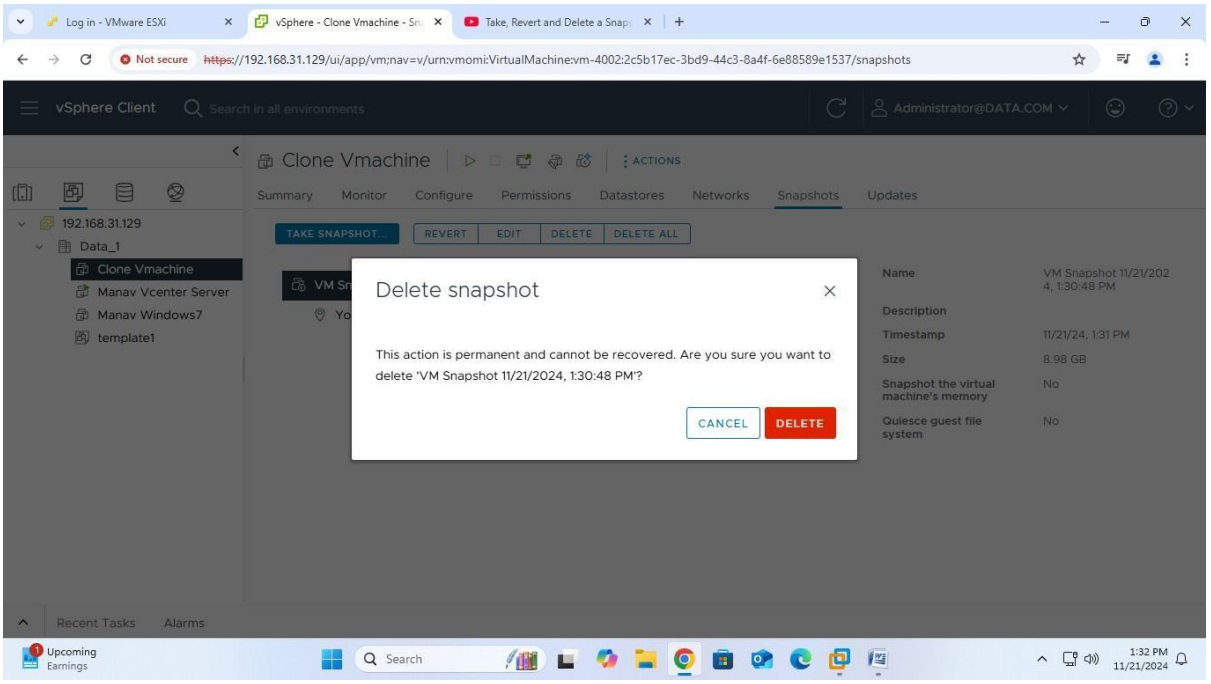
- Now go to VM again and Snapshots->Manage Snapshots.



- Click on Revert.



- Click on Delete.



- Here we Successfully Deleted Snapshots.